

Chapter 1: Vector Spaces

Linear Algebra Done Right (4th Edition), by Sheldon Axler

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Contents

1A: $\mathbb{R}^n$ and $\mathbb{C}^n$	2
1B: Definition of Vector Space	4
1C: Subspaces	6

1A: $\mathbb{R}^n$ and $\mathbb{C}^n$

Problem 1

Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbb{C}$.

Proof. Let $\alpha = a + bi, \beta = c + di$. Then we have that

$$\begin{aligned}\alpha + \beta &= (a + bi) + (c + di) \\ &= (c + di) + (a + bi) \\ &= \beta + \alpha\end{aligned}$$

□

Problem 3

Show that $(\alpha\beta)\gamma = \alpha(\beta\gamma)$ for all $\alpha, \beta, \gamma \in \mathbb{C}$.

Proof. Choose arbitrary $\alpha, \beta, \gamma \in \mathbb{C}$. Denote $\alpha = a + bi, \beta = c + di, \gamma = e + fi$. Then we have that

$$\begin{aligned}(\alpha\beta)\gamma &= ((ac - bd) + (ad + bc)i)(e + fi) \\ &= (ace - bde - adf - bcf) + (ade + bce + acf - bdf)i\end{aligned}$$

At the same time, we have

$$\begin{aligned}\alpha(\beta\gamma) &= (a + bi)((ce - df) + (cf + de)i) \\ &= (ace - adf - bcf - bde) + (ade + acf + bce - bdf)i\end{aligned}$$

Hence, $(\alpha\beta)\gamma = \alpha(\beta\gamma)$.

□

Problem 5

Show that for any $\alpha \in \mathbb{C}$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha + \beta = 0$.

Proof. Denote $\alpha = a + bi$. By the property of a field, we know there exists unique $c = -a$ and $d = -b$ such that $\beta = c + di$ and $\alpha + \beta = 0$. Suppose for the sake of contradiction that β is not unique, then there exists $c' + d'i$ such that $(a + c') + (b + d')i = 0$ while $c' \neq a$ or $d' \neq d$, contradicting the uniqueness of additive inverse property. □

Problem 8

Find two distinct squared roots of i .

Proof. Suppose $\alpha = a + bi$'s square equals one.

$$(a + bi)^2 = a^2 - b^2 + 2abi = 1$$

Then $|a| = |b|$, $2ab = 1$. So we get the solution to be

$$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i, -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$$

□

Problem 10

Show that $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z}) \forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{F}^n$.

Proof.

$$\begin{aligned} (\mathbf{x} + \mathbf{y}) + \mathbf{z} &= (x_1 + y_1, \dots, x_n + y_n) + (z_1, \dots, z_n) \\ &= (x_1, \dots, x_n) + (y_1 + z_1, \dots, y_n + z_n) \\ &= \mathbf{x} + (\mathbf{y} + \mathbf{z}) \end{aligned}$$

□

Problem 14

Show that $\gamma(\mathbf{x} + \mathbf{y}) = \gamma\mathbf{x} + \gamma\mathbf{y} \forall \gamma \in \mathbb{F}, \mathbf{x}, \mathbf{y} \in \mathbb{F}^n$.

Proof.

$$\begin{aligned} \gamma(\mathbf{x} + \mathbf{y}) &= \gamma(x_1 + y_1, \dots, x_n + y_n) \\ &= (\gamma(x_1 + y_1), \dots, \gamma(x_n + y_n)) \\ &= (\gamma x_1 + \gamma y_1, \dots, \gamma x_n + \gamma y_n) \\ &= \gamma(x_1, \dots, x_n) + \gamma(y_1, \dots, y_n) \\ &= \gamma\mathbf{x} + \gamma\mathbf{y} \end{aligned}$$

□

1B: Definition of Vector Space

Theorem 1. *A vector space is a set that is closed under vector addition and scalar multiplication. It also has the following properties:*

- commutativity
- associativity
- additive identity
- additive inverse
- multiplicative identity
- (scalar) distributive

Notation: $\mathbb{F}^S$.

Explanation: If S is a set, then $\mathbb{F}^S$ denotes the set of functions from S to $\mathbb{F}$ (scalar function). e.g. $f \in \mathbb{F}^S$.

Comment: $\mathbb{F}^S$ is a vector space; One can think of $f \in \mathbb{F}^n$ as $f: \{1, 2, \dots, n\} \rightarrow \mathbb{F}$.

Problem 1

Prove that $-(-\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in V$.

Proof. We know $-(-\mathbf{v})$ is the unique additive inverse of $-\mathbf{v}$. At same time by definition, $\mathbf{v} + (-\mathbf{v}) = 0$ and thus by commutativity $(-\mathbf{v}) + \mathbf{v} = 0$. This shows that $\mathbf{v}$ is the unique additive inverse of $(-\mathbf{v})$, and such that $-(-\mathbf{v}) = \mathbf{v}$. $\square$

Problem 2

Suppose $a \in \mathbb{F}$, $\mathbf{v} \in V$, and $a\mathbf{v} = 0$. Prove that $a = 0$ or $\mathbf{v} = 0$.

Proof. Suppose for the sake of contradiction that $a \neq 0$ and $\mathbf{v} \neq 0$ but $a\mathbf{v} = 0$.

$$\begin{aligned}\mathbf{v} &= 1\mathbf{v} \\ \mathbf{v} &= \frac{1}{a} \cdot a\mathbf{v} \\ \mathbf{v} &= \frac{1}{a}0\end{aligned}$$

This forms a contradiction. $\square$

Problem 3

Suppose $\mathbf{v}, \mathbf{w} \in V$. Explain why there exists a unique $\mathbf{x} \in V$ such that $\mathbf{v} + 3\mathbf{x} = \mathbf{w}$.

Proof. Suppose there exists $\mathbf{x}'$ which also satisfies the condition. Then we have

$$\mathbf{v} + 3\mathbf{x} = \mathbf{w} \quad \mathbf{v} + 3\mathbf{x}' = \mathbf{w} \quad (1)$$

This gives that $\mathbf{x} = (\mathbf{w} - \mathbf{v})/3 = \mathbf{x}'$ which shows $\mathbf{x}$ is unique. $\square$

Problem 4

The empty set is not a vector space, why?

Proof. There is no additive identity in the empty set. $\square$

Problem 7

Suppose S is a nonempty set. Let V^S denotes the set of functions from S to V . Define a natural addition and scalar multiplication on V^S , and show that V^S is a vector space with these definitions.

Proof. Let $f, g \in V^S: S \rightarrow V$. Define the addition and multiplication to be that

$$f + g(x) = f(x) + g(x)$$

We have that

- commutativity: $f + g(x) = f(x) + g(x) = g(x) + f(x) = g + f(x)$
- associativity: $(f + g) + h(x) = f(x) + g(x) + h(x) = f + (g + h)(x)$
- additive identity: Define $0: S \rightarrow 0 \in V$, then $f + 0(x) = 0 + f(x) = f(x)$
- additive inverse: for every $f \in V^S$, define $g(x) = -f(x)$ which exists by the property of vector space and thus we have that $g + f = 0$ for every x and thus that it exists.
- multiplicative identity: same as above
- (scalar) distributive: $a(f + g)(x) = a(f(x) + g(x)) = af(x) + ag(x)$

$\square$

1C: Subspaces

Definition 2 (subspace). A subset $\mathcal{U}$ of V is called subspace of V if $\mathcal{U}$ is also a vector space with the same additive identity, addition, and scalar multiplication as on V .

Remark 3. The set $\{0\}$ is the smallest subspace of V , and V itself is the largest subspace of V .

Remark 4. The subspace of $\mathbb{R}^2$ are precisely $\{0\}$, all lines in $\mathbb{R}^2$ containing the origin, $\mathbb{R}^2$.

Definition 5 (Sum of subspace). Suppose $V_1, \dots, V_m$ are subspaces. The sum of them, denoted by $V_1 + \dots + V_m$, is the set of all possible sums of element of $V_1, \dots, V_m$. Specifically,

$$V_1 + \dots + V_m = \{v_1 + \dots + v_m : v_1 \in V_1, \dots, v_m \in V_m\}$$

Lemma 6. Suppose $V_1, \dots, V_m$ are subspaces of V . Then $V_1 + \dots + V_m$ is the smallest subspace of V containing $V_1, \dots, V_m$.

Definition 7 (Direct Sum). Suppose $V_1, \dots, V_m$ are subspaces of V .

- The sum $V_1 + \dots + V_m$ is called a direct sum if each element of $V_1 + \dots + V_m$ can be written only as a sum of $v_1, \dots, v_m$, where each $v_k \in V_k$.
- If $V_1 + \dots + V_m$ is a direct sum, then $V_1 \oplus \dots \oplus V_m$ denotes $V_1 + \dots + V_m$, with $\oplus$ serving as the indication that this is a direct sum.

Example. Suppose V_k is a subspace of $\mathbb{F}^n$ of those vectors whose coordinates are all zero but k -th coordinate. Then we have

$$\mathbb{F}^n = V_1 \oplus \dots \oplus V_m$$

Example (Sum that is not a direct sum). Suppose

$$V_1 = \{(x, y, 0) \in \mathbb{F}^3 : x, y \in \mathbb{F}\}$$

$$V_2 = \{(0, 0, z) \in \mathbb{F}^3 : z \in \mathbb{F}\}$$

$$V_3 = \{(0, y, y) \in \mathbb{F}^3 : y \in \mathbb{F}\}$$

Then $F^3 = V_1 + V_2 + V_3$ because for every $(x, y, z) \in \mathbb{F}^3$,

$$(x, y, z) = (x, y, 0) + (0, 0, z) + (0, 0, 0)$$

However, $F^3 \neq V_1 \oplus V_2 \oplus V_3$ since

$$\begin{aligned} (0, 0, 0) &= (0, -1, 0) + (0, 0, -1) + (0, 1, 1) \\ &= (0, 0, 0) + (0, 0, 0) + (0, 0, 0) \end{aligned}$$

Theorem 8. Suppose $V_1, \dots, V_m$ are subspaces of V . Then $V_1 + \dots + V_m$ is a direct sum if and only if the only way to write 0 as a sum of $v_1 + \dots + v_m$, where $v_k \in V_k$, is by taking each v_k to equal 0 .

Theorem 9. Suppose that $\mathcal{U}$ and $\mathcal{W}$ are subspaces of V . Then

$$\mathcal{U} + \mathcal{W} \text{ is a direct sum} \iff \mathcal{U} \cap \mathcal{W} = \{0\}$$

Problem 1

Verify the following examples to be valid subspaces:

1. If $b \in \mathbb{F}$, then

$$\{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 : x_3 = 5x_4 + b\}$$

is a subspace of $\mathbb{F}^4$ if and only if $b = 0$.

2. The set of continuous real-valued functions on the interval $[0, 1]$ is a subspace of $\mathbb{R}^{[0,1]}$.
3. The set of differentiable real-valued functions on $\mathbb{R}$ is a subspace of $\mathbb{R}^{\mathbb{R}}$.
4. The set of differentiable real-valued functions f on the interval $(0, 3)$ such that $f'(2) = b$ is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.
5. The set of all sequences of complex numbers with limit 0 is a subspace of $\mathbb{C}^\infty$.

Proof. 1. $\Rightarrow (0, 0, 0, 0)$ is an element and thus $0 = 0 + b$, $b = 0 \Leftarrow$ Easy to verify.

2. 0 function is cts; cts functions are closed under addition and scalar multiplication.
3. 0 function is differentiable; differentiable functions are closed under addition and scalar multiplication.
4. For this to be closed under addition, one needs to restrict that $f'(2) + g'(2) = b + b = b$ and thus $b = 0$.
5. $\lim_{n \rightarrow \infty} a(S_1 + S_2) = \lim_{n \rightarrow \infty} aS_1 + \lim_{n \rightarrow \infty} aS_2 = 0 + 0 = 0$. At the same time, the 0 sequence has limit 0 .

□

Problem 4

Suppose $b \in \mathbb{R}$. Show that the set of continuous real-valued functions f on the interval $[0, 1]$ such that $\int_0^1 f = b$ is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$.

Proof. $\Rightarrow \int_0^1 f + g = \int_0^1 f + \int_0^1 g = 2b = b$ so $b = 0$.
 $\Leftarrow 0$ is in the set; closed under addition/multiplication. $\square$

Problem 5

Prove that $\mathbb{R}^2$ is not a subspace of $\mathbb{C}^2$ over the field $\mathbb{C}$.

Proof. This does not hold for scalar multiplication since we've defined scalar to be complex numbers. To see this, Let $a = (x + yi) \in \mathbb{C}$ and $\mathbf{z} = (z_1, z_2) \in \mathbb{R}^2$. We have that $a\mathbf{z} = (z_1(x + yi), z_2(x + yi)) \notin \mathbb{R}^2$. $\square$

Problem 7

Prove or disprove: If $\mathcal{U}$ is a nonempty subset of $\mathbb{R}^2$ such that $\mathcal{U}$ is closed under addition and under taking additive inverse ($-u \in \mathcal{U}$), then $\mathcal{U}$ is a subspace in $\mathbb{R}^2$.

Proof. No. $\mathcal{U} = \{(x_1, x_2) : x_1, x_2 \in \mathbb{Z}\}$. Then $\frac{1}{2}(1, 1) \notin \mathcal{U}$. $\square$

Problem 9

A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is called *periodic* if there exists a positive number p s.t. $f(x) = f(x + p)$ for all $x \in \mathbb{R}$. Is the set of periodic functions from $\mathbb{R}$ to $\mathbb{R}$ a subspace of $\mathbb{R}^{\mathbb{R}}$?

Proof. No, problem occurs with the addition. Suppose we have $f(x) = f(x + p)$ and $g(x) = g(x + q)$. Then $(f + g)(x) = f(x) + g(x) = f(x + p) + g(x + q) \neq (f + g)(x + l)$ for some fixed l for all p, q . $\square$

Problem 11

Prove that the intersection of every collection of subspaces of V is a subspace of V .

Proof. Let $\bigcap_i V_i$ denote the collection of subspaces of V . Then we know $0 \in \bigcap_i V_i$. Let $a \in \mathbb{F}, \mathbf{x}, \mathbf{y} \in \bigcap_i V_i$. We have that $a(\mathbf{x} + \mathbf{y}) \in \bigcap_i V_i$ and thus finish the proof. $\square$

Problem 12

Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.

Proof. Let V_1, V_2 be two subspaces of V .

$\Rightarrow$ Suppose for the sake of contradiction that there exists $v_1 \in V_1$ s.t. $v_1 \notin V_2$ and $v_2 \in V_2$ s.t. $v_2 \notin V_1$. Then by assumption we have that $v_1 + v_2 \in V_1 \cup V_2$. Here we can also show that $v_1 + v_2 \notin V_1$ because if it does, $v_1 + v_2 + (-v_1) = v_2 \in V_1$. Similarly, $v_1 + v_2 \notin V_2$. Thus we've reached a contradiction.

$\Leftarrow$ This direction is trivial. $\square$

Problem 14

Suppose

$$\mathcal{U} = \{(x, -x, 2x) \in \mathbb{F}^3 : x \in \mathbb{F}\} \text{ and } \mathcal{W} = \{(x, x, 2x) \in \mathbb{F}^3 : x \in \mathbb{F}\}$$

Describe $\mathcal{U} + \mathcal{W}$.

Proof.

$$(x, -x, 2x) + (y, y, 2y) = (x + y, -x + y, 2(x + y))$$

One can think of this as $\mathcal{U} + \mathcal{W} = \{(a, b, 2a) : a, b \in \mathbb{F}\}$. $\square$

Problem 15

Suppose $\mathcal{U}$ is a subspace of V , what is $\mathcal{U} + \mathcal{U}$?

Proof. $\mathcal{U} + \mathcal{U} = \mathcal{U}$.

Take $\mathbf{u}_1 + \mathbf{u}_2 \in \mathcal{U} + \mathcal{U}$, then $\mathbf{u}_1 + \mathbf{u}_2 \in \mathcal{U}$. Conversely, take $\mathbf{u} \in \mathcal{U}$, then $\mathbf{u} = \mathbf{u} + 0 \in \mathcal{U} + \mathcal{U}$. $\square$

Problem 16

Is the operation of addition on the subspaces of V commutative ($\mathcal{U} + \mathcal{W} = \mathcal{W} + \mathcal{U}$)?

Proof. Take $u \in \mathcal{U}, w \in \mathcal{W}$, then $u + w = w + u$, implying the conclusion. $\square$

Problem 18

Does the operation of addition on the subspaces of V have an additive identity? Which subspaces have additive inverses?

Proof. Yes, the zero subspace *i.e.* $\{0\}$ is the additive identity. The subspace that have additive inverses is only $\{0\}$. $\square$

Problem 19

Prove or disprove: If $V_1, V_2, \mathcal{U}$ are subspaces of V such that

$$V_1 + \mathcal{U} = V_2 + \mathcal{U}$$

then $V_1 = V_2$.

Proof. Counterexample: Consider when $\mathcal{U} = V_1 \sqcup V_2$, then the relation holds while $V_1 \neq V_2$. $\square$

Problem 20

Suppose

$$\mathcal{U} = \{(x, x, y, y) \in \mathbb{F}^4 : x, y \in \mathbb{F}\}$$

Find a subspace $\mathcal{W} \in \mathbb{F}^4$ s.t. $\mathbb{F}^4 = \mathcal{U} \oplus \mathcal{W}$.

Proof. Define $\mathcal{W} = \{(0, a, b, 0) \in \mathbb{F}^4 : a, b \in \mathbb{F}\}$ to be a subspace of $\mathbb{F}^4$.

Then first $\mathcal{W} + \mathcal{U} \subseteq \mathbb{F}^4$. Take $(q, w, e, r) \in \mathbb{F}^4$, we can have $(q, w, e, r) = (q, q, r, r) + (0, w - q, e - r, r) \in \mathcal{U} + \mathcal{W}$. We have $\mathbb{F}^4 = \mathcal{U} + \mathcal{W}$. Furthermore, take $(x, x, y, y) \in \mathcal{U}, (0, a, b, 0) \in \mathcal{W}$. For the element to be in the intersection, we need to have $(x, x, y, y) = (0, a, b, 0)$ which implies that $x = y = a = b = 0$ and thus $\mathcal{W} \cap \mathcal{U} = \{0\}$. $\square$

Problem 21

Suppose

$$\mathcal{U} = \{x, y, x + y, x - y, 2x\} \in \mathbb{F}^5 : x, y \in \mathbb{F}$$

Find a subspace $\mathcal{W} \in \mathbb{F}^5$ s.t. $\mathbb{F}^5 = \mathcal{U} \oplus \mathcal{W}$.

Proof. Define

$$\mathcal{W} = \{(0, 0, m, n, z) : m, n, z \in \mathbb{F}\}$$

Then $(a, b, c, d, e) = (a, b, a + b, a - b, 2a) + (0, 0, c - (a + b), d - (a - b), e - 2a)$. The rest follows exactly as in P20. $\square$

Problem 23

Prove or disprove: If $V_1, V_2, \mathcal{U}$ are subspaces of V s.t.

$$V = V_1 \oplus \mathcal{U} \text{ and } V = V_2 \oplus \mathcal{U},$$

then $V_1 = V_2$.

Proof. Counterexample: Let $\mathcal{U} = \{(x, x) : x \in \mathbb{F}\}, V_1 = \{(x, 0) : x \in \mathbb{F}\}, V_2 = \{(0, x) : x \in \mathbb{F}\}$. $\square$

Problem 24

A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is called *even* if $f(-x) = f(x)$ and *odd* if $f(-x) = -f(x)$ for all $x \in \mathbb{R}$. Let V_e denote the set of real-valued even functions on $\mathbb{R}$ and let V_o denote the set of real-valued odd functions on $\mathbb{R}$. Show that $\mathbb{R}^{\mathbb{R}} = V_e \oplus V_o$.

Proof. $\Leftarrow$ Trivial direction.

$$\Rightarrow f(x) = \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2}$$

$\square$

Chapter 2: Finite-Dimensional Vector Spaces

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Contents

2A: Span and Linear Independence	2
2A Problem Sets	3
2B: Bases	6
2B Problem Sets	7
2C: Dimension	9
2C Problem Sets	10

2A: Span and Linear Independence

Definition 1 (Linear Combination). A linear combination of a list $v_1, \dots, v_m$ of vectors in V is a vector of the form

$$a_1v_1 + \dots + a_mv_m$$

where $a_1, \dots, a_m \in \mathbb{F}$.

Definition 2 (Span). The set of all linear combinations of list of vectors $v_1, \dots, v_m$ in V is called the span of $v_1, \dots, v_m$, denoted by $\text{span}(v_1, \dots, v_m)$. In other words,

$$\text{span}(v_1, \dots, v_m) = \{a_1v_1 + \dots + a_mv_m : a_1, \dots, a_m \in \mathbb{F}\}$$

The span of the empty list $()$ is defined to be $\{0\}$.

Theorem 3. The span of a list of vectors in V is the smallest subspace of V containing all vectors in the list.

Definition 4 (Spans). If $\text{span}(v_1, \dots, v_m)$ equals V , we say the list $v_1, \dots, v_m$ spans V .

Definition 5 (Finite-dimensional vector space). A vector space is called finite-dimensional if some list of vectors in it spans the space.

Definition 6 (polynomial). A function $p: \mathbb{F} \rightarrow \mathbb{F}$ is called a polynomial with coefficients in $\mathbb{F}$ if there exist $a_0, \dots, a_m \in \mathbb{F}$ s.t.

$$(z) = a_0 + a_1z + a_2z^2 + \dots + a_mz^m$$

for all $z \in \mathbb{F}$.

$\mathcal{P}(\mathbb{F})$ is the set of all polynomials with coefficients in $\mathbb{F}$.

Definition 7 (Linear independence). A list of vectors $v_1, \dots, v_m$ in V is called linearly independent if the only choice of $a_1, \dots, a_m \in \mathbb{F}$ that makes

$$a_1v_1 + \dots + a_mv_m = 0$$

is $a_1 = \dots = a_m = 0$.

The empty list $()$ is also declared to be linearly independent.

Lemma 8. Suppose $v_1, \dots, v_m$ is a linearly dependent list in V . Then there exists $k \in \{1, 2, \dots, m\}$ s.t.

$$v_k \in \text{span}(v_1, \dots, v_{k-1})$$

Furthermore, if k satisfies the condition above and the k^{th} term is removed from $v_1, \dots, v_m$, then the span of the remaining list equals $\text{span}(v_1, \dots, v_m)$.

Lemma 9 (length of linearly independent list $\leq$ length of spanning list). In a finite-dimensional vector space, the length of every linearly independent list of vectors is less than or equal to the length of every spanning list of vectors.

Problem 1

Find a list of four distinct vectors in $\mathbb{F}^3$ whose span equals

$$\{(x, y, z) \in \mathbb{F}^3 : x + y + z = 0\}$$

Proof. Example: $(1, 0, -1), (0, -1, 1), (1, -1, 0), (1, -2, 1)$. $\square$

Problem 2

Prove or give a counterexample: If v_1, v_2, v_3, v_4 spans V , then the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

also spans V .

Proof. Take any $v \in V$, then we have $v = a_1v_1 + a_2v_2 + a_3v_3 + a_4v_4 = a_1(v_1 - v_2) + (a_1 + a_2)(v_2 - v_3) + (a_1 + a_2 + a_3)(v_3 - v_4) + (a_1 + a_2 + a_3 + a_4)v_4$. Conversely, any linear combination of these vectors still belong to V . $\square$

Problem 3

Suppose $v_1, \dots, v_m$ is a list of vectors in V . For $k \in \{1, \dots, m\}$, let

$$w_k = v_1 + \dots + v_k$$

Show that $\text{span}(v_1, \dots, v_m) = \text{span}(w_1, \dots, w_m)$

Proof. Take $v = \sum_{i=1}^m a_i v_i$ from l.h.s, then we can write

$$\begin{aligned} v &= a_1 v_1 + \dots + a_m v_m \\ &= (a_1 - a_m) v_1 + \dots + (a_{m-1} - a_m) v_{m-1} + a_m w_m \\ &= (a_1 - a_m - a_{m-1}) v_1 + \dots + (a_{m-2} - a_{m-1} - a_m) v_{m-2} + (a_{m-1} - a_m) w_{m-1} + a_m w_m \\ &= \sum_{i=1}^m \left(a_i - \sum_{j=i+1}^m a_j \right) w_i \in \text{r.h.s} \end{aligned}$$

Conversely, take $w = \sum_{i=1}^m b_i w_i = \sum_{i=1}^m (b_i \sum_{j=1}^i c_j) v_j \in \text{l.h.s.}$ $\square$

Problem 4

- (a) Show that a list of length one in a vector space is linearly independent if and only if the vector in the list is not 0.
- (b) Show that a list of length two in a vector space is linearly independent if and only if neither of the two vectors in the list is a scalar multiple of the other.

Proof. (a) In order for the only way to write $av = 0$ is to ensure $v \neq 0$.

(b) $av_1 + bv_2 = 0$. $\Rightarrow$ the only solution is $a = b = 0$ so v_1 cannot be a multiple of v_2 ; vice versa. $\Leftarrow$ same reason. $\square$

Problem 8

Suppose v_1, v_2, v_3, v_4 is linearly independent in V . Prove that the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

is also linearly independent.

Proof.

$$a_1(v_1 - v_2) + a_2(v_2 - v_3) + a_3(v_3 - v_4) + a_4v_4 = a_1v_1 + (-a_1 + a_2)v_2 + (-a_2 + a_3)v_3 + (-a_3 + a_4)v_4$$

The only solution is $a_1 = 0, -a_1 + a_2 = 0, -a_2 + a_3 = 0, -a_3 + a_4 = 0$. $\square$

Problem 9

Prove or give a counter example: If $v_1, \dots, v_m$ is a linearly independent list of vectors in V , then

$$5v_1 - 4v_2, v_2, v_3, \dots, v_m$$

is also linearly independent.

Proof.

$$\begin{aligned} a_1(5v_1 - 4v_2) + a_2v_2 + \dots + a_mv_m \\ = 5a_1v_1 + (-4a_1 + a_2)v_2 + \dots + a_mv_m \end{aligned}$$

The only solution is $5a_1 = -4a_1 + a_2 = \dots = a_m = 0$. $\square$

Problem 10

Prove or give a counterexample: If $v_1, \dots, v_m$ is a linearly independent list of vectors in V and $\lambda \in \mathbb{F}$ with $\lambda \neq 0$, then $\lambda v_1, \dots, \lambda v_m$ is linearly independent.

Proof.

$$\sum_i^m \lambda a_i v_i = 0$$

The only solution is $a_i = 0$ for all i . $\square$

Problem 12

Suppose $v_1, \dots, v_m$ is linearly independent in V and $w \in V$. Prove that if $v_1 + w, \dots, v_m + w$ is linearly dependent, then $w \in \text{span}(v_1, \dots, v_m)$.

Proof. We know the only solution to $\sum_i^m a_i v_i$ is all $a_i = 0$. Now we have that non-zero a_i for solving $\sum_i^m a_i v_i + \sum_i^m a_i w = 0$. Thus $w = \frac{\sum_i^m a_i v_i}{\sum_i^m a_i}$ which completes the proof. $\square$

Problem 13

Suppose $v_1, \dots, v_m$ is linearly independent in V and $w \in V$. Show that

$$v_1, \dots, v_m, w \text{ is linearly independent} \iff w \notin \text{span}(v_1, \dots, v_m)$$

Proof. $\Rightarrow$ By contradiction, if w in the span then it can be written as linear combination for some nonzero a_i and thus they are linearly dependent.

$\Leftarrow$ Similarly. $\square$

Problem 17

Prove that V is infinite-dimensional if and only if there is a sequence $v_1, v_2, \dots$ of vectors in V such that $v_1, \dots, v_m$ is linearly independent for every positive integer m .

Proof. $\Rightarrow$ There doesn't exist any finite list of vectors that span the space. For the sake of contradicting assumes for every sequence $v_1, \dots$ of vectors in $V \exists m$ such that $v_1, \dots, v_m$ is linearly dependent. Then this means we can construct the basis for the vector space as follows: select $v_i \in V$ s.t. v_i and $v_1, \dots, v_{i-1}$ are linearly independent. This means that there exists m s.t. $v_1, \dots, v_m$ that spans V , forming a contradiction.

$\Leftarrow$ Suppose for contradiction. Then there exists a span, contradicting the linear independence claim for every m . $\square$

Problem 18

Prove $\mathbb{F}^\infty$ is infinite-dimensional.

Proof. We apply Problem 17. Construct v_i to be the vector that has 1 on the i -th coordinate and 0 elsewhere. clearly $v_1, v_2, \dots$ are s.t. $v_1, \dots, v_m$ is linearly independent for every positive integer m . $\square$

2B: Bases

Definition 10 (basis). *A basis of V is a list of vectors in V that is linearly independent and spans V .*

Theorem 11 (criterion for basis). *A list $v_1, \dots, v_m$ of vectors in V is a basis of V if and only if every $v \in V$ can be written uniquely in the form*

$$v = \sum_i^m a_i v_i$$

where $a_i \in \mathbb{F}$.

Lemma 12 (every spanning list contains a basis). *Every spanning list in a vector space can be reduced to a basis of the vector space.*

Lemma 13. *Every finite-dimensional vector space has a basis.*

Lemma 14. *Every linearly independent list of vectors in a finite-dimensional vector space can be extended to a basis of the vector space.*

Lemma 15. *Suppose V is finite-dimensional and $\mathcal{U}$ is a subspace of V . Then there is a subspace W of V such that $V = \mathcal{U} \oplus W$.*

Problem 1

Find all vector spaces that have exactly one basis.

Proof. The only answer is $\{0\}$. Otherwise, for any basis v one can get av for $a \neq 0, a \neq 1$. $\square$

Problem 4

1. Let $\mathcal{U}$ be the subspace of $\mathbb{C}^5$ defined by $\mathcal{U} = \{(z_1, z_2, z_3, z_4, z_5) \in \mathbb{C}^5 : 6z_1 = z_2, z_3 + 2z_4 + 3z_5 = 0\}$ Find a basis of $\mathcal{U}$.
2. Extend the basis to a basis in $\mathbb{C}^5$.
3. Find a subspace $\mathcal{W}$ of $\mathbb{C}^5$ s.t. $\mathbb{C}^5 = \mathcal{U} \oplus \mathcal{W}$.

Proof. 1. $(z_1, 6z_1, -2z_4 - 3z_5, z_4, z_5) \Rightarrow \{(1, 6, 0, 0, 0), (0, 0, -2, 1, 0), (0, 0, -3, 0, 1)\}$
 2. $\{(1, 6, 0, 0, 0), (0, 0, -2, 1, 0), (0, 0, -3, 0, 1), (0, 1, 0, 0, 0), (0, 0, 1, 0, 0)\}$
 3. $\mathcal{W} = \text{span}(\{(0, 1, 0, 0, 0), (0, 0, 1, 0, 0)\})$

$\square$

Problem 5

Suppose V is finite-dimensional and $\mathcal{U}, \mathcal{W}$ are subspaces of V such that $V = \mathcal{U} + \mathcal{W}$. Prove that there exists a basis of V consisting of vectors in $\mathcal{U} \cup \mathcal{W}$.

Proof. Let $\{v_i\}_{i=1}^m$ denote the basis for the vector space V . By definition we have $v_i = u_i + w_i$ for some u_i, w_i . Then we have the spanning set of the vector space V $\sum_i^m a_i(u_i + w_i)$, which can be reduced to a basis by the lemma. $\square$

Problem 7

Suppose v_1, v_2, v_3, v_4 is a basis of V . Prove that

$$v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$$

is also a basis of V .

Proof. We know v_1, v_2, v_3, v_4 is linearly independent and spans V .

$$a_1(v_1 + v_2) + a_2(v_2 + v_3) + a_3(v_3 + v_4) + a_4 v_4 = a_1 v_1 + (a_1 + a_2)v_2 + (a_2 + a_3)v_3 + (a_3 + a_4)v_4$$

which shows the linear independence. For proving spanning, let $v \in V$ then

$$v = \sum_{i=1}^4 a_i v_i = a_1(v_1 + v_2) + (a_2 - a_1)(v_2 + v_3) + (a_3 - a_2)(v_3 + v_4) + (a_4 - a_3)v_4$$

$\square$

Problem 8

Prove or give a counterexample: If v_1, v_2, v_3, v_4 is a basis of V and $\mathcal{U}$ is a subspace of V such that $v_1, v_2 \in \mathcal{U}$ and $v_3 \notin \mathcal{U}$ and $v_4 \notin \mathcal{U}$, then v_1, v_2 is a basis of $\mathcal{U}$.

Proof. Take $V = \mathbb{R}^4$ and the standard basis. Consider $\mathcal{U} = \{(x_1, x_2, x_3, kx_3)\}$, then we disprove the claim. $\square$

Problem 10

Suppose $\mathcal{U}$ and $\mathcal{W}$ are subspaces of V s.t. $V = \mathcal{U} \oplus \mathcal{W}$. Suppose also that $u_1, \dots, u_m$ is a basis of $\mathcal{U}$ and $w_1, \dots, w_n$ is a basis of $\mathcal{W}$. Prove that

$$u_1, \dots, u_m, w_1, \dots, w_n$$

is a basis of V .

Proof. We know that this set is linearly independent (otherwise violating the direct sum assumption) so it suffices to prove the spanning. Let $v \in V$, then $v = u + w = \sum_{i=1}^m a_i u_i + \sum_{j=1}^n b_j w_j$. $\square$

2C: Dimension

Lemma 16 (basis length does not depend on basis). *Any two bases of a finite-dimensional vector space have the same length.*

Definition 17 (dimension). • *The dimension of a finite-dimensional vector space is the length of any basis of the vector space.*

• *The dimension of a finite-dimensional vector space V is denoted by $\dim V$.*

Corollary 18. *If V is finite-dimensional and $\mathcal{U}$ is a subspace of V , then $\dim \mathcal{U} \leq \dim V$.*

Corollary 19. *Suppose V is finite-dimensional. Then every linearly independent list of vectors in V of length $\dim V$ is a basis of V .*

Corollary 20. *Suppose that V is finite-dimensional and $\mathcal{U}$ is a subspace of V such that $\dim \mathcal{U} = \dim V$. Then $\mathcal{U} = V$.*

Corollary 21. *Suppose that V is finite-dimensional. Then every spanning list of vectors in V of length $\dim V$ is a basis of V .*

Theorem 22 (dimension of a sum). *If V_1 and V_2 are subspaces of a finite-dimensional vector space, then*

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2)$$

Problem 1

Show that the subspaces of $\mathbb{R}^2$ are precisely $\{0\}$, all lines in $\mathbb{R}^2$ containing the origin and $\mathbb{R}^2$.

Proof. We know $\dim(\mathbb{R}^2) = 2$ so the subspace dimension is either 0 ($\{0\}$) or 1 (then this means it has to be lines crossing the origin). $\square$

Problem 5

- (a) Let $\mathcal{U} = \{p \in \mathcal{P}_4(\mathbb{F}) : p(2) = p(5)\}$. Find a basis of $\mathcal{U}$.
- (b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{F})$.
- (c) Find a subspace $\mathcal{W}$ of $\mathcal{P}_4(\mathbb{F})$ s.t. $\mathcal{P}_4(\mathbb{F}) = \mathcal{U} \oplus \mathcal{W}$.

Proof. (a) This means that

$$a_0 + 2a_1 + 4a_2 + 8a_3 + 16a_4 = a_0 + 5a_1 + 25a_2 + 125a_3 + 625a_4$$

Solving this gives that $a_1 = -7a_2 - 39a_3 - 203a_4$. So we can write that

$$\begin{aligned} p(x) &= a_0 + (-7a_2 - 39a_3 - 203a_4)x + a_2x^2 + a_3x^3 + a_4x^4 \\ &= a_0 + a_2(x^2 - 7x) + a_3(x^3 - 39x) + a_4(x^4 - 203x) \end{aligned}$$

The basis now becomes $\{1, x^2 - 7x, x^3 - 39x, x^4 - 203x\}$

(b) $\{1, x, x^2 - 7x, x^3 - 39x, x^4 - 203x\}$

(c) $\mathcal{W} = \{x\}$ $\square$

Problem 8

Suppose $v_1, \dots, v_m$ is linearly independent in V and $w \in V$. Prove that

$$\dim \text{span}(v_1 + w, \dots, v_m + w) \geq m - 1$$

Proof. We claim that

$$v_{i+1} - v_i \in \text{span}(v_1 + w, \dots, v_m + w) \text{ for all } m \geq i \geq 2$$

as $v_{i+1} - v_i = (v_{i+1} + w) - (v_i + w)$. Thus $v_2 - v_1, \dots, v_m - v_{m-1}$ is in the $\text{span}(v_1 + w, \dots, v_m + w)$. We've proved that $v_2 - v_1, \dots, v_m - v_{m-1}$ is linearly independent and this list has $m - 1$ vectors and thus we've proved the claim. $\square$

Problem 9

Suppose m is a positive integer and $p_0, p_1, \dots, p_m \in \mathcal{P}(\mathbb{F})$ are such that each p_k has degree k . Prove that $p_0, p_1, \dots, p_m$ is a basis of $\mathcal{P}_m(\mathbb{F})$.

Proof. It's easy to see that this list is linearly independent. Take any element in $\mathcal{P}_m(\mathbb{F})$, we can decompose that by degrees and get each component to be some multiple of p'_i s. $\square$

Problem 10

Suppose m is a positive integer. For $0 \leq k \leq m$, let

$$p_k(x) = x^k(1-x)^{m-k}$$

Show that $p_0, \dots, p_m$ is a basis of $\mathcal{P}_m(\mathbb{F})$.

Proof. It suffices to prove that $p_k(x)$ are linearly independent. We know from binomial theorem that

$$(x+y)^m = \sum_{i=0}^m \binom{m}{i} x^i y^{m-i}$$

Applying this identity here we get that

$$p_k(x) = x^k \sum_{i=0}^{m-k} \binom{m-k}{i} 1^{m-i} (-1)^i x^i = \sum_{i=0}^{m-k} a_i x^{k+i}$$

where $a_i = \binom{m-k}{i} (-1)^i$. For the polynomial to be identically 0 ($\sum_{k=0}^m c_k p_k(x) = 0$), for each power x^i we need the coefficient to be 0:

$$\sum_{k=0}^{\min(i,m)} c_k \binom{m-k}{i-k} (-1)^{i-k} = 0$$

We can prove the only solution for this is all $c_k = 0$ by induction on m .

The base case is trivial. Assume the statement holds for $k = m-1$, then we try to prove for $p_m(x)$, where we have $P(x) - c_m p_m(x) = \sum_{k=0}^{m-1} c_k p_k(x) = 0$. This means that $c_m = 0$ and we've proved the linear independence. $\square$

Problem 11

Suppose $\mathcal{U}$ and $\mathcal{W}$ are both four-dimensional subspaces of $\mathbb{C}^6$. Prove that there exist two vectors in $\mathcal{U} \cap \mathcal{W}$ such that neither of these vectors is a scalar multiple of the other.

Proof. $\dim(\mathcal{U} \cap \mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{U} + \mathcal{W}) \geq 8 - 6 = 2$

So there must exist two linearly independent vectors in the intersection. $\square$

Problem 12

Suppose that $\mathcal{U}$ and $\mathcal{W}$ are subspaces of $\mathbb{R}^8$ such that $\dim \mathcal{U} = 3$, $\dim \mathcal{W} = 5$ and $\mathcal{U} + \mathcal{W} = \mathbb{R}^8$. Prove that $\mathbb{R}^8 = \mathcal{U} \oplus \mathcal{W}$.

Proof. Similar to problem 11, we can get that $\dim(\mathcal{U} \cap \mathcal{W}) = 0$. $\square$

Problem 14

Suppose V is a ten-dimensional vector space and V_1, V_2, V_3 are subspaces of V with $\dim V_1 = \dim V_2 = \dim V_3 = 7$. Prove that $V_1 \cap V_2 \cap V_3 \neq \{0\}$.

Proof.

$$\dim((V_1 \cap V_2) + V_3) = \dim(V_1 \cap V_2) + \dim V_3 - \dim(V_1 \cap V_2 \cap V_3)$$

and also that

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2)$$

This gives that

$$\dim(V_1 \cap V_2 \cap V_3) = \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_1 + V_2) - \dim((V_1 \cap V_2) + V_3)$$

Note that from question we know $\dim V_1 + \dim V_2 + \dim V_3 = 21 > 20 = 2 \dim V$

Hence we know

$$\dim(V_1 \cap V_2 \cap V_3) > (\dim(V) - \dim(V_1 + V_2)) + (\dim(V) - \dim((V_1 \cap V_2) + V_3)) > 0$$

We've thus proved the claim. $\square$

Problem 16

Suppose V is finite-dimensional and $\mathcal{U}$ is a subspace of V with $\mathcal{U} \neq V$. Let $n = \dim V$ and $m = \dim \mathcal{U}$. Prove that there exist $n - m$ subspaces of V , each of dimension $n - 1$, whose intersection equals $\mathcal{U}$.

Proof. To show existence, we can start with the basis for $\mathcal{U} : u_1, \dots, u_m$. We extend the basis to $V : \{u_1, \dots, u_m, v_1, \dots, v_{n-m}\} := K$. Construct the subspace $V_i = \text{span}\{K \setminus \{v_i\}\}$. Then we know that $\bigcap_i V_i = \text{span}\{u_1, \dots, u_m\}$. $\square$

Problem 17

Suppose that $V_1, \dots, V_m$ are finite-dimensional subspaces of V . Prove that $V_1 + \dots + V_m$ is finite-dimensional and

$$\dim(V_1 + \dots + V_m) \leq \dim V_1 + \dots + \dim V_m$$

Proof. We prove by induction on m . The base case is trivial. Assume the statement hold for k , then for $k + 1$, we have that (Denote $V_1 + \dots + V_k = M_k$)

$$\dim(M_k + V_{k+1}) \leq \dim(V_1) + \dots + \dim(V_{k+1})$$

which is finite. $\square$

Problem 18

Suppose V is finite-dimensional with $\dim V = n \geq 1$. Prove that there exist one-dimensional subspaces $V_1, \dots, V_n$ of V such that

$$V = V_1 \oplus \dots \oplus V_n$$

Proof. We know there are basis $\{v_1, \dots, v_n\}$ for V . Hence we can construct

$$V_i = \text{span}\{v_i\}$$

□

Problem 19

Prove or give a counter example:

$$\begin{aligned} \dim(V_1 + V_2 + V_3) &= \dim V_1 + \dim V_2 + \dim V_3 \\ &\quad - \dim(V_1 \cap V_2) - \dim(V_1 \cap V_3) - \dim(V_2 \cap V_3) \\ &\quad + \dim(V_1 \cap V_2 \cap V_3) \end{aligned}$$

Proof. Consider $V_1 = \langle e_1 \rangle, V_2 = \langle e_2 \rangle, V_3 = \langle e_1 + e_2 \rangle$

then L.H.S = 2 ,but R.H.S = 3.

□

Problem 20

Prove that if V_1, V_2 , and V_3 are subspaces of a finite-dimensional vector space, then

$$\begin{aligned} &\dim(V_1 + V_2 + V_3) \\ &= \dim V_1 + \dim V_2 + \dim V_3 \\ &\quad - \frac{\dim(V_1 \cap V_2) + \dim(V_1 \cap V_3) + \dim(V_2 \cap V_3)}{3} \\ &\quad - \frac{\dim((V_1 + V_2) \cap V_3) + \dim((V_1 + V_3) \cap V_2) + \dim((V_2 + V_3) \cap V_1)}{3}. \end{aligned}$$

Proof. We know that

$$\begin{aligned} \dim((V_1 + V_2) + V_3) &= \dim(V_1 + V_2) + \dim(V_3) - \dim((V_1 + V_2) \cap V_3) \\ &= \dim(V_1) + \dim(V_2) + \dim(V_3) - \dim(V_1 \cap V_2) - \dim((V_1 + V_2) \cap V_3) \end{aligned}$$

then, by considering $3 \dim(V_1 + V_2 + V_3)$, and using above equality by pairing V_1, V_2, V_3 , we obtain the desired equality.

□

Chapter 3: Linear Maps

Linear Algebra Done Right (4th Edition), by Sheldon Axler

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Contents

3A: Vector Space of Linear Maps	2
3A Problem Sets	3
3B: Null Spaces and Ranges	7
3B Problem Sets	8
3C: Matrices	13
3C Problem Sets	16
3D: Invertibility and Isomorphisms	23
3D Problem Sets	25
3E: Products and Quotients of Vector Spaces	30
3E Problem Sets	33
3F: Duality	40
3F Problem Sets	43

3A: Vector Space of Linear Maps

Definition 1 (Linear Map). A linear map from V to W is a function $T: V \rightarrow W$ with the following properties:

- **Additivity:** $T(u + v) = Tu + Tv$ for all $u, v \in V$.
- **Homogeneity:** $T(\lambda v) = \lambda T(v)$ for all $\lambda \in \mathbb{F}$ and $v \in V$.

Notation: $\mathcal{L}(V, W), \mathcal{L}(V)$

- The set of linear maps from V to W is denoted by $\mathcal{L}(V, W)$.
- The set of linear maps from V to V is denoted by $\mathcal{L}(V)$.

Lemma 2 (linear map basis lemma). Suppose $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_n \in W$. Then there exists a unique linear map $T: V \rightarrow W$ such that

$$Tv_k = w_k$$

for each $k = 1, \dots, n$.

Definition 3 (additional and scalar multiplication on $\mathcal{L}(V, W)$). Suppose $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$. The sum $S + T$ and the product λT are the linear maps from V to W defined by

$$(S + T)(v) = Sv + Tv \text{ and } (\lambda T)(v) = \lambda(Tv)$$

for all $v \in V$.

Remark 4. $\mathcal{L}(V, W)$ is a vector space.

Definition 5 (product of linear maps). If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then the product $ST \in \mathcal{L}(U, W)$ is defined by

$$(ST)(u) = S(Tu)$$

for all $u \in U$.

Remark 6 (algebraic properties of product of linear maps). We have associativity, identity, and distributive properties whenever such properties are defined.

Theorem 7 (linear maps take 0 to 0). Suppose T is a linear map from V to W . Then $T(0) = 0$.

Problem 1

Suppose $b, c \in \mathbb{R}$. Define $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by

$$T(x, y, z) = (2x - 4y + 3z + b, 6x + cxyz)$$

Show that T is linear if and only if $b = c = 0$.

Proof. T is linear $\iff T(x, y, z = 0) \iff b = 0$

In addition, let's only consider the second coordinate, then we have

$$T(x_1 + x_2, y_1 + y_2, z_1 + z_2)_2 = 6(x_1 + x_2) + c(x_1 + x_2)(y_1 + y_2)(z_1 + z_2)$$

which only equals $T(x_1, y_1, z_1) + T(x_2, y_2, z_2)$ if $c = 0$. $\square$

Problem 3

Suppose that $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$. Show that there exist scalars $A_{j,k} \in \mathbb{F}$ for $j = 1, \dots, m$ and $k = 1, \dots, n$ such that

$$T(x_1, \dots, x_n) = (A_{1,1}x_1 + \dots + A_{1,n}x_n, \dots, A_{m,1}x_1 + \dots + A_{m,n}x_n)$$

for every $(x_1, \dots, x_n) \in \mathbb{F}^n$.

Proof. Let $\{u_1, \dots, u_n\}$ denote the standard basis of $\mathbb{F}^n$. We have that

$$Tu_i = (A_{1,i}, \dots, A_{m,i})$$

Take arbitrary $(x_1, \dots, x_n) \in \mathbb{F}^n$, we have that

$$Tx_i u_i = x_i(A_{1,i}, \dots, A_{m,i})$$

thus we have that

$$T(x_1, \dots, x_n) = (A_{1,1}x_1 + \dots + A_{1,n}x_n, \dots, A_{m,1}x_1 + \dots + A_{m,n}x_n)$$

$\square$

Problem 4

Suppose $T \in \mathcal{L}(V, W)$ and $v_1, \dots, v_m$ is a list of vectors in V such that $Tv_1, \dots, Tv_m$ is a linearly independent list in W . Prove that $v_1, \dots, v_m$ is linearly independent.

Proof. This means that the only solution to $\sum_{i=1}^m a_i T(v_i) = \sum_{i=1}^m T(a_i v_i) = 0$ is all $a_i = 0$. However, as we know $T(0) = 0$ so $a_i v_i = 0$ and thus we've proved the claim. $\square$

Problem 7

Show that every linear map from a one-dimensional vector space to itself is multiplication by some scalar. More precisely, prove that if $\dim V = 1$ and $T \in \mathcal{L}(V)$, then there exists $\lambda \in \mathbb{F}$ such that $Tv = \lambda v$ for all $v \in V$.

Proof. Since $\dim V = 1$, every $v \in V$ can be expressed as $\lambda v'$ for some other $v' \in V$. As $Tv \in V$, we have $Tv = \beta v$. Then $Tv = T\lambda v' = \beta\lambda v' = \beta v$ $\square$

Problem 8

Give an example of a function $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}$ such that

$$\phi(av) = a\phi(v)$$

for all $a \in \mathbb{R}$ and all $v \in \mathbb{R}^2$ but ϕ is not linear.

Proof. Consider $f(x_1, x_2) = x_1^2/x_2$ if $x_2 \neq 0$ o.w. 0, then $f(a(x_1, x_2)) = ax_1^2 = af(x_1, x_2)$. However, $f((x_1 + y_1, x_2 + y_2)) = (x_1 + y_1)^2/(x_2 + y_2) \neq f(x_1, x_2) + f(y_1, y_2)$. $\square$

Problem 11

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T is a scalar multiple of the identity if and only if $ST = TS$ for all $S \in \mathcal{L}(V)$.

Proof. $\Rightarrow$ We can express $T = \lambda I$. Then $S(\lambda I) = \lambda S = \lambda IS = TS$.

$\Leftarrow$ Let $v_1, \dots, v_m$ be a basis of V . Pick S_i such that $S_i(\sum_{i=1}^m a_i v_i) = a_i v_i$, which is clearly a linear operator. Then we have that

$$S_i T(v) = T S_i(v)$$

$$S_i \sum_{j=1}^m b_j v_j = T(a_i v_i)$$

$$b_i v_i = a_i T(v_i)$$

This shows that for all v_i , there exists λ_i such that $T(v_i) = \lambda_i v_i$. Next we show that such λ_i does not depend on i . Construct S_{ij} subtly such that $S_{ij} \sum_{k=1}^n a_k v_k = a_j v_i + a_i v_j$.

Then we have that

$$S_{ij} T v = T S_{ij} v$$

$$S_{ij} \left(\sum_{k=1}^n \lambda_k a_k v_k \right) = T(a_j v_i + a_i v_j)$$

$$\lambda_j a_j v_i + \lambda_i a_i v_j = \lambda_i a_j v_i + \lambda_j a_i v_j$$

This shows that $\lambda_i = \lambda_j$ for all i, j and thus we've shown that $T = \lambda I$ for some λ . $\square$

Problem 12

Suppose U is a subspace of V with $U \neq V$. Suppose $S \in \mathcal{L}(U, W)$ and $S \neq 0$. Define $T: V \rightarrow W$ by

$$Tv = \begin{cases} Sv & \text{if } v \in U \\ 0 & \text{if } v \in V \text{ and } v \notin U \end{cases}$$

Prove that T is not a linear map on V .

Proof. Take $u \in U$ such that $u \notin \text{Null}(S)$. Take $v \in V \setminus U$, then $u + v \in V \setminus U$. This means that

$$T(u + v) = S(u + v) = 0 \neq T(u) + T(v) = Su$$

□

Problem 13

Suppose V is finite-dimensional. Prove that every linear map on a subspace of V can be extended to a linear map on V . In other words, show that if U is a subspace of V and $S \in \mathcal{L}(U, W)$, then there exists $T \in \mathcal{L}(V, W)$ such that $Tu = Su$ for all $u \in U$.

Proof. Note that there exists subspace P s.t. $V = P \oplus U$. For all $v \in V$, $v = p + u$ for some p, u . Then define $T(v) = Su + p$. Clearly we have that $T(u) = Su$ for all $u \in U$. It now only suffices to prove T is a linear map. Homogeneity is trivial to show. For additivity, $T(v_1 + v_2) = T(u_1 + u_2 + p_1 + p_2) = S(u_1 + u_2) + p_1 + p_2 = (Su_1 + p_1) + (Su_2 + p_2) = Tv_1 + Tv_2$. □

Problem 14

Suppose V is finite-dimensional with $\dim V > 0$, and suppose W is infinite-dimensional. Prove that $\mathcal{L}(V, W)$ is infinite-dimensional.

Proof. Recall the definition of infinite-dimension from ch2 P17:

V is infinite-dimensional if and only if there is a sequence $v_1, v_2, \dots$ of vectors in V such that $v_1, \dots, v_m$ is linearly independent for every positive integer m .

W being infinite-dimensional implies this. Denote the sequence by $w_1, w_2, \dots$. Let $v_1, \dots, v_m$ be the basis for V . Define a sequence of linear operators as follows: $T_k \in \mathcal{L}(V, W)$ such that $T_k(v) = v_k$. Then we have that for every positive integer m , the solution for the following equation is all $a_i = 0$.

$$a_1 T_1(v) + \dots + a_m T_m(v) = a_1 v_1 + \dots + a_m v_m = 0$$

This shows that $\mathcal{L}(V, W)$ is infinite-dimensional. □

Problem 15

Suppose $v_1, \dots, v_m$ is a linearly dependent list of vectors in V . Suppose also that $W \neq \{0\}$. Prove that there exist $w_1, \dots, w_m \in W$ such that no $T \in \mathcal{L}(V, W)$ satisfies $Tv_k = w_k$ for each $k = 1, \dots, m$.

Proof. Take $w_1, \dots, w_m$ to be linearly independent list of vectors in W . Then we have that

$$Ta_1v_1 + \dots + Ta_mv_m = a_1w_1 + \dots + a_mw_m = 0$$

The only solution is that $a_i = 0$ for all i , but this contradicts that $v_1, \dots, v_m$ is linearly dependent. $\square$

Problem 16

Suppose V is finite-dimensional with $\dim V > 1$. Prove that there exist $S, T \in \mathcal{L}(V)$ such that $ST \neq TS$.

Proof. Let $v_1, \dots, v_m$ be the basis of V . Define $S(v) = S(\sum_{i=1}^m a_i v_i) = \sum_{i=1}^m a_{m-i} v_i$ and $T(v) = a_1 v_1$. Then we have that

$$STv = Sa_1v_1 = a_1v_1$$

but

$$TSv = T \sum_{i=1}^m a_{m-i} v_i = a_m v_1$$

$\square$

Problem 17

Suppose V is finite-dimensional. Show that the only two-sided ideals of $\mathcal{L}(V)$ are $\{0\}$ and $\mathcal{L}(V)$, where we define that a subspace $\mathcal{E}$ of $\mathcal{L}(V)$ is called two-sided ideal if $TE \in \mathcal{E}$ and $ET \in \mathcal{E}$ for all $E \in \mathcal{E}$ and all $T \in \mathcal{L}(V)$.

Proof. It's easy to verify that $\{0\}$ and $\mathcal{L}(V)$ are two-sided ideal of $\mathcal{L}(V)$. Suppose for the sake of contradiction that such $\mathcal{E}$ exists. Let $e_1, \dots, e_m$ be its basis and let $e_1, \dots, e_m, e_{m+1}, \dots, e_n$ be the basis for V . Define $T(v) = T(\sum_{i=1}^n a_i e_i) = \sum_{i=1}^n a_i e_{n-i}$ to be a linear map in V (ez to verify), and $E_j(u) = E_j(\sum_{k=1}^m b_k e_k) = b_j e_j$. We have reached the contradicting example such that $TE_j(v) = a_j e_{n-j} \in V \setminus \mathcal{E}$. $\square$

3B: Null Spaces and Ranges

Definition 8 (null space). For $T \in \mathcal{L}(V, W)$, the null space of T , denoted by $\text{null } T$, is the subset of V consisting of those vectors that T maps to 0:

$$\text{null } T = \{v \in V : Tv = 0\}$$

Corollary 9. Suppose $T \in \mathcal{L}(V, W)$, then $\text{null } T$ is a subspace of V .

Definition 10 (injective). A function $T: V \rightarrow W$ is called **injective** if $Tu = Tv$ implies $u = v$.

Theorem 11. Let $T \in \mathcal{L}(V, W)$. Then T is injective if and only if $\text{null } T = \{0\}$.

Definition 12 (range). For $T \in \mathcal{L}(V, W)$, the **range** of T is the subset of W consisting of those vectors that are equal to Tv for some $v \in V$:

$$\text{range } T = \{Tv : v \in V\}$$

Corollary 13. Suppose $T \in \mathcal{L}(V, W)$, then $\text{range } T$ is a subspace of W .

Definition 14 (surjective). A function $T: V \rightarrow W$ is called **surjective** if its range equals W .

Theorem 15 (fundamental theorem of linear map). Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then $\text{range } T$ is finite-dimensional and

$$\dim V = \dim \text{null } T + \dim \text{range } T$$

Proof. Let $u_1, \dots, u_m$ be a basis of $\text{null } T$ and let $u_1, \dots, u_m, v_1, \dots, v_n$ be a basis of V . It now suffices to prove $\dim \text{range } T = n$. Let $v \in T$, then

$$v = \sum_{i=1}^m a_i u_i + \sum_{j=1}^n b_j v_j$$

We apply T on both sides to get that

$$Tv = \sum_{j=1}^n b_j Tv_j$$

which shows that $Tv_1, \dots, Tv_n$ spans $\text{range } T$ and thus it's finite-dimensional. To show they are linearly independent, we have

$$\sum_{j=1}^n b_j Tv_j = T \sum_{j=1}^n b_j v_j = 0$$

The only solution is that all $b_j = 0$. □

Corollary 16 (linear map to a lower-dimen space is not injective). Suppose V and W are finite-dimensional vector spaces such that $\dim V > \dim W$. Then no linear map from V to W is injective.

Corollary 17 (linear map to a higher-dimen space is not surjective). Suppose V and W are finite-dimensional vector spaces such that $\dim V < \dim W$. Then no linear map from V to W is surjective.

Application Consider the system of linear equation defined by the map $T: \mathbb{F}^n \rightarrow \mathbb{F}^m$:

$$T(x_1, \dots, x_n) = \left(\sum_{k=1}^n A_{1,k} x_k, \dots, \sum_{k=1}^n A_{m,k} x_k \right)$$

Corollary 18 (homogeneous systems of linear equations). *A homogeneous system of linear equations with more variables than equations has nonzero solutions.*

Corollary 19 (inhomogeneous system of linear equations). *An inhomogeneous system of linear equations with more equations than variables has no solution for some choice of the constant terms.*

Problem 1

Give an example of a linear map T with $\dim \text{null } T = 3$ and $\dim \text{range } T = 2$.

Proof. Consider $T(x_1, x_2, x_3, x_4, x_5) = (0, 0, 0, x_4, x_5)$.

It's easy to verify that $\dim \text{null } T = 3$ and applying the theorem of linear map solves the problem. $\square$

Problem 2

Suppose $S, T \in \mathcal{L}(V)$ are such that $\text{range } S \subseteq \text{null } T$. Prove that $(ST)^2 = 0$.

Proof. This means that take $x \in V$, $T(S(x)) = 0$. We have $(ST)^2 = STST = 0$. $\square$

Problem 3

Suppose $v_1, \dots, v_m$ is a list of vectors in V . Define $T \in \mathcal{L}(\mathbb{F}^m, V)$ by

$$T(z_1, \dots, z_m) = z_1 v_1 + \dots + z_m v_m$$

- (a) what property of T corresponds to $v_1, \dots, v_m$ spanning V ?
- (b) what property of T corresponds to $v_1, \dots, v_m$ being linearly independent?

Proof. (a) surjective ($\text{range} = V$)

(b) injective ($z_1, \dots, z_m$ is identically zero if and only if $z_1 v_1 + \dots + z_m v_m = 0$) $\square$

Problem 4

Show that $\{T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4): \dim \text{null } T > 2\}$ is not a subspace of $\mathcal{L}(\mathbb{R}^5, \mathbb{R}^4)$.

Proof. Try to come up with a counterexample. Consider $f(e_1, e_2, e_3, e_4, e_5) = (e_1, 0, 0, e_4)$ and $g(e_1, e_2, e_3, e_4, e_5) = (0, e_2, e_3, 0)$. Then we have that $\text{null } f = \{0, e_2, e_3, 0, e_5\}$ and $\text{null } g = \{e_1, 0, 0, e_4, e_5\}$ so both in $\{T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4) : \dim \text{null } T > 2\}$. However, $f + g(e_1, e_2, e_3, e_4, e_5) = \{(e_1, e_2, e_3, e_4)\}$ and this means their $\dim \text{null} = 5 - 4 = 1 \nmid 3$. $\square$

Problem 7

Suppose V and W are finite-dimensional and $2 \leq \dim V \leq \dim W$. Show that $\{T \in \mathcal{L}(V, W) : T \text{ is not injective}\}$ is not a subspace of $\mathcal{L}(V, W)$.

Proof. T is not injective means that $\dim V > \dim W$, which contradicts the assumption in the question. $\square$

Problem 9

Suppose $T \in \mathcal{L}(V, W)$ is injective and $v_1, \dots, v_n$ is linearly independent in V . Prove that $Tv_1, \dots, Tv_n$ is linearly independent in W .

Proof.

$$\sum_{i=1}^n z_i v_i = 0 \iff \sum_{i=1}^n z_i T(v_i) = 0$$

$\square$

Problem 10

Suppose $v_1, \dots, v_n$ spans V and $T \in \mathcal{L}(V, W)$. Show that $Tv_1, \dots, Tv_n$ spans $\text{range } T$.

Proof. Take v in V , then we know $v = \sum_{i=1}^n a_i v_i$. Take w in $\text{range } T$. Then we know there exists $v \in V$ such that $w = T(v)$ and thus $w = T(\sum_{i=1}^n a_i v_i) = \sum_{i=1}^n a_i T(v_i)$. $\square$

Problem 11

Suppose that V is finite-dimensional and that $T \in \mathcal{L}(V, W)$. Prove that there exists a subspace $\mathcal{U}$ of V such that

$$\mathcal{U} \cap \text{null } T = \{0\} \text{ and } \text{range } T = \{Tu : u \in \mathcal{U}\}$$

Proof. Since we know $\text{null } T$ is a subspace of V , then there exists $\mathcal{U}$ such that $\mathcal{U} \oplus \text{null } T = V$. We can define $\mathcal{U}$ to be such case. To finish the proof, let $v = u + t, u \in \mathcal{U}, t \in \text{null } T$, then $\text{range } T = \{T(u + t) = Tu : u \in \mathcal{U}\}$. $\square$

Problem 16

Suppose V and W are both finite-dimensional. Prove that there exists an injective linear map from V to W if and only if $\dim V \leq \dim W$.

Proof. $\Rightarrow T \in \mathcal{L}(V, W)$ is injective. Then this means that $\dim V = \dim \text{range } T \leq \dim W$.

$\Leftarrow$ By assumption, we can define a linear map T such that $Tv_i = w_i$ where v_i, w_i are the respective basis of V and W . Then we have that

$$\sum_{i=1}^n a_i Tv_i = \sum_{i=1}^n a_i w_i = 0 \iff a_i \text{ are identically zero}$$

This means that $\text{null } T$ is $\{0\}$ and thus it is injective. $\square$

Problem 17

Suppose V and W are both finite-dimensional. Prove that there exists a surjective linear map from V onto W if and only if $\dim V \geq \dim W$.

Proof. $\Rightarrow \dim V \geq \text{range } T = \dim W$.

$\Leftarrow$ By assumption we can define a linear map T such that $Tv_i = w_i$ for $1 \leq i \leq \dim W$ and $Tv_i = 0$ for $\dim W \leq i \leq \dim V$. This means that take $w \in W$, then $w = \sum_{i=1}^{\dim W} a_i w_i$. At the same time, we know that for all w_i , there exists v_i s.t. $Tv_i = w_i$, therefore we have $w = \sum_{i=1}^{\dim W} a_i Tv_i$ and thus T is surjective. $\square$

Problem 18

Suppose V and W are finite-dimensional and that $\mathcal{U}$ is a subspace of V . Prove that there exists $T \in \mathcal{L}(V, W)$ such that $\text{null } T = \mathcal{U}$ if and only if $\dim \mathcal{U} \geq \dim V - \dim W$.

Proof. We know $\dim V = \text{null } T + \text{range } T$ and $\text{range } T \leq \dim W$. Therefore, $\dim \mathcal{U} = \dim \text{null } T \geq \dim V - \dim W$. $\square$

Problem 19

Suppose W is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that T is injective if and only if there exists $S \in \mathcal{L}(W, V)$ such that ST is the identity operator on V .

Proof. T is injective $\iff T(v)$ is unique for $v \in V$. $\iff$ We can define $S: STv = v$. $\square$

Problem 21

Suppose V is finite-dimensional, $T \in \mathcal{L}(V, W)$, and $\mathcal{U}$ is a subspace of W . Prove that $\mathcal{T} = \{v \in V: Tv \in \mathcal{U}\}$ is a subspace of V and

$$\dim \mathcal{T} = \dim \text{null } T + \dim(\mathcal{U} \cap \text{range } T)$$

Proof. To prove subspace, we can simply follow by definition. We can define $S \in \mathcal{L}(\mathcal{T}, \mathcal{U})$ such that $Sv = Tv$ for all $v \in \mathcal{T}$. Then we have that $\text{range}(S) \in \mathcal{U} \cap \text{range}(T)$ and that $\text{null } T = \text{null } S$. Hence we have proved the claim. $\square$

Problem 22

Suppose $\mathcal{U}$ and V are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(\mathcal{U}, V)$. Prove that

$$\dim \text{null } ST \leq \dim \text{null } S + \dim \text{null } T$$

Proof. We can write that

$$\text{null } ST = \{u \in \mathcal{U} : T(u) \in \text{null } (S)\}$$

By the previous question, we know that

$$\dim \text{null } ST = \dim \text{null } T + \dim \text{null } S \cap \text{range } T \leq \dim \text{null } T + \dim \text{null } S$$

□

Problem 23

Suppose $\mathcal{U}$ and V are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(\mathcal{U}, V)$. Prove that

$$\dim \text{range } ST \leq \min\{\dim \text{range } S, \dim \text{range } T\}$$

Proof. First note that $\dim \text{range } ST \leq \dim \text{range } S$. This because $\text{range } ST \subseteq \text{range } S$.

To prove $\dim \text{range } ST \leq \dim \text{range } T$, we have that $\dim \mathcal{U} = \dim \text{null } T + \dim \text{range } T = \dim \text{null } ST + \dim \text{range } ST$. Since we know $\dim \text{null } T \geq \dim \text{null } ST$, $\dim \text{range } ST \leq \dim \text{range } T$. □

Problem 24

- (a) Suppose $\dim V = 5$ and $S, T \in \mathcal{L}(V)$ are such that $ST = 0$. Prove that $\dim \text{range } TS \leq 2$.
- (b) Give an example of $S, T \in \mathcal{L}(\mathbb{F}^5)$ with $ST = 0$ and $\dim \text{range } TS = 2$.

Proof. (a) $\dim \text{null } ST = 5 \leq \dim \text{null } T + \dim \text{null } S$

We also know that $5 = \dim \text{null } T + \dim \text{range } T = \dim \text{range } S + \dim \text{null } S$ and thus we have that $\dim \text{range } T + \dim \text{range } S \leq 5$. This implies that $\min\{\dim \text{range } T, \dim \text{range } S\} \leq 2$. Hence, by applying P23, we finish the proof.

(b) Consider $T(x_1, x_2, x_3, x_4, x_5) = (x_3, x_4, 0, 0, 0)$, $S(x_1, x_2, x_3, x_4, x_5) = (0, 0, x_3, x_4, x_5)$. Then we have that $ST(x_1, x_2, x_3, x_4, x_5) = S(x_3, x_4, 0, 0, 0) = (0, 0, 0, 0, 0)$ while $TS(x_1, x_2, x_3, x_4, x_5) = T(0, 0, x_3, x_4, x_5) = (x_3, x_4, 0, 0, 0)$. □

Problem 25

Suppose that W is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\text{null } S \subseteq \text{null } T$ if and only if there exists $E \in \mathcal{L}(W)$ such that $T = ES$.

Proof. $\text{null } S \subseteq \text{null } T \Rightarrow \dim \text{range } T \leq \dim \text{range } S \Rightarrow$ let $s_1, \dots, s_n$ be basis of $\text{range } S$ and $t_1, \dots, t_m$ be basis of $\text{range } T$ where $m \leq n$. We can always define $E \in \mathcal{L}(W)$ such that $E(s_i) = t_i$ for all $1 \leq i \leq m$ and 0 otherwise.

The other direction is trivial. $\square$

Problem 26

Suppose that V is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\text{range } S \subseteq \text{range } T$ if and only if there exists $E \in \mathcal{L}(V)$ such that $S = TE$.

Proof. $\text{range } S \subseteq \text{range } T \Rightarrow$ Take $v_1, \dots, v_m$ to be the basis of V , define the linear map $Sw_i = s_i$ for $s_i \in \text{range } S \subseteq \text{range } T$. Then there exists $u_1, \dots, u_m$ such that $Tu_i = s_i$ for all i . Then we can define the linear map $Ev_i = u_i$ for all i such that $S = TE$.

The other direction is trivial. $\square$

Problem 27

Suppose $P \in \mathcal{L}(V)$ and $P^2 = P$. Prove that $V = \text{null } P \oplus \text{range } P$.

Proof. Take $v \in \text{null } (P) \cap \text{range } (P)$. Then this means that $P(v) = 0$ and there exists u s.t. $v = P(u)$. At the same time, $P(v) = P(P(u)) = P(u) = 0 = v$. So we have that $\text{null } P \cap \text{range } P = \{0\}$. Since P is defined on $\mathcal{L}(P)$, we have that $V = \text{null } P \oplus \text{range } P$. $\square$

3C: Matrices

Definition 20 (matrix, $\mathbf{A}_{j,k}$). Suppose m and n are nonnegative integers. An m -by- n matrix $\mathbf{A}$ is a rectangular array of elements of $\mathbb{F}$ with m rows and n columns:

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{1,1} & \cdots & \mathbf{A}_{1,n} \\ \vdots & & \vdots \\ \mathbf{A}_{m,1} & \cdots & \mathbf{A}_{m,n} \end{pmatrix}$$

where the notation $\mathbf{A}_{j,k}$ denotes the entry in row j and column k .

Definition 21 (matrix of a linear map, $\mathcal{M}(T)$). Suppose $T \in \mathcal{L}(V, W)$ and $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ is a basis of W . The matrix of T with respect to these bases is the m -by- n matrix $\mathcal{M}(T)$ whose entries $\mathbf{A}_{j,k}$ are defined by

$$Tv_k = \mathbf{A}_{1,k}w_1 + \cdots + \mathbf{A}_{m,k}w_m$$

If the bases $v_1, \dots, v_n$ and $w_1, \dots, w_m$ are not clear from the context, then the notation $\mathcal{M}(T, (v_1, \dots, v_n), (w_1, \dots, w_m))$ is used.

Remark 22. The k -th column of $\mathcal{M}(T)$ consists of the scalars needed to write Tv_k as a linear combination of $w_1, \dots, w_m$:

$$Tv_k = \sum_{j=1}^m \mathbf{A}_{j,k}w_j$$

Remark 23. If T is a linear map from n -dimensional vector space to an m -dimensional vector space, then $\mathcal{M}(T)$ is an m -by- n matrix.

Corollary 24 (Matrix addition and scalar multiplication). Suppose $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$. Then $\mathcal{M}(\lambda S + T) = \lambda \mathcal{M}(S) + \mathcal{M}(T)$.

For m and n positive integers, the set of all m -by- n matrices with entries in F is denoted by $\mathbb{F}^{m,n}$.

Theorem 25. Suppose m and n are positive integers. $\mathbb{F}^{m,n}$ is a vector space of dimension mn .

Definition 26 (matrix multiplication). Suppose $\mathbf{A}$ is an m -by- n matrix and $\mathbf{B}$ is an n -by- p matrix. Then $\mathbf{AB}$ is defined to be the m -by- p matrix whose entry in row j , column k , is given by the equation

$$(\mathbf{AB})_{j,k} = \sum_{r=1}^n \mathbf{A}_{j,r} \mathbf{B}_{r,k}$$

In words, the entry in row j , column k of $\mathbf{AB}$ is computed by taking row j of $\mathbf{A}$ and column k of $\mathbf{B}$.

Motivations. Let $v_1, \dots, v_n$ to be the basis of V , $w_1, \dots, w_m$ to be the basis of W , and $u_1, \dots, u_p$ to be the basis of U . Consider linear maps $T: U \rightarrow V$ and $S: V \rightarrow W$. Suppose $\mathcal{M}(T) = \mathbf{A}$ and $\mathcal{M}(S) = \mathbf{B}$. For $1 \leq k \leq p$, we have

$$\begin{aligned} (ST)u_k &= S \left(\sum_{r=1}^n \mathbf{B}_{r,k} v_r \right) \\ &= \sum_{r=1}^n \mathbf{B}_{r,k} (Sv_r) \\ &= \sum_{r=1}^n \mathbf{B}_{r,k} \sum_{j=1}^m \mathbf{A}_{j,r} w_j \\ &= \sum_{j=1}^m \left(\sum_{r=1}^n \mathbf{A}_{j,r} \mathbf{B}_{r,k} \right) w_j \end{aligned}$$

Theorem 27 (matrix of product of linear maps). *If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$.*

Definition 28 ($\mathbf{A}_{j,\cdot}, \mathbf{A}_{\cdot,k}$). *Suppose $\mathbf{A}$ is an m -by- n matrix*

- *If $1 \leq j \leq m$, then $\mathbf{A}_{j,\cdot}$ denotes the 1-by- n matrix consisting of row j of $\mathbf{A}$.*
- *If $1 \leq k \leq n$, then $\mathbf{A}_{\cdot,k}$ denotes the m -by-1 matrix consisting of column k of $\mathbf{A}$.*

Corollary 29 (entry of matrix product equals row times column). *Suppose $\mathbf{A}$ is an m -by- n matrix and $\mathbf{B}$ is an n -by- p matrix. Then*

$$(\mathbf{AB})_{j,k} = \mathbf{A}_{j,\cdot} \mathbf{B}_{\cdot,k}$$

if $1 \leq j \leq m$ and $1 \leq k \leq p$. In other words, the entry in row j , column k , of $\mathbf{AB}$ equals (row j of $\mathbf{A}$) times (column k of $\mathbf{B}$).

Corollary 30 (column of matrix product equals matrix times column). *Suppose $\mathbf{A}$ is an m -by- n matrix and $\mathbf{B}$ is an n -by- p matrix. Then*

$$(\mathbf{AB})_{\cdot,k} = \mathbf{A} \mathbf{B}_{\cdot,k}$$

if $1 \leq k \leq p$. In other words, column k of $\mathbf{AB}$ equals $\mathbf{A}$ times column k of $\mathbf{B}$.

Corollary 31 (linear combination of columns). *Suppose $\mathbf{A}$ is an m -by- n matrix*

and $\mathbf{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ is an n -by-1 matrix. Then

$$\mathbf{Ab} = \mathbf{b}_1 \mathbf{A}_{\cdot,1} + \dots + \mathbf{b}_n \mathbf{A}_{\cdot,n}.$$

In other words, $\mathbf{A}\mathbf{b}$ is a linear combination of the columns of $\mathbf{A}$, with the scalars that multiply the columns coming from $\mathbf{b}$.

Theorem 32 (matrix multiplication as linear combination of columns). *Suppose C is an m -by- c matrix and R is a c -by- n matrix.*

- If $k \in \{1, \dots, n\}$, then column k of CR is a linear combination of the columns of C , with the coefficients of this linear combination coming from column k of R .
- If $j \in \{1, \dots, m\}$, then row j of CR is a linear combination of the rows of R , with the coefficients of this linear combination coming from row j of C .

Definition 33 (column rank, row rank). *Suppose $\mathbf{A}$ is an m -by- n matrix with entries in $\mathbb{F}$.*

- The column rank of $\mathbf{A}$ is the dimension of the span of the columns of $\mathbf{A}$ in $\mathbb{F}^{m,1}$.
- The row rank of $\mathbf{A}$ is the dimension of the span of the rows of $\mathbf{A}$ in $\mathbb{F}^{1,n}$.

Definition 34 (transpose, $\mathbf{A}^\top$). *The transpose of a matrix $\mathbf{A}$, denoted by $\mathbf{A}^\top$, is the matrix obtained from $\mathbf{A}$ by interchanging rows and columns. Specifically, if $\mathbf{A}$ is an m -by- n matrix, then $\mathbf{A}^\top$ is the n -by- m matrix whose entries are given by the equation*

$$(\mathbf{A}^\top)_{k,j} = \mathbf{A}_{j,k}$$

Lemma 35 (column-row factorization). *Suppose $\mathbf{A}$ is an m -by- n matrix with entries in $\mathbb{F}$ and column rank $c \geq 1$. Then there exists an m -by- c matrix $\mathbf{C}$ and a c -by- n matrix $\mathbf{R}$, both with entries in $\mathbb{F}$, such that $\mathbf{A} = \mathbf{C}\mathbf{R}$.*

Proof. Each column of $\mathbf{A}$ is an m -by-1 matrix. The list $\mathbf{A}_{\cdot,1}, \dots, \mathbf{A}_{\cdot,n}$ of columns of $\mathbf{A}$ can be reduced to a basis of the span of the columns of $\mathbf{A}$. This basis has length c . The c columns in this basis can be put together to form an m -by- c matrix $\mathbf{C}$.

If $k \in \{1, \dots, n\}$, then column k of $\mathbf{A}$ is a linear combination of the columns of $\mathbf{C}$. Make the coefficients of this linear combination into column k of a c -by- n matrix that we call $\mathbf{R}$. Then $\mathbf{A} = \mathbf{C}\mathbf{R}$. $\square$

Theorem 36 (column rank equals row rank). *Suppose $\mathbf{A} \in \mathbb{F}^{m,n}$. Then the column rank of $\mathbf{A}$ equals the row rank of $\mathbf{A}$.*

Proof. Let c denote the column rank of $\mathbf{A}$. Let $\mathbf{A} = \mathbf{C}\mathbf{R}$ be the column-row factorization of $\mathbf{A}$ given by the proof before, where $\mathbf{C}$ is m -by- c and $\mathbf{R}$ is c -by- n . Then the column-row factorization lemma tells us that every row of $\mathbf{A}$ is a linear combination of the rows of $\mathbf{R}$. Because $\mathbf{R}$ has c rows, this implies that the row of $\mathbf{A}$ is less than or equal to the column rank c of $\mathbf{A}$.

To prove the other direction, we can do the same thing to $\mathbf{A}^\top$ and then we can get that column rank of $\mathbf{A} = \text{row rank of } \mathbf{A}^\top \leq \text{column rank of } \mathbf{A}^\top = \text{row rank of } \mathbf{A}$ which we proved above. $\square$

Problem 1

Suppose $T \in \mathcal{L}(V, W)$. Show that with respect to each choice of bases of V and W , the matrix of T has at least $\dim \text{range } T$ nonzero entries.

Proof. Let $v_1, \dots, v_n$ be the basis of V and $w_1, \dots, w_m$ be the basis of W . Suppose there exists a matrix $\mathbf{A} = \mathcal{M}(T)$ of T has less than $\dim \text{range } T$ nonzero entries. This means that $\mathbf{A}$ has at most $\dim \text{range } T - 1$ nonzero columns so this implies that $\dim \text{range } T < \dim \text{range } T - 1$ which forms a contradiction. $\square$

Problem 2

Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that $\dim \text{range } T = 1$ if and only if there exists a basis of V and a basis of W such that with respect to these bases, all entries of $\mathcal{M}(T)$ equal 1.

Proof. $\Rightarrow$ We solve this problem through careful construction of the basis. That is, we will construct basis $v_1, \dots, v_n$ of V and $w_1, \dots, w_m$ of W such that $Tv_i = w_1 + \dots + w_m$ for all i . We can achieve this again mainly because $\dim \text{range } T = 1$. let $u_1, \dots, u_m$ be a set of arbitrary basis of W . Take $w \in \text{range } T$ so we have $w = \sum_{i=1}^m a_i u_i$. We consider all the coefficients a_i as follows: take an index set I such that for all $i \in I, a_i \neq 0$ and we have $|I| = c$. Take arbitrary $j \in I$ and we construct the basis as follows: let $w_j = (a_j - (m - r))u_j, w_k = a_k u_k$ for all $k \in I, k \neq j$, and $w_k = u_k + u_j$ for all $k \notin I$. Then we have that $w = \sum_{i=1}^m w_i$. Since we know $w \in \text{range } T$, then there exists $v_1 \in V$ s.t. $T(v_1) = w$. Let $v_2, \dots, v_n$ be the basis of $\text{null } T$. Since we know $\dim V = \dim \text{null } T + 1$ so $v_1, v_2, \dots, v_n$ constitutes a basis of V . In this way, we successfully constructs the basis such that all entries of $\mathcal{M}(T)$ equal 1.

$\Leftarrow$ There exists basis $v_1, \dots, v_n$ of V and $w_1, \dots, w_m$ of W such that all entries of $\mathcal{M}(T)$ equal 1. This means that $Tv_1 = \dots = Tv_n = w_1 + \dots + w_m$. So we first have that $\dim \text{range } T \neq 0$. To prove $\dim \text{range } T = 1$, take arbitrary $u_1, u_2 \in \text{range } T$. Then we know that $u_1 = T(\sum_{i=1}^n a_i v_i) = \sum_{i=1}^n a_i (w_1 + \dots + w_m)$ and similarly $u_2 = \sum_{i=1}^n b_i (w_1 + \dots + w_m)$. Here, we have that

$$u_1 = \left(\sum_i^n a_i \right) / \left(\sum_i^n b_i \right) u_2$$

and thus $\dim \text{range } T$ has to be 1-dimensional since every two arbitrary vector is simply a scalar multiple of each other. $\square$

Problem 3

Suppose $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ is a basis of W . Show that if $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$, then $\mathcal{M}(\lambda S + T) = \lambda \mathcal{M}(S) + \mathcal{M}(T)$.

Proof.

$$\begin{aligned} (\lambda S + T)v_k &= \lambda S(v_k) + T(v_k) \\ &= \lambda \sum_{i=1}^m \mathbf{A}_{i,k} w_i + \sum_{i=1}^m \mathbf{B}_{i,k} w_i \end{aligned}$$

□

Problem 5

Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that there exists a basis of V and a basis of W such that with respect to these bases, all entries of $\mathcal{M}(T)$ are 0 except that the entries in row k , column k equal 1 if $1 \leq k \leq \dim \text{range } T$.

Proof. Let $\dim V = n, \dim W = m, \dim \text{range } T = k$. The main idea of the proof is to construct basis $v_1, \dots, v_n$ and $w_1, \dots, w_m$ such that $Tv_i = w_i$ for all $1 \leq i \leq k$ and zero otherwise. (So the constructed T satisfies the requirement).

We proceed with obtaining the basis $w_1, \dots, w_k$ of range T and extend this $w_1, \dots, w_k, \dots, w_m$ to the basis of W . We know for all $1 \leq i \leq k$, there exists $v_i \in V$ s.t. $Tv_i = w_i$. We claim that $v_1, \dots, v_k$ are linearly independent. To prove this, see $\sum_{j=1}^k a_j v_j = 0 \Rightarrow \sum_{j=1}^k a_j T(v_j) = \sum_{j=1}^k a_j w_j = 0$. Similar to the proof in the fundamental theorem of linear map, we extend the basis to V through considering the null space. We further claim that (let $K = \text{span}(v_1, \dots, v_k)$) $V = K \oplus \text{null } T$ (which is easy to prove). Hence, extending the basis from K with the basis from null T completes the proof. □

Problem 6

Suppose $v_1, \dots, v_m$ is a basis of V and W is finite-dimensional. Suppose $T \in \mathcal{L}(V, W)$. Prove that there exists a basis $w_1, \dots, w_n$ of W such that all entries in the first column of $\mathcal{M}(T)$ [with respect to these bases] are 0 except for possibly a 1 in the first row, first column.

Proof. Let $w_1 = Tv_1$ and extend w_1 to basis of W . Then we automatically obtains T that gets the desired property. □

Problem 7

Suppose $w_1, \dots, w_n$ is a basis of W and V is finite-dimensional. Suppose $T \in \mathcal{L}(V, W)$. Prove that there exists a basis $v_1, \dots, v_m$ of V such that all entries in the first of $\mathcal{M}(T)$ [wrt. these bases] are 0 except for possibly a 1 in the first row, first column.

Proof. Take arbitrary $u_1 \in V$ and we have that $Tu_1 = \sum_{i=1}^n a_{i,1} w_i$. Take $v_1 = u_1/a_{1,1}$, then $Tv_1 = w_1 + \sum_{j=2}^n b_{j,1} w_j$. We can extend $v_1, u_2, \dots, u_m$ to be

the basis of V . Then consider

$$Tu_j = \sum_{k=1}^n b_{k,j} w_k$$

Consider to let $b_{1,j} = 0$ for all $2 \leq j \leq n$. To do this, let $v_j = u_j - b_{1,j}v_1$, then we have that $T(v_j) = T(u_j - b_{1,j}v_1) = \sum_{k=2}^n c_{k,j} w_k$. It now left to verify that $v_1, v_2, \dots, v_m$ is the basis.

$$\begin{aligned} q_1 v_1 + \dots + q_m v_m &= 0 \\ q_1 v_1 + q_2(u_2 - b_{1,2}v_1) + \dots + q_m(u_m - b_{1,m}v_1) &= 0 \\ (q_1 - (q_2 b_{1,2} + \dots + q_m b_{1,m}))v_1 + q_2 u_2 + \dots + q_m u_m &= 0 \end{aligned}$$

The only solution is that $q_1 - (q_2 b_{1,2} + \dots + q_m b_{1,m}) = q_2 = \dots = q_m = 0$ and thus we have completed the proof. $\square$

Problem 8

Suppose $\mathbf{A}$ is an m -by- n matrix and $\mathbf{B}$ is an n -by- p matrix. Prove that

$$(\mathbf{AB})_{j,\cdot} = \mathbf{A}_{j,\cdot} \mathbf{B}$$

for each $1 \leq j \leq m$. In other words, show that row j of $\mathbf{AB}$ equals (row j of $\mathbf{A}$) times $\mathbf{B}$.

Proof. We know that

$$(\mathbf{AB})_{j,k} = \sum_{i=1}^n \mathbf{A}_{j,i} \mathbf{B}_{i,k}$$

For instance $(\mathbf{AB})_{j,1} = \sum_{i=1}^n \mathbf{A}_{j,i} \mathbf{B}_{i,1} = \mathbf{A}_{j,\cdot} \mathbf{B}_{\cdot,1}$. Similarly, $(\mathbf{AB})_{j,k} = \mathbf{A}_{j,\cdot} \mathbf{B}_{\cdot,k}$ and thus by treating this we have

$$(\mathbf{AB})_{j,\cdot} = \mathbf{A}_{j,\cdot} \mathbf{B}$$

$\square$

Problem 9

Suppose $\mathbf{a} = (a_1, \dots, a_n)$ is a 1-by- n matrix and $\mathbf{B}$ is an n -by- p matrix. Prove that

$$\mathbf{aB} = a_1 \mathbf{B}_{1,\cdot} + \dots + a_n \mathbf{B}_{n,\cdot}$$

In other words, show that $\mathbf{aB}$ is a linear combination of the rows of $\mathbf{B}$, with the scalars that multiply the rows coming from $\mathbf{a}$.

Proof. By definition of matrix multiplication, the entry in the column k of the l.h.s. equals

$$(\mathbf{aB})_{1,k} = a_1 \mathbf{B}_{1,k} + \dots + a_n \mathbf{B}_{n,k}$$

which equals the right-hand side. $\square$

Problem 11

Prove that the distributive property holds for matrix addition and multiplication. In other words, suppose $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}, \mathbf{E}, \mathbf{F}$ are matrices whose sizes are such that $\mathbf{A}(\mathbf{B} + \mathbf{C})$ and $(\mathbf{D} + \mathbf{E})\mathbf{F}$ make sense. Explain this and prove that

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC} \quad (\mathbf{D} + \mathbf{E})\mathbf{F} = \mathbf{DF} + \mathbf{EF}$$

Proof. We let $\mathbf{A}, \mathbf{D}, \mathbf{E}$ to be m -by- n matrix and $\mathbf{B}, \mathbf{C}, \mathbf{F}$ to be n -by- p matrix. Then all the matrix multiplication make sense. We proceed to prove the equality:

$$\begin{aligned} (\mathbf{A}(\mathbf{B} + \mathbf{C}))_{j,k} &= \sum_{i=1}^n \mathbf{A}_{j,i}(\mathbf{B} + \mathbf{C})_{i,k} \\ &= \sum_{i=1}^n \mathbf{A}_{j,i}(\mathbf{B}_{i,k} + \mathbf{C}_{i,k}) \\ &= \sum_{i=1}^n (\mathbf{A}_{j,i}\mathbf{B}_{i,k}) + (\mathbf{A}_{j,i}\mathbf{C}_{i,k}) \\ &= \mathbf{AB} + \mathbf{AC} \end{aligned}$$

For the other one, we have that

$$\begin{aligned} ((\mathbf{D} + \mathbf{E})\mathbf{F})_{j,k} &= \sum_{i=1}^n (\mathbf{D} + \mathbf{E})_{j,i}\mathbf{F}_{i,k} \\ &= \sum_{i=1}^n (\mathbf{D}_{j,i}\mathbf{F}_{i,k}) + (\mathbf{E}_{j,i}\mathbf{F}_{i,k}) \\ &= \mathbf{DF} + \mathbf{EF} \end{aligned}$$

□

Problem 13

Suppose $\mathbf{A}$ is an n -by- n matrix and $1 \leq j, k \leq n$. Show that the entry in row j , column k of $\mathbf{A}^3$ (which is defined to mean $\mathbf{AAA}$) is

$$\sum_{p=1}^n \sum_{r=1}^n \mathbf{A}_{j,p} \mathbf{A}_{p,r} \mathbf{A}_{r,k}$$

Proof.

$$\begin{aligned}
(\mathbf{A}(\mathbf{A}\mathbf{A}))_{j,k} &= \sum_p^n \mathbf{A}_{j,p}(\mathbf{A}\mathbf{A})_{p,k} \\
&= \sum_p^n \mathbf{A}_{j,p} \sum_{r=1}^n \mathbf{A}_{p,r} \mathbf{A}_{r,k} \\
&= \sum_p^n \sum_r^n \mathbf{A}_{j,p} \mathbf{A}_{p,r} \mathbf{A}_{r,k}
\end{aligned}$$

□

Problem 14

Suppose m and n are positive integers. Prove that the function $\mathbf{A} \mapsto \mathbf{A}^\top$ is a linear map from $\mathbf{F}^{m,n}$ to $\mathbf{F}^{n,m}$.

Proof. Let T denote the linear map such that $T(\mathbf{A}) = \mathbf{A}^\top$. Then we have that

$$T(\lambda\mathbf{A} + \mathbf{B}) = (\lambda\mathbf{A} + \mathbf{B})^\top = \lambda\mathbf{A}^\top + \mathbf{B}^\top = \lambda T(\mathbf{A}) + T(\mathbf{B})$$

□

Problem 15

Prove that if $\mathbf{A}$ is an m -by- n matrix and $\mathbf{C}$ is an n -by- p matrix, then

$$(\mathbf{AC})^\top = \mathbf{C}^\top \mathbf{A}^\top$$

Proof.

$$\begin{aligned}
((\mathbf{AC})^\top)_{j,k} &= (\mathbf{AC})_{k,j} \\
&= \sum_{i=1}^n \mathbf{A}_{k,i} \mathbf{C}_{i,j}
\end{aligned}$$

On the other hand,

$$\begin{aligned}
(\mathbf{C}^\top \mathbf{A}^\top)_{j,k} &= \sum_{i=1}^n \mathbf{C}_{j,i}^\top \mathbf{A}_{i,k}^\top \\
&= \sum_{i=1}^n \mathbf{A}_{k,i} \mathbf{C}_{i,j}
\end{aligned}$$

□

Problem 16

Suppose $\mathbf{A}$ is an m -by- n matrix with $\mathbf{A} \neq 0$. Prove that the rank of $\mathbf{A}$ is 1 if and only if there exist $(c_1, \dots, c_m) \in \mathbb{F}^m$ and $(d_1, \dots, d_n) \in \mathbb{F}^n$ such that $\mathbf{A}_{j,k} = c_j d_k$ for every $j = 1, \dots, m$ and every $k = 1, \dots, n$.

Proof. $\Rightarrow$ If rank of $\mathbf{A}$ is 1, then this means that the span of the columns of $\mathbf{A}$ is 1-dimensional, take $\mathbf{A}_{\cdot,1} \in \mathbb{F}^m$ from the span. (Note that here we assume $\mathbf{A}_{\cdot,1} \neq 0$ ow. take the first nonzero one) Then we can let $(c_1, \dots, c_m) = \text{vec}(\mathbf{A}_{\cdot,1})$. and let For every other columns, they are the scalar multiple of the the first one, so there exists $d_2, \dots, d_n$ s.t. $\mathbf{A}_{j,k} = c_j d_k$ (we take $d_1 = 1$).

$\Leftarrow$ If we denote $\mathbf{c} = (c_1, \dots, c_m)$, then

$$\mathbf{A} = \begin{pmatrix} d_1 \mathbf{c} & \cdots & d_n \mathbf{c} \end{pmatrix}$$

so each column is a scalar multiple of the other and thus $\mathbf{A}$ has rank 1. $\square$

Problem 17

Suppose $T \in \mathcal{L}(V)$, and $u_1, \dots, u_n$ and $v_1, \dots, v_n$ are bases of V . Prove that the following are equivalent:

- (a) T is injective.
 - (b) The column of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{n,1}$.
 - (c) The columns of $\mathcal{M}(T)$ span $\mathbb{F}^{n,1}$.
 - (d) The rows of $\mathcal{M}(T)$ span $\mathbb{F}^{1,n}$.
 - (e) The rows of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{1,n}$.
- Here $\mathcal{M}(T)$ means $\mathcal{M}(T, (u_1, \dots, u_n), (v_1, \dots, v_n))$.

Proof. Let $\mathbf{A}$ represents the matrix of $\mathcal{M}(T)$.

(a) $\Rightarrow$ (b): $a_1 \mathbf{A}_{\cdot,1} + \dots + a_n \mathbf{A}_{\cdot,n} = 0$. We also know that $Tv_i = \sum_{j=1}^n \mathbf{A}_{j,i} u_j$. Then we have that

$$T(a_1 v_1 + \dots + a_n v_n) = \sum_{i=1}^n a_i T(v_i) = \sum_{i=1}^n a_i \sum_{j=1}^n \mathbf{A}_{j,i} u_j = \sum_{j=1}^n \left(\sum_{i=1}^n a_i \mathbf{A}_{j,i} \right) u_j = 0$$

To prove all a_i is zero, we note that $a_1 v_1 + \dots + a_n v_n \in \text{null } T$ and since T is injective, the only possible case is that all $a_i = 0$ and thus we prove the claim.

(b) $\Rightarrow$ (c) The columns of $\mathcal{M}(T)$ forms a basis and thus span the space.

(c) $\Rightarrow$ (d) Column rank of $\mathcal{M}(T)$ is n and thus row rank is also n and thus the rows span $\mathbb{F}^{1,n}$.

(d) $\Rightarrow$ (e) Same as above.

(e) $\Rightarrow$ (a) The only solution to $a_1 \mathbf{A}_{1,\cdot} + \dots + a_n \mathbf{A}_{n,\cdot} = 0$ is all $a_i = 0$. Suppose for the sake of contradiction that T is not injective so there exists nonzero $v \in \text{null } T$. We know that $v = \sum_{i=1}^n a_i v_i$ and that $Tv_i = \sum_{j=1}^n \mathbf{A}_{j,i} u_j$. Then

$$Tv = \sum_{i=1}^n a_i T(v_i) = \sum_{i=1}^n a_i \sum_{j=1}^n \mathbf{A}_{j,i} u_j = \sum_{j=1}^n \left(\sum_{i=1}^n a_i \mathbf{A}_{j,i} \right) u_j = 0$$

and thus $\sum_{j=1}^n a_i \mathbf{A}_{j,i} = 0$ for all j . We know not all $a_i = 0$ since v is non-trivial and thus the solution contradicts that the rows are all linearly independent. Proof completed. $\square$

3D: Invertibility and Isomorphisms

Definition 37 (invertible, inverse). • A linear map $T \in \mathcal{L}(V, W)$ is called invertible if there exists a linear map $S \in \mathcal{L}(W, V)$ such that ST equals the identity operator on V and TS equals the identity operator on W .

- A linear map $S \in \mathcal{L}(W, V)$ satisfying $ST = I$ and $TS = I$ is called an inverse of T .

Theorem 38. An invertible linear map has a unique inverse.

If T is invertible, then its inverse is denoted by T^{-1} .

Theorem 39. A linear map is invertible if and only if it is injective and surjective.

Proof. $\Rightarrow$ We have an invertible linear map T that $Tu = Tv$. Then we have

$$u = T^{-1}Tu = T^{-1}Tv = v$$

To show T is surjective, we have that for any $w \in W, w = T(T^{-1}w)$.

$\Leftarrow$ Define S such that for each $w \in W$, $S(w)$ is the unique element s.t. $T(S(w)) = w$. (we can do this due to injectivity and surjectivity). Then we have that $T(ST)v = (TS)Tv = Tv$ and thus $STv = v$ and thus $ST = I$. So we suffice the identity operator condition. It's easy to show that S is a linear map. $\square$

Theorem 40. Suppose that V and W are finite-dimensional vector spaces, $\dim V = \dim W$, and $T \in \mathcal{L}(V, W)$. Then

$$T \text{ is invertible} \iff T \text{ is injective} \iff T \text{ is surjective.}$$

Corollary 41. Suppose V and W are finite-dimensional vector spaces of the same dimension, $S \in \mathcal{L}(W, V)$, and $T \in \mathcal{L}(V, W)$. Then $ST = I$ if and only if $TS = I$.

Definition 42 (Isomorphism). An **isomorphism** is an invertible linear map. Two vector spaces are called **isomorphic** if there is an isomorphism from one vector space onto the other one.

Theorem 43 (dimension shows whether vector spaces are isomorphic). Two finite-dimensional vector spaces over $\mathbb{F}$ are isomorphic if and only if they have the same dimension.

Theorem 44. Suppose $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ is a basis of W . Then $\mathcal{M}$ is an isomorphism between $\mathcal{L}(V, W)$ and $\mathbb{F}^{m,n}$.

Corollary 45. Suppose V and W are finite-dimensional. Then $\mathcal{L}(V, W)$ is finite-dimensional and

$$\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$$

Definition 46 (matrix of a vector, $\mathcal{M}(v)$). Suppose $v \in V$ and $v_1, \dots, v_n$ is a basis of V . Then **matrix of v** with respect to the basis is the n -by-1 matrix

$$\mathcal{M}(v) = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix},$$

where $b_1, \dots, b_n$ are the scalars such that

$$v = b_1 v_1 + \dots + b_n v_n$$

Remark 47. The matrix $\mathcal{M}(v)$ of a vector $v \in V$ depends on the basis $v_1, \dots, v_n$ and v . We can think of elements of V as relabeled to be n -by-1 matrices, i.e. $V \mapsto \mathbb{F}^{n,1}$.

Corollary 48. Suppose $T \in \mathcal{L}(V, W)$ and $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ is a basis of W . Let $1 \leq k \leq n$. Then k th column of $\mathcal{M}(T)$, which is denoted by $\mathcal{M}(T)_{\cdot, k}$, equals $\mathcal{M}(Tv_k)$

Theorem 49 (linear maps act like matrix multiplication). Suppose $T \in \mathcal{L}(V, W)$ and $v \in V$. Suppose $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ is a basis of W . Then

$$\mathcal{M}(Tv) = \mathcal{M}(T)\mathcal{M}(v)$$

Proof.

$$\begin{aligned} \mathcal{M}(Tv) &= b_1 \mathcal{M}(Tv_1) + \dots + b_n \mathcal{M}(Tv_n) \\ &= b_1 \mathcal{M}(T)_{\cdot, 1} + \dots + b_n \mathcal{M}(T)_{\cdot, n} \\ &= \mathcal{M}(T)\mathcal{M}(v) \end{aligned}$$

□

Remark 50. Each m -by- n matrix $\mathbf{A}$ induces a linear map from $\mathbb{F}^{n,1}$ to $\mathbb{F}^{m,1}$, namely the matrix multiplication function that takes $x \in \mathbb{F}^{n,1}$ to $\mathbf{A}x \in \mathbb{F}^{m,1}$. We can think of every linear map (from a finite vector space to another) as a matrix multiplication map after suitable relabeling via the isomorphisms given by $\mathcal{M}$. Specifically, if $T \in \mathcal{L}(V, W)$ and we identify $v \in V$ with $\mathcal{M}(v) \in \mathbb{F}^{n,1}$, then the result above says that we can identify Tv with $\mathcal{M}(T)\mathcal{M}(v)$.

Theorem 51 (dimension of range T equals column rank of $\mathcal{M}(T)$). Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then $\dim \text{range } T$ equals the column rank of $\mathcal{M}(T)$.

Theorem 52 (change-of-basis-formula). Suppose $T \in \mathcal{L}(V)$. Suppose $u_1, \dots, u_n$ and $v_1, \dots, v_n$ are bases of V . Let

$$\mathbf{A} = \mathcal{M}(T, (u_1, \dots, u_n)) \text{ and } \mathbf{B} = \mathcal{M}(T, (v_1, \dots, v_n))$$

and $\mathbf{C} = \mathcal{M}(I, (u_1, \dots, u_n), (v_1, \dots, v_n))$. Then

$$\mathbf{A} = \mathbf{C}^{-1}\mathbf{B}\mathbf{C}.$$

Proof.

□

Problem 1

Suppose $T \in \mathcal{L}(V, W)$ is invertible. Show that T^{-1} is invertible and

$$(T^{-1})^{-1} = T.$$

Proof. T^{-1} is invertible because there exists T such that $TT^{-1} = T^{-1}T = I$.

$$T^{-1}T = TT^{-1} = I$$

so $(T^{-1})^{-1} = T$.

□

Problem 2

Suppose $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$ are both invertible linear maps. Prove that $ST \in \mathcal{L}(U, W)$ is invertible and that $(ST)^{-1} = T^{-1}S^{-1}$.

Proof. $(ST)(T^{-1}S^{-1}) = S(TT^{-1})S^{-1} = I = T^{-1}S^{-1}ST$.

□

Problem 3

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent.

- (a) T is invertible.
- (b) $Tv_1, \dots, Tv_n$ is a basis of V for every basis $v_1, \dots, v_n$ of V .
- (c) $Tv_1, \dots, Tv_n$ is a basis of V for some basis $v_1, \dots, v_n$ of V .

Proof. (a) $\Rightarrow$ (b) It only suffice to prove linear independence. We can show this

$$a_1Tv_1 + \dots + a_nTv_n = 0 \iff a_1v_1 + \dots + a_nv_n = 0$$

since T is injective and thus the only solution is all a_i are identically zero.

(b) $\Rightarrow$ (c) Trivial.

(c) $\Rightarrow$ (a) By the linear map lemma, there exists $S \in \mathcal{L}(V)$ such that $S(Tv_i) = v_i$ for all i . Such S is the inverse of T (one can verify) and thus T is invertible. □

Problem 5

Suppose V is finite-dimensional, U is a subspace of V , and $S \in \mathcal{L}(U, V)$. Prove that there exists an invertible linear map T from V to itself such that $Tu = Su$ for every $u \in U$ if and only if S is injective.

Proof. $\Rightarrow$ Since T is invertible, we can define T^{-1} restricted to U , where $S^{-1}u = T^{-1}u$ for all $u \in U$. Then S has an inverse and thus is injective.

$\Leftarrow$ Let $v_1, \dots, v_m$ be the basis of U and $v_1, \dots, v_m, x_1, \dots, x_n$ be the extended basis of V . Define $Tv_i = Sv_i$ and $Tx_i = x_i$. Then T is injective and thus invertible. $\square$

Problem 6

Suppose that W is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\text{null } S = \text{null } T$ if and only if there exists an invertible $E \in \mathcal{L}(W)$ such that $S = ET$.

Proof. $\Rightarrow$ Suppose V is finite-dimensional for simplicity (one can do more work for relaxing this assumption). Let $V = \text{null } S \oplus C = \text{null } T \oplus C$ for some C . Then we know that $T|_C: C \rightarrow \text{range } (T(C))$ and $S|_C: C \rightarrow \text{range } (S(C))$ is invertible. Then there exists an invertible map $\hat{E}: \text{range } (T(C)) \rightarrow \text{range } (S(C))$ as $\dim \text{range } T(C) = \text{range } (S(C)) = \dim V$. Extending this map to W solves the problem (take bases from $\text{range } T(C)$ and extend to W with “identity” map).

$\Leftarrow$ Take $v \in \text{null } T$, then $S(v) = E(Tv) = E(0) = 0$ and thus $v \in \text{null } S$. Conversely, take $v \in \text{null } S$, then $T(v) = E^{-1}S(v) = 0$ so $v \in \text{null } T$. Thus $\text{null } S = \text{null } T$. $\square$

Problem 7

Suppose that V is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\text{range } S = \text{range } T$ if and only if there exists an invertible $E \in \mathcal{L}(V)$ such that $S = TE$.

Proof. $\Rightarrow$ By 3B P26, we know that there exists $\tilde{E} \in \mathcal{L}(V)$ s.t. $S = T\tilde{E}$ as $\text{range } S \subseteq \text{range } T$. $\text{range } S = \text{range } T$ implies that $\dim \text{null } S = \dim \text{range } T$. Then we can define an invertible map $\bar{E}$ from $\text{null } S$ to $\text{null } T$. We can define the map E as follows: Denote $V = \text{null } S \oplus P$, then $E(v) = \tilde{E}(v_1) + \bar{E}(v_2)$ for $v = v_1 + v_2$, where $v_1 \in P, v_2 \in \text{null } S$ and now $\tilde{E} = \tilde{E}|_P$. Then we have that $TE(v) = T(\tilde{E}(v_1) + \bar{E}(v_2)) = S(v_1) + S(v_2) = S(v)$. Now, it left to verify that our proposed E is invertible.

To show this, we mainly aim at proving $\text{range } \tilde{E} \cap \text{range } \bar{E} = \{0\}$, as if we have this, we know that $E(v) = \tilde{E}(v_1) + \bar{E}(v_2) = 0 \Rightarrow \tilde{E}(v_1) = \bar{E}(v_2) = 0$. We know $\bar{E}$ is invertible so $v_2 = 0$. To see $\tilde{E}$ is injective, we can see that for any $v_1 \in \text{null } \tilde{E}$, we have $Sv_1 = T\tilde{E}v_1 = T0 = 0$ and thus $v_1 \in \text{null } S \cap P = 0$, so $v_1 = 0$ and thus $\text{null } (\tilde{E}) = \{0\}$, E is injective and therefore invertible.

To complete the proof, let $u \in \text{range } \tilde{E} \cap \text{range } \bar{E}$, then we know there exists $v_1 \in P, v_2 \in \text{null } S$ s.t. $u = \tilde{E}(v_1) = \bar{E}(v_2)$. We also know that $u \in \text{null } T$ as $\text{range } \bar{E} = \text{null } T$. Now, we have that $S(v_1) = T\tilde{E}(v_1) = T\bar{E}(v_2) = 0$ and thus $v_1 \in \text{null } S$. So $v_1 \in P \cap \text{null } S = \{0\}$ and thus we have that $u = \{0\}$, finishing the proof.

$\Leftarrow$ Take any $s \in \text{range } S$, then there exists $E(s)$ s.t. $T(E(s)) = s$. Conversely, take any $t \in \text{range } T$, then there exists $v \in V$ s.t. $t = Tv = TEE^{-1}v = S(E^{-1}v)$.

Thus $\text{range } S = \text{range } T$. $\square$

Problem 8

Suppose V and W are finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that there exist invertible $E_1 \in \mathcal{L}(V)$ and $E_2 \in \mathcal{L}(W)$ such that $S = E_2 T E_1$ if and only if $\dim \text{null } S = \dim \text{null } T$.

Proof. $\Rightarrow$ We know from 3B exercises that $\dim \text{range } ST \leq \min(\dim \text{range } S, \dim \text{range } T)$. Applying here gets that $\dim \text{range } S \leq \dim \text{range } T$. Consider that $T = E_2^{-1} S E_1^{-1}$, then we can also derive $\dim \text{range } T \leq \dim \text{range } S$ and thus $\dim \text{range } S = \dim \text{range } T$. By the fundamental theorem of linear map, we can get that $\dim \text{null } S = \dim \text{null } T$.

$\Leftarrow$ We know $\dim \text{null } S = \dim \text{null } T$, so there exists an isomorphism $\tilde{E}_1: \text{null } S \rightarrow \text{null } T$. Extending $\tilde{E}_1: \text{null } S \rightarrow V$ still preserves its injectivity. Therefore, by P5, we have an invertible $E_1: V \rightarrow V$ such that $E_1 u = \tilde{E}_1 u$ for all $u \in \text{null } S$. Now we intend to show that $\text{null } S = \text{null } T E_1$, as proving so would imply there exists an invertible $E_2 \in \mathcal{L}(W)$ such that $S = E_2 T E_1$ by P6. To see this, take $s \in \text{null } S$, then $T E_1(s) = T(\tilde{E}_1 s) = 0$ and thus $s \in \text{null } T E_1$. To see the other direction, take $t \in \text{null } T E_1$, then we know $E_1 t \in \text{null } T = \text{range } \tilde{E}_1$. Thus there exists $v \in \text{null } S$ s.t. $E_1 t = \tilde{E}_1 v = E_1 v$. As E_1 is invertible, $t = v \in \text{null } S$ and thus we have shown $\text{null } S = \text{null } T E_1$, completing the proof. $\square$

Problem 9

Suppose V is finite-dimensional and $T: V \rightarrow W$ is a surjective linear map of V onto W . Prove that there is a subspace U of V such that $T|_U$ is an isomorphism of U onto W .

Proof. T being surjective means that $\dim V \geq \dim W$. Take any subspace of V that has $\dim W$, then there is an isomorphism between this subspace and W . $\square$

Problem 10

Suppose V and W are finite-dimensional and U is a subspace of V . Let

$$\mathcal{E} = \{T \in \mathcal{L}(V, W) : U \subseteq \text{null } T\}.$$

- (a) Show that $\mathcal{E}$ is a subspace of $\mathcal{L}(V, W)$.
- (b) Find a formula for $\dim \mathcal{E}$ in terms of $\dim V$, $\dim W$, and $\dim U$.

Proof. (a) Take $T_1, T_2 \in \mathcal{E}$ and $\lambda \in \mathbb{F}$, then consider $\text{null } (\lambda T_1 + T_2)$. Let $u \in U$, then $\lambda T_1 + T_2(u) = \lambda T_1(u) + T_2(u) = 0 + 0 = 0$ and thus we have $\lambda T_1 + T_2 \in \mathcal{E}$ and thus $\mathcal{E}$ is a subspace of $\mathcal{L}(V, W)$.

(b) Following the hint, define $\Phi: \mathcal{L}(V, W) \rightarrow \mathcal{L}(U, W)$ by $\Phi(T) = T|_U$. Then $\Phi \in \mathcal{L}(\mathcal{L}(V, W), \mathcal{L}(U, W))$. We try to show the range and null space of Φ .

We first claim $\text{null } \Phi = \mathcal{E}$. To see this, take $T \in \text{null } \Phi$, then we know $\Phi(T) = T|_U = 0 \in \mathcal{L}(U, W)$, meaning that U is the subset of $\text{null } T$, so $\text{null } \Phi \subseteq \mathcal{E}$. Conversely, take $T \in \mathcal{E}$, then we have that for all $u \in U$, $Tu = 0$, then we have that $T|_U = 0$. This shows that $\mathcal{E} \subseteq \text{null } \Phi$.

Next we claim $\text{range } \Phi = \mathcal{L}(U, W)$. Take any $S \in \mathcal{L}(U, W)$, then we can naturally always extend S to $T \in \mathcal{L}(V, W)$ by setting $Tu = Su$ for all $u \in U$. The converse direction is trivial.

Thus we have that $\dim \mathcal{E} = \dim \text{null } \Phi$ and $\dim \text{range } \Phi = \dim \mathcal{L}(U, W) = \dim U \dim W$ and $\dim \mathcal{L}(V, W) = \dim V \dim W = \dim \text{null } \Phi + \dim \text{range } \Phi$, thus we derive that

$$\dim V \dim W = \dim \mathcal{E} + \dim U \dim W.$$

□

Problem 14

Prove or give a counterexample: If V is finite-dimensional and $R, S, T \in \mathcal{L}(V)$ are such that RST is surjective, then S is injective.

Proof. We know that

$$\dim \text{range } RST = \dim \text{range } V \leq \min\{\dim \text{range } R, \dim \text{range } S, \dim \text{range } T\}$$

This means that $\dim RST \leq \dim \text{range } S$. If $\dim \text{null } S > 0$, then $\dim \text{range } S < \dim V$ and thus contradicts the assumption that RST is surjective. □

Problem 15

Suppose $T \in \mathcal{L}(V)$ and $v_1, \dots, v_m$ is a list in V such that $Tv_1, \dots, Tv_m$ spans V . Prove that $v_1, \dots, v_m$ spans V .

Proof. We can reduce the list to basis and use previous conclusions. Since $Tv_1, \dots, Tv_m$ spans V , we can reduce the list to $Tv_1, \dots, Tv_n$ such that the list is the basis of V . By previous results, $v_1, \dots, v_n$ is also linearly independent and its dimension equals the $\dim V$ and thus is the basis of V . It thus spans V . Extending the list to $v_1, \dots, v_m$ also spans V . □

Problem 16

Prove that every linear map from $\mathbb{F}^{n,1}$ to $\mathbb{F}^{m,1}$ is given by a matrix multiplication. In other words, prove that if $T \in \mathcal{L}(\mathbb{F}^{n,1}, \mathbb{F}^{m,1})$, then there exists an m -by- n matrix $\mathbf{A}$ such that $Tx = \mathbf{A}x$ for every $x \in \mathbb{F}^{n,1}$.

Proof. We can simply consider the standard basis of $\mathbb{F}^{n,1}$ and $\mathbb{F}^{m,1}$. Then Let $\mathbf{A}$ be the matrix of T wrt these bases. Then we have that by definition,

$$Tx = \mathcal{M}(Tx) = \mathcal{M}(T)\mathcal{M}(x) = \mathbf{A}x$$

□

Problem 17

Suppose V is finite-dimensional and $S \in \mathcal{L}(V)$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by

$$\mathcal{A}(T) = ST$$

for $T \in \mathcal{L}(V)$.

- (a) Show that $\dim \text{null } \mathcal{A} = \dim V \dim \text{null } S$.
- (b) Show that $\dim \text{range } \mathcal{A} = \dim V \dim \text{range } S$.

Proof. (a) We prove this by showing $\text{null } \mathcal{A} = \mathcal{L}(V, \text{null } S)$. Take $A \in \text{null } \mathcal{A}$, then we know that $\mathcal{A}(A) = SA = 0$ and thus $Av \in \text{null } S$ for all $v \in V$. For the other direction, take $A \in \mathcal{L}(V, \text{null } S)$, then $SA(v) = \mathcal{A}(v) = 0$ and thus $A \in \text{null } \mathcal{A}$.

- (b) We know $\dim(\mathcal{L}(V)) = \dim \text{null } \mathcal{A} + \dim \text{range } \mathcal{A}$. Hence

$$\dim \text{range } \mathcal{A} = \dim V(\dim V - \dim \text{null } S) = \dim V \dim \text{range } S.$$

□

Problem 18

Show that V and $\mathcal{L}(\mathbb{F}, V)$ are isomorphic vector spaces.

Proof. For simplicity, assume V is finite-dimensional, then $\dim V = \dim \mathbb{F} \dim V = \dim \mathcal{L}(\mathbb{F}, V)$ and thus they are isomorphic. One might do a more careful construction for infinite-dimensional V . □

Problem 19

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has the same matrix with respect to every basis of V if and only if T is a scalar multiple of the identity operator.

Proof. $\Rightarrow$ We know $Tv_k = \sum_{j=1}^n A_{j,k}v_j = \sum_{j=1}^n 2A_{j,k}v_j = T(2v_k)$. This means that $A_{j,k} = 0$ for $j \neq k$. When $j = k$, we have that $Tv_k = A_{k,k}v_k$ for some arbitrary ordering of the same basis. Then this means that $A_{i,i} = A_{j,j}$ for all i, j . Thus, T is a scalar multiple of the identity operator.

$\Leftarrow$ This follows by the definition of the identity operator. □

3E: Products and Quotients of Vector Spaces

Definition 53 (product of vector spaces). *Suppose $V_1, \dots, V_m$ are vector spaces over $\mathbb{F}$.*

- The **product** $V_1 \times \dots \times V_m$ is defined by

$$V_1 \times \dots \times V_m = \{(v_1, \dots, v_m) : v_1 \in V_1, \dots, v_m \in V_m\}.$$

- Addition on $V_1 \times \dots \times V_m$ is defined by

$$(u_1, \dots, u_m) + (v_1, \dots, v_m) = (u_1 + v_1, \dots, u_m + v_m).$$

- Scalar multiplication on $V_1 \times \dots \times V_m$ is defined by

$$\lambda(v_1, \dots, v_m) = (\lambda v_1, \dots, \lambda v_m).$$

Theorem 54 (product of vector spaces is a vector space). *Suppose $V_1, \dots, V_m$ are vector spaces over $\mathbb{F}$. Then $V_1 \times \dots \times V_m$ is a vector space over $\mathbb{F}$.*

Theorem 55 (dimension of a product is the sum of dimensions). *Suppose $V_1, \dots, V_m$ are finite-dimensional vector spaces. Then $V_1 \times \dots \times V_m$ is finite-dimensional and*

$$\dim(V_1 \times \dots \times V_m) = \dim V_1 + \dots + \dim V_m.$$

Lemma 56 (products and direct sums). *Suppose that $V_1, \dots, V_m$ are subspaces of V . Define a linear map $\Gamma : V_1 \times \dots \times V_m \rightarrow V_1 + \dots + V_m$ by*

$$\Gamma(v_1, \dots, v_m) = v_1 + \dots + v_m.$$

Then $V_1 + \dots + V_m$ is a direct sum if and only if Γ is injective.

Theorem 57 (a sum is a direct sum if and only if dimensions add up). *Suppose V is finite-dimensional and $V_1, \dots, V_m$ are subspaces of V . Then $V_1 + \dots + V_m$ is a direct sum if and only if*

$$\dim(V_1 + \dots + V_m) = \dim V_1 + \dots + \dim V_m.$$

Definition 58 ($v + U$, translation). *Suppose $v \in V$ and $U \subset V$. Then $v + U$ is the subset of V defined by*

$$v + U = \{v + u : u \in U\}.$$

*The set $v + U$ is said to be a **translate** of U .*

Definition 59 (quotient space, V/U). *Suppose U is a subspace of V . Then the **quotient space** V/U is the set of all translates of U . Thus*

$$V/U = \{v + U : v \in V\}.$$

Remark 60. If $U = \{(x, 2x) \in \mathbb{R}^2 : x \in R\}$, then $\mathbb{R}^2/U$ is the set of all lines in $\mathbb{R}^2$ that have slope 2.

Lemma 61 (two translates of a subspace are equal or disjoint). Suppose U is a subspace of V and $v, w \in V$. Then

$$v - w \in U \iff v + U = w + U \iff (v + U) \cap (w + U) = \emptyset$$

Definition 62 (addition and scalar multiplication on V/U). Suppose U is a subspace of V . Then **addition** and **scalar multiplication** are defined on V/U by

$$\begin{aligned} (v + U) + (w + U) &= (v + w) + U \\ \lambda(v + U) &= (\lambda v) + U \end{aligned}$$

for all $v, w \in V$ and all $\lambda \in \mathbb{F}$.

Theorem 63 (quotient space is a vector space). Suppose U is a subspace of V . Then V/U , with the operations of addition and scalar multiplication as defined above, is a vector space.

Definition 64 (quotient map). Suppose U is a subspace of V . The **quotient map** $\pi: V \rightarrow V/U$ is the linear map defined by

$$\pi(v) = v + U$$

for each $v \in V$.

Theorem 65 (dimension of quotient space). Suppose V is finite-dimensional and U is a subspace of V . Then

$$\dim V/U = \dim V - \dim U$$

Definition 66. Suppose $T \in \mathcal{L}(V, W)$. Define $\tilde{T}: V/(\text{null } T) \rightarrow W$ by

$$\tilde{T}(v + \text{null } T) = Tv.$$

Another interpretation of quotient space is on **congruence of subspaces**. The definition goes as follows:

If Y is a subspace of X , then two vectors $x_1, x_2 \in X$ are **congruent modulo** Y , denoted $x_1 \cong x_2 \pmod{Y}$ if $x_1 - x_2 \in Y$.

The quotient space X/Y denotes the set of equivalence classes in X , modulo Y by defining that

$$\{x\} + \{z\} := \{x + z\} \quad a\{x\} := \{ax\}$$

So here one can think of the zero vector of $v + U$ defined as the equivalence class that contains that zero vector $0 \in V$:

$$0 + U = \{0 + u : u \in U\} = U.$$

This means that the zero element in the quotient space V/U is simply the subspace U , and this equivalence class contains all vectors differ from 0 by an element of U , meaning it is precisely U itself, so $\pi(v) = 0$ if and only if $v \in U$.

Remark 67. The output of $\pi(v)$ is the equivalence class $v + U$, which is the set of all vectors in V that are equivalent to v under the subspace U .

For $\tilde{T}$, the input to it is an equivalent class $v + \text{null } (T)$ in the quotient space $V/\text{null } T$. It represents all vectors in V that differ from v by a vector in $\text{null } T$. Its output is Tv , which is an element in W .

The null space is

$$\text{null } (\tilde{T}) = \{v + \text{null } (T) : \tilde{T}(v + \text{null } (T)) = 0\}.$$

Since we know that $\tilde{T}(v + \text{null } T) = Tv$, we have that

$$\tilde{T}(v + \text{null } T) = 0 \iff Tv = 0.$$

so $v \in \text{null } T$ and thus $\text{null } \tilde{T} = \{\text{null } T\}$. We have that $\text{range } \tilde{T} = \text{range } T$.

Theorem 68. Suppose $T \in \mathcal{L}(V, W)$. Then

- (a) $\tilde{T} \circ \pi = T$, where π is the quotient map of V onto $V/\text{null } T$;
- (b) $\tilde{T}$ is injective;
- (c) $\text{range } \tilde{T} = \text{range } T$;
- (d) $V/\text{null } T$ and $\text{range } T$ are isomorphic vector spaces.

Remark 69. One might think of the domain of $\tilde{T}$ as the non-trivial domain of T and the range as the normal image of T .

Problem 1

Suppose T is function from V to W . The *graph* of T is the subset of $V \times W$ defined by

$$\text{graph of } T = \{(v, Tv) \in V \times W : v \in V\}.$$

Prove that T is a linear map if and only if the graph of T is a subspace of $V \times W$.

Proof. The two properties closely relate to each other:

$$(v_1, Tv_1) + (v_2, Tv_2) \in V \times W : v \in V \iff Tv_1 + Tv_2 = T(v_1 + v_2)$$

and

$$(\lambda v, T\lambda v) \in V \times W : v \in V \iff T\lambda v = \lambda Tv$$

□

Problem 2

Suppose that $V_1, \dots, V_m$ are vector spaces such that $V_1 \times \dots \times V_m$ is finite-dimensional. Prove that V_k is finite-dimensional for each $k = 1, \dots, m$.

Proof. Recall the definition of infinite-dimension from ch2 P17:

V is infinite-dimensional if and only if there is a sequence $v_1, v_2, \dots$ of vectors in V such that $v_1, \dots, v_m$ is linearly independent for every positive integer m .

Suppose for the sake of contradiction that there exists V_k to be infinite-dimensional. Then this means that there is a sequence $v_{k,1}, v_{k,2}, \dots$ of vectors in V_k such that $v_{k,1}, \dots, v_{k,m}$ is linearly independent for every positive integer m . We can build such sequence for the product space as well. More specifically, consider the sequence $0, \dots, v_{k,j}, \dots, 0$ where we fill 0 for each of other V_j 's and only leave the sequence from V_k . In this case, we can see that the product space is infinite-dimensional, contradicting the claim. □

Problem 3

Suppose $V_1, \dots, V_m$ are vector spaces. Prove that $\mathcal{L}(V_1 \times \dots \times V_m, W)$ and $\mathcal{L}(V_1, W) \times \dots \times \mathcal{L}(V_m, W)$ are isomorphic vector spaces.

Proof. We define a permutation invariant set function $S: \mathcal{L}(V_1, W) \times \dots \times \mathcal{L}(V_m, W) \rightarrow \mathcal{L}(V_1 \times \dots \times V_m, W)$ such that

$$S(T_1, \dots, T_m) = T_1 + \dots + T_m$$

We will show that S is an isomorphism (it's linear). For injectivity, null S means that for any arbitrary input $T_1 + \dots + T_m = 0$, which implies that $T_i = 0$ for all i . For surjectivity, take $\phi \in \mathcal{L}(V_1 \times \dots \times V_m, W)$, we can define

$T_i(v_i) = \phi((0, \dots, v_i, \dots, 0))$ where i means for the space V_i . Then we can naturally get that $\phi((v_1, \dots, v_m)) = \phi(\sum_{i=1}^m e_i^\top v_i) = \sum_{i=1}^m T_i(v_i)$ and thus we finish the proof. $\square$

Problem 4

Suppose $W_1, \dots, W_m$ are vector spaces. Prove that $\mathcal{L}(V, W_1 \times \dots \times W_m)$ and $\mathcal{L}(V, W_1) \times \dots \times \mathcal{L}(V, W_m)$ are isomorphic vector spaces.

We can simply define a concatenate operator $S: \mathcal{L}(V, W_1) \times \dots \times \mathcal{L}(V, W_m) \rightarrow \mathcal{L}(V, W_1 \times \dots \times W_m)$ such that

$$S(T_1, \dots, T_m) = (T_1, \dots, T_m)$$

We prove this linear map is an isomorphism. The injective property can be proved using the same logic as above. For surjectivity, take $\phi \in \mathcal{L}(V, W_1 \times \dots \times W_m)$, then we can naturally obtain $T_k = \pi_k \phi$ where π_k means projecting the k -th component of the input. Note that $T_k \in \mathcal{L}(V, W_k)$ and therefore we finish the proof.

Problem 5

For m a positive integer, define V^m by

$$V^m = \underbrace{V \times \dots \times V}_{m \text{ times}}.$$

Prove that V^m and $\mathcal{L}(\mathbb{F}^m, V)$ are isomorphic vector spaces.

Proof.

$$\dim(V^m) = m \dim V = \dim(\mathcal{L}(\mathbb{F}^m, V)).$$

$\square$

Problem 6

Suppose that v, x are vectors in V and that U, W are subspaces of V such that $v + U = x + W$. Prove that $U = W$.

Proof. Note that $v = x + w$ for some $w \in W$ if we take $u = 0 \in U$ and similarly $x = v + u$ for some $u \in U$. Hence $v - x \in W$ and $x - v \in U$. Then we have that take $u \in U, u = (x + w) - v = (x - v) + w \in W$. Similarly, take $w \in W, w = (v + u) - x = (v - x) + u \in U$. Hence $W = U$. $\square$

Problem 9

Prove that a nonempty subset A of V is a translate of some subspace of V if and only if $\lambda v + (1 - \lambda)w \in A$ for all $v, w \in A$ and all $\lambda \in \mathbb{F}$.

Proof. $\Rightarrow$ Suppose $A = x + U$ for some subspace $U \subseteq V$. Then take $a_1 = x + u_1 \in A$ and $a_2 = x + u_2 \in A$. We have that

$$\lambda a_1 + (1 - \lambda)a_2 = \lambda x + \lambda u_1 + (1 - \lambda)x + (1 - \lambda)u_2 = x + (\lambda u_1 + (1 - \lambda)u_2) \in A$$

$\Leftarrow$ take any $x \in A$ and define the subspace such that

$$U := (-x) + A$$

first $0 = -x + x \in U$. Next, take $v_1 - x$ and $v_2 - x \in U$. We have that $v_1 - x + v_2 - x = (-x) + (2v_1 - x)/2 + (2v_2 - x)/2 \in A$ as $2v_i - x \in A$ by taking $\lambda = 2$. Finally, $\lambda(-x + v) = (-x) + (\lambda v + (1 - \lambda)x) \in A$. $\square$

Problem 10

Suppose $A_1 = v + U_1$ and $A_2 = w + U_2$ for some $v, w \in V$ and some subspaces U_1, U_2 of V . Prove that the intersection $A_1 \cap A_2$ is either a translate of some subspace of V or is the empty set.

Proof. If $A_1 \cap A_2$ is not empty, then $S = A_1 \cap A_2$ is a subspace such that $v + U_1|_S = w + U_2|_S$, so we know that $U_1|_S = U_2|_S$ by Problem 6. Hence it is a translate of some subspace of V . Otherwise, it is the empty set. $\square$

Problem 11

Suppose $U = \{(x_1, x_2, \dots) \in \mathbb{F}^\infty : x_k \neq 0 \text{ for only finitely many } k\}$.

- Show that U is a subspace of $\mathbb{F}^\infty$.
- Prove that $\mathbb{F}^\infty/U$ is infinite-dimensional.

Proof. (a) all zero, where $x_k \neq 0$ for zero k 's (finitely many) is an element of U . Next, take $(x_1, x_2, \dots)$ and $(y_1, y_2, \dots) \in U$. Then $(x_1 + y_1, x_2 + y_2, \dots)$ will have finitely many nonzero since each of them has only finitely many nonzero entries. Same holds for $\lambda(x_1, x_2, \dots)$.

(b) Let's consider a "standard basis" $\{v_i\}_{i=1}^\infty$ that only has 0 on i -th spot and all 1 otherwise. We can see that for all m , $v_1, \dots, v_m$ is linearly independent. And so does $v_1 + U, \dots, v_m + U$ since each $v_i \notin U$. $\square$

Problem 12

Suppose $v_1, \dots, v_m \in V$. Let

$$A = \{\lambda_1 v_1 + \dots + \lambda_m v_m : \lambda_1, \dots, \lambda_m \in F \text{ and } \lambda_1 + \dots + \lambda_m = 1\}.$$

- Prove that A is a translate of some subspace of V .
- Prove that if B is a translate of some subspace of V and $\{v_1, \dots, v_m\} \subseteq B$, then $A \subseteq B$.
- Prove that A is a translate of some subspace of V of dimension less than m .

Proof. (a) We try to show this through using the conclusion from Problem 9. Let $a_1 = \sum_{i=1}^m \alpha_i v_i$ and $a_2 = \sum_{i=1}^m \gamma_i v_i$ s.t. $\sum_{i=1}^m \alpha_i = \sum_{i=1}^m \gamma_i = 1$. We have that

$$\lambda a_1 + (1 - \lambda) a_2 = \sum_{i=1}^m (\lambda \alpha_i + (1 - \lambda) \gamma_i) v_i$$

We can show that

$$\sum_{i=1}^m (\lambda \alpha_i + (1 - \lambda) \gamma_i) = \lambda \sum_{i=1}^m \alpha_i + (1 - \lambda) \sum_{i=1}^m \gamma_i = 1$$

Thus $\lambda a_1 + (1 - \lambda) a_2 \in A$ and thus A is a translate of some subspace of V .

(b) Let $B = w + Y$ for some subspace $Y \subseteq V$. Then we have that $v_k = w + y_k$. Take $x \in A$, then we know that

$$x = \sum_{i=1}^m \lambda_i v_i = \sum_{i=1}^m \lambda_i (w + y_i) = w + \sum_{i=1}^m \lambda_i y_i \in B$$

(c) Write $A = w + U$. Denote $B = \text{span}\{v_1, \dots, v_m\}$. Then by (b) we know that $A \subseteq B$. We will show that $\dim U < m$. The statement is trivial if $v_1, \dots, v_m$ is not linearly independent. Let's consider the case that it's indeed linearly independent. Here we claim that $w \notin U$ and thus $\dim U < m$. To see this, the only way to write $\sum_{i=1}^m \lambda_i v_i = 0$ is to let all $\lambda_i = 0$, but in this case the vectors will not be in A , so $0 \notin A$. Suppose for the sake of contradiction that $w \in U$. This implies that $0 = w + (-w) \in w + U = A$, forming a contradiction. Hence we finish the proof. $\square$

Problem 13

Suppose U is a subspace of V such that V/U is finite-dimensional. Prove that V is isomorphic to $U \times (V/U)$.

Proof.

$$\dim V = \dim U + (\dim V - \dim U) = \dim U + \dim(V/U).$$

$\square$

Problem 14

Suppose U and W are two subspaces of V and $V = U \oplus W$. Suppose $w_1, \dots, w_m$ is a basis of W . Prove that $w_1 + U, \dots, w_m + U$ is a basis of V/U .

Proof. First we show that the list is linearly independent. We consider the following system:

$$0 + U = \lambda_1(w_1 + U) + \cdots + \lambda_m(w_m + U) = (\lambda_1 w_1 + \cdots + \lambda_m w_m) + U$$

This means that $\lambda_1 w_1 + \cdots + \lambda_m w_m \in U$ and since we know that $w_1, \dots, w_m \notin U$, we have that the only solution is $\lambda_i = 0$ for all i .

Next we show that the list spans V/U . Take arbitrary $v + U \in V/U$, then we have that $v = u + w$ for some $w = \sum_{i=1}^m \lambda_i w_i$. Then we have that

$$v + U = \sum_{i=1}^m \lambda_i w_i + U = \sum_{i=1}^m \lambda_i (w_i + U).$$

Thus the list spans the quotient space. $\square$

Problem 15

Suppose U is a subspace of V and $v_1 + U, \dots, v_m + U$ is a basis of V/U and $u_1, \dots, u_n$ is a basis of U . Prove that $v_1, \dots, v_m, u_1, \dots, u_n$ is a basis of V .

Proof. We first know that

$$\dim V = \dim V/U + \dim U.$$

So it only suffices to prove the list either spans V or is linearly independent. Take any $v \in V$. Then we have that

$$v + U = \lambda_1 v_1 + \cdots + \lambda_m v_m + U.$$

Thus $v - \sum_{i=1}^m \lambda_i v_i \in U$. So

$$v = \sum_{i=1}^m \lambda_i v_i + \sum_{j=1}^n \alpha_j u_j.$$

Hence, we finish the proof. $\square$

Problem 16

Suppose $\varphi \in \mathcal{L}(V, \mathbb{F})$ and $\varphi \neq 0$. Prove that $\dim V/(\text{null } \varphi) = 1$.

Proof. We know that $\dim \text{range } \varphi = 1$ and thus

$$\dim V/\text{null } \varphi = \dim V - \dim \text{null } \varphi = \dim \text{range } \varphi = 1.$$

$\square$

Problem 17

Suppose that U is a subspace of V such that $\dim V/U = 1$. Prove that there exists $\varphi \in \mathcal{L}(V, \mathbb{F})$ such that $\text{null } \varphi = U$.

Proof. We make construction as follows: (1) there exists a natural isomorphism $T: V/U \rightarrow \mathbb{F}$. We further define $S: V \rightarrow V/U$ by $S(v) = v + U$. Then we show the map

$$\varphi = TS$$

satisfies the requirement.

First take $u \in U$. Then we have that

$$\varphi(u) = TS(u) = T(u + U) = T(0) = 0$$

and thus $u \in \text{null } \varphi$.

Conversely take $u \in \text{null } \varphi$, then

$$0 = \varphi(u) = T(S(u)) = T^{-1}T(S(u)) = S(u).$$

Thus $u \in \text{null } S = U$. We've finished the proof. $\square$

Problem 18

Suppose U is a subspace of V such that V/U is finite-dimensional.

(a) Show that if W is a finite-dimensional subspace of V and $V = U + W$, then $\dim W \geq \dim V/U$.

(b) Prove that there exists a finite-dimensional subspace W of V such that $\dim W = \dim V/U$ and $V = U \oplus W$.

Proof. (a) We know that

$$\dim W = \dim V + \dim(U \cap W) - \dim U \geq \dim V - \dim U = \dim V/U.$$

(b) We will construct the space W through some manipulation with the basis. Since we are given V/U , we start with the basis $w_1 + U, \dots, w_m + U$ of that. Note that $\{w_1, \dots, w_m\}$ is linearly independent as the only solution is all-zero. We can now define

$$W = \text{span}\{w_1, \dots, w_m\}.$$

Then we have that $\dim W = \dim V/U$ and now it suffices to verify $V = U \oplus W$. To see this, one only needs to show $W \cap U = \{0\}$ as we've proved that $V = W + U$ in previous questions (of similar construction). Take arbitrary $v \in W \cap U$. Then we have that

$$v = \sum_{i=1}^m \lambda_i w_i$$

and that

$$v + U = \sum_{i=1}^m \lambda_i w_i + U = 0 + U$$

The only solution is all $\lambda_i = 0$. $\square$

Problem 19

Suppose $T \in \mathcal{L}(V, W)$ and U is a subspace of V . Let π denote the quotient map from V onto V/U . Prove that there exists $S \in \mathcal{L}(V/U, W)$ such that $T = S \circ \pi$ if and only if $U \subseteq \text{null } T$.

Proof. $\Rightarrow$ Take $u \in U$, then we have that

$$T(u) = S\pi(u) = S(u + U) = S(0) = 0$$

and thus $u \in \text{null } T$.

$\Leftarrow$ We define $S: V/U \rightarrow W$ by

$$S(v + U) = T(v)$$

It now only suffices to show that this map is valid for the equivalence mapping. Take $v_1 + U = v_2 + U$. Then we have that $v_1 - v_2 \in U \subseteq \text{null } T$ and thus $T(v_1 - v_2) = 0 = Tv_1 - Tv_2$ and thus $S(v_1 + U) = Tv_1 = Tv_2 = S(v_2 + U)$. This map is clearly linear and thus we have $T = S \circ \pi$. $\square$

3F: Duality

Definition 70 (linear functional). A **linear functional** on V is a linear map from V to $\mathbb{F}$. In other words, a linear functional is an element of $\mathcal{L}(V, \mathbb{F})$.

Definition 71 (dual space, V'). The **dual space** of V , denoted by V' , is the vector space of all linear functionals on V . In other words, $V' = \mathcal{L}(V, \mathbb{F})$.

Lemma 72. Suppose V is finite-dimensional. Then V' is also finite-dimensional and

$$\dim V' = \dim V.$$

Definition 73 (dual basis). If $v_1, \dots, v_n$ is a basis of V , then the **dual basis** of $v_1, \dots, v_n$ is the list $\varphi_1, \dots, \varphi_n$ of elements of V' , where each φ_j is the linear functional on V such that

$$\varphi_j(v_k) = \begin{cases} 1 & \text{if } k = j, \\ 0 & \text{if } k \neq j. \end{cases}$$

Theorem 74 (dual basis gives coefficients for linear combination). Suppose $v_1, \dots, v_n$ is a basis of V and $\varphi_1, \dots, \varphi_n$ is the dual basis. Then

$$v = \varphi_1(v)v_1 + \dots + \varphi_n(v)v_n$$

for each $v \in V$.

Theorem 75 (dual basis is a basis of the dual space). Suppose V is finite-dimensional. Then the dual basis of a basis of V is a basis of V' .

Definition 76 (dual map, T'). Suppose $T \in \mathcal{L}(V, W)$. The **dual map** of T is the linear map $T' \in \mathcal{L}(W', V')$ defined for each $\varphi \in W'$ by

$$T'(\varphi) = \varphi \circ T.$$

Corollary 77 (algebraic properties of dual map). Suppose $T \in \mathcal{L}(V, W)$. Then

- (a) $(S + T)' = S' + T'$ for all $S \in \mathcal{L}(V, W)$;
- (b) $(\lambda T)' = \lambda T'$ for all $\lambda \in \mathbb{F}$;
- (c) $(ST)' = T'S'$ for all $S \in \mathcal{L}(W, U)$.

The goal of this section is to describe null T' and range T' in terms of range T and null T .

Definition 78 (annihilator, U^0). For $U \subseteq V$, the **annihilator** of U , denoted by U^0 , is defined by

$$U^0 = \{\varphi \in V' : \varphi(u) = 0 \text{ for all } u \in U\}.$$

Remark 79. U^0 is a subspace of V' .

Theorem 80 (dimension of the annihilator). *Suppose V is finite-dimensional and U is a subspace of V . Then*

$$\dim U^0 = \dim V - \dim U.$$

Lemma 81. *Suppose V is finite-dimensional and U is a subspace of V . Then*

- (a) $U^0 = 0 \iff U = V$;
- (b) $U^0 = V' \iff U = \{0\}$.

Lemma 82 (the null space of T'). *Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then*

- (a) $\text{null } T' = (\text{range } T)^0$;
- (b) $\dim \text{null } T' = \dim \text{null } T + \dim W - \dim V$.

Theorem 83 (T surjective is equivalent to T' injective). *Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then*

$$T \text{ is surjective} \iff T' \text{ is injective}.$$

Lemma 84 (the range of T'). *Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then*

- (a) $\dim \text{range } T' = \dim \text{range } T$;
- (b) $\text{range } T' = (\text{null } T)^0$.

Theorem 85 (T surjective is equivalent to T' injective). *Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then*

$$T \text{ is surjective} \iff T' \text{ is injective}.$$

Theorem 86 (matrix of T' is transpose of matrix of T). *Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then*

$$\mathcal{M}(T') = (\mathcal{M}(T))^\top$$

Proof. Let $v_1, \dots, v_n$ be basis of V and $w_1, \dots, w_m$ be basis of W . Let $\varphi_1, \dots, \varphi_n$ be the dual basis of V' and $\psi_1, \dots, \psi_m$ be the dual basis of W' .

Let $A = \mathcal{M}(T)$ and $C = \mathcal{M}(T')$. Suppose $1 \leq j \leq m$ and $1 \leq k \leq n$. From the definition of $\mathcal{M}(T')$ we have

$$T'(\psi_j) = \sum_{r=1}^n C_{r,j} \varphi_r$$

Then applying both sides of the equation above to v_k gives that

$$\begin{aligned} (\psi_j \circ T)(v_k) &= \sum_{r=1}^n C_{r,j} \varphi_r(v_k) \\ &= C_{k,j} \end{aligned}$$

At the same time, we have that

$$\begin{aligned}
(\psi_j \circ T)(v_k) &= \psi_j(Tv_k) \\
&= \psi_j\left(\sum_{r=1}^m A_{r,k}w_r\right) \\
&= \sum_{r=1}^m A_{r,k}\psi_j(w_r) \\
&= A_{j,k}
\end{aligned}$$

Here we have that $C_{k,j} = A_{j,k}$ and thus $C = A^\top$. Hence, $\mathcal{M}(T') = (\mathcal{M}(T))^\top$, as desired. $\square$

We can use duality to provide an alternative proof for the rank of matrix:

Theorem 87 (column rank equals row rank). *Suppose $\mathbf{A} \in \mathbb{F}^{m,n}$. Then the column rank of $\mathbf{A}$ equals the row rank of $\mathbf{A}$.*

Problem 3

Suppose V is finite-dimensional and $v \in V$ with $v \neq 0$. Prove that there exists $\varphi \in V'$ such that $\varphi(v) = 1$.

Proof. The main requirement is to ensure that the constructed φ is valid. Note that there exists subspace $W \subseteq V$ such that $V = W \oplus \text{span}\{v\}$. Thus we can define $\varphi(u) = i$ where $u = w + iv$ for $u \in V, w \in W, i \in \mathbb{F}$. Here φ is a valid linear map and $\varphi(v) = 1$. $\square$

Problem 6

Suppose $\varphi, \beta \in V'$. Prove that $\text{null } \varphi \subseteq \text{null } \beta$ if and only if there exists $c \in \mathbb{F}$ such that $\beta = c\varphi$.

Proof. $\Rightarrow$ If $\varphi = 0$, then it trivially holds. If $\varphi \neq 0$, then we can first take $v_0 \in V$ s.t. $\varphi(v_0) \neq 0$, then define $c = \beta(v_0)/\varphi(v_0)$, then we have that $\beta(v) = \beta(v_0 + u) = \beta(v_0) = c\varphi(v_0) = c\varphi(v)$ for some $u \in \text{null } \varphi$.

$\Leftarrow$ Take $v \in \text{null } \varphi$, then $\beta(v) = c\varphi(v) = 0$. Thus $\text{null } \varphi \subseteq \text{null } \beta$. $\square$

Problem 8

Suppose $v_1, \dots, v_n$ is a basis of V and $\varphi_1, \dots, \varphi_n$ is the dual basis of V' . Define $\Gamma: V \rightarrow \mathbb{F}^n$ and $\Lambda: \mathbb{F}^n \rightarrow V$ by

$$\Gamma(v) = (\varphi_1(v), \dots, \varphi_n(v)) \text{ and } \Lambda(a_1, \dots, a_n) = a_1v_1 + \dots + a_nv_n.$$

Explain why Γ and Λ are inverses of each other.

Proof. Take $a_1, \dots, a_n \in \mathbb{F}^n$, then

$$\begin{aligned} \Gamma(\Lambda(a_1, \dots, a_n)) &= \Gamma(a_1v_1 + \dots + a_nv_n) \\ &= (\varphi_1(a_1v_1), \dots, \varphi_n(a_nv_n)) \\ &= (a_1, \dots, a_n) \end{aligned}$$

Similarly, take $v = a_1v_1 + \dots + a_nv_n$, we can get that

$$\Lambda(\Gamma(v)) = v$$

$\square$

Problem 9

Suppose m is a positive integer. Show that the dual basis of the basis $1, x, \dots, x^m$ of $\mathcal{P}_m(\mathbb{R})$ is $\varphi_0, \varphi_1, \dots, \varphi_m$, where

$$\varphi_k(p) = \frac{p^{(k)}(0)}{k!}.$$

Proof. We consider different cases. If $j = k$, then

$$\varphi_k(x^j) = \frac{(x^j)^{(k)}}{k!} = \frac{k!}{k!} = 1$$

If $j < k$, then

$$\varphi_k(x^j) = \frac{(x^j)^{(k)}}{k!} = \frac{0^{j-k} j! / (j-k)!}{k!} = 0$$

If $j > k$, then

$$\varphi_k(x^j) = \frac{(x^j)^{(k)}}{k!} = \frac{0}{k!} = 0$$

Hence, we have that

$$\varphi_k(x_j) = \begin{cases} 1 & \text{if } j = k \\ 0 & \text{o.w.} \end{cases}$$

□

Problem 11

Suppose $v_1, \dots, v_n$ is a basis of V and $\varphi_1, \dots, \varphi_n$ is the corresponding dual basis of V' . Prove that

$$\psi = \psi(v_1)\varphi_1 + \dots + \psi(v_n)\varphi_n.$$

Proof. By property of dual basis, we have that

$$\psi = a_1\varphi_1 + \dots + a_n\varphi_n$$

Apply v_k on both sides give that

$$\begin{aligned} \psi(v_k) &= a_1\varphi_1(v_k) + \dots + a_n\varphi_n(v_k) \\ &= a_k \end{aligned}$$

Substitute this back gives the desired equality.

□

Problem 13

Show that the dual map of the identity operator on V is the identity operator on V' .

Proof. Take arbitrary $f \in V'$, then we have that

$$I'(f)(v) = f \circ I(v) = f(v)$$

for all $v \in V$.

□

Problem 16

Suppose W is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$T' = 0 \iff T = 0.$$

Proof. This is obvious from $\dim \text{range } T' = \dim \text{range } T$. $\square$

Problem 19

Suppose $U \subseteq V$. Explain why

$$U^0 = \{\varphi \in V' : U \subseteq \text{null } \varphi\}$$

Proof. This follows from definition. $\square$

Problem 20

Suppose V is finite-dimensional and U is a subspace of V . Show that

$$U = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in U^0\}.$$

Proof. Denote $K = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in U^0\}$.

Take $u \in U$ and $\varphi \in U^0$, then by definition $\varphi(u) = 0$, and thus $u \in K$.

Conversely, take $\varphi \in U^0$. Suppose for the sake of contradiction that there exist $v \notin U$ but $v \in K$, then $\varphi(v) \neq 0$, contradicting that $v \in K$, thus completing the proof. $\square$

Problem 21

Suppose V is finite-dimensional and U and W are subspaces of V .

- (a) Prove that $W^0 \subseteq U^0$ if and only if $U \subseteq W$.
- (b) Prove that $W^0 = U^0$ if and only if $U = W$.

Proof. (a) $\Rightarrow$ We know that there exists $\psi \in V'$ such that $\text{null } \psi = W$. Then $\psi \in W^0$ and thus $\psi \in U^0$. This means that $U \subseteq W = \text{null } \psi$.

$\Leftarrow$ Take $\psi \in W^0$. So we know $\psi(w) = 0$ for all $w \in W$. Suppose for the sake of contradiction that $\psi \notin U^0$, then there exists $u \in U$ s.t. $\psi(u) \neq 0$. However, as $u \in W$, we have reached a contradiction.

(b) $W^0 = U^0 \iff W_0 \subseteq U_0$ and $U_0 \subseteq W_0 \iff U \subseteq W$ and $W \subseteq U \iff U = W$. $\square$

Problem 22

Suppose V is finite-dimensional and U and W are subspaces of V .

- (a) Show that $(U + W)^0 = U^0 \cap W^0$.
- (b) Show that $(U \cap W)^0 = U^0 + W^0$.

Proof. Note that we have

$$\begin{aligned}(U + W)^0 &= \{\varphi \in V' : \varphi(v) = 0 \text{ for every } v \in U + W\} \\ (U \cap W)^0 &= \{\varphi \in V' : \varphi(v) = 0 \text{ for every } v \in U \cap W\}\end{aligned}$$

(a) Take $\varphi \in (U + W)^0$, then we know that $\varphi(u) = 0$ and $\varphi(w) = 0$ for all $u \in U, w \in W$. Thus $(U + W)^0 \subseteq U^0 \cap W^0$. Conversely, take $\varphi \in U^0 \cap W^0$, then we know that for every $v = u + w, \varphi(v) = \varphi(u) + \varphi(w) = 0$, therefore $U^0 \cap W^0 \subseteq (U + W)^0$.

(b) Take $\varphi_1 \in U^0, \varphi_2 \in W^0$, then we have that $\varphi_1 + \varphi_2(v) = \varphi_1(v) + \varphi_2(v) = 0$ for $v \in U \cap W$, therefore $U^0 + W^0 \subseteq (U \cap W)^0$. It suffices now to show that the dimension equal each other.

$$\dim((U \cap W)^0) = \dim(V) - \dim(U \cap W)$$

Note that

$$\begin{aligned}\dim(U^0 + W^0) &= \dim(U^0) + \dim(W^0) - \dim(U^0 \cap W^0) \\ &= (\dim(V) - \dim(U)) + (\dim(V) - \dim(W)) - \dim((U + W)^0) \\ &= 2 \dim(V) - (\dim(U) + \dim(W)) - (\dim(V) - \dim(U + W)) \\ &= \dim(V) - (\dim(U) + \dim(W) - \dim(U + W)) \\ &= \dim(V) - \dim(U \cap W)\end{aligned}$$

Thus we have completed the proof. $\square$

Problem 23

Suppose V is finite-dimensional and $\varphi_1, \dots, \varphi_m \in V'$. Prove that the following three sets are equal to each other.

- (a) $\text{span}(\varphi_1, \dots, \varphi_m)$.
- (b) $((\text{null } \varphi_1) \cap \dots \cap (\text{null } \varphi_m))^0 := A$.
- (c) $\{\varphi \in V' : (\text{null } \varphi_1) \cap \dots \cap (\text{null } \varphi_m) \subseteq \text{null } \varphi\} := B$.

Proof. (a) $\Rightarrow$ (b) take $\varphi = \sum_{i=1}^m a_i \varphi_i \in \text{span}(\varphi_1, \dots, \varphi_m)$. Then we know that for every $v \in (\text{null } \varphi_1) \cap \dots \cap (\text{null } \varphi_m)$, $\varphi(v) = \sum_{i=1}^m a_i \varphi_i(v) = 0$.

(b) $\Rightarrow$ (c) Take $\varphi \in A$, then by P19 this holds.

(c) $\Rightarrow$ (a) Take $\varphi \in B$, then we know by P3 there exists $c \in \mathbb{F}$ such that $\varphi = c\varphi_i$ for all $1 \leq i \leq m$. This means that $\varphi \in \text{span}(\varphi_1, \dots, \varphi_m)$. $\square$

Problem 24

Suppose V is finite-dimensional and $v_1, \dots, v_m \in V$. Define a linear map $\Gamma: V' \rightarrow \mathbb{F}^m$ by $\Gamma(\varphi) = (\varphi(v_1), \dots, \varphi(v_m))$.

- (a) Prove that $v_1, \dots, v_m$ spans V if and only if Γ is injective.
- (b) Prove that $v_1, \dots, v_m$ is linearly independent if and only if Γ is surjective.

Proof. (a) $\Rightarrow$ Take $v = \sum_{i=1}^m a_i v_i$ and $\varphi \in \text{null } \Gamma$. We aim to prove that $\varphi = 0$. We have that $\Gamma(\varphi) = (\varphi(v_1), \dots, \varphi(v_m)) = 0$ so $\varphi(v_1) = \dots = \varphi(v_m) = 0$. So for all $v \in V$, $\varphi(v) = \sum_{i=1}^m a_i \varphi(v_i) = 0$. Thus φ is the zero map and thus Γ is injective.

$\Leftarrow$ Suppose for the sake of contradiction that $v_1, \dots, v_m$ does not span V , then this means there exists some subspace W such that $\text{span}(v_1, \dots, v_m) \oplus W = V$. Then there exists nonzero $\varphi \in V'$ such that $\varphi(v_k) = 0$ for all k (one can set the basis in W to be nonzero mapping). Then this means that $\varphi \in \text{null } \Gamma \neq 0$, contradicting that Γ is injective.

(b) $\Rightarrow$ Let K be the subspace spanned by the linearly independent list of vectors $v_1, \dots, v_m$. Then by the linear map lemma, for all $(a_1, \dots, a_m) \in \mathbb{F}^m$, there exists a unique mapping $T': K \rightarrow \mathbb{F}$ such that $T'v_i = a_i$. Extending T' to V ensures that Γ is surjective.

$\Leftarrow$ Suppose for the sake of contradiction that $v_1, \dots, v_m$ are not linearly independent, then we know there exists nonzero a'_i 's such that $a_1 v_1 + \dots + a_m v_m = 0$. Applying any linear function $\varphi \in V'$ to this linear combination yields that

$$\varphi\left(\sum_{i=1}^m a_i v_i\right) = \varphi(0) = 0$$

This equals that

$$\sum_{i=1}^m a_i \varphi(v_i) = \sum_{i=1}^m a_i \Gamma(\varphi)_i = 0$$

Since there is nonzero a_i 's, this means that the image of Γ is a strict subspace of $\mathbb{F}^m$, therefore contradicting the hypothesis that it's surjective. $\square$

Problem 25

Suppose V is finite-dimensional and $\varphi_1, \dots, \varphi_m \in V'$. Define a linear map $\Gamma: V \rightarrow \mathbb{F}^m$ by $\Gamma(v) = (\varphi_1(v), \dots, \varphi_m(v))$.

(a) Prove that $\varphi_1, \dots, \varphi_m$ spans V' if and only if Γ is injective.

(b) Prove that $\varphi_1, \dots, \varphi_m$ is linearly independent if and only if Γ is surjective.

Proof. (a) $\Rightarrow$ Take any $\varphi \in V'$, then $\varphi = \sum_{i=1}^m a_i \varphi_i$. Take any $v \in V$ and let $\Gamma(v) = 0$. Then this means that

$$\Gamma(v) = (\varphi_1(v), \dots, \varphi_m(v)) = 0$$

So we have that $\varphi_i(v) = 0$ for all i . By the spanning assumption, we have that for any $\varphi \in V'$, $\varphi(v) = \sum_{i=1}^m a_i \varphi_i(v) = 0$. Thus $v = 0$ and then $\text{null } (\Gamma) = \{0\}$.

$\Leftarrow$ Suppose $\varphi_1, \dots, \varphi_m$ does not span V' , then this means there exists some subspace W such that $\text{span}(\varphi_1, \dots, \varphi_m) \oplus W = V'$. Let $\varphi_1, \dots, \varphi_n$ be the basis of $\text{span}(\varphi_1, \dots, \varphi_m)$, $\varphi_{n+1}, \dots, \varphi_l$ to be the basis of W . Let $v_1, \dots, v_n, v_{n+1}, \dots, v_l$ be the corresponding basis for V . Then there exists nonzero $v \in V$ such that

$\varphi_k(v) = 0$ for all k as $v = \sum_{i=1}^n \varphi_i(v)v_i + \sum_{j=n+1}^l \varphi_j(v)v_j$, where one can obtain nonzero $\varphi_j(v)$ for some $n+1 \leq j \leq l$ but zero $\varphi_i(v)$ for all $1 \leq i \leq n$. Such $v \in \text{null } \Gamma$, contradicting the hypothesis that Γ is injective.

(b) $\Rightarrow$ Let K be the subspace spanned by the linearly independent list of vectors $\varphi_1, \dots, \varphi_m$. Then for all $(a_1, \dots, a_m) \in \mathbb{F}^m$, take $v = \sum_{i=1}^m a_i v_i$ for the corresponding basis $v_1, \dots, v_m$. Then we have that $\varphi_i(v) = a_i$ so Γ is surjective.

$\Leftarrow$ Suppose for the sake of contradiction that $\varphi_1, \dots, \varphi_m$ are not linearly independent, then we know there exists nonzero a_i 's such that $a_1\varphi_1 + \dots + a_m\varphi_m = 0$. Applying any vector $v \in V$ to this linear functional yields that

$$\sum_{i=1}^m a_i \varphi_i(v) = 0(v) = 0$$

This equals that

$$\sum_{i=1}^m a_i \varphi_i(v) = \sum_{i=1}^m a_i \Gamma(v)_i = 0$$

Since there is nonzero a_i 's, this means that the image of Γ is a strict subspace of $\mathbb{F}^m$, therefore contradicting the hypothesis that it's surjective. $\square$

Problem 26

Suppose V is finite-dimensional and Ω is a subspace of V' . Prove that

$$\Omega = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in \Omega\}^0$$

Proof. Denote $U = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in \Omega\}$. So we have that

$$U^0 = \{\psi \in V' : \psi(v) = 0 \text{ for every } v \in U\}$$

$\Rightarrow$ Take $\varphi \in \Omega$ and $v \in U$, then by definition $\varphi(v) = 0$ and thus $\varphi \in U^0$.

$\Leftarrow$ Take $\psi \in U^0$, suppose $\psi \notin \Omega$. Take $v \in U$, then we know that for every $\varphi \in \Omega$, $\varphi(v) = 0$. Since $\psi \notin \Omega$, there exists v , s.t. $\psi(v) \neq 0$. However, this contradicts that $\psi(v) = 0$ for all $v \in U$, the proof is completed. $\square$

Problem 28

Suppose V is finite-dimensional and $\varphi_1, \dots, \varphi_m$ is a linearly independent list in V' . Prove that

$$\dim((\text{null } \varphi_1) \cap \dots \cap (\text{null } \varphi_m)) = (\dim V) - m$$

Proof. Note that $(\bigcap_{i=1}^m \text{null } \varphi_i)^0 = \text{span}(\varphi_1, \dots, \varphi_m)$ by P23. Then we have that $\dim \bigcap_{i=1}^m \text{null } \varphi_i = \dim V - \dim \text{span}(\varphi_1, \dots, \varphi_m) = \dim V - m$. $\square$

Problem 30

Suppose V is finite-dimensional and $\varphi_1, \dots, \varphi_n$ is a basis of V' . Show that there exists a basis of V whose dual basis is $\varphi_1, \dots, \varphi_n$.

Proof. We know that $\dim V' = n$ and that $\dim \text{null } \varphi_i + 1(\dim \text{range } \varphi_i) = n$. Thus we can construct that $V = \text{null } \varphi_i \oplus U_i$ for all $1 \leq i \leq n$. Here $\dim U_i = 1$. Here we extract all v_i 's from those U_i . By linear map lemma, there exists $v_i \in U_i$ such that $\varphi_i(v_i) = 1$. We claim that $v_1, \dots, v_n$ is the corresponding basis of V .

We first verify that it is the "corresponding basis". Consider $\varphi_j(v_i)$ for $j \neq i$. Suppose for contradiction that there exists nonzero $w \in U_i \cap U_k$. Since we know all U_i are 1-d, $U_i = U_k$. Here we have that $\text{null } \varphi_i = \text{null } \varphi_k$, then this means that $\varphi_i = c\varphi_k$ for some $c \in \mathbb{F}$, but this contradicts the assumption that $\varphi_1, \dots, \varphi_n$ is the basis of V' .

Next we verify it is basis. It only suffices to verify that the list is linearly independent. To see this, $0 = \sum_{i=1}^m a_i v_i$. Apply φ_i on both side yield that $0 = \varphi_i(0) = \varphi(\sum_{i=1}^m a_i v_i) = a_i$. Thus all coefficients are zero. $\square$

Problem 32

The *double dual* space of V , denoted by V'' , is defined to be the dual space of V' . In other words, $V'' = (V')'$. Define $\Lambda: V \rightarrow V''$ by

$$(\Lambda v)(\varphi) = \varphi(v)$$

for each $v \in V$ and $\varphi \in V'$.

- (a) Show that Λ is a linear map from V to V'' .
- (b) Show that if $T \in \mathcal{L}(V)$, then $T'' \circ \Lambda = \Lambda \circ T$, where $T'' = (T')'$.
- (c) Show that if V is finite-dimensional, then Λ is an isomorphism from V onto V'' .

Proof. (a) First, $\Lambda(0)(\varphi) = \varphi(0) = 0$. Next, take $v_1, v_2 \in V$, then $\Lambda(\lambda v_1 + v_2)(\varphi) = \lambda\varphi(v_1) + \varphi(v_2) = \lambda\Lambda(v_1) + \Lambda(v_2)$.

(b) By definition, the double dual map $T'': V'' \rightarrow W''$ is defined as $T''(\psi) = \psi(T'(\varphi))$ for $\psi \in V''$ and for all $\varphi \in W'$. This further equals that $T''(\psi) = \psi \circ T(v)$ for $v \in V$.

$$\begin{aligned} T'' \circ (\Lambda v)(\varphi) &= (\Lambda v) \circ T'(\varphi) \\ &= (\Lambda v)\varphi \circ T \\ &= \Lambda \circ T \end{aligned}$$

(c) First we have that $\dim V = \dim V' = \dim V''$. Note that if we take $v \in \text{null } \Lambda$, then this means for all linear functional $\varphi \in V'$, we have that $\Lambda v(\varphi) = \varphi(v) = 0$. So $v = 0$ and thus Λ is injective and therefore an isomorphism. $\square$

Chapter 4: Polynomials

Linear Algebra Done Right (4th Edition), by Sheldon Axler

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Definition 1 ($\operatorname{Re} z$, $\operatorname{Im} z$). Suppose $z = a + bi$, where a and b are real numbers.

- The **real part** of z , denoted by $\operatorname{Re} z$, is defined by $\operatorname{Re} z = a$.
- The **imaginary part** of z , denoted by $\operatorname{Im} z$, is defined by $\operatorname{Im} z = b$.

Definition 2 (complex conjugate, $\bar{z}$, absolute value, $|z|$). Suppose $z \in \mathbb{C}$.

- The **complex conjugate** of $z \in \mathbb{C}$, denoted by $\bar{z}$, is defined by

$$\bar{z} = \operatorname{Re} z - i \operatorname{Im} z$$

- The **absolute value** of a complex number z , denoted by $|z|$, is defined by

$$|z| = \sqrt{(\operatorname{Re}(z))^2 + (\operatorname{Im}(z))^2}$$

A function $p: \mathbb{F} \rightarrow \mathbb{F}$ is called a polynomial of degree m if there exists $a_0, \dots, a_m \in \mathbb{F}$ with $a_m \neq 0$ such that

$$p(z) = a_0 + a_1 z + \dots + a_m z^m$$

for all $z \in \mathbb{F}$.

Definition 3 (zero of a polynomial). A number $\lambda \in \mathbb{F}$ is called a **zero** (or **root**) of a polynomial $p \in \mathcal{P}(\mathbb{F})$ if

$$p(\lambda) = 0.$$

Lemma 4 (each zero of a polynomial corresponds to a degree-one factor). Suppose m is a positive integer and $p \in \mathcal{P}(\mathbb{F})$ is a polynomial of degree m . Suppose $\lambda \in \mathbb{F}$. Then $p(\lambda) = 0$ if and only if there exists a polynomial $q \in \mathcal{P}(\mathbb{F})$ of degree $m - 1$ such that

$$p(z) = (z - \lambda)q(z)$$

for every $z \in \mathbb{F}$.

Theorem 5 (degree m implies at most m zeros). Suppose m is a positive integer and $p \in \mathcal{P}(\mathbb{F})$ is a polynomial of degree m . Then p has at most m zeros in $\mathbb{F}$.

Theorem 6 (division algorithm for polynomials). Suppose that $p, s \in \mathcal{P}(\mathbb{F})$ with $s \neq 0$. Then there exist unique polynomials $q, r \in \mathcal{P}(\mathbb{F})$ such that

$$p = sq + r$$

and $\deg r < \deg s$.

Theorem 7 (fundamental theorem of algebra, first version). *Every nonconstant polynomial with complex coefficients has a zero in $\mathbb{C}$.*

Theorem 8 (fundamental theorem of algebra, second version). *If $p \in \mathcal{P}(\mathbb{C})$ is a nonconstant polynomial, then p has a unique factorization (except for the order of the factors) of the form*

$$p(z) = c(z - \lambda_1) \cdots (z - \lambda_m),$$

where $c, \lambda_1, \dots, \lambda_m \in \mathbb{C}$.

Lemma 9 (polynomials with real coefficients have nonreal zeros in pairs). *Suppose $p \in \mathcal{P}(\mathbb{C})$ is a polynomial with real coefficients. If $\lambda \in \mathbb{C}$ is a zero of p , then so is $\bar{\lambda}$.*

Theorem 10 (factorization of a quadratic polynomial). *Suppose $b, c \in \mathbb{R}$. Then there is a polynomial factorization of the form*

$$x^2 + bx + c = (x - \lambda_1)(x - \lambda_2)$$

with $\lambda_1, \lambda_2 \in \mathbb{R}$ if and only if $b^2 \geq 4c$.

Theorem 11 (factorization of a polynomial over $\mathbb{R}$). *Suppose $p \in \mathcal{P}(\mathbb{R})$ is a nonconstant polynomial. Then p has a unique factorization (except for the order of the factors) of the form*

$$p(x) = c(x - \lambda_1) \cdots (x - \lambda_m)(x^2 + b_1x + c_1) \cdots (x^2 + b_Mx + c_M)$$

where $c, \lambda_1, \dots, \lambda_m, c_1, \dots, c_M \in \mathbb{R}$, with $b_k^2 < 4c_k$ for each k .

Problem 1

Suppose $w, z \in \mathbb{C}$. Verify the following equalities and inequalities.

- (a) $z + \bar{z} = 2\Re z$
- (b) $z - \bar{z} = 2(\Im z)i$
- (c) $z\bar{z} = |z|^2$
- (d) $\overline{w + z} = \bar{w} + \bar{z}$ and $\overline{wz} = \bar{w}\bar{z}$
- (e) $\bar{\bar{z}} = z$
- (f) $|\Re z| \leq |z|$ and $|\Im z| \leq |z|$
- (g) $|\bar{z}| = |z|$
- (h) $|wz| = |w||z|$

Proof. Suppose $z = a + bi, w = c + di$. We verify these as follows:

- (a) $z + \bar{z} = (a + bi) + (a - bi) = 2a = 2\Re z$
- (b) $z - \bar{z} = (a + bi) - (a - bi) = 2bi = 2(\Im z)i$
- (c) $z\bar{z} = (a + bi)(a - bi) = a^2 + b^2 = |z|^2$
- (d) $\overline{w + z} = \overline{(a + c) + (b + d)i} = (a + c) - (b + d)i = (a - bi) + (c - di) = \bar{z} + \bar{w}$
- (e) $\bar{\bar{z}} = \overline{a - bi} = a + bi = z$
- (f) $|\Re z| = a^2 \leq a^2 + b^2 = |z|^2$; $|\Im z| = b^2 \leq a^2 + b^2 = |z|^2$
- (g) $|\bar{z}| = |a - bi| = |a + bi| = |z|$
- (h) $|wz| = |(c + di)(a + bi)| = |(ca - db) + (cb + da)i| = (ca - db)^2 + (cd + ba)^2$
 This further equals that $|wz| = c^2a^2 + d^2b^2 + c^2d^2 + b^2a^2$. At the same time $|w||z| = (c^2 + d^2)(a^2 + b^2) = c^2a^2 + c^2b^2 + d^2a^2 + d^2b^2$. Therefore they equal each other.

□

Problem 2

Prove the **reverse triangle inequality**: if $w, z \in \mathbb{C}$, then $||w| - |z|| \leq |w - z|$.

Proof. Suppose $z = a + bi, w = c + di$. We have that

$$\begin{aligned}
 |w - z|^2 &= (w - z)(\overline{w} - \overline{z}) \\
 &= w\overline{w} + z\overline{z} - w\overline{z} - z\overline{w} \\
 &= |w|^2 + |z|^2 - w\overline{z} - \overline{w}z \\
 &= |w|^2 + |z|^2 - 2\Re(w\overline{z}) \\
 &\leq |w|^2 + |z|^2 - 2|w\overline{z}| \\
 &= |w|^2 + |z|^2 - 2|w||z| \\
 &= (|w| - |z|)^2
 \end{aligned}$$

Taking the square root gets the desired equality. $\square$

Problem 3

Suppose V is a complex vector space and $\varphi \in V'$. Define $\sigma: V \rightarrow \mathbb{R}$ by $\sigma(v) = \Re\varphi(v)$ for each $v \in V$. Show that

$$\varphi(v) = \sigma(v) - i\sigma(iv)$$

for all $v \in V$.

Proof.

$$\begin{aligned}
 \varphi(v) &= \Re\varphi(v) + \Im\varphi(v) \\
 &= \sigma(v) + i\Re(-iiv) \\
 &= \sigma(v) - i\sigma(iv)
 \end{aligned}$$

$\square$

Problem 6

Suppose that m and n are positive integers with $m \leq n$, and suppose $\lambda_1, \dots, \lambda_m \in \mathbb{F}$. Prove that there exists a polynomial $p \in \mathcal{P}(\mathbb{F})$ with $\deg p = n$ such that $0 = p(\lambda_1) = \dots = p(\lambda_m)$ and such that p has no other zeros.

Proof. We can first construct $p(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_m)$ for $x \in \mathbb{F}$. This is a polynomial with degree m and has exactly m zeros. To increase the degree without introducing additional roots, we can make that $p(x) = (x - \lambda_1) \cdots (x - \lambda_m)^{n-m+1}$. This polynomial doesn't have additional roots, as the proof procedure follows from the second version of fundamental theorem of algebra. $\square$

Problem 7

Suppose that m is a nonnegative integer, $z_1, \dots, z_{m+1}$ are distinct elements of $\mathbb{F}$, and $w_1, \dots, w_{m+1} \in \mathbb{F}$. Prove that there exists a unique polynomial $p \in \mathcal{P}_m(\mathbb{F})$ such that

$$p(z_k) = w_k$$

for each $k = 1, \dots, m+1$.

Proof. Define a linear map $T: \mathcal{P}_m(\mathbb{F}) \rightarrow \mathbb{F}^{m+1}$ by

$$T(p) = (p(z_1), \dots, p(z_{m+1}))$$

We can tell that T is injective because if there exists nonzero $p \in \text{null } T$, then this means there are $m+1$ roots for the m -deg polynomial, forming a contradiction. Therefore, T is an isomorphism as the dimension matches.

This implies the existence of the unique solution as the question asks. $\square$

Problem 9

Prove that every polynomial of odd degree with real coefficients has a real zero.

Proof. Suppose for the sake of contradiction that there is no real root. Then the factorization would become

$$p(x) = (x^2 + b_1x + c_1) \cdots (x^2 + b_Mx + c_M)$$

but this contradicts that the degree is odd. $\square$

Problem 11

Suppose $p \in \mathcal{P}(\mathbb{C})$. Define $q: \mathbb{C} \rightarrow \mathbb{C}$ by

$$q(z) = p(z)\overline{p(\bar{z})}$$

Prove that q is a polynomial with real coefficients.

Proof. Suppose $p(z) = \sum_{i=0}^n a_i z^i$ for $a_i \in \mathbb{C}, z \in \mathbb{C}$. Then we can find that

$$p(\bar{z}) = \sum_{i=0}^n a_i \bar{z}^i$$

and thus

$$\overline{p(\bar{z})} = \overline{\sum_{i=0}^n a_i \bar{z}^i} = \sum_{i=0}^n \overline{a_i} z^i$$

Finally, we can get that

$$q(z) = p(z)\overline{p(\bar{z})} = \left(\sum_{i=0}^n a_i z^i\right) \left(\sum_{i=0}^n \bar{a}_i z^i\right)$$

We can consider the general form of a coefficient of z^k :

$$c_k = \sum_{i+j=k} a_i \bar{a}_j = \sum_{j+i=k} \bar{a}_j a_i = \overline{c_k}$$

Thus the coefficients are real.

□

Chapter 5: Eigenvalues and Eigenvectors

Linear Algebra Done Right (4th Edition), by Sheldon Axler

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Contents

5A: Invariant Subspaces	2
5A Problem Sets	4
5B: The Minimal Polynomial	14
5B Problem Sets	15
5C: Upper-Triangular Matrices	21
5C Problem Sets	22
5D: Diagonalizable Operators	26
5D Problem Sets	28
5E: Commuting Operators	34
5E Problem Sets	35

5A: Invariant Subspaces

Definition 1 (operator). A linear map from a vector space to itself is called an **operator**.

Definition 2 (invariant subspace). Suppose $T \in \mathcal{L}(V)$. A subspace U of V is called **invariant** under T if $Tu \in U$ for every $u \in U$.

Remark 3. Four common invariant subspaces of $T \in \mathcal{L}(V)$ are: $\{0\}$, V , $\text{null } T$, $\text{range } T$. To discover invariant subspaces other than $\{0\}$, V (the simplest possible nontrivial one is the invariant subspace of dimension one) motivates the concept of eigenspaces.

Definition 4 (eigenvalue). Suppose $T \in \mathcal{L}(V)$. A number $\lambda \in \mathbb{F}$ is called an **eigenvalue** of T if there exists $v \in V$ such that $v \neq 0$ and $Tv = \lambda v$.

Remark 5. Note that here we can think of $U = \{\lambda v : \lambda \in \mathbb{F}\} = \text{span}(v)$ as the one-dimensional invariant subspace spanned by the eigenvector v .

Lemma 6 (equivalent conditions to be an eigenvalue). Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$. Then the following are equivalent:

- (a) λ is an eigenvalue of T .
- (b) $T - \lambda I$ is not injective.
- (c) $T - \lambda I$ is not surjective.
- (d) $T - \lambda I$ is not invertible.

Definition 7 (eigenvector). Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$ is an eigenvalue of T . A vector $v \in V$ is called an **eigenvector** of T corresponding to λ if $v \neq 0$ and $Tv = \lambda v$.

Remark 8. A vector $v \in V$ with $v \neq 0$ is an eigenvector of T corresponding to λ if and only if $v \in \text{null}(T - \lambda I)$.

$$Tv = \lambda v \iff (T - \lambda I)v = 0$$

Lemma 9 (linearly independent eigenvectors). Suppose $T \in \mathcal{L}(V)$. Then every list of eigenvectors of T corresponding to distinct eigenvalues of T is linearly independent.

Theorem 10 (operator cannot have more eigenvalues than dimension of vector space). Suppose V is finite-dimensional. Then each operator on V has at most $\dim V$ distinct eigenvalues.

Definition 11 (T^m). Suppose $T \in \mathcal{L}(V)$ and m is a positive integer.

- $T^m \in \mathcal{L}(V)$ is defined by $T^m = \underbrace{T \cdots T}_{m \text{ times}}$.

- T^0 is defined to be the identity operator I on V .
- If T is invertible with inverse T^{-1} , then $T^{-m} \in \mathcal{L}(V)$ is defined by

$$T^{-m} = (T^{-1})^m.$$

Definition 12 ($p(T)$). Suppose $T \in \mathcal{L}(V)$ and $p \in \mathcal{P}(\mathbb{F})$ is a polynomial given by

$$p(z) = a_0 + a_1z + a_2z^2 + \cdots + a_mz^m$$

for all $z \in \mathbb{F}$. Then $p(T)$ is the operator on V defined by

$$p(T) = a_0I + a_1T + a_2T^2 + \cdots + a_mT^m.$$

Remark 13. If we fix an operator $T \in \mathcal{L}(V)$, then the function from $\mathcal{P}(\mathbb{F})$ to $\mathcal{L}(V)$ given by $p \mapsto p(T)$ is linear.

Definition 14 (product of polynomials). If $p, q \in \mathcal{P}(\mathbb{F})$, then $pq \in \mathcal{P}(\mathbb{F})$ is the polynomial defined by

$$(pq)(z) = p(z)q(z)$$

for all $z \in \mathbb{F}$.

Theorem 15. Suppose $p, q \in \mathcal{P}(\mathbb{F})$ and $T \in \mathcal{L}(V)$. Then

- (a) $(pq)(T) = p(T)q(T)$;
- (b) $p(T)q(T) = q(T)p(T)$.

Definition 16 (null space and range of $p(T)$ are invariant under T). Suppose $T \in \mathcal{L}(V)$ and $p \in \mathcal{P}(\mathbb{F})$. Then null $p(T)$ and range $p(T)$ are invariant under T .

Problem 1

Suppose $T \in \mathcal{L}(V)$ and U is a subspace of V .

- (a) Prove that if $U \subseteq \text{null } T$, then U is invariant under T .
- (b) Prove that if $\text{range } T \subseteq U$, then U is invariant under T .

Proof. (a) Suppose for contradiction that U is not invariant under T . Then this means that there exists $u \in U$ such that $Tu \notin U$. We know that $Tu = 0 \in \text{null } T$. As $U \in \text{null } T$, this forms a contradiction.

(b) Take $u \in U$, then $Tu \in \text{range } T \subseteq U$, and thus $Tu \in U$. □

Problem 3

Suppose $T \in \mathcal{L}(V)$. Prove that the intersection of every collection of subspaces of V invariant under T is invariant under T .

Proof. Suppose $\{V_i\}_{i=1}^m$ for positive integer m is a collection of invariant subspaces. We hope to prove that $\bigcap_{i=1}^m V_i$ is invariant.

Take any $v \in \bigcap_{i=1}^m V_i$, then we know that $Tv \in V_i$ for each i and thus $Tv \in \bigcap_{i=1}^m V_i$. □

Problem 5

Suppose $T \in \mathcal{L}(\mathbb{R}^2)$ is defined by $T(x, y) = (-3y, x)$. Find the eigenvalues of T .

Proof. We have that

$$T(x, y) = (-3y, x) = \lambda(x, y)$$

Then this means that $-3y = \lambda x$, $x = \lambda y$. Solving this gives that $\lambda^2 = -3$ and thus $\lambda = \pm\sqrt{3}i$. □

Problem 7

Define $T \in \mathcal{L}(\mathbb{F}^3)$ by $T(z_1, z_2, z_3) = (2z_2, 0, 5z_3)$. Find all eigenvalues and eigenvectors of T .

Proof.

$$T(z_1, z_2, z_3) = (2z_2, 0, 5z_3) = \lambda(z_1, z_2, z_3)$$

Then $\lambda z_1 = 2z_2$, $\lambda z_2 = 0$, $\lambda z_3 = 5z_3$. This gives that $\lambda = 5$ or 0 . For $\lambda = 0$, the eigenvector would be $\{(z_1, 0, z_3) : z_1, z_3 \in \mathbb{F}\}$. For $\lambda = 5$, we have that $\{(0, 0, z_3) : z_3 \in \mathbb{F}\}$. □

Problem 9

Define $T: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$ by $Tp = p'$. Find all eigenvalues and eigenvectors of T .

Proof.

$$Tp = p' = \lambda p$$

This means that $\deg p' = \deg p$ so $\deg p = 0$. In this case, the only satisfying solution is $\lambda = 0$ and correspondingly $v = \{c\}: c \in \mathbb{F}\}$. $\square$

Problem 11

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $\alpha \in \mathbb{F}$. Prove that there exists $\delta > 0$ such that $T - \lambda I$ is invertible for all $\lambda \in \mathbb{F}$ such that $0 < |\alpha - \lambda| < \delta$.

Proof. Let λ^* be the eigenvalue of T which is closest to α , i.e. $\lambda^* = \min_{\lambda} \{|\lambda - \alpha|\}$. We can take $\delta = |\alpha - \lambda^*|/2$, then we know that under the condition $T - \lambda I$ is injective and thus reaching the desired result. $\square$

Problem 13

Suppose $T \in \mathcal{L}(V)$. Suppose $S \in \mathcal{L}(V)$ is invertible.

- (a) Prove that T and $S^{-1}TS$ have the same eigenvalues.
- (b) What is the relationship between the eigenvectors of T and the eigenvectors of $S^{-1}TS$?

Proof. (a) Notice that $S^{-1}v$ is an eigenvector of $S^{-1}TS$ for $v \in V$. For the forward direction, take λ to be the eigenvalue of T . Then we know that $Tv = \lambda v$ for all $v \in V$. Let's consider $S^{-1}TS(S^{-1}u) = S^{-1}Tu = S^{-1}\lambda u = \lambda(S^{-1}u)$. Therefore, λ is also an eigenvalue for $S^{-1}TS$. Conversely, take λ to be the eigenvalue of $S^{-1}TS$. Then we have that $S^{-1}TS(S^{-1}v) = S^{-1}Tv = \lambda S^{-1}v$. Multiplying S on both sides yield that $Tv = \lambda v$. Thus λ is also an eigenvalue of T .

- (b) v is an eigenvector of $S^{-1}TS$ iff $u = Sv$ is an eigenvector of T .

$\square$

Problem 15

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbb{F}$. Show that λ is an eigenvalue of T if and only if λ is an eigenvalue of the dual operator $T' \in \mathcal{L}(V')$.

Proof. Forward direction, take λ to be an eigenvalue of T , then we have that $Tv = \lambda v$. Then take any $\varphi \in V'$, we have that $T'(\varphi)(v) = \varphi \circ Tv = \lambda\varphi(v)$.

Backward direction, take λ to be an eigenvalue of T' , then we know there exists nonzero $\varphi \in V'$ such that $T'\varphi(v) = \varphi T(v) = \lambda\varphi(v)$. This implies that

$$\varphi(Tv - \lambda v) = 0$$

Suppose for contradiction that λ is not an eigenvalue of T , then this means that $T - \lambda I$ is invertible and therefore its image equals the whole space V . Then this means that φ is a zero functional, forming a contradiction. Hence, $Tv = \lambda v$ and thus λ is an eigenvalue of T as well. □

Problem 16

Suppose $v_1, \dots, v_n$ is a basis of V and $T \in \mathcal{L}(V)$. Prove that if λ is an eigenvalue of T , then

$$|\lambda| \leq n \max\{|\mathcal{M}(T)_{j,k}| : 1 \leq j, k \leq n\},$$

where $\mathcal{M}(T)_{j,k}$ denotes the entry in row j , column k of the matrix of T with respect to the basis $v_1, \dots, v_n$.

Proof. We know that

$$Tv_k = \sum_{j=1}^n \mathcal{M}(T)_{j,k} v_j = \lambda v_k$$

and that by Triangular inequality we have

$$|Tv_k| \leq \sum_{j=1}^n |\mathcal{M}(T)_{j,k} v_j| \leq M \sum_{j=1}^n |v_j|$$

where we take $M = \max\{|\mathcal{M}(T)_{j,k}| : 1 \leq j, k \leq n\}$. Here we take k to be the index such that maximizes the norm, i.e. $k = \max_k |v_k|$. Then we have that

$$|Tv_k| = |\lambda| |v_k| \leq nM |v_k|$$

which gives us that

$$|\lambda| \leq nM$$

□

Problem 18

Suppose $\mathbb{F} = \mathbb{R}$, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbb{C}$. Prove that λ is an eigenvalue of the complexification T_C if and only if $\bar{\lambda}$ is an eigenvalue of T_C , where $T_C: \mathcal{L}(V_C)$ is defined by

$$T_C(u + iv) = Tu + iTv$$

for all $u, v \in V$.

Proof. Let $v_1, \dots, v_n$ be basis of V_C . Suppose λ is an eigenvalue of T_C and $v = \sum_{i=1}^n a_i v_i$ is the corresponding eigenvector. Then we have that

$$\lambda v = Tv = \sum_{i=1}^n a_i T v_i$$

Taking conjugates yields that

$$\bar{\lambda} \bar{v} = \overline{\sum_{i=1}^n a_i T v_i} = \sum_{i=1}^n \bar{a}_i \overline{T v_i}$$

Let M be the matrix of T with real entries, and then we have

$$\overline{T v_i} = \overline{\sum_{j=1}^n M_{j,i} v_j} = \sum_{j=1}^n M_{j,i} \bar{v}_j = T \bar{v}_i$$

Substitute this back to the previous equation gives that

$$\bar{\lambda} \bar{v} = T \left(\overline{\sum_{i=1}^n a_i v_i} \right) = T \bar{v}$$

Hence $\bar{\lambda}$ is also an eigenvalue. □

Problem 19

Show that the forward shift operator $T \in \mathcal{L}(\mathbb{F}^\infty)$ defined by

$$T(z_1, z_2, \dots) = (0, z_1, z_2, \dots)$$

has no eigenvalues.

Proof. We have that

$$(0, z_1, z_2, \dots) = \lambda(z_1, z_2, \dots)$$

So we either have $\lambda = 0$ or $z_1 = 0 = z_2 = \dots$. In the first case, we have that $z_1 = 0 = z_2 = \dots$ and thus there is no nonzero eigenvector. In the second case, there is also no nonzero eigenvectors. □

Problem 21

Suppose $T \in \mathcal{L}(V)$ is invertible.

- (a) Suppose $\lambda \in \mathbb{F}$ with $\lambda \neq 0$. Prove that λ is an eigenvalue of T if and only if $\frac{1}{\lambda}$ is an eigenvalue of T^{-1} .
- (b) Prove that T and T^{-1} have the same eigenvectors.

Proof.

$$Tv = \lambda v \iff T^{-1}Tv = T^{-1}\lambda v \iff \frac{1}{\lambda}v = T^{-1}v$$

□

Problem 23

Suppose V is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that ST and TS have the same eigenvalues.

Proof. $\Rightarrow$ Let λ, v be the eigenvalue and eigenvector of ST . Then take $u = Tv$, we have that

$$TS(u) = TS(Tv) = T(ST)(v) = T(\lambda v) = \lambda(Tv) = \lambda u$$

Thus λ is also the eigenvalue of TS .

Conversely, let λ, v be eigenvalue and eigenvector of TS . Then take $u = Sv$, we have that

$$ST(u) = ST(Sv) = S(TS)(v) = S(\lambda v) = \lambda(Sv) = \lambda u$$

Thus λ is also the eigenvalue of ST .

□

Problem 24

Suppose A is an n -by- n matrix with entries in $\mathbb{F}$. Define $T \in \mathcal{L}(\mathbb{F}^n)$ by $Tx = Ax$, where elements of $\mathbb{F}^n$ are thought of as n -by-1 column vectors.

- (a) Suppose the sum of the entries in each row of A equals 1. Prove that 1 is an eigenvalue of T .
- (b) Suppose the sum of entries in each column of A equals 1. Prove that 1 is an eigenvalue of T .

Proof. Let $x = (1, \dots, 1)^\top$ be the all-one vector. Then we have that

$$(Tx)_{i,:} = (Ax)_{i,:} = \sum_{j=1}^n A_{i,j}x_j = \sum_{j=1}^n A_{i,j} = 1$$

Therefore, $Tx = x$ and thus 1 is an eigenvalue of T .

For the column case, we can simply take $A = A^\top$ and we know the eigenvalues of A equal $A^\top$. □

Problem 25

Suppose $T \in \mathcal{L}(V)$ and u, w are eigenvectors of T such that $u + w$ is also an eigenvector of T . Prove that u and w are eigenvectors of T corresponding to the same eigenvalue.

Proof. Suppose not. Then there exists λ_u and λ_w such that

$$Tu = \lambda_u u \quad Tw = \lambda_w w$$

We also know that

$$T(u + w) = Tu + Tw = \lambda_u u + \lambda_w w = \lambda(u + w)$$

for some λ . Solving this gives that $\lambda = \lambda_u = \lambda_w$. $\square$

Problem 26

Suppose $T \in \mathcal{L}(V)$ is such that every nonzero vector in V is an eigenvector of T . Prove that T is a scalar multiple of the identity operator.

Proof. We claim that $Tv = \lambda v$ for all nonzero $v \in V$, i.e. $T = \lambda I$. Let's get $v_1, v_2 \in V$. First we consider the case that they are linearly dependent, meaning that $v_1 = cv_2$ for some c . We have that

$$\lambda_1 v_1 = Tv_1 = cTv_2 = c\lambda_2 v_2 = \lambda_2 v_1$$

Therefore $\lambda_1 = \lambda_2$. Next we consider the case that they are linearly independent, then we have that

$$\lambda_{1+2}(v_1 + v_2) = T(v_1 + v_2) = Tv_1 + Tv_2 = \lambda_1 v_1 + \lambda_2 v_2$$

Therefore $\lambda_{1+2} = \lambda_1 = \lambda_2$, completing the proof. $\square$

Problem 28

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has at most $1 + \dim \text{range } T$ distinct eigenvalues.

Proof. Suppose T has m distinct eigenvalues. For nonzero λ_k , we have that

$$T\left(\frac{1}{\lambda_k}v_k\right) = v_k$$

Thus at most $m - 1$ distinct eigenvectors are in $\text{range } T$, so $m - 1 \leq \dim \text{range } T$ and $m = 1 + \dim \text{range } T$ at most. $\square$

Problem 33

Suppose $T \in \mathcal{L}(V)$ and m is a positive integer.

- (a) Prove that T is injective if and only if T^m is injective.
- (b) Prove that T is surjective if and only if T^m is surjective.

Proof. (a) $\Rightarrow$ Given T is injective, take $v \in \text{null } T^m$. Then we know that $T^m(v) = T(T^{m-1}v) = 0$, which implies that $T^{m-1}v = 0$. This can be recursively deduced to that $Tv = 0$ and thus $v = 0$. Therefore T^m is injective.

$\Leftarrow$ Conversely, take $v \in \text{null } T$, then we have $T^m(v) = T^{m-1}(Tv) = T^{m-1}(0) = 0$. Since T^m is injective, $v = 0$.

(b) $\Rightarrow$ Given T is surjective, then $\dim \text{range } T = \dim V$. We also have $\dim \text{range } T^2 = \dim V$ as T is surjective. Therefore, T^m is surjective.

$\Leftarrow$ Let $w \in V$, then there exists $v \in V$ such that $T^m(v) = w$. Then we have that let $u = T^{m-1}(v)$, then $Tu = w$. Thus T is also surjective. $\square$

Problem 34

Suppose V is finite-dimensional and $v_1, \dots, v_m \in V$. Prove that the list $v_1, \dots, v_m$ is linearly independent if and only if there exists $T \in \mathcal{L}(V)$ such that $v_1, \dots, v_m$ are eigenvectors of T corresponding to distinct eigenvalues.

Proof. $\Rightarrow$ We can define $Tv_k = kv_k$ and extend T to V .

$\Leftarrow$ This is proved by Theorem in the book. $\square$

Problem 35

Suppose that $\lambda_1, \dots, \lambda_n$ is a list of distinct real numbers. Prove that the list $e^{\lambda_1 x}, \dots, e^{\lambda_n x}$ is linearly independent in the vector space of real-valued functions on $\mathbb{R}$.

Proof. Following the hint, let $V = \text{span}(e^{\lambda_1 x}, \dots, e^{\lambda_n x})$. Define an operator $D \in \mathcal{L}(V)$ by $Df = f'$. Then we have that

$$De^{\lambda_k x} = \lambda_k e^{\lambda_k x}$$

Here naturally λ_k is a distinct eigenvalue with corresponding eigenvector $e^{\lambda_k x}$. By P34, we complete the proof. $\square$

Problem 37

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by

$$\mathcal{A}(S) = TS$$

for each $S \in \mathcal{L}(V)$. Prove that the set of eigenvalues of T equals the set of eigenvalues of $\mathcal{A}$.

Proof. $\Rightarrow$ Let λ be an eigenvalue of T and v be the corresponding eigenvector of T . Suppose $v_1, \dots, v_n$ is the basis of V , we can define $S \in \mathcal{L}(V)$ such that

$$Sv_i = v$$

for all i . Here we have that

$$\begin{aligned} \mathcal{A}(S)(v) &= TS(v) \\ &= TS \sum_{i=1}^n a_i v_i \\ &= T \sum_{i=1}^n a_i Sv_i \\ &= \sum_{i=1}^n a_i Tv \\ &= \lambda \sum_{i=1}^n a_i v \\ &= \lambda \sum_{i=1}^n S(a_i v_i) \\ &= (\lambda S)(v) \end{aligned}$$

Hence λ is also an eigenvalue of $\mathcal{A}(S)$.

$\Leftarrow$ Consider

$$\mathcal{A}(S) = TS = \lambda S$$

Applying v on both sides yields that

$$T(Sv) = \lambda(Sv)$$

Therefore, λ is also an eigenvalue of T . □

Problem 38

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and U is a subspace of V invariant under T . The *quotient operator* $T/U \in \mathcal{L}(V/U)$ is defined by

$$(T/U)(v + U) = Tv + U$$

for each $v \in V$.

- (a) Show that the definition of T/U makes sense and show that T/U is an operator on V/U .
- (b) Show that each eigenvalue of T/U is an eigenvalue of T .

Proof. (a) Suppose $v - w \in U$. As U is invariant under T , $T(v - w) = Tv - Tw \in U$. So we have that $Tv + U = Tw + U$ and thus it is well-defined. To show that T/U is an operator, we can just check its linearity. $(T/U)((\lambda v_1 + U) + (v_2 + U)) = (T/U)((\lambda v_1 + v_2) + U) = \lambda_1 T v_1 + U + T v_2 + U = \lambda_1 T/U(v_1 + U) + T/U(v_2 + U)$.
(b) Let λ be the eigenvalue of T/U , then we have

$$(T/U)(v + U) = Tv + U = \lambda v + U$$

This means that

$$Tv = \lambda v + u$$

for some $u \in U$. Now suppose we also have $u' \in U$, then

$$T(v + u') = \lambda v + u + Tu'$$

We hope to find u' s.t. $u + Tu' = \lambda u'$, equivalently $(\lambda I - T)u' = u$.

Since U is invariant under T , it is also invariant under $T - \lambda I$. Here if $T - \lambda I$ is not invertible on U , then we are done (λ would be an eigenvalue). If it is, then we can define $u' = -(T - \lambda I)^{-1}u$, completing the proof. $\square$

Problem 39

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has an eigenvalue if and only if there exists a subspace of V of dimension $\dim V - 1$ that is invariant under T .

Proof. $\Rightarrow$ By the hypothesis, there exists $\lambda \in \mathbb{F}$, $v \in V$ with $v \neq 0$ s.t. $Tv = \lambda v$. We claim that

$$v^\perp = \{u \in V : v^\top u = 0\}$$

is invariant under T of dimension $n - 1$. Take any $u \in v^\perp$, then $v^\top T(u) = (v^\top T)(u) = \lambda v^\top u = 0$. Hence $T(u) \in v^\perp$. This set has dimension $\dim V - 1$ as it is the orthogonal complement of a dimensional 1 subspace.

$\Leftarrow$ Let U be the the invariant subspace of $\dim V - 1$ under T . Let $\{w_1, \dots, w_{n-1}\}$ be its basis. Then we can extend the basis to $\{w_1, \dots, w_{n-1}, v\}$ for some $v \in V \setminus U$. Hence, we have that

$$T(v) = \sum_{i=1}^{n-1} a_i w_i + cv$$

for some scalar $c \in \mathbb{F}$. We claim that c is the eigenvalue of T . Suppose not, then we can get that

$$(T - cI)(v) = \sum_{i=1}^{n-1} a_i w_i$$

Then this means that $v \in U$, reaching a contradiction. $\square$

Problem 40

Suppose $S, T \in \mathcal{L}(V)$ and S is invertible. Suppose $p \in \mathcal{P}(\mathbb{F})$ is a polynomial. Prove that

$$p(STS^{-1}) = Sp(T)S^{-1}$$

Proof. We note that

$$(STS^{-1})^m = (STS^{-1})(STS^{-1}) \cdots (STS^{-1}) = ST^m S^{-1}$$

Therefore,

$$\begin{aligned} p(STS^{-1}) &= a_0 I + a_1 (STS^{-1}) + \cdots + a_m (STS^{-1})^m \\ &= a_0 I + a_1 (STS^{-1}) + \cdots + a_m (ST^m S^{-1}) \\ &= S(a_0 I + a_1 T + \cdots + a_m T^m)S^{-1} \\ &= Sp(T)S^{-1} \end{aligned}$$

□

Problem 41

Suppose $T \in \mathcal{L}(V)$ and U is a subspace of V invariant under T . Prove that U is invariant under $p(T)$ for every polynomial $p \in \mathcal{L}(\mathbb{F})$.

Proof. Take $u \in U$, then

$$p(T)(u) = a_0 I(u) + a_1 T(u) + \cdots + a_m T^m(u)$$

Now it suffices to prove that $T^m(u) \in U$ for all positive integer. To see this, we can make an inductive argument on m and gets the desired conclusion. □

5B: The Minimal Polynomial

Theorem 17 (existence of eigenvalues). *Every operator on a finite-dimensional nonzero complex vector space has an eigenvalue.*

Definition 18 (monic polynomial). A **monic polynomial** is a polynomial whose highest-degree coefficient equals 1.

Theorem 19 (existence, uniqueness, and degree of minimal polynomial). *Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then there is a unique monic polynomial $p \in \mathcal{P}(\mathbb{F})$ of smallest degree such that $p(T) = 0$. Furthermore, $\deg p \leq \dim V$.*

Definition 20 (minimal polynomial). *Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then the **minimal polynomial** of T is the unique monic polynomial $p \in \mathcal{P}(\mathbb{F})$ of smallest degree such that $p(T) = 0$.*

Remark 21. *This means to find the smallest positive integer m such that*

$$c_0I + c_1T + \cdots + c_{m-1}T^{m-1} = -T^m$$

has a solution $c_0, c_1, \dots, c_{m-1} \in \mathbb{F}$. A way to solve this is to pick $v \in V$ with $v \neq 0$ and consider that

$$c_0v + c_1Tv + \cdots + c_{\dim V-1}T^{\dim V-1}v = -T^{\dim V}v.$$

Theorem 22 (eigenvalues are the zeros of the minimal polynomial). *Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$.*

- (a) *The zeros of the minimal polynomial of T are the eigenvalues of T .*
- (b) *If V is a complex vector space, then the minimal polynomial of T has the form*

$$(z - \lambda_1) \cdots (z - \lambda_m).$$

where $\lambda_1, \dots, \lambda_m$ is a list of all eigenvalues of T , possibly with repetitions.

Theorem 23 ($q(T) = 0 \iff q$ is a polynomial multiple of the minimal polynomial). *Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $q \in \mathcal{P}(\mathbb{F})$. Then $q(T) = 0$ if and only if q is a polynomial multiple of the minimal polynomial of T .*

Theorem 24 (minimal polynomial of a restriction operator). *Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and U is a subspace of V that is invariant under T . Then the minimal polynomial of T is a polynomial multiple of the minimal polynomial of $T|_U$.*

Corollary 25. *Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then T is not invertible if and only if the constant term of the minimal polynomial of T is 0.*

Theorem 26 (even-dimensional null space). *Suppose $\mathbb{F} = \mathbb{R}$ and V is finite-dimensional. Suppose also that $T \in \mathcal{L}(V)$ and $b, c \in \mathbb{R}$ with $b^2 < 4c$. Then $\dim \text{null}(T^2 + bT + cI)$ is an even number.*

Theorem 27. *Every operator on an odd-dimensional vector space has an eigenvalue.*

Problem 1

Suppose $T \in \mathcal{L}(V)$. Prove that 9 is an eigenvalue of T^2 if and only if 3 or -3 is an eigenvalue of T .

Proof. $\Rightarrow$ Given 9 is an eigenvalue of T^2 , then this means that $(T^2 - 9I) = (T - 3I)(T + 3I) = 0$. So either 3 or -3 is an eigenvalue of T .

$\Leftarrow$ We know there exists nonzero $v \in V$ s.t. $Tv = \pm 3v$, so $T(Tv) = T(\pm 3v) = \pm 3T(v) = (\pm 3)^2 v = 9v$. $\square$

Problem 2

Suppose V is a complex vector space and $T \in \mathcal{L}(V)$ has no eigenvalues. Prove that every subspace of V invariant under T is either $\{0\}$ or infinite-dimensional.

Proof. Suppose for contradiction that there exists finite-dimensional nontrivial subspace U of V such that is invariant under T . So we know that by theorem 5.31 and 5.22 $T|_U$ has eigenvalue and so does T . This forms a contradiction to the hypothesis. $\square$

Problem 3

Suppose n is a positive integer and $T \in \mathcal{L}(\mathbb{F}^n)$ is defined by

$$T(x_1, \dots, x_n) = (x_1 + \dots + x_n, \dots, x_1 + \dots + x_n)$$

- (a) Find all eigenvalues and eigenvectors of T .
- (b) Find the minimal polynomial of T .

Proof. (a) Let v be such that

$$(x_1 + \dots + x_n, \dots, x_1 + \dots + x_n) = (\lambda x_1, \dots, \lambda x_n)$$

This means $\sum_{i=1}^n x_i = \lambda x_j = \lambda x_k$ for all j, k . If $\lambda = 0$, then $\sum_{i=1}^n x_i = 0$ and the eigenvector would be $\{(x_1, \dots, x_n) : \sum_{i=1}^n x_i = 0\}$. If $\lambda \neq 0$, then we have that $x_i = x_j$ for all i, j and thus $\lambda = n$, the corresponding eigenvector is that $\{(a, \dots, a) : a \in \mathbb{F}\}$.

- (b) We know the eigenvalues are 0; n , so it must be of the form

$$p(\lambda) = \lambda(\lambda - n)$$

$\square$

Problem 4

Suppose $\mathbb{F} = \mathbb{C}$, $T \in \mathcal{L}(V)$, $p \in \mathcal{P}(\mathbb{C})$, and $\alpha \in \mathbb{C}$. Prove that α is an eigenvalue of $p(T)$ if and only if $\alpha = p(\lambda)$ for some eigenvalue λ of T .

Proof. $\Rightarrow$ Given α is an eigenvalue of $p(T)$, then we know that

$$p(z) - \alpha = c(z - \lambda_1) \cdots (z - \lambda_m)$$

for some λ_i 's. Note that since p is nonconstant $c \neq 0$. Then this means that one of $z - \lambda_i = 0 \iff T - \lambda_i I = 0$. Then this means $p(\lambda_i) - \alpha = 0$ so $p(\lambda_i) = \alpha$.

$\Leftarrow$ Conversely, we have that for some nonzero v to be the eigenvector of T

$$\begin{aligned} p(T)(v) &= (a_0 I + a_1 T + \cdots + a_m T^m)(v) \\ &= (a_0 I + a_1 \lambda + \cdots + a_m \lambda^m)(v) \\ &= \alpha(v) \end{aligned}$$

□

Problem 6

Suppose $T \in \mathcal{L}(\mathbb{F}^2)$ is defined by $T(w, z) = (-z, w)$. Find the minimal polynomial of T .

Proof. Pick $v = e_1$ then $Tv = (0, 1)$ and $T^2(v) = (-1, 0)$. We have that

$$(1 + T^2)(v) = 0$$

□

Problem 8

Suppose $T \in \mathcal{L}(\mathbb{R}^2)$ is the operator of counterclockwise rotation by 1° . Find the minimal polynomial of T .

Proof. We know that the matrix represented by the operator is that

$$T = \begin{bmatrix} \cos(1^\circ) & -\sin(1^\circ) \\ \sin(1^\circ) & \cos(1^\circ) \end{bmatrix}$$

Let $v = e_1, Tv = (\cos 1^\circ, \sin 1^\circ)$ and $T^2 v = (\cos^2 1^\circ - \sin^2 1^\circ, 2 \cos 1^\circ \sin 1^\circ)$. Hence, we have that

$$\begin{aligned} (-1)1 + (2 \cos 1^\circ) \cos 1^\circ &= \cos^2 1^\circ - \sin^2 1^\circ \\ (-1)0 + (2 \cos 1^\circ) \sin 1^\circ &= 2 \cos 1^\circ \sin 1^\circ \end{aligned}$$

Then the minimal polynomial is

$$p(T) = T^2 - 2 \cos 1^\circ T - I$$

□

Problem 9

Suppose $T \in \mathcal{L}(V)$ is such that with respect to some basis of V , all entries of the matrix of T are rational numbers. Explain why all coefficients of the minimal polynomial of T are rational numbers.

Proof. Let M be the matrix of T and d be the degree of its minimal polynomial. Consider

$$\mathbf{A} = \left(\text{vec}(I), \text{vec}(M), \dots, \text{vec}(M^d) \right)$$

Then we know that $\mathbf{A}$ is rational and we can partition $\mathbf{A}$ into first d column and last 1 column where the last column is a linear combination of the first d linearly independent ones. This further implies that we can find a solution using Gaussian elimination, where its entries are therefore all rational as well, and it is indeed the coefficients of the minimal polynomial. $\square$

Problem 10

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $v \in V$. Prove that

$$\text{span}(v, Tv, \dots, T^m v) = \text{span}(v, Tv, \dots, T^{\dim V - 1} v)$$

for all integers $m \geq \dim V - 1$.

Proof. Denote $A = \text{span}(v, Tv, \dots, T^m v)$ and $B = \text{span}(v, Tv, \dots, T^{\dim V - 1} v)$.

$\Rightarrow$ Since the list A has length greater than $\dim V$, the elements after $T^{\dim V - 1} v$ can be written as linear combination of the previous terms (aka terms in B). Therefore any element in A can be written as linear combination of elements in B so $A \subseteq B$.

$\Leftarrow$ This direction is trivial. $\square$

Problem 12

Define $T \in \mathcal{L}(\mathbb{F}^n)$ by $T(x_1, x_2, x_3, \dots, x_n) = (x_1, 2x_2, 3x_3, \dots, nx_n)$. Find the minimal polynomial of T .

Proof. Let $v = (1, 0, 0, \dots, 0)$. Then $T^k v = e_1$ for all $1 \leq k \leq n$. Thus we have that

$$p(T) = T^n - I$$

$\square$

Problem 13

Suppose $T \in \mathcal{L}(V)$ and $p \in \mathcal{P}(\mathbb{F})$. Prove that there exists a unique $r \in \mathcal{P}(\mathbb{F})$ such that $p(T) = r(T)$ and $\deg r$ is less than the degree of the minimal polynomial of T .

Proof. Let $m_T(x)$ be the minimal polynomial of T . By the division algorithm, there exists polynomial $q(x)$ and $r(x)$ with $\deg r < \deg m_T$ such that

$$p(x) = q(x)m_T(x) + r(x)$$

Evaluating at T then yields that

$$p(T) = r(T)$$

Here we prove the uniqueness of r . Suppose for contradiction that there exists another $r'(x)$ such that

$$p(T) = r(T) = r'(T)$$

If we now define $r''(x) = r(x) - r'(x) = 0 = r''(T)$, then it must be a multiple of $m_T(x)$ by the definition of the minimal polynomial. However, as its degree is less than m_T , this forms a contradiction, completing the proof. $\square$

Problem 15

Suppose V is a finite-dimensional complex vector space with $\dim V > 0$ and $T \in \mathcal{L}(V)$. Define $f: \mathbb{C} \rightarrow \mathbb{R}$ by

$$f(\lambda) = \dim \text{range } (T - \lambda I).$$

Prove that f is not a continuous function.

Proof. Suppose λ is an eigenvalue of T and consider a sequence $\{\lambda_k\}$ converging to λ , where λ_k 's are not eigenvalues of T (you may verify such a sequence exists). For λ_k near λ , we have that $f(\lambda_k) = \dim V - \dim \text{null } (T - \lambda_k I) = \dim V$, where $f(\lambda) < \dim V$. Hence, f is discontinuous at λ and therefore not a continuous function. $\square$

Problem 17

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and p is the minimal polynomial of T . Suppose $\lambda \in \mathbb{F}$. Show that the minimal polynomial of $T - \lambda I$ is the polynomial q defined by $q(z) = p(z + \lambda)$.

Proof. First we can see that $q(T - \lambda I) = p(T) = 0$. Now it suffices to prove the minimality condition. Suppose there exists another monic polynomial $r(z)$ of lesser degree than $q(z)$ such that $r(T - \lambda I) = 0$. We can now define $s(x) = r(x - \lambda)$ and then $s(T) = r(T - \lambda I) = 0$ with $\deg s < \deg p$, contradicting the minimal polynomial assumption of p . Therefore, we complete the proof. $\square$

Problem 19

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Let $\mathcal{E}$ be the subspace of $\mathcal{L}(V)$ defined by

$$\mathcal{E} = \{q(T) : q \in \mathcal{P}(\mathbb{F})\}.$$

Prove that $\dim \mathcal{E}$ equals the degree of the minimal polynomial of T .

Proof. Note that $\{a_0I, a_1T, a_2T^2, a_3T^3, \dots, a_{n-1}T^{n-1}\} := A$ is a basis of $\mathcal{E}$ with dimension n . Suppose $\deg m_T = m$ for notational purposes. We have that

A is linearly independent $\iff$ Express T^m as linear combination of terms in A
 $\iff$ Minimal degree of m_T is n

□

Problem 21

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial has degree at most $1 + \dim \text{range } T$.

Proof. Let $k = \dim \text{range } T, n = \dim V$. We note that $T(\text{range } T) \subseteq \text{range } T$, so T naturally induces a well-defined linear operator $S \in \mathcal{L}(\text{range } T)$ by restriction ($Sv = Tv \forall v \in \text{range } T$). Consider the minimal polynomial $m_S(x)$ of S . We know that $\deg m_S \leq \dim \text{range } T = k$. Under the construction, we also have $T^2 = TS$, which implies that T behaves like S on $\text{range } T$. Note for any $v \in \text{null } T, T(v) = 0$.

Now we can construct a polynomial that annihilates T on V . Define $q(x) = x \cdot m_S(x)$. We claim that $q(T) = 0$. This is true since for $v \in \text{null } T, q(T)(v) = T \cdot m_S(T)(v) = T(0) = 0$. For $v \in \text{range } T$, we have $m_S(T)(v) = m_S(S)(v) = 0$ and therefore $q(T)(v) = 0$. Note that under this construction, we have $\deg q \leq k + 1$. Since the minimal polynomial divides q , its degree is also upper bounded by $1 + \dim \text{range } T$.

□

Problem 22

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T is invertible if and only if $I \in \text{span}(T, T^2, \dots, T^{\dim V})$.

Proof. $\Rightarrow$ This means that

$$I = a_0T + a_1T^2 + \dots + a_nT^n$$

with $n = \dim V$. Rewriting the equation gives that (for the largest nonzero degree), we have

$$-I + \frac{a_0}{a_m}T + \frac{a_1}{a_m}T^2 + \dots + T^m = 0$$

Therefore the constant term of minimal polynomial of T is not 0 and thus T is invertible.

$\Leftarrow$ We know that

$$I = a_0T + a_1T^2 + \dots + a_nT^n = T(a_0I + a_1T + \dots + a_nT^{n-1})$$

Define $S = (a_0I + a_1T + \dots + a_nT^{n-1})$ so $TS = I$. At the same time,

$$ST = (a_0I + a_1T + \dots + a_nT^{n-1})T = a_0T + a_1T^2 + \dots + a_nT^n = I$$

So T is invertible. $\square$

Problem 23

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Let $n = \dim V$. Prove that if $v \in V$, then $\text{span}(v, Tv, \dots, T^{n-1}v)$ is invariant under T .

Proof. We know that $Tv, \dots, T^{n-1}v \in \text{span}(v, Tv, \dots, T^{n-1}v)$, so it suffices to prove that $T^n v \in \text{span}(v, Tv, \dots, T^{n-1}v)$. To see this, notice that the list

$$v, Tv, \dots, T^{n-1}v, T^n v$$

has length more than n and can be reduced to at least n terms. Therefore, we complete the proof. $\square$

Problem 28

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial of $T' \in \mathcal{L}(V')$ equals the minimal polynomial of T .

Let p' and p be the minimal polynomial of T' and T respectively. We know that $T'(\varphi) = \varphi \circ T$. We have that

$$p'(\varphi) = (c_0I + c_1T' + \dots + c_k(T')^k)(\varphi) = \varphi \circ (c_0I + c_1T + \dots + c_kT^k) = \varphi \circ p = 0$$

which essentially shows that the minimality condition is iff for p' and p , otherwise it would reach a contradiction that one is the minimal polynomial.

Problem 29

Show that every operator on a finite-dimensional vector space of dimension at least two has an invariant subspace of dimension two.

Proof. We prove this claim through the induction on $n = \dim V$. For base case, we have $\dim V = 2$, then as V is invariant, the claim is satisfied.

For inductive case, suppose the statement holds for V with dimension $n - 1$. Then consider V with dimension n . Take any nonzero $v \in V$ and consider $V \setminus \text{span}(v)$, which is a space of dimension $n - 1$, so it has an invariant subspace of dimension two, which is still a subspace of V , completing the proof. $\square$

5C: Upper-Triangular Matrices

Definition 28 (matrix of an operator, $\mathcal{M}(T)$). Suppose $T \in \mathcal{L}(V)$. The **matrix of T** with respect to a basis $v_1, \dots, v_n$ of V is the n -by- n matrix

$$\mathcal{M}(T) = \begin{bmatrix} A_{1,1} & \cdots & A_{1,n} \\ \vdots & & \vdots \\ A_{n,1} & \cdots & A_{n,n} \end{bmatrix}$$

whose entries $A_{j,k}$ are defined by

$$Tv_k = A_{1,k}v_1 + \cdots + A_{n,k}v_n.$$

The notation $\mathcal{M}(T, (v_1, \dots, v_n))$ is used if the basis is not clear from the context.

Remark 29. Operators have square matrices.

Definition 30 (diagonal of a matrix). The **diagonal** of a square matrix consists of the entries on the line from the upper left corner to the bottom right corner.

Definition 31 (upper-triangular matrix). A square matrix is called **upper triangular** if all entries below the diagonal are 0.

Theorem 32 (conditions for upper-triangular matrix). Suppose $T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of V . Then the following are equivalent.

- (a) The matrix of T with respect to $v_1, \dots, v_n$ is upper triangular.
- (b) $\text{span}(v_1, \dots, v_k)$ is invariant under T for each $k = 1, \dots, n$.
- (c) $Tv_k \in \text{span}(v_1, \dots, v_k)$ for each $k = 1, \dots, n$.

Lemma 33. Suppose $T \in \mathcal{L}(V)$ and V has a basis with respect to which T has an upper-triangular matrix with diagonal matrix with diagonal entries $\lambda_1, \dots, \lambda_n$. Then

$$(T - \lambda_1 I) \cdots (T - \lambda_n I) = 0$$

Theorem 34 (determination of eigenvalues from upper-triangular matrix). Suppose $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of V . Then the eigenvalues of T are precisely the entries on the diagonal of that upper-triangular matrix.

Remark 35. Main proof technique: $(T - \lambda_k)Iv_k \in \text{span}(v_1, \dots, v_{k-1})$ for the matrix of T to be upper-triangular.

Lemma 36 (necessary and sufficient condition to have an upper-triangular matrix). Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then T has an upper-triangular matrix with respect to some basis of V if and only if the minimal polynomial equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some $\lambda_1, \dots, \lambda_m \in \mathbb{F}$.

Theorem 37 (if $\mathbb{F} = \mathbb{C}$, then every operator on V has an upper-triangular matrix). Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Then T has an upper-triangular matrix with respect to some basis of V .

Problem 1

Prove or give a counter example: If $T \in \mathcal{L}(V)$ and T^2 has an upper-triangular matrix with respect to some basis of V , then T has an upper-triangular matrix with respect to some basis of V .

Proof. Since T^2 has an upper triangular matrix wrt. some basis of V , its minimal polynomial equals $p_{T^2}(x) = (x - \lambda_1) \cdots (x - \lambda_m)$ for some $\lambda_1, \dots, \lambda_m \in \mathbb{F}$. This means that $p_{T^2}(T^2) = (T^2 - \lambda_1 I) \cdots (T^2 - \lambda_m I) = (T - \sqrt{\lambda_1} I)(T + \sqrt{\lambda_1} I) \cdots (T - \sqrt{\lambda_m} I)(T + \sqrt{\lambda_m} I) = 0$. This means that the minimal polynomial of T can also be in the form of an upper-triangular matrix. □

Problem 3

Suppose $T \in \mathcal{L}(V)$ is invertible and $v_1, \dots, v_n$ is a basis of V with respect to which the matrix of T is upper triangular, with $\lambda_1, \dots, \lambda_n$ on the diagonal. Show that the matrix of T^{-1} is also upper triangular with respect to the basis $v_1, \dots, v_n$, with

$$\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_n}$$

on the diagonal.

Proof. We first show that T^{-1} is also upper triangular wrt. the same basis. To see this, we know that $Tv_k \in \text{span}(v_1, \dots, v_k)$ for each k , or equivalently,

$$T(v_k) = \lambda_k v_k + \sum_{j=1}^{k-1} A_{j,k} v_j$$

Let $w_k = T^{-1}v_k$, then we know that $Tw_k = v_k \in \text{span}(v_1, \dots, v_k)$. So this implies that $w_k \in \text{span}(v_1, \dots, v_k)$ by property of T . Thus this completes the first part of the proof. For the second part, notice that $TT^{-1} = I$ and since the two matrices are both upper-triangular, by rule of matrix multiplication, $T_{ii}T_{ii}^{-1} = 1$ and thus the entries on the diagonal of T^{-1} is the inverse of T 's, completing the proof. □

Problem 6

Suppose $\mathbb{F} = \mathbb{C}$, V is finite-dimensional, and $T \in \mathcal{L}(V)$. Prove that if $k \in \{1, \dots, \dim V\}$, then V has a k -dimensional subspace invariant under T .

Proof. We have proved in the book that T has an upper-triangular matrix with respect to some basis of V . Then by equivalent conditions of upper-triangular matrix, we know that for some basis $v_1, \dots, v_n$, the $\text{span}(v_1, \dots, v_k)$ is invariant under T for each $k = 1, \dots, n$, completing the proof. □

Problem 7

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $v \in V$.

- (a) Prove that there exists a unique monic polynomial p_v of smallest degree such that $p_v(T)v = 0$.
- (b) Prove that the minimal polynomial of T is a polynomial multiple of p_v .

Proof. (a) First we prove the existence. Consider the list

$$v, Tv, \dots, T^n v$$

which is a linearly dependent list in $\dim V = n$ space. Therefore we can always reduce this list to a smallest degree such that

$$T^m v = \sum_{k=0}^{m-1} \frac{a_k}{a_m} T^k v$$

Next we prove uniqueness. Suppose there exists another monic polynomial of minimal degree s such that $s(T)(v) = 0$. Then we can consider

$$0 = p(T)(v) - s(T)(v) = (p - s)(T)(v)$$

So this polynomial $(p - s)$ also satisfies the required condition and it has smaller degree than p or s , reaching a contradiction.

(b) Let m be the minimal polynomial of T . By the division algorithm, we have

$$m(z) = q(z)p(z) + r(z)$$

for some polynomial q and r with $\deg r < \deg p$. We claim that $r = 0$. This is true as

$$0 = m(T)(v) = qp(T)(v) + r(T)(v) = r(T)(v)$$

Thus $r(T)(v) = 0$. If r is non-zero map, then this means there exists a monic polynomial of even lesser degree than p that also maps v to 0, contradicting that p is the unique smallest monic polynomial. Therefore we complete the proof. $\square$

Problem 9

Suppose B is a square matrix with complex entries. Prove that there exists an invertible square matrix A with complex entries such that $A^{-1}BA$ is an upper-triangular matrix.

Proof. Consider the linear map T represented by the matrix B . Since the field is taken over $\mathbb{C}$, we know that T has an upper-triangular matrix with respect to some basis of V . Then let A be such change-of-basis matrix we can indeed find the desired matrix. $\square$

Problem 10

Suppose $T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of V . Show that the following are equivalent.

- (a) The matrix of T with respect to $v_1, \dots, v_n$ is lower triangular.
- (b) $\text{span}(v_k, \dots, v_n)$ is invariant under T for each $k = 1, \dots, n$.
- (c) $Tv_k \in \text{span}(v_k, \dots, v_n)$ for each $k = 1, \dots, n$.

Proof. (a) $\Rightarrow$ (b) This means that $Tv_j \in \text{span}(v_j, \dots, v_n)$. If $j \geq k$, then we have that

$$Tv_j \in \text{span}(v_k, \dots, v_n)$$

(b) $\Rightarrow$ (c) This holds by definition.

(c) $\Rightarrow$ (a) This means when writing each Tv_k as a linear combination of the basis vectors $v_1, \dots, v_n$, we need to use only the vectors $v_k, \dots, v_n$. Hence, all entries above the diagonal of $\mathcal{M}(T)$ are 0 and thus it is an lower-triangular matrix. $\square$

Problem 12

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of V , and U is a subspace of V that is invariant under T .

- (a) Prove that $T|_U$ has an upper-triangular matrix with respect to some basis of U .
- (b) Prove that the quotient operator T/U has an upper-triangular matrix with respect to some basis of V/U .

Proof. We know that there exists a basis $v_1, \dots, v_n$ such that $Tv_k \in \text{span}(v_1, \dots, v_k)$ for each k .

(a) We can always find a subset of the basis such that $v_1, \dots, v_m$ is the basis of U (WLOG we make the numbering easy). Then for the list we always have that $Tv_k \in \text{span}(v_1, \dots, v_k)$ for each k and thus $T|_U$ has an upper-triangular matrix.

(b) Consider $\{v_{m+1} + U, \dots, v_n + U\}$, which is the basis of V/U . Note that we still have $T(v_k + U) \in \text{span}(v_{m+1} + U, \dots, v_k + U)$, hence T/U also has an upper-triangular matrix. $\square$

Problem 14

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has an upper-triangular matrix with respect to some basis of V if and only if the dual operator T' has an upper-triangular matrix with respect to some basis of the dual space.

Proof. From 5B problem 28 we know that the minimal polynomial of T' equals the minimal polynomial of T . Then this means that they have the same roots and therefore the same form of $(z - \lambda_1) \cdots (z - \lambda_m)$ if they having upper-triangular matrix. The iff is directly deduced from the iff of the relation between each one's minimal polynomial. $\square$

5D: Diagonalizable Operators

Definition 38 (diagonal matrix). A **diagonal matrix** is a square matrix that is 0 everywhere except possibly on the diagonal.

Remark 39. The entries on the diagonal are precisely the eigenvalue of the operator.

Definition 40 (diagonalizable). An operator on V is called **diagonalizable** if the operator has a diagonal matrix with respect to some basis of V .

Remark 41. Diagonalization may require a different basis.

Definition 42 (eigenspace, $E(\lambda, T)$). Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$. The **eigenspace** of T corresponding to λ is the subspace $E(\lambda, T)$ of V defined by

$$E(\lambda, T) = \text{null}(T - \lambda I) = \{v \in V : Tv = \lambda v\}.$$

Hence $E(\lambda, T)$ is the set of all eigenvectors of T corresponding to λ , along with the 0 vector.

Remark 43. λ is an eigenvalue of T if and only if $E(\lambda, T) \neq \{0\}$.

Theorem 44. Suppose $T \in \mathcal{L}(V)$ and $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues of T . Then

$$E(\lambda_1, T) + \dots + E(\lambda_m, T)$$

is a direct sum. Furthermore, if V is finite-dimensional, then

$$\dim E(\lambda_1, T) + \dots + \dim E(\lambda_m, T) \leq \dim V$$

Theorem 45 (conditions equivalent to diagonalizability). Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Let $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T . Then the following are equivalent.

- (a) T is diagonalizable.
- (b) V has a basis consisting of eigenvectors of T .
- (c) $V = E(\lambda_1, T) \oplus \dots \oplus E(\lambda_m, T)$.
- (d) $\dim V = \dim E(\lambda_1, T) + \dots + \dim E(\lambda_m, T)$.

Corollary 46 (enough eigenvalues implies diagonalizability). Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues. Then T is diagonalizable.

Theorem 47 (necessary and sufficient condition for diagonalizability). Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then T is diagonalizable if and only if the minimal polynomial of T equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some list of **distinct** numbers $\lambda_1, \dots, \lambda_m \in \mathbb{F}$.

Corollary 48. Suppose $T \in \mathcal{L}(V)$ is diagonalizable and U is a subspace of V that is invariant under T . Then $T|_U$ is a diagonalizable operator on U .

Definition 49 (Gershgorin disks). Suppose $T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of V . Let A denote the matrix of T with respect to this basis. A **Gershgorin disk** of T with respect to the basis $v_1, \dots, v_n$ is a set of the form

$$\{z \in \mathbb{F} : |z - A_{j,j}| \leq \sum_{k=1, k \neq j}^n |A_{j,k}|\},$$

where $j \in \{1, \dots, n\}$.

Remark 50. Intuition: if the nondiagonal entries of A are small, then each eigenvalue of T is near a diagonal entry of A .

Theorem 51 (Gershgorin disk theorem). Suppose $T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of V . Then each eigenvalue of T is contained in some Gershgorin disk of T with respect to the basis $v_1, \dots, v_n$.

Problem 1

Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$.

- (a) Prove that if $T^4 = I$, then T is diagonalizable.
- (b) Prove that if $T^4 = T$, then T is diagonalizable.
- (c) Give an example of an operator $T \in \mathcal{L}(\mathbb{C}^2)$ such that $T^4 = T^2$ and T is not diagonalizable.

Proof. (a) The polynomial $p(x) = x^4 - 1$ annihilates T ($p(T) = 0$) has four distinct roots. The minimal polynomial of T divides p and therefore T is diagonalizable.

(b) The polynomial $p(x) = x^4 - x = x(x-1)(x^2+1)$ annihilates T has four distinct roots. The minimal polynomial of T divides p and therefore T is diagonalizable.

(c) Consider

$$T = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Then we have that

$$T^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = T^4$$

□

Problem 3

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that if the operator T is diagonalizable, then $V = \text{null } T \oplus \text{range } T$.

Proof. By equivalent characterizations, there exists $v_1, \dots, v_n$ to be the eigenbasis of V . Let $\lambda_1, \dots, \lambda_n$ be the corresponding eigenvalues. We can partition these vectors into corresponding basis for $\text{null } T$ and $\text{range } T$. Naturally, we can partition these eigenbasis into two parts: let $\lambda_1, \dots, \lambda_m$ denotes the eigenbasis with 0 eigenvalues and $\lambda_{m+1}, \dots, \lambda_n$ denotes the eigenbasis with nonzero eigenvalues. We relabel the eigenvectors accordingly. We claim that $\text{null } T = \text{span}(v_1, \dots, v_m)$ and $\text{range } T = \text{span}(v_{m+1}, \dots, v_n)$. The proof is complete once we complete proving our claim.

For proving the null space, $\Rightarrow$ take $v \in \text{null } T$, then we have that

$$\begin{aligned} Tv &= T \sum_{i=1}^m a_i v_i + T \sum_{i=m+1}^n a_i v_i \\ &= \sum_{i=m+1}^n (a_i \lambda_i) v_i = 0 \end{aligned}$$

By linear independence of eigenbasis (as $\lambda_i \neq 0$ for $m+1 \leq i \leq n$), we have that $a_i = 0$ for all $m+1 \leq i \leq n$. Thus $v \in \text{span}(v_1, \dots, v_m)$. $\Leftarrow$ for the other direction, it follows naturally by construction.

For proving the range space, $\Rightarrow$ take $v \in \text{range } T$, then there exists $u \in V$ s.t.

$$v = Tu = \sum_{i=m+1}^n (a_i \lambda_i) v_i$$

Therefore, $v \in \text{span}(v_{m+1}, \dots, v_n)$. $\Leftarrow$ for the other direction, it follows by construction. $\square$

Problem 4

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent.

- (a) $V = \text{null } T \oplus \text{range } T$.
- (b) $V = \text{null } T + \text{range } T$.
- (c) $\text{null } T \cap \text{range } T = \{0\}$.

Proof. (a) $\Rightarrow$ (b) Trivial.

(b) $\Rightarrow$ (c) We know that

$$\dim V = \dim(\text{null } T + \text{range } T) = \dim \text{null } T + \dim \text{range } T - \dim \text{null } T \cap \text{range } T$$

At the same time, $\dim V = \dim \text{null } T + \dim \text{range } T$, hence

$$\dim \text{null } T \cap \text{range } T = 0$$

(c) $\Rightarrow$ (a) This can be proved similarly as above. $\square$

Problem 5

Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Prove that T is diagonalizable if and only if

$$V = \text{null } (T - \lambda I) \oplus \text{range } (T - \lambda I)$$

for every $\lambda \in \mathbb{C}$.

Proof. $\Rightarrow$ Given T is diagonalizable, then we know that $T - \lambda I$ is also diagonalizable (the matrix of that is diagonal). Hence, by P3, we get the desired result.

$\Leftarrow$ The minimal polynomial of T can be written as $(z - \lambda_1)^{n_1} \dots (z - \lambda_m)^{n_m}$ for some distinct $\lambda_1, \dots, \lambda_m \in \mathbb{C}$. If $n_1 = \dots = n_m = 1$, then we are done. WLOG suppose $n_1 > 1$. Then take arbitrary $v \in V$, we have

$$\prod_{k=1}^m (T - \lambda_k I)^{n_k}(v) = (T - \lambda_1 I)(T - \lambda_1 I)^{n_1-1} \prod_{k=2}^m (T - \lambda_k I)^{n_k}(v) = 0$$

This implies that $(T - \lambda_1 I)^{n_1-1} \prod_{k=2}^m (T - \lambda_k I)^{n_k} \in \text{null } (T - \lambda_1 I) \cap \text{range } (T - \lambda_1 I) = \{0\}$. Therefore, this reach a contradiction to the minimal polynomial we previously get. $\square$

Problem 6

Suppose $T \in \mathcal{L}(\mathbb{F}^5)$ and $\dim E(8, T) = 4$. Prove that $T - 2I$ or $T - 6I$ is invertible.

Proof. We know that

$$\dim E(8, T) + \dim E(2, T) + \dim E(6, T) \leq 5$$

Since $\dim(8, T) = 4$, then we have that

$$\dim E(2, T) = 0 \quad \text{or} \quad \dim E(6, T) = 0$$

In other words, $T - 2I$ or $T - 6I$ is invertible. $\square$

Problem 7

Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

$$E(\lambda, T) = E\left(\frac{1}{\lambda}, T^{-1}\right)$$

for every $\lambda \in \mathbb{F}$ with $\lambda \neq 0$.

Proof. We know that

$$Tv = \lambda v \iff v = T^{-1}\lambda v \iff \frac{1}{\lambda}v = T^{-1}v$$

$\square$

Problem 8

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Let $\lambda_1, \dots, \lambda_m$ denote the distinct nonzero eigenvalues of T . Prove that

$$\dim E(\lambda_1, T) + \dots + \dim E(\lambda_m, T) \leq \dim \text{range } T.$$

Proof. It suffices to show that

$$E' = E(\lambda_1, T) + \dots + E(\lambda_m, T) = E(\lambda_1, T) \oplus \dots \oplus E(\lambda_m, T) \subseteq \text{range } (T)$$

To see this, let $v \in E(\lambda_k, T)$, then we know that $Tv = \lambda_k v$ and that $v = T \frac{1}{\lambda_k} v \in \text{range } T$. Hence, we complete the proof. $\square$

Problem 9

Suppose $R, T \in \mathcal{L}(\mathbb{F}^3)$ each have 2, 6, 7 as eigenvalues. Prove that there exists an invertible operator $S \in \mathcal{L}(\mathbb{F}^3)$ such that $R = S^{-1}TS$.

Proof. This means that R, T are both diagonalizable and thus invertible. Thus we can define a change-of-basis operator S that is invertible (one may provide more details here). $\square$

Problem 12

Suppose $T \in \mathcal{L}(\mathbb{C}^3)$ is such that 6 and 7 are eigenvalues of T . Furthermore, suppose T does not have a diagonal matrix with respect to any basis of $\mathbb{C}^3$. Prove that there exists $(z_1, z_2, z_3) \in \mathbb{C}^3$ such that

$$T(z_1, z_2, z_3) = (6 + 8z_1, 7 + 8z_2, 13 + 8z_3).$$

Proof. This means that T only has 6, 7 as the eigenvalues and so 8 is not an eigenvalue of T . Equivalently, $T - 8I$ is invertible and thus there exists $(z_1, z_2, z_3) \in \mathbb{C}^3$ such that $(T - 8I)(z_1, z_2, z_3) = (6, 7, 13)$. Rewriting the equation gives the desired result. $\square$

Problem 13

Suppose A is a diagonal matrix with distinct entries on the diagonal and B is a matrix of the same size as A . Show that $AB = BA$ if and only if B is a diagonal matrix.

Proof. $\Rightarrow$ We can simply examine each entry of AB and BA , note that

$$(AB)_{ij} = \sum_{k=1}^n A_{i,k} B_{k,j} = A_{i,i} B_{i,j}$$

and

$$(BA)_{ij} = \sum_{k=1}^n B_{i,k} A_{k,j} = A_{j,j} B_{i,j}$$

Since $A_{i,i} \neq A_{j,j}$, $B_{i,j} = 0$ for $i \neq j$. Hence we have B is diagonal.

$\Leftarrow$ This naturally holds by matrix multiplication. $\square$

Problem 14

- (a) Give an example of a finite-dimensional complex vector space and an operator T on that vector space such that T^2 is diagonalizable but T is not diagonalizable.
- (b) Suppose $\mathbb{F} = \mathbb{C}$, k is a positive integer, and $T \in \mathcal{L}(V)$ is invertible. Prove that T is diagonalizable if and only if T^k is diagonalizable.

Proof. (a) Consider the matrix of T to be

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

and let $\mathbb{F} = \mathbb{R}$. Then $T^2 = 0$ which is diagonalizable.

(b) $\Rightarrow$ By P13, we know that since T is diagonal, T^2 is diagonal, and $T(T^2)$ is diagonal. Recursively applying the argument yields that T^m is diagonal and therefore diagonalizable.

$\Leftarrow$ We know that $p(x) = (x^n - \lambda_1) \cdots (x^n - \lambda_k)$ annihilates T where $\lambda_1, \dots, \lambda_k$ are distinct eigenvalues of T^m . Since all λ_i are distinct, $x^n - \lambda_i$ and $x^n - \lambda_j$ do not share any common root for $i \neq j$. Then since the minimal polynomial of T divides $p(x)$, it also does not have any repeated roots and therefore T is also diagonalizable. $\square$

Problem 16

Suppose $T \in \mathcal{L}(V)$ is diagonalizable. Let $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T . Prove that a subspace U of V is invariant under T if and only if there exist subspaces $U_1, \dots, U_m$ of V such that $U_k \subseteq E(\lambda_k, T)$ for each k and $U = U_1 \oplus \cdots \oplus U_m$.

Proof. $\Leftarrow$ Take any $u = u_1 + \cdots + u_m \in U$, then $Tu = Tu_1 + \cdots + Tu_m = \lambda_1 u_1 + \cdots + \lambda_m u_m \in U$.

$\Rightarrow$ Define $U_k = E(\lambda_k, T) \cap U$. Clearly $U_k \subseteq E(\lambda_k, T)$. Since U is invariant under T , U_k is also invariant. By 5.65 we know that $T|_{U_k}$ is also diagonalizable. Since we know that $E(\lambda_i, T) \cap E(\lambda_j, T) = \{0\}$ for $i \neq j$, $U_j \cap U_i = \{0\}$ for $i \neq j$ as well. Now it suffices to show that $U \subseteq U_1 \oplus \cdots \oplus U_m$. Take $u \in U \subset V$, since we know that $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$, $u \in E_k$ for some k , therefore completing the proof. $\square$

Problem 18

Suppose that $T \in \mathcal{L}(V)$ is diagonalizable and U is a subspace of V that is invariant under T . Prove that the quotient operator T/U is a diagonalizable operator on V/U .

Proof. First we know that there exists eigenbasis $v_1, \dots, v_n$. We can first partition these basis into $v_1, \dots, v_m$ for U and then $v_{m+1} + U, \dots, v_n + U$ would be the basis for V/U . Notice that here we have $(T/U)(v_j + U) = T(v_j) + U = \lambda_j v_j + U$, with the same eigenbasis so preserving the diagonalizability. $\square$

Problem 20

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T is diagonalizable if and only if the dual operator T' is diagonalizable.

Proof. WLOG we go from the forward direction $\Rightarrow$. Let $v_1, \dots, v_n$ be the eigenbasis and $\varphi_1, \dots, \varphi_n$ be the corresponding dual basis. Then we have that

$$(T'\varphi_i)(v_j) = \varphi_i \circ Tv_j = \lambda_j \varphi_i(v_j) = \lambda_i \delta_{ij}$$

Hence, we have that $T'\varphi_i = \lambda_i \varphi_i$ for each i and thus T' is diagonalizable. $\square$

Problem 22

Suppose $T \in \mathcal{L}(V)$ and A is an n -by- n matrix that is the matrix of T with respect to some basis of V . Prove that if

$$|A_{j,j}| > \sum_{k=1, k \neq j}^n |A_{j,k}|$$

for each $j \in \{1, \dots, n\}$, then T is invertible.

In other words, the implication is that if the diagonal entries of the matrix of T are large enough compared to non-diagonal ones, then T is invertible.

Proof. Equivalently, we aim to prove that 0 is not an eigenvalue of A . From the Gershgorin disk theorem, we know that every eigenvalue of A lies within at least one of the Gershgorin disk

$$\left\{ z \in \mathbb{F} : |z - A_{j,j}| \leq \sum_{k=1, k \neq j}^n |A_{j,k}| \right\}$$

for $j \in \{1, \dots, n\}$. From the question, we can see that none of the eigenvalues are 0 and therefore A is strictly diagonally dominant and hence invertible. $\square$

5E: Commuting Operators

Definition 52 (commute). • Two operators S and T on the same vector space **commute** if $ST = TS$.

- Two square matrices A and B of the same size **commute** if $AB = BA$.

Lemma 53 (commuting operators correspond to commuting matrices). Suppose $S, T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of V . Then S and T commute if and only if $\mathcal{M}(S, (v_1, \dots, v_n))$ and $\mathcal{M}(T, (v_1, \dots, v_n))$ commute.

Lemma 54 (eigenspace is invariant under commuting operator). Suppose $S, T \in \mathcal{L}(V)$ commute and $\lambda \in \mathbb{F}$. Then $E(\lambda, S)$ is invariant under T .

Theorem 55 (simultaneous diagonalizability $\iff$ commutativity). Two diagonalizable operators on the same vector space have diagonal matrices with respect to the same basis if and only if the two operators commute.

Lemma 56 (common eigenvector for commuting operators). Every pairs of commuting operators on a finite-dimensional nonzero complex vector space has a common eigenvector.

Lemma 57 (commuting operators are simultaneously upper triangular). Suppose V is a finite-dimensional complex vector space and S, T are commuting operators on V . Then there is a basis of V with respect to which both S and T have upper-triangular matrices.

Theorem 58 (eigenvalues of sum and product of commuting operators). Suppose V is a finite-dimensional complex vector space and S, T are commuting operators on V . Then

- Every eigenvalue of $S + T$ is an eigenvalue of S plus an eigenvalue of T ,
- Every eigenvalue of ST is an eigenvalue of S times an eigenvalue of T .

Problem 2

Suppose $\mathcal{E}$ is a subset of $\mathcal{L}(V)$ and every element of $\mathcal{E}$ is diagonalizable. Prove that there exists a basis of V with respect to which every element of $\mathcal{E}$ has a diagonal matrix if and only if every pair of element of $\mathcal{E}$ commutes.

Proof. $\Rightarrow$ This follows by products of diagonal matrices.

$\Leftarrow$ We are given that every pair of element of $\mathcal{E}$ commutes. Pick any $A \in \mathcal{E}$ and let $E(\lambda) = \{v \in V : Av = \lambda v\}$ be the λ -eigenspace of A . We know that as A is diagonalizable,

$$V = \bigoplus_{\lambda \in \mathbb{F}} E(\lambda)$$

Take another element $B \in \mathcal{E}$, then we know that $E(\lambda)$ is invariant under B and thus $B|_{E(\lambda)}$ is diagonalizable by 5.75. This means that we can further get that $E(\lambda, \mu) = \{v \in E(\lambda) : Bv = \mu v\} = \{v \in V : Av = \lambda v, Bv = \mu v\}$, and we have that

$$V = \bigoplus_{\lambda, \mu \in \mathbb{F}} E(\lambda, \mu)$$

Applying the argument recursively yields that

$$V = \bigoplus_{\{\lambda_i\}_{i=1}^{\infty} : \lambda_i \in \mathbb{F}} E(\{\lambda_i\}_{i=1}^{\infty})$$

where $E(\{\lambda_i\}_{i=1}^{\infty})$ consist of all vectors v such that $T_i v = \lambda_i v$ for commuting operators $T_i \in \mathcal{E}$. Note that even notionally we make ∞ this is still a finite set since the collection of finite(ly nonzero eigenvalues) is still finite. Hence, we derive a common eigenbasis and finish the proof. $\square$

Problem 3

Suppose $S, T \in \mathcal{L}(V)$ are such that $ST = TS$. Suppose $p \in \mathcal{P}(\mathbb{F})$.

- (a) Prove that $\text{null } p(S)$ is invariant under T .
- (b) Prove that $\text{range } p(S)$ is invariant under T .

Proof. (a) Let $p(S) = a_0 I + a_1 S + a_2 S^2 + \cdots + a_m S^m$. Take $v \in \text{null } p(S)$, then this means that $p(S)(v) = 0$. We have that

$$p(S)(Tv) = T(p(S)(v)) = T(0) = 0$$

Hence, $Tv \in \text{null } p(S)$.

(b) Take $v \in \text{range } p(S)$, then this means there exists $u \in V$ such that $p(S)(u) = v$. Consider that

$$p(S)(Tu) = T(p(S)u) = Tv$$

Therefore, $Tv \in \text{range } p(S)$. $\square$

Problem 5

Prove that a pair of operators on a finite-dimensional vector space commute if and only if their dual operators commute.

Proof. Take $\phi \in V'$ and $v \in V$. Then $\Rightarrow$ assume $TS = ST$, we have that

$$(T^*S^*\phi)(v) = S^*\phi(Tv) = \phi(STv)$$

At the same time

$$(S^*T^*\phi)(v) = T^*\phi(Sv) = \phi(TSv)$$

Hence, we have that

$$T^*S^* = S^*T^*$$

.

$\Leftarrow$ Assume $T^*S^* = S^*T^*$, then

$$(T^*S^*\phi)(v) = \phi(STv) = \phi(TSv) = (S^*T^*\phi)(v)$$

Since this holds for all $\phi \in V', v \in V$, $ST = TS$. □

Problem 6

Suppose V is a complex vector space, $S, T \in \mathcal{L}(V)$ commute. Prove that there exist $\alpha, \lambda \in \mathbb{C}$ such that

$$\text{range}(S - \alpha I) + \text{range}(T - \lambda I) \neq V.$$

Proof. Let $v_1, \dots, v_n$ be the same basis such that $\mathcal{M}(S), \mathcal{M}(T)$ are diagonal. Then we let $\alpha = \mathcal{M}(S)_{1,1}$ and $\lambda = \mathcal{M}(T)_{1,1}$. Then this means that

$$\text{range}(S - \lambda I) + \text{range}(T - \lambda I) = V \setminus \text{span}(v_1) \neq V.$$

□

Chapter 6: Inner Product Spaces

Linear Algebra Done Right (4th Edition), by Sheldon Axler

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Contents

6A: Inner Products and Norms	2
6A Problem Sets	4
6B: Orthonormal Bases	14
6B Problem Sets	16
6C: Orthogonal Complements and Minimization Problems	23
6C Problem Sets	26

6A: Inner Products and Norms

Definition 1 (dot product). For $x, y \in \mathbb{R}^n$, the **dot product** of x and y , denoted by $x \cdot y$, is defined by

$$x \cdot y = x_1 y_1 + \cdots + x_n y_n,$$

where $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$.

Definition 2 (inner product). An **inner product** on V is a function that takes each ordered pair (u, v) of elements of V to a number $\langle u, v \rangle \in \mathbb{F}$ and has the following properties:

- (a) **positivity**: $\langle v, v \rangle \geq 0$ for all $v \in V$.
- (b) **definiteness**: $\langle v, v \rangle = 0$ if and only if $v = 0$.
- (c) **additivity in first slot**: $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ for all $u, v, w \in V$.
- (d) **homogeneity in first slot**: $\langle \lambda u, v \rangle = \lambda \langle u, v \rangle$ for all $\lambda \in \mathbb{F}$ and all $u, v \in V$.
- (e) **conjugate symmetry**: $\langle u, v \rangle = \overline{\langle v, u \rangle}$ for all $u, v \in V$.

Definition 3 (inner product space). An **inner product space** is a vector space V along with an inner product on V .

Corollary 4 (basic properties of an inner product). (a) For each fixed $v \in V$, the function that takes $u \in V$ to $\langle u, v \rangle$ is a linear map from V to $\mathbb{F}$.

- (b) $\langle 0, v \rangle = 0$ for every $v \in V$.
- (c) $\langle v, 0 \rangle = 0$ for every $v \in V$.
- (d) $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$ for all $u, v, w \in V$.
- (e) $\langle u, \lambda v \rangle = \overline{\lambda} \langle u, v \rangle$ for all $\lambda \in \mathbb{F}$ and $u, v \in V$.

Definition 5 (norm, $\|v\|$). For $v \in V$, the **norm** of v , denoted by $\|v\|$, is defined by

$$\|v\| = \sqrt{\langle v, v \rangle}$$

Corollary 6 (basic properties of the norm). Suppose $v \in V$.

- (a) $\|v\| = 0$ if and only if $v = 0$.
- (b) $\|\lambda v\| = |\lambda| \|v\|$ for all $\lambda \in \mathbb{F}$.

Remark 7. Working with norms squared is usually easier than working directly with norms.

Definition 8 (orthogonal). Two vectors $u, v \in V$ are called **orthogonal** if $\langle u, v \rangle = 0$.

Corollary 9 (orthogonality and 0). (a) 0 is orthogonal to every vector in V .

(b) 0 is the only vector in V that is orthogonal to itself.

Theorem 10 (Pythagorean Theorem). Suppose $u, v \in V$. If u and v are orthogonal, then

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2$$

Lemma 11 (orthogonal decomposition). Suppose $u, v \in V$, with $v \neq 0$. Set $c = \frac{\langle u, v \rangle}{\|v\|^2}$ and $w = u - \frac{\langle u, v \rangle}{\|v\|^2}v$. Then

$$u = cv + w \quad \text{and} \quad \langle w, v \rangle = 0.$$

Theorem 12 (Cauchy-Schwarz inequality). Suppose $u, v \in V$. Then

$$|\langle u, v \rangle| \leq \|u\| \|v\|$$

This inequality is an equality if and only if one of u, v is a scalar multiple of the other.

Theorem 13 (triangle inequality). Suppose $u, v \in V$. Then

$$\|u + v\| \leq \|u\| + \|v\|.$$

This inequality is an equality if and only if one of u, v is a nonnegative real multiple of the other.

Theorem 14 (parallelogram equality). Suppose $u, v \in V$. Then

$$\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2).$$

Problem 1

Prove or give a counter example: If $v_1, \dots, v_m \in V$, then

$$\sum_{j=1}^m \sum_{k=1}^m \langle v_j, v_k \rangle \geq 0.$$

Proof. By linearity of inner products,

$$\sum_{j=1}^m \sum_{k=1}^m \langle v_j, v_k \rangle = \left\langle \sum_{j=1}^m v_j, \sum_{k=1}^m v_k \right\rangle \geq 0$$

since the two terms equal other and the conclusion follows by positivity of inner products. $\square$

Problem 2

Suppose $S \in \mathcal{L}(V)$. Define $\langle \cdot, \cdot \rangle_1$ by

$$\langle u, v \rangle_1 = \langle Su, Sv \rangle$$

for all $u, v \in V$. Show that $\langle \cdot, \cdot \rangle_1$ is an inner product on V if and only if S is injective.

Proof. $\langle \cdot, \cdot \rangle_1$ is inner product $\iff \langle v, v \rangle_1 = \langle Sv, Sv \rangle = 0$ if and only if $v = 0$ $\iff S$ is injective. (Other properties are omitted for checking) $\square$

Problem 3

- (a) Show that the function taking an ordered pair $((x_1, x_2), (y_1, y_2))$ of elements of $\mathbb{R}^2$ to $|x_1 y_1| + |x_2 y_2|$ is not an inner product on $\mathbb{R}^2$.
- (b) Show that the function taking an ordered pair $((x_1, x_2, x_3), (y_1, y_2, y_3))$ of elements of $\mathbb{R}^3$ to $x_1 y_1 + x_3 y_3$ is not an inner product on $\mathbb{R}^3$.

Proof. (a) Consider $x = (2, -2)$ and $y = (-2, 2)$ and $z = (1, 1)$. Then $\langle x, z \rangle = \langle y, z \rangle = 4$, but $\langle x + y, z \rangle = 0$.

(b) We have $\langle (0, 1, 0), (0, 1, 0) \rangle = 0$ but the element is nonzero. $\square$

Problem 4

Suppose $T \in \mathcal{L}(V)$ is such that $\|Tv\| \leq \|v\|$ for every $v \in V$. Prove that $T - \sqrt{2}I$ is injective.

Proof. Suppose for contradiction that $T - \sqrt{2}I$ is not injective and therefore $\sqrt{2}$ is an eigenvalue of T , so $Tv = \sqrt{2}v$ for some nonzero v . Taking the norm yields that

$$\|Tv\| = \sqrt{2}\|v\|$$

which violates the assumption that $\|Tv\| \leq \|v\|$. $\square$

Problem 5

Suppose V is a real inner product space.

- (a) Show that $\langle u + v, u - v \rangle = \|u\|^2 - \|v\|^2$ for every $u, v \in V$.
- (b) Show that if $u, v \in V$ have the same norm, then $u + v$ is orthogonal to $u - v$.
- (c) Use (b) to show that the diagonals of a rhombus are perpendicular to each other.

Proof. (a) We have that

$$\begin{aligned}\langle u + v, u - v \rangle &= \langle u, u \rangle - \langle v, v \rangle - \langle u, v \rangle + \langle v, u \rangle \\ &= \|u\|^2 - \|v\|^2\end{aligned}$$

(b) We know $\|u\| = \|v\|$, then

$$\langle u + v, u - v \rangle = \|u\|^2 - \|v\|^2 = 0$$

which shows that they are orthogonal.

(c) omitted. $\square$

Problem 6

Suppose $u, v \in V$. Prove that $\langle u, v \rangle = 0 \iff \|u\| \leq \|u + av\|$ for all $a \in \mathbb{F}$.

Proof. $\Rightarrow$ Given $\langle u, v \rangle = 0$, then

$$\begin{aligned}\|u + av\|^2 &= \langle u + av, u + av \rangle \\ &= \|u\|^2 + \bar{a}\langle u, v \rangle + |a|^2\langle v, v \rangle \\ &= \|u\|^2 + |a|^2\langle v, v \rangle \\ &\geq \|u\|^2\end{aligned}$$

$\Leftarrow$ If $v = 0$, then it's trivial. Consider $v \neq 0$. Let $a = \frac{\langle u, v \rangle}{\|v\|^2}$. Then we have that

$$\begin{aligned} \left\| u - \frac{\langle u, v \rangle}{\|v\|^2} v \right\|^2 &= \left\langle u - \frac{\langle u, v \rangle}{\|v\|^2} v, u - \frac{\langle u, v \rangle}{\|v\|^2} v \right\rangle \\ &= \|u\|^2 - \frac{\overline{\langle u, v \rangle}}{\|v\|^2} \langle u, v \rangle - \frac{\langle u, v \rangle}{\|v\|^2} \langle v, u \rangle + \left| \frac{\langle u, v \rangle}{\|v\|^2} \right|^2 \|v\|^2 \\ &= \|u\|^2 - 2 \frac{|\langle u, v \rangle|^2}{\|v\|^2} + \frac{|\langle u, v \rangle|^2}{\|v\|^2} \\ &= \|u\|^2 - \frac{|\langle u, v \rangle|^2}{\|v\|^2} \geq \|u\|^2 \end{aligned}$$

This implies that

$$\frac{|\langle u, v \rangle|^2}{\|v\|^2} = 0$$

Since $v \neq 0$, $\langle u, v \rangle = 0$. □

Problem 7

Suppose $u, v \in V$. Prove that $\|au + bv\| = \|bu + av\|$ for all $a, b \in \mathbb{R}$ if and only if $\|u\| = \|v\|$.

Proof. Notice that

$$\begin{aligned} \|au + bv\|^2 &= \langle au + bv, au + bv \rangle \\ &= |a|^2 \|u\|^2 + a\bar{b} \langle u, v \rangle + b\bar{a} \langle v, u \rangle + |b|^2 \|v\|^2 \\ &= |a|^2 \|u\|^2 + |b|^2 \|v\|^2 + ab(\langle u, v \rangle + \langle v, u \rangle) \end{aligned}$$

At the same time we have

$$\|bu + av\|^2 = |b|^2 \|u\|^2 + |a|^2 \|v\|^2 + ab(\langle u, v \rangle + \langle v, u \rangle)$$

Then this means $\|au + bv\| = \|bu + av\|$ for all $a, b \in \mathbb{R}$ iff $|a|^2 \|u\|^2 + |b|^2 \|v\|^2 = |b|^2 \|u\|^2 + |a|^2 \|v\|^2$ for all $a, b \in \mathbb{R}$ iff $\|u\| = \|v\|$. □

Problem 8

Suppose $a, b, c, x, y \in \mathbb{R}$ and $a^2 + b^2 + c^2 + x^2 + y^2 \leq 1$. Prove that $a + b + c + 4x + 9y \leq 10$.

Proof. Let

$$u = (a, b, c, x, y) \quad v = (1, 1, 1, 4, 9)$$

and consider the standard real euclidean inner product. Then we can apply the Cauchy-Schwarz:

$$|\langle u, v \rangle|^2 = \left(\sum_{i=1}^5 u_i v_i \right)^2 \leq \left(\sum_{i=1}^5 u_i^2 \right) \left(\sum_{i=1}^5 v_i^2 \right) = \|u\|^2 \|v\|^2$$

Expanding this gives that

$$(a + b + c + 4x + 9y)^2 \leq (a^2 + b^2 + c^2 + x^2 + y^2)(1 + 1 + 1 + 16 + 81) \leq 100$$

Therefore, we have that

$$a + b + c + 4x + 9y \leq 10$$

□

Problem 9

Suppose $u, v \in V$ and $\|u\| = \|v\| = 1$ and $\langle u, v \rangle = 1$. Prove that $u = v$.

Proof. Suppose for contradiction that $u \neq v$, then $u - v \neq 0$. Then

$$\|u - v\|^2 = \langle u - v, u - v \rangle = \|u\|^2 + \|v\|^2 - \langle u, v \rangle - \langle v, u \rangle = 2 - 2 = 0$$

forming a contradiction. Therefore, $u = v$.

□

Problem 10

Suppose $u, v \in V$ and $\|u\| \leq 1$ and $\|v\| \leq 1$. Prove that

$$\sqrt{1 - \|u\|^2} \sqrt{1 - \|v\|^2} \leq 1 - |\langle u, v \rangle|.$$

Proof. Notice that by the Cauchy-Schwarz $|\langle u, v \rangle| \leq \|u\| \|v\| = 1$. Hence, we have that

$$1 - \|u\| \|v\| \leq 1 - |\langle u, v \rangle|$$

So now it suffices to show that

$$(1 - \|u\|^2)(1 - \|v\|^2) \leq (1 - \|u\| \|v\|)^2$$

This is not hard to see, as r.h.s - l.h.s = $(\|u\| - \|v\|)^2 \geq 0$.

□

Problem 12

Suppose a, b, c, d are positive numbers.

- Prove that $(a + b + c + d)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}\right) \geq 16$.
- For which positive numbers a, b, c, d is the inequality above an equality?

Proof. (a) Let $u = (\sqrt{a}, \sqrt{b}, \sqrt{c}, \sqrt{d})$ and $v = (\frac{1}{\sqrt{a}}, \frac{1}{\sqrt{b}}, \frac{1}{\sqrt{c}}, \frac{1}{\sqrt{d}})$. Then applying the Cauchy-Schwarz yields that

$$\langle u, v \rangle^2 = 16 \leq (a + b + c + d) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right) = \|u\|^2 \|v\|^2$$

(b) By the Cauchy-Schwarz, this is an equality iff $u = cv$, i.e. $(\sqrt{a}, \sqrt{b}, \sqrt{c}, \sqrt{d}) = c(\frac{1}{\sqrt{a}}, \frac{1}{\sqrt{b}}, \frac{1}{\sqrt{c}}, \frac{1}{\sqrt{d}})$, which holds if $a = b = c = d$. $\square$

Problem 13

Show that the square of an average is less than or equal to the average of the squares. More precisely, show that if $a_1, \dots, a_n \in \mathbb{R}$, then the square of the average of $a_1, \dots, a_n$ is less than or equal to the average of $a_1^2, \dots, a_n^2$.

Proof. We try to prove

$$\left(\frac{1}{n} \sum_{i=1}^n a_i \right)^2 \leq \frac{1}{n} \sum_{i=1}^n a_i^2$$

Take $u = (a_1, \dots, a_n)$ and $v = (\frac{1}{n}, \dots, \frac{1}{n})$. Then applying the Cauchy-Schwarz yields that

$$\langle u, v \rangle^2 = \left(\frac{1}{n} \sum_{i=1}^n a_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \frac{1}{n} = \|u\|^2 \|v\|^2$$

$\square$

Problem 15

Suppose u, v are nonzero vectors in $\mathbb{R}^2$. Prove that

$$\langle u, v \rangle = \|u\| \|v\| \cos \theta,$$

where θ is the angle between u and v .

Proof. By law of cosines, we have that

$$\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\| \|v\| \cos \theta$$

This means that

$$\begin{aligned} 2\|u\| \|v\| \cos \theta &= \|u\|^2 + \|v\|^2 - \|u - v\|^2 \\ &= \|u\|^2 + \|v\|^2 + 2\langle u, v \rangle - \|u\|^2 - \|v\|^2 \\ &= 2\langle u, v \rangle \end{aligned}$$

$\square$

Problem 17

Prove that

$$\left(\sum_{k=1}^n a_k b_k \right)^2 \leq \left(\sum_{k=1}^n k a_k^2 \right) \left(\sum_{k=1}^n \frac{b_k^2}{k} \right)$$

Proof. Consider $u = (a_1, \sqrt{2}a_2, \dots, \sqrt{n}a_n)$ and $v = (b_1, \frac{b_2}{\sqrt{2}}, \dots, \frac{b_n}{\sqrt{n}})$. Applying the Cauchy-Schwarz solves the problem. $\square$

Problem 19

Suppose $v_1, \dots, v_n$ is a basis of V and $T \in \mathcal{L}(V)$. Prove that if λ is an eigenvalue of T , then

$$|\lambda|^2 \leq \sum_{j=1}^n \sum_{k=1}^n |\mathcal{M}(T)_{j,k}|^2,$$

where $\mathcal{M}(T)_{j,k}$ denotes the entry in row j , column k of the matrix of T wrt. the basis $v_1, \dots, v_n$.

Proof.

$$|\lambda|^2 \|v\|^2 = \|\mathcal{M}(T)v\|^2 \leq \|\mathcal{M}(T)\|_F^2 \|v\|^2$$

for nonzero eigenvector v . Then expanding the Frobenius norm of $\mathcal{M}(T)$ gets the desired inequality. $\square$

Problem 20

Prove the **reverse triangular inequality**: if $u, v \in V$, then $|||u| - |v|| \leq \|u - v\|$.

Proof.

$$\begin{aligned} \|u - v\|^2 &= \langle u - v, u - v \rangle \\ &= \|u\|^2 + \|v\|^2 - (\langle u, v \rangle + \langle v, u \rangle) \\ &\geq \|u\|^2 + \|v\|^2 - 2\|u\|\|v\| \\ &= (\|u\| - \|v\|)^2 \end{aligned}$$

Taking off the square yields the expected solution. $\square$

Problem 21

Suppose $u, v \in V$ such that

$$\|u\| = 3, \quad \|u + v\| = 4, \quad \|u - v\| = 6.$$

What number does $\|v\|$ equal?

Proof. We know that

$$\begin{aligned} \|v\| &\geq \|u + v\| - \|u\| = 1 \\ \|v\|^2 &= (\|u + v\|^2 + \|u - v\|^2)/2 - \|u\|^2 = (16 + 36)/2 - 9 = 17 \end{aligned}$$

So $\|v\| = \sqrt{17}$. □

Problem 22

Show that if $u, v \in V$, then

$$\|u + v\| \|u - v\| \leq \|u\|^2 + \|v\|^2.$$

Proof. Let $a = \|u + v\|$, $b = \|u - v\|$, then we know that

$$a^2 + b^2 = 2(\|u\|^2 + \|v\|^2)$$

We have that

$$(a - b)^2 \geq 0$$

Expanding it gives that

$$(a - b)^2 = (a^2 + b^2) - 2ab \geq 0$$

equivalently,

$$\|u\|^2 + \|v\|^2 \geq \|u + v\| \|u - v\|$$
□

Problem 23

Suppose $v_1, \dots, v_m \in V$ are such that $\|v_k\| \leq 1$ for each $k = 1, \dots, m$. Show that there exists $a_1, \dots, a_m \in \{1, -1\}$ such that

$$\|a_1 v_1 + \dots + a_m v_m\| \leq \sqrt{m}.$$

Proof. We consider a probabilistic approach: Let $a_1, \dots, a_m$ be the iid Rademacher variables such with $a_i = 1$ w.p. $1/2$ and $a_i = -1$ w.p. $1/2$. Then we can define a random vector

$$X = \sum_{i=1}^m a_i v_i$$

and we can compute the expected value

$$\mathbb{E}[\|X\|^2] = \mathbb{E}\left[\left\|\sum_{i=1}^m a_i v_i\right\|^2\right] = \mathbb{E}\left[\left(\sum_{i=1}^m a_i v_i \cdot \sum_{j=1}^m a_j v_j\right)\right] = \sum_{i=1}^m \sum_{j=1}^m (v_i \cdot v_j) \mathbb{E}[a_i a_j]$$

Note that here $\mathbb{E}[a_i a_j] = \delta_{ij}$ and that

$$\mathbb{E}[\|X\|^2] = \sum_{k=1}^m \mathbb{E}[a_k^2] (v_k \cdot v_k) = \sum_{k=1}^m \|v_k\|^2 \leq m$$

which gives that

$$\mathbb{E}[\|X\|] \leq \sqrt{m}$$

and shows the existence proof. $\square$

Problem 25

Suppose $p > 0$. Prove that there is an inner product on $\mathbb{R}^2$ such that the associated norm is given by

$$\|(x, y)\| = (|x|^p + |y|^p)^{1/p}$$

for all $(x, y) \in \mathbb{R}^2$ if and only if $p = 2$.

Proof. $\Leftarrow$ Given $p = 2$, the natural euclidean dot product induces a well-defined norm, e.g. $\|(x, y)\| = (x^2 + y^2)^{1/2}$.

$\Rightarrow$ Note that the parallelogram equalities need to hold. Thus pick $u = (1, 0)$, $v = (0, 1)$, and then

$$\|u + v\|^2 + \|u - v\|^2 = 2 \cdot 4^{1/p}$$

and

$$2(\|u\|^2 + \|v\|^2) = 4$$

They only equal each other when

$$2 \cdot 4^{1/p} = 4$$

which holds only if $p = 2$. $\square$

Problem 26

Suppose V is a real inner product space. Prove that

$$\langle u, v \rangle = \frac{\|u + v\|^2 - \|u - v\|^2}{4}$$

for all $u, v \in V$.

Proof.

$$\begin{aligned}
\|u + v\|^2 - \|u - v\|^2 &= \langle u + v, u + v \rangle - \langle u - v, u - v \rangle \\
&= (\|u\|^2 + 2\langle u, v \rangle + \|v\|^2) - (\|u\|^2 - 2\langle u, v \rangle + \|v\|^2) \\
&= 4\langle u, v \rangle
\end{aligned}$$

□

Problem 29

Suppose $V_1, \dots, V_m$ are inner product spaces. Show that the equation

$$\langle (u_1, \dots, u_m), (v_1, \dots, v_m) \rangle = \langle u_1, v_1 \rangle + \dots + \langle u_m, v_m \rangle$$

defines an inner product on $V_1 \times \dots \times V_m$.

Proof. We check this by definition. Let $u, v, w \in V_1 \times \dots \times V_m$.

positivity: $\langle v, v \rangle = \langle v_1, v_1 \rangle + \dots + \langle v_m, v_m \rangle \geq 0$ as each of them ≥ 0 .

definiteness: Suppose that $\langle v, v \rangle = \langle v_1, v_1 \rangle + \dots + \langle v_m, v_m \rangle = 0$. Then as each of the individual element ≥ 0 , the only solution is $v = 0$. Conversely, if $v = 0$, then $\langle v, v \rangle = 0$.

additivity in first slot: $\langle u + v, w \rangle = \langle u_1 + v_1, w_1 \rangle + \dots + \langle u_m + v_m, w_m \rangle = (\langle u_1, w_1 \rangle + \dots + \langle u_m, w_m \rangle) + (\langle v_1, w_1 \rangle + \dots + \langle v_m, w_m \rangle) = \langle u, w \rangle + \langle v, w \rangle$

homogeneity in first slot: follows similarly as above.

conjugate symmetry: follows similarly as above.

□

Problem 31

Suppose $u, v, w \in V$. Prove that

$$\left\| w - \frac{1}{2}(u + v) \right\|^2 = \frac{\|w - u\|^2 + \|w - v\|^2}{2} - \frac{\|u - v\|^2}{4}.$$

Proof. Let $w - u = a$ and $w - v = b$, then we have

$$\begin{aligned}
\text{l.h.s} &= \|a/2 + b/2\|^2 \\
&= 2(\|a/2\|^2 + \|b/2\|^2) - \|a/2 - b/2\|^2 \\
&= \frac{\|a\|^2 + \|b\|^2}{2} - \frac{\|a - b\|^2}{4} = \text{r.h.s}
\end{aligned}$$

Substituting a and b gets the desired result.

□

Problem 32

Suppose that E is a subset of V with the property that $u, v \in E$ implies $\frac{1}{2}(u + v) \in E$. Let $w \in V$. Show that there is at most one point in E that is closest to w . In other words, show that there is at most one $u \in E$ such that

$$\|w - u\| \leq \|w - x\|$$

for all $x \in E$.

Proof. Suppose for contradiction that there is another $v \in E, v \neq u$ such that

$$\|w - v\| \leq \|w - x\|$$

for all $x \in E$. Then we have that

$$\left\| w - \frac{1}{2}(u + v) \right\| = \frac{\|w - u\|^2 + \|w - v\|^2}{2} - \frac{\|u - v\|^2}{4}$$

by problem 31. Notice that

$$\frac{\|w - u\|^2 + \|w - v\|^2}{2} - \frac{\|u - v\|^2}{4} \leq \|w - x\| - \frac{\|u - v\|^2}{4} \leq \|w - x\|$$

for all $x \in E$, reaching a contradiction ($u = v$). □

6B: Orthonormal Bases

Definition 15 (orthonormal). A list of vectors is called **orthonormal** if each vector in the list has norm 1 and is orthogonal to all the other vectors in the list.

Corollary 16. Suppose $e_1, \dots, e_m$ is an orthonormal list of vectors in V . Then

$$\|a_1 e_1 + \dots + a_m e_m\|^2 = |a_1|^2 + \dots + |a_m|^2$$

for all $a_1, \dots, a_m \in \mathbb{F}$.

Corollary 17. Every orthonormal list of vectors is linearly independent.

Theorem 18 (Bessel's inequality). Suppose $e_1, \dots, e_m$ is an orthonormal list of vectors in V . If $v \in V$ then

$$|\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_m \rangle|^2 \leq \|v\|^2$$

Definition 19 (orthonormal basis). An **orthonormal basis** of V is an orthogonal list of vectors in V that is also a basis of V .

Corollary 20. Suppose V is finite-dimensional. Then every orthonormal list of vectors in V of length $\dim V$ is an orthonormal basis of V .

Remark 21. Usually we write $v = \sum_{i=1}^n a_i v_i$, but with orthonormal basis we can just take $a_k = \langle v, e_k \rangle$.

Lemma 22 (writing a vector as a linear combination of an orthonormal basis). Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $u, v \in V$. Then

- (a) $v = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_n \rangle e_n$,
- (b) $\|v\|^2 = |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_n \rangle|^2$,
- (c) $\langle u, v \rangle = \langle u, e_1 \rangle \overline{\langle v, e_1 \rangle} + \dots + \langle u, e_n \rangle \overline{\langle v, e_n \rangle}$.

Theorem 23 (Gram-Schmidt procedure). Suppose $v_1, \dots, v_m$ is a linearly independent list of vectors in V . Let $f_1 = v_1$. For $k = 2, \dots, m$, define f_k inductively by

$$f_k = v_k - \frac{\langle v_k, f_1 \rangle}{\|f_1\|^2} f_1 - \dots - \frac{\langle v_k, f_{k-1} \rangle}{\|f_{k-1}\|^2} f_{k-1}.$$

For each $k = 1, \dots, m$, let $e_k = \frac{f_k}{\|f_k\|}$. Then $e_1, \dots, e_m$ is an orthonormal list of vectors in V such that

$$\text{span}(v_1, \dots, v_k) = \text{span}(e_1, \dots, e_k)$$

for each $k = 1, \dots, m$.

Corollary 24. Every finite-dimensional inner product space has an orthonormal basis.

Corollary 25. *Suppose V is finite-dimensional. Then every orthonormal list of vectors in V can be extended to an orthonormal basis of V .*

Lemma 26 (upper-triangular matrix with respect to some orthonormal basis). *Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then T has an upper-triangular matrix with respect to some orthonormal basis of V if and only if the minimal polynomial of T equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some $\lambda_1, \dots, \lambda_m \in \mathbb{F}$.*

Theorem 27 (Schur's theorem). *Every operator on a finite-dimensional complex inner product space has an upper-triangular matrix with respect to some orthonormal basis.*

Theorem 28 (Riesz representation theorem). *Suppose V is finite-dimensional and φ is a linear functional on V . Then there is a unique vector $v \in V$ such that*

$$\varphi(u) = \langle u, v \rangle$$

for every $u \in V$.

Problem 1

Suppose $e_1, \dots, e_m$ is a list of vectors in V such that

$$\|a_1 e_1 + \dots + a_m e_m\|^2 = |a_1|^2 + \dots + |a_m|^2$$

for all $a_1, \dots, a_m \in \mathbb{F}$. Show that $e_1, \dots, e_m$ is an orthonormal list.

Proof. We have that

$$\begin{aligned} \left\| \sum_{i=1}^m a_i e_i \right\|^2 &= \left\langle \sum_{i=1}^m a_i e_i, \sum_{i=1}^m a_i e_i \right\rangle \\ &= \sum_{i=1}^m \sum_{j=1}^m a_i \overline{a_j} \langle e_i, e_j \rangle \\ &= \sum_{i=1}^m |a_i|^2 \end{aligned}$$

For this holds for arbitrary choices of $a_1, \dots, a_m \in \mathbb{F}$, we need to have that

$$\langle e_i, e_j \rangle = \delta_{ij}$$

which shows the vectors are orthogonal to each other. To see each of them is norm 1, we can set $a_k = 1$ and $a_j = 0$ for all $j \neq k$, which gives that $\|e_k\|^2 = |a_k|^2 = 1$, and thus each of the vector is normalized, completing the proof. $\square$

Problem 3

Suppose $e_1, \dots, e_m$ is an orthonormal list in V and $v \in V$. Prove that

$$\|v\|^2 = |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_m \rangle|^2 \iff v \in \text{span}(e_1, \dots, e_m)$$

Proof. $\Rightarrow$ We can decompose v into two parts, one is $v_{proj} = \sum_{i=1}^m \langle v, e_i \rangle e_i$, which is the orthogonal projection of v onto the subspace spanned by $e_1, \dots, e_m$. We claim that $v - v_{proj}$ is orthogonal to v_{proj} . This can be seen as

$$\begin{aligned} \langle v_{proj}, v - v_{proj} \rangle &= \left\langle \sum_{i=1}^m \langle v, e_i \rangle e_i, v - \sum_{j=1}^m \langle v, e_j \rangle e_j \right\rangle \\ &= \sum_{i=1}^m |\langle v, e_i \rangle|^2 - \sum_{i=1}^m |\langle v, e_i \rangle|^2 = 0 \end{aligned}$$

Then by Pythagorean theorem we have

$$\|v\|^2 = \|v_{proj}\|^2 + \|v - v_{proj}\|^2$$

where $\|v\|^2 = \|v_{proj}\|^2$ and thus $v = v_{proj}$. Equivalently, $v \in \text{span}(e_1, \dots, e_m)$.

$\Leftarrow$ This means that $v = \sum_{i=1}^m a_i e_i$. However, we know that $a_i = \langle v, e_i \rangle$, so $\|v\|^2 = \sum_{j=1}^m |\langle v, e_j \rangle|^2$ by repeatedly applying the Pythagorean theorem. $\square$

Problem 4

Suppose n is a positive integer. Prove that

$$\frac{1}{\sqrt{2\pi}}, \frac{\cos x}{\sqrt{\pi}}, \frac{\cos 2x}{\sqrt{\pi}}, \dots, \frac{\cos nx}{\sqrt{\pi}}, \frac{\sin x}{\sqrt{\pi}}, \frac{\sin 2x}{\sqrt{\pi}}, \dots, \frac{\sin nx}{\sqrt{\pi}}$$

is an orthonormal list of vectors in $C[-\pi, \pi]$, the vector space of continuous real-valued functions on $[-\pi, \pi]$ with inner product

$$\langle f, g \rangle = \int_{-\pi}^{\pi} fg.$$

Proof. First, we show each of the element has norm 1.

$$\begin{aligned} \left\| \frac{1}{\sqrt{2\pi}} \right\| &= \sqrt{\int_{-\pi}^{\pi} \frac{1}{2\pi} dx} = 1 \\ \left\| \frac{\cos nx}{\sqrt{\pi}} \right\| &= \sqrt{\int_{-\pi}^{\pi} \frac{\cos^2 nx}{\pi} dx} = \sqrt{\frac{1}{\pi} \left[\frac{x}{2} + \frac{\sin(2nx)}{4n} \right]_{-\pi}^{\pi}} = 1 \\ \left\| \frac{\sin nx}{\sqrt{\pi}} \right\| &= \sqrt{\int_{-\pi}^{\pi} \frac{\sin^2 nx}{\pi} dx} = \sqrt{\frac{1}{\pi} \left[\frac{x}{2} - \frac{\cos(2nx)}{4n} \right]_{-\pi}^{\pi}} = 1 \end{aligned}$$

Next, we show that each element is orthogonal to each other, there are many different cases, we begin examine here:

$$\begin{aligned} \left\langle \frac{1}{\sqrt{2\pi}}, \frac{\cos nx}{\sqrt{\pi}} \right\rangle &= \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} \cos nx dx = \frac{1}{\sqrt{2\pi}} \left[\frac{\sin nx}{n} \right]_{-\pi}^{\pi} = 0 \\ \left\langle \frac{1}{\sqrt{2\pi}}, \frac{\sin nx}{\sqrt{\pi}} \right\rangle &= \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} \sin nx dx = \frac{1}{\sqrt{2\pi}} \left[-\frac{\cos nx}{n} \right]_{-\pi}^{\pi} = 0 \end{aligned}$$

Similarly, one can derive between every different pairs of element, their inner product is 0 for different index. The derivation is omitted. $\square$

Problem 6

Suppose $e_1, \dots, e_n$ is an orthonormal basis of V .

(a) Prove that if $v_1, \dots, v_n$ are vectors in V such that

$$\|e_k - v_k\| < \frac{1}{\sqrt{n}}$$

for each k , then $v_1, \dots, v_n$ is a basis of V .

(b) Show that there exist $v_1, \dots, v_n \in V$ such that

$$\|e_k - v_k\| \leq \frac{1}{\sqrt{n}}$$

for each k , but $v_1, \dots, v_n$ is not linearly independent.

Proof. (a) Suppose for contradiction that $v_1, \dots, v_n$ is not a basis and thus linearly dependent. Then there exists scalars $a_1, \dots, a_n \in \mathbb{F}$ not all zero such that $\sum_{i=1}^n a_i v_i = 0$. Then we have that

$$\sum_{i=1}^n a_i (v_i - e_i) + \sum_{i=1}^n a_i e_i = 0$$

which means that

$$\left\| \sum_{i=1}^n a_i (v_i - e_i) \right\| = \left\| \sum_{i=1}^n a_i e_i \right\|$$

Note that

$$\left\| \sum_{i=1}^n a_i (v_i - e_i) \right\| \leq \sum_{i=1}^n \|a_i (v_i - e_i)\| = \sum_{i=1}^n |a_i| \|v_i - e_i\| < \sum_{i=1}^n \frac{|a_i|}{\sqrt{n}} \leq \left(\sum_{i=1}^n |a_i|^2 \right)^{1/2}$$

where the last inequality is shown by the Cauchy-Schwarz. This reaches a contradiction, as

$$\left\| \sum_{i=1}^n a_i e_i \right\| = \left(\sum_{i=1}^n |a_i|^2 \right)^{1/2} < \left(\sum_{i=1}^n |a_i|^2 \right)^{1/2}$$

(b) Suppose $v_1 = e_1 + \frac{1}{\sqrt{n}} e_2$ and $v_j = e_j$ for $2 \leq j \leq n$. Hence we have

$$\|e_1 - v_1\| = \left\| \frac{1}{\sqrt{n}} e_2 \right\| = \frac{1}{\sqrt{n}}$$

where other conditions hold trivially. However, we can clearly tell that $v_1, \dots, v_n$ is not linearly independent. $\square$

Problem 9

Suppose $e_1, \dots, e_m$ is the result of applying the Gram-Schmidt procedure to a linearly independent list $v_1, \dots, v_m$ in V . Prove that $\langle v_k, e_k \rangle > 0$ for each $k = 1, \dots, m$.

Proof. In the Gram-Schmidt process, we decompose v_k into v_{proj} and f_k where v_{proj} is the Orthogonal projection of v_k onto the $\text{span}(v_1, \dots, v_{k-1}) = \text{span}(e_1, \dots, e_{k-1})$. To show $\langle v_k, e_k \rangle > 0$, it's equivalent to show $\langle v_k, f_k \rangle > 0$, which naturally holds as

$$\langle v_k, f_k \rangle = \langle f_k + v_{proj}, f_k \rangle = \langle f_k, f_k \rangle > 0$$

□

Problem 11

Find a polynomial $q \in \mathcal{P}_2(\mathbb{R})$ such that $p(\frac{1}{2}) = \int_0^1 pq$ for every $p \in \mathcal{P}_2(\mathbb{R})$.

Proof. Define $\varphi \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ to be $\varphi(p) = p(\frac{1}{2})$ and consider the inner product $\langle p, q \rangle = \int_0^1 pq$. Following the Riesz representation theorem, we can derive that

$$q = \overline{\varphi(e_1)}e_1 + \overline{\varphi(e_2)}e_2 + \overline{\varphi(e_3)}e_3$$

where we can consider the orthonormal basis $\sqrt{\frac{1}{2}}, \sqrt{\frac{3}{2}}x, \sqrt{\frac{45}{8}}(x^2 - \frac{1}{3})$. Then

$$\begin{aligned} q(x) &= \sqrt{\frac{1}{2}}\sqrt{\frac{1}{2}} + \sqrt{\frac{3}{2}}\frac{1}{2}\sqrt{\frac{3}{2}}x + \sqrt{\frac{45}{8}}\left(\frac{1}{4} - \frac{1}{3}\right)\sqrt{\frac{45}{8}}\left(x^2 - \frac{1}{3}\right) \\ &= \frac{1}{2} + \frac{3}{4}x + \frac{5}{32} - \frac{15}{32}x^2 \\ &= -\frac{15}{32}x^2 + \frac{3}{4}x + \frac{21}{32} \end{aligned}$$

□

Problem 13

Show that a list $v_1, \dots, v_m$ of vectors in V is linearly dependent if and only if the Gram-Schmidt formula produces $f_k = 0$ for some $k \in \{1, \dots, m\}$.

Proof. At each step k , the formula aims at decomposes $v_k = v_{proj} + f_k$, where v_{proj} is the orthogonal projection of v_k onto $\text{span}(e_1, \dots, e_{k-1})$. $f_k = 0$ equivalently means that $v_k = v_{proj}$, which means that $v \in \text{span}(e_1, \dots, e_{k-1}) = \text{span}(v_1, \dots, v_{k-1})$ and therefore renders the list to be linearly dependent. □

Problem 14

Suppose V is a real inner product space and $v_1, \dots, v_m$ is a linearly independent list of vectors in V . Prove that there exist exactly 2^m orthonormal lists $e_1, \dots, e_m$ of vectors in V such that

$$\text{span}(v_1, \dots, v_k) = \text{span}(e_1, \dots, e_k)$$

for all $k \in \{1, \dots, m\}$.

Proof. We prove this statement through induction on m . For base case, consider $\text{span}(v_1)$ for nonzero $v_1 \in V$, there are only two nonzero vectors in $\text{span}(v_1)$: $\pm \frac{v_1}{\|v_1\|}$. So there are exactly $2^1 = 1$ orthonormal list of vectors.

For induction, assume that for $v_1, \dots, v_{k-1}$ linearly independent list of vectors in V , there exist exactly 2^{k-1} orthonormal lists $e_1, \dots, e_{k-1}$ of vectors in V such that

$$\text{span}(v_1, \dots, v_{k-1}) = \text{span}(e_1, \dots, e_{k-1})$$

For k , by the Gram-Schmidt, we have the e_k such that

$$\text{span}(v_1, \dots, v_k) = \text{span}(e_1, \dots, e_k)$$

Suppose such choice of e_k is not unique and there's other e'_k also satisfies

$$\text{span}(e_1, \dots, e'_k) = \text{span}(e_1, \dots, e_k)$$

which indicates that $e'_k = \sum_{i=1}^k \langle e'_k, e_i \rangle e_i = \langle e'_k, e_k \rangle e_k$ and that

$$1 = \|e'_k\| = |\langle e'_k, e_k \rangle|$$

so $e'_k = \pm e_k$ and this gives $2 * 2^{m-1} = 2^m$ choices of orthonormal lists of vectors. $\square$

Problem 15

Suppose $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are inner products on V such that $\langle u, v \rangle_1 = 0$ if and only if $\langle u, v \rangle_2 = 0$. Prove that there is a positive number c such that $\langle u, v \rangle_1 = c \langle u, v \rangle_2$ for every $u, v \in V$.

Proof. It suffices to prove that $c = \frac{\langle u, v \rangle_1}{\langle u, v \rangle_2}$ for every $u, v \in V$ is a constant number.

First, pick nonzero $u \in V$. Then we know that $\langle u, u \rangle_1 > 0, \langle u, u \rangle_2 > 0$. Pick $v \in V$ s.t. $\langle u, v \rangle_1 \neq 0, \langle u, v \rangle_2 \neq 0$ (i.e. they are not orthogonal). So we have that

$$\left\langle u - \frac{\langle u, v \rangle_2}{\langle v, v \rangle_2} v, v \right\rangle_1 = 0 = \left\langle u - \frac{\langle u, v \rangle_2}{\langle v, v \rangle_2} v, v \right\rangle_2$$

by the orthogonal decomposition of u . This gives that

$$\frac{\langle u, v \rangle_1}{\langle u, v \rangle_2} = \frac{\langle v, v \rangle_1}{\langle v, v \rangle_2}$$

Similarly, we have

$$\left\langle u, v - \frac{\langle u, v \rangle_2}{\langle v, v \rangle_2} u \right\rangle_1 = 0 = \left\langle u, v - \frac{\langle u, v \rangle_2}{\langle v, v \rangle_2} u \right\rangle_2$$

and that

$$\frac{\langle u, v \rangle_1}{\langle u, v \rangle_2} = \frac{\langle u, u \rangle_1}{\langle u, u \rangle_2} = \frac{\langle v, v \rangle_1}{\langle v, v \rangle_2} = c$$

which yields the desired solution. $\square$

Problem 16

Suppose V is finite-dimensional. Suppose $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are inner products on V with corresponding norms $\|\cdot\|_1$ and $\|\cdot\|_2$. Prove that there exists a positive number c such that $\|v\|_1 \leq c\|v\|_2$ for every $v \in V$.

Proof. Let $e_1, \dots, e_n$ be an orthonormal basis of V wrt. $\langle \cdot, \cdot \rangle_1$ and $f_1, \dots, f_n$ be an orthonormal basis of V wrt. $\langle \cdot, \cdot \rangle_2$. Pick nonzero $v \in V$. Then there exists $\varphi \in V'$ such that

$$\|v\|_1^2 = \sum_{i=1}^n |\langle v, e_i \rangle_1|^2 \leq |\varphi(v)|^2$$

We can proceed with that

$$\begin{aligned} \|v\|_1^2 &\leq |\varphi(v)|^2 \\ &= |\langle v, \overline{\varphi(f_1)}f_1 + \dots + \overline{\varphi(f_n)}f_n \rangle_2|^2 \\ &\leq \left\| \sum_{i=1}^n \overline{\varphi(f_i)}f_i \right\|_2^2 \|v\|_2^2 \end{aligned}$$

which completes the proof. $\square$

Problem 17

Suppose $\mathbb{F} = \mathbb{C}$ and V is finite-dimensional. Prove that if T is an operator on V such that 1 is the only eigenvalue of T and $\|Tv\| \leq \|v\|$ for all $v \in V$, then T is the identity operator.

Proof. By Schur's theorem, there exists an orthonormal basis $e_1, \dots, e_n$ such that the matrix of T is upper-triangular. Then 1 is the only component on the diagonal entries. Hence,

$$\|Te_k\| \leq \|e_k\| = 1$$

Note that $Te_k = \sum_{i=1}^{k-1} a_i e_i + e_k$ since we know the upper-triangular matrix has diagonal term to be 1. This gives that

$$\left\| \sum_{i=1}^{k-1} a_i e_i + e_k \right\| = \|e_k\| + \sum_{i=1}^{k-1} |a_i| \|e_i\| \leq \|e_k\|$$

so for each e_k , the off-diagonal entries a_i are all 0 and thus the matrix of T is the identity matrix, and T is the identity operator. $\square$

Problem 18

Suppose $u_1, \dots, u_m$ is a linearly independent list in V . Show that there exists $v \in V$ such that $\langle u_k, v \rangle = 1$ for all $k \in \{1, \dots, m\}$.

Proof. Define $\varphi \in V'$ s.t. $\varphi(u_k) = 1$ for all k . By Riesz representation theorem, there is a unique $v \in V$ s.t.

$$\varphi(u_k) = \langle u_k, v \rangle = 1$$

$\square$

6C: Orthogonal Complements and Minimization Problems

Definition 29 (orthogonal complement, $U^\perp$). If U is a subset of V , then the **orthogonal complement** of U , denoted by $U^\perp$, is the set of all vectors in V that are orthogonal to every vector in U :

$$U^\perp = \{v \in V : \langle u, v \rangle = 0 \text{ for every } u \in U\}.$$

Corollary 30. Properties of orthogonal complement:

- (a) If U is a subset of V , then $U^\perp$ is a subspace of V .
- (b) $\{0\}^\perp = V$.
- (c) $V^\perp = \{0\}$.
- (d) If U is a subset of V , then $U \cap U^\perp \subseteq \{0\}$.
- (e) If G and H are subsets of V and $G \subseteq H$, then $H^\perp \subseteq G^\perp$.

Corollary 31. Suppose U is a finite-dimensional subspace of V . Then

$$V = U \oplus U^\perp$$

and thus $\dim U^\perp = \dim V - \dim U$. In addition,

$$U = (U^\perp)^\perp$$

Corollary 32. Suppose U is a finite-dimensional subspace of V . Then

$$U^\perp = \{0\} \iff U = V.$$

Definition 33 (orthogonal projection, P_U). Suppose U is a finite-dimensional subspace of V . The **orthogonal projection** of V onto U is the operator $P_U \in \mathcal{L}(V)$ defined as follows: for each $v \in V$, write $v = u + w$, where $u \in U$ and $w \in U^\perp$. Then let $P_U v = u$.

Remark 34. Suppose $u \in V$ with $u \neq 0$ and $U = \text{span}(u)$. If $v \in V$, then

$$v = \frac{\langle v, u \rangle}{\|u\|^2} u + \left(v - \frac{\langle v, u \rangle}{\|u\|^2} u \right).$$

Then this implies that

$$P_U v = \frac{\langle v, u \rangle}{\|u\|^2} u$$

Corollary 35 (properties of orthogonal projection P_U). Suppose U is a finite-dimensional subspace of V . Then

- (a) $P_U \in \mathcal{L}(V)$;
- (b) $P_U u = u$ for every $u \in U$;

- (c) $P_U w = 0$ for every $w \in U^\perp$;
- (d) $\text{range } P_U = U$;
- (e) $\text{null } P_U = U^\perp$;
- (f) $v - P_U v \in U^\perp$ for every $v \in V$;
- (g) $P_U^2 = P_U$;
- (h) $\|P_U v\| \leq \|v\|$ for every $v \in V$;
- (i) if $e_1, \dots, e_m$ is an orthonormal basis of U and $v \in V$, then

$$P_U v = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m$$

Theorem 36 (Riesz representation theorem, revisited). *Suppose V is finite-dimensional. For each $v \in V$, define $\varphi_v \in V'$ by*

$$\varphi_v(u) = \langle u, v \rangle$$

for each $u \in V$. Then $v \mapsto \varphi_v$ is a one-to-one function from V to V' .

Theorem 37 (minimizing distance to a subspace). *Suppose U is a finite-dimensional subspace of V , $v \in V$, and $u \in U$. Then*

$$\|v - P_U v\| \leq \|v - u\|.$$

Furthermore, the inequality above is an equality if and only if $u = P_U v$.

Lemma 38 (restriction of a linear map to obtain a one-to-one and onto map). *Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then $T|_{(\text{null } T)^\perp}$ is an injective map of $(\text{null } T)^\perp$ onto $\text{range } T$.*

Definition 39 (pseudoinverse, $T^\dagger$). *Suppose that V is finite-dimensional and $T \in \mathcal{L}(V, W)$. The **pseudoinverse** $T^\dagger \in \mathcal{L}(W, V)$ of T is the linear map from W to V defined by*

$$T^\dagger w = (T|_{(\text{null } T)^\perp})^{-1} P_{\text{range } T} w$$

for each $w \in W$.

Corollary 40 (algebraic properties of the pseudoinverse). *Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$.*

- (a) *If T is invertible, then $T^\dagger = T^{-1}$.*
- (b) *$TT^\dagger = P_{\text{range } T} =$ the orthogonal projection of W onto $\text{range } T$.*
- (c) *$T^\dagger T = P_{(\text{null } T)^\perp} =$ the orthogonal projection of V onto $(\text{null } T)^\perp$.*

Theorem 41 (pseudoinverse provides best approximate solution or best solution). *Suppose V is finite-dimensional, $T \in \mathcal{L}(V, W)$, and $w \in W$.*

(a) If $v \in V$, then

$$\left\| T(T^\dagger w) - w \right\| \leq \|Tv - w\|$$

with equality if and only if $v \in T^\dagger w + \text{null } T$.

(b) If $v \in T^\dagger w + \text{null } T$, then

$$\left\| T^\dagger w \right\| \leq \|v\|,$$

with equality if and only if $v = T^\dagger w$.

Problem 1

Suppose $v_1, \dots, v_m \in V$. Prove that

$$\{v_1, \dots, v_m\}^\perp = (\text{span}(v_1, \dots, v_m))^\perp.$$

Proof. Denote $A = \{v_1, \dots, v_m\}^\perp$ and $B = \text{span}(v_1, \dots, v_m)^\perp$
 $\Rightarrow$ Let $v \in A$, then $\langle v, v_i \rangle = 0$ for all i . So we have

$$\left\langle v, \sum_{i=1}^m a_i v_i \right\rangle = \sum_{i=1}^m \overline{a_i} \langle v, v_i \rangle = 0$$

which means that $v \in B$.

$\Leftarrow$ Conversely, let $v \in B$, then naturally by definition $v \in A$. $\square$

Problem 4

Suppose $e_1, \dots, e_n$ is a list of vectors in V with $\|e_k\| = 1$ for each $k = 1, \dots, n$ and

$$\|v\|^2 = |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_n \rangle|^2$$

for all $v \in V$. Prove that $e_1, \dots, e_n$ is an orthonormal basis of V .

Proof. It now suffices to prove that $\langle e_i, e_j \rangle = \delta_{ij}$. To see this, take $v = e_i$, then we have that

$$\|v\|^2 = \|e_i\|^2 = 1 = \sum_{j \neq i} |\langle e_i, e_j \rangle|^2 + 1$$

This gives that $\langle e_i, e_j \rangle = 0$ for all $i \neq j$, completing the proof. $\square$

Problem 5

Suppose that V is finite-dimensional and U is a subspace of V . Show that $P_{U^\perp} = I - P_U$, where I is the identity operator on V .

Proof. Take $v \in V$, then we know $v = u + w$ for $u \in U, w \in U^\perp$. We have that

$$P_U v = u \quad P_{U^\perp} v = w = v - u = (I - P_U)v$$

$\square$

Problem 6

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Show that

$$T = TP_{(\text{null } T)^\perp} = P_{\text{range } T}T.$$

Proof. Take arbitrary $v \in V$, then $v = u + w$ for $u \in \text{null } T$ and $w \in (\text{null } T)^\perp$. We have

$$Tv = T(u + w) = Tw = TP_{(\text{null } T)^\perp}v$$

Furthermore, since $Tv \in \text{range } T$, $P_{\text{range } T}$ acts as an identity operator for Tv , thus we have the second equality. $\square$

Problem 7

Suppose that X and Y are finite-dimensional subspaces of V . Prove that $P_X P_Y = 0$ if and only if $\langle x, y \rangle = 0$ for all $x \in X$ and all $y \in Y$.

Proof. $\Rightarrow$ take arbitrary $y \in Y$, then $P_X(y) = 0$, which means that $y = 0 + (y - 0)$ where $0 \in X$ and thus all $y \in Y$ are orthogonal to $x \in X$, completing this direction.

$\Leftarrow$ Take $v \in V$, then $v = y + y'$ for $y \in Y, y' \in Y^\perp$ and we further have $y = 0 + y$ for $0 \in X$ and $y \in X^\perp$. We now have that

$$P_X P_Y(v) = P_X(y) = 0$$

$\square$

Problem 9

Suppose V is finite-dimensional. Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$ and every vector in $\text{null } P$ is orthogonal to every vector in $\text{range } P$. Prove that there exists a subspace U of V such that $P = P_U$.

Proof. We can simply take $U = \text{range } P$. Note that $V = \text{null } P \oplus \text{range } P$ as

$$v = Pv + (v - Pv)$$

where $P(v - Pv) = 0$ so $v - Pv \in \text{null } P$.

Then take $v = v_1 + v_2$ where $v_1 \in \text{null } P, v_2 \in \text{range } P$, then we have

$$Pv = P(v_1 + v_2) = Pv_2 = P_U v$$

$\square$

Problem 11

Suppose $T \in \mathcal{L}(U)$ and U is a finite-dimensional subspace of V . Prove that

$$U \text{ is invariant under } T \iff P_U T P_U = T P_U$$

Proof. U is invariant under $T \iff Tu \in U$ for all $u \in U \iff$ for $v = u + u^\perp \in V$, $TP_U(v) = Tu = P_U(Tu) = P_U T P_U(v)$ $\square$

Problem 13

Suppose $\mathbb{F} = \mathbb{R}$ and V is finite-dimensional. For each $v \in V$, let φ_v denote the linear functional on V defined by

$$\varphi_v(u) = \langle u, v \rangle$$

for all $u \in V$.

- (a) Show that $v \mapsto \varphi_v$ is an injective linear map from V to V' .
- (b) Use (a) and a dimension-counting argument to show that $v \mapsto \varphi_v$ is an isomorphism from V onto V' .

Proof. (a) denote this map $v \mapsto \varphi_v$ to be T . Then take $v \in \text{null } T$, we have $T(v) = \varphi_v = 0$. By definition, since this holds for all $u \in V$, $v = 0$ and thus T is injective. To show it's also linear, $T(\lambda v_1 + v_2)(u) = \varphi_{\lambda v_1 + v_2}(u) = \langle u, \lambda v_1 + v_2 \rangle = \lambda \langle u, v_1 \rangle + \langle u, v_2 \rangle = \lambda \varphi_{v_1} + \varphi_{v_2} = \lambda T(v_1) + T(v_2)$.

(b) We know that $\dim V = \dim V'$ and combining this with (a) yields the solution. $\square$

Problem 15

In $\mathbb{R}^4$, let

$$U = \text{span}((1, 1, 0, 0), (1, 1, 1, 2))$$

Find $u \in U$ such that $\|u - (1, 2, 3, 4)\|$ is as small as possible.

Proof. We first find the orthonormal basis of U and apply the formula, i.e. $P_U(v) = \sum_{i=1}^n \langle v, e_i \rangle e_i$. Using the Gram-Schmidt, we can find that

$$\left\{ \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, 0 \right), \left(0, 0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right) \right\}$$

Then we can get that

$$\begin{aligned} u &= \left\langle (1, 2, 3, 4), \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, 0 \right) \right\rangle \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, 0 \right) + \\ &\quad \left\langle (1, 2, 3, 4), \left(0, 0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right) \right\rangle \left(0, 0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right) \\ &= \left(\frac{3}{2}, \frac{3}{2}, \frac{11}{5}, \frac{22}{5} \right) \end{aligned}$$

$\square$

Problem 19

Suppose V is finite-dimensional and $P \in \mathcal{L}(V)$ is an orthogonal projection of V onto some subspace of V . Prove that $P^\dagger = P$.

Proof. Suppose the subspace is U . Take $u \in U$, then we know that $u \in \text{range } P$, and thus

$$P^\dagger u = (P|_{(\text{null } P)^\perp})^{-1} P_{\text{range } P} u = (P|_{(\text{null } P)^\perp})^{-1} u = u = Pu$$

Take $u \in U^\perp$, then we have $Pu = 0$ and also $P^\dagger u = 0$ by definition. Thus these two operators equal each other. $\square$

Problem 20

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Show that

$$\text{null } T^\dagger = (\text{range } T)^\perp \text{ and } \text{range } T^\dagger = (\text{null } T)^\perp.$$

Proof. We know that $T^\dagger = (T|_{(\text{null } T)^\perp})^{-1} P_{\text{range } T}$ and the first part $T|_{(\text{null } T)^\perp}$ we've shown it's bijective with the restriction in book's lemma. So for $v \in \text{null } T^\dagger$, $P_{\text{range } T} v = 0$ and thus $v \in (\text{range } T)^\perp$. Conversely, it holds by definition.

For the other equality, take $v \in \text{range } T^\dagger$, then there exists $u \in \text{range } T$ s.t. $T|_{(\text{null } T)^\perp} v = u$, so $v \in (\text{null } T)^\perp$. Conversely, take $v \in (\text{null } T)^\perp$, then there exists $u \in \text{range } T$ s.t. $Tv = u$, and we have $T^\dagger u = v$ so $v \in \text{range } T^\dagger$. $\square$

Problem 22

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$TT^\dagger T = T \text{ and } T^\dagger TT^\dagger = T^\dagger.$$

Proof. For the first equality, take $v \in \text{null } T$, then $TT^\dagger Tv = Tv = 0$. Take $v \in (\text{null } T)^\perp$, then $TT^\dagger(Tv) = T(v)$ by definition.

For the second equality, take $w \in (\text{range } T)^\perp$, then $T^\dagger TT^\dagger w = 0 = T^\dagger w$. Take nonzero $w \in \text{range } T$, then there exists $v \in (\text{null } T)^\perp$ such that $Tv = w$, hence $T^\dagger TT^\dagger w = T^\dagger Tv = v = T^\dagger w$. $\square$

Problem 23

Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$(T^\dagger)^\dagger = T.$$

Proof. Denote $S = T^\dagger$, we have that

$$(T^\dagger)^\dagger = S^\dagger = (S|_{(\text{null } S)^\perp})^{-1} P_{\text{range } S} = (S|_{\text{range } T})^{-1} P_{(\text{null } T)^\perp}$$

$\square$

where we use the conclusion from problem 20. Note that if $v \in \text{null } T$, then naturally $(T^\dagger)^\dagger v = Tv = 0$. If $v \in (\text{null } T)^\perp$, then first note that

$$(S|_{\text{range } T})^{-1}P_{(\text{null } T)^\perp} = (S|_{\text{range } T})^{-1}v$$

Expanding the definition gives that

$$(S|_{\text{range } T})^{-1}v = ((T|_{(\text{null } T)^\perp})^{-1}P_{\text{range } T})|_{\text{range } T})^{-1}v = T|_{(\text{null } T)^\perp}v = Tv$$

Therefore, we complete the proof.

Chapter 7: Operators on Inner Product Spaces

Linear Algebra Done Right (4th Edition), by Sheldon Axler

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Contents

7A: Self-Adjoint and Normal Operators	2
7A Problem Sets	4
7B: Spectral Theorem	10
7B Problem Sets	11
7C: Positive Operators	16
7C Problem Sets	17
7D: Isometries, Unitary Operators, and Matrix Factorization	21
7D Problem Sets	23
7E: Singular Value Decomposition	28
7E Problem Sets	30
7F: Consequences of Singular Value Decomposition	33
7F Problem Sets	36

7A: Self-Adjoint and Normal Operators

Definition 1 (adjoint, T^*). Suppose $T \in \mathcal{L}(V, W)$. The **adjoint** of T is the function $T^*: W \rightarrow V$ such that

$$\langle Tv, w \rangle = \langle v, T^*w \rangle$$

for every $v \in V$ and every $w \in W$.

Corollary 2. If $T \in \mathcal{L}(V, W)$, then $T^* \in \mathcal{L}(W, V)$. In other words, the adjoint of a linear map is a linear map.

Corollary 3 (properties of the adjoint). Suppose $T \in \mathcal{L}(V, W)$. Then

- (a) $(S + T)^* = S^* + T^*$ for all $S \in \mathcal{L}(V, W)$;
- (b) $(\lambda T)^* = \bar{\lambda}T^*$ for all $\lambda \in \mathbb{F}$;
- (c) $(T^*)^* = T$;
- (d) $(ST)^* = T^*S^*$ for all $S \in \mathcal{L}(W, U)$ (here U is a finite-dimensional inner product space over $\mathbb{F}$).
- (e) $I^* = I$;
- (f) if T is invertible, then T^* is invertible and $(T^*)^{-1} = (T^{-1})^*$.

Theorem 4 (null space and range of T^*). Suppose $T \in \mathcal{L}(V, W)$. Then

- (a) $\text{null } T^* = (\text{range } T)^\perp$;
- (b) $\text{range } T^* = (\text{null } T)^\perp$;
- (c) $\text{null } T = (\text{range } T^*)^\perp$;
- (d) $\text{range } T = (\text{null } T^*)^\perp$.

Definition 5 (conjugate transpose, A^*). The **conjugate transpose** of an m -by- n matrix A is the n -by- m matrix A^* obtained by interchanging the rows and columns and then taking the complex conjugate of each entry. In other words, if $j \in \{1, \dots, n\}$ and $k \in \{1, \dots, m\}$, then

$$(A^*)_{j,k} = \overline{A_{k,j}}$$

Remark 6. We denote $A^\top$ (transpose) when we know the matrix is real. Note that wrt. nonorthonormal bases, the matrix of T^* does not necessarily equal the conjugate transpose of the matrix of T .

Theorem 7 (matrix of T^* equals conjugate transpose of matrix of T). Let $T \in \mathcal{L}(V, W)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $f_1, \dots, f_m$ is an orthonormal basis of W . Then $\mathcal{M}(T^*, (f_1, \dots, f_m), (e_1, \dots, e_n))$ is the conjugate transpose of $\mathcal{M}(T, (e_1, \dots, e_n), (f_1, \dots, f_m))$. In other words,

$$\mathcal{M}(T^*) = (\mathcal{M}(T))^*$$

Remark 8. *Orthogonal complements and adjoints are concepts that are easier to work with than annihilators and dual maps in the context of inner product spaces.*

Definition 9 (self-adjoint). *An operator $T \in \mathcal{L}(V)$ is called **self-adjoint** if $T = T^*$. In other words, an operator $T \in \mathcal{L}(V)$ is self-adjoint if and only if*

$$\langle Tv, w \rangle = \langle v, Tw \rangle$$

for all $v, w \in V$.

Corollary 10. *Every eigenvalue of a self-adjoint operator is real.*

Corollary 11. *Suppose V is a **complex** inner product space and $T \in \mathcal{L}(V)$. Then*

$$\langle Tv, v \rangle = 0 \text{ for every } v \in V \iff T = 0.$$

Corollary 12. *Suppose V is a **complex** inner product space and $T \in \mathcal{L}(V)$. Then*

$$T \text{ is self-adjoint} \iff \langle Tv, v \rangle \in \mathbb{R} \text{ for every } v \in V$$

Remark 13. *The above two corollaries do not hold for real inner product spaces.*

Theorem 14. *Suppose T is a self-adjoint operator on V . Then*

$$\langle Tv, v \rangle = 0 \text{ for every } v \in V \iff T = 0$$

Definition 15 (normal). *An operator on an inner product space is called **normal** if it commutes with its adjoint. In other words, $T \in \mathcal{L}(V)$ is normal if $TT^* = T^*T$.*

Remark 16. *Every self-adjoint operator is normal, but not vice versa.*

Theorem 17. *Suppose $T \in \mathcal{L}(V)$. Then*

$$T \text{ is normal} \iff \|Tv\| = \|T^*v\| \text{ for every } v \in V$$

Corollary 18 (range, null space, and eigenvectors of a normal operator). *Suppose $T \in \mathcal{L}(V)$ is normal. Then*

- (a) $\text{null } T = \text{null } T^*$;
- (b) $\text{range } T = \text{range } T^*$;
- (c) $V = \text{null } T \oplus \text{range } T$;
- (d) $T - \lambda I$ is normal for every $\lambda \in \mathbb{F}$;
- (e) if $v \in V$ and $\lambda \in \mathbb{F}$, then $Tv = \lambda v$ if and only if $T^*v = \bar{\lambda}v$.

Theorem 19. *Suppose $T \in \mathcal{L}(V)$ is normal. Then eigenvectors of T corresponding to distinct eigenvalues are orthogonal.*

Theorem 20. *Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then T is normal if and only if there exist commuting self-adjoint operators A and B such that $T = A + iB$.*

Problem 1

Suppose n is a positive integer. Define $T \in \mathcal{L}(\mathbb{F}^n)$ by

$$T(z_1, \dots, z_n) = (0, z_1, \dots, z_{n-1})$$

Find a formula for $T^*(z_1, \dots, z_n)$.

Proof. We have that

$$\begin{aligned} \langle T(x_1, \dots, x_n), (y_1, \dots, y_n) \rangle &= \langle (0, x_1, \dots, x_{n-1}), (y_1, \dots, y_n) \rangle \\ &= x_1 y_2 + \dots + x_{n-1} y_n \\ &= \langle (x_1, \dots, x_n), (y_2, \dots, y_n, 0) \rangle \\ &= \langle (x_1, \dots, x_n), T^*(y_1, \dots, y_n) \rangle \end{aligned}$$

where $T^*(z_1, \dots, z_n) = (z_2, \dots, z_n, 0)$. □

Problem 2

Suppose $T \in \mathcal{L}(V, W)$. Prove that

$$T = 0 \iff T^* = 0 \iff T^*T = 0 \iff TT^* = 0$$

Proof.

$$\begin{aligned} T = 0 &\iff \langle Tv, v \rangle = 0 \text{ for all } v \\ &\iff \langle v, T^*v \rangle = 0 \text{ for all } v \\ &\iff T^* = 0 \\ &\iff T^*T = T^*(0) = 0 \\ &\iff TT^* = 0 \end{aligned}$$

□

Problem 4

Suppose $T \in \mathcal{L}(V)$ and U is a subspace of V . Prove that

$$U \text{ is invariant under } T \iff U^\perp \text{ is invariant under } T^*$$

Proof. Let $u \in U$ and $w \in U^\perp$, then we know that

$$\langle Tu, w \rangle = \langle u, T^*w \rangle$$

Thus

$$\langle Tu, w \rangle = 0 \iff \langle u, T^*w \rangle = 0$$

which implies the desired result. □

Problem 5

Suppose $T \in \mathcal{L}(V, W)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $f_1, \dots, f_m$ is an orthonormal basis of W . Prove that

$$\|Te_1\|^2 + \dots + \|Te_n\|^2 = \|T^*f_1\|^2 + \dots + \|T^*f_m\|^2$$

Proof. Note that first we have

$$\|Te_k\|^2 = \langle Te_k, Te_k \rangle = \left\langle Te_k, \sum_{j=1}^m \langle Te_k, f_j \rangle f_j \right\rangle = \sum_{j=1}^m |\langle Te_k, f_j \rangle|^2$$

At the same time, we have

$$\langle Te_k, f_j \rangle = \langle e_k, T^*f_j \rangle$$

Combining the two equations yields that

$$\sum_{i=1}^n \|Te_i\|^2 = \sum_{i=1}^n \sum_{j=1}^m |\langle Te_i, f_j \rangle|^2 = \sum_{j=1}^m \sum_{i=1}^n |\langle e_i, T^*f_j \rangle|^2 = \sum_{j=1}^m \|T^*f_j\|^2$$

□

Problem 9

Prove that the product of two self-adjoint operators on V is self-adjoint if and only if the two operators commute.

Proof. Let T and S be two self-adjoint operators on V . First if $ST = TS$, then $(ST)^* = (TS)^*$. It suffices to show the forward direction. Let's suppose WLOG that ST is self-adjoint. Then we know that $(ST)^* = T^*S^* = S^*T^* = (TS)^*$. This implies that for arbitrary $v, w \in V$, we have

$$\begin{aligned} \langle v, T^*S^*w \rangle &= \langle v, S^*T^*w \rangle = \langle TSv, w \rangle \\ &= \langle STv, w \rangle \end{aligned}$$

This implies that $TS = ST$.

□

Problem 10

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Prove that T is self-adjoint if and only if

$$\langle Tv, v \rangle = \langle T^*v, v \rangle$$

for all $v \in V$.

Proof.

$$T \text{ self-adjoint} \iff T = T^* \iff \langle Tv, v \rangle = \langle T^*v, v \rangle$$

□

Problem 12

An operator $B \in \mathcal{L}(V)$ is called **skew** if

$$B^* = -B$$

Suppose that $T \in \mathcal{L}(V)$. Prove that T is normal if and only if there exist commuting operators A and B such that A is self-adjoint, B is a skew operator, and $T = A + B$.

Proof. $\Rightarrow$ Take $A = \frac{T+T^*}{2}$ and $B = \frac{T-T^*}{2}$, so it holds that $T = A + B$. We first verify that A is self-adjoint.

$$\begin{aligned} 4AA^* &= (T + T^*)(T + T^*)^* = (T + T^*)(T^* + T) = TT^* + TT^* + T^*T^*T^*T \\ 4A^*A &= (T + T^*)^*(T + T^*) = (T^* + T)(T + T^*) = T^*T + T^*T^*T^*T + TT^* \end{aligned}$$

Therefore $AA^* = A^*A$ and A is self-adjoint. For B , we have that

$$B^* = \frac{T^* - T}{2} = -B$$

$\Leftarrow$ We have that $T^* = (A + B)^* = A^* - B = A - B$

$$TT^* = (A + B)(A - B) = A^2 - AB + BA - B^2 = A^2 - B^2$$

$$T^*T = (A - B)(A + B) = A^2 + AB - BA - B^2 = A^2 - B^2$$

since A and B commute. $\square$

Problem 15

Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

- (a) T is self-adjoint $\iff T^{-1}$ is self-adjoint;
- (b) T is normal $\iff T^{-1}$ is normal.

Proof. (a) T is self-adjoint $\iff T = T^* \iff T^{-1} = (T^*)^{-1} = (T^{-1})^* \iff T^{-1}$ is self-adjoint.

(b) T is normal $\iff TT^* = T^*T \iff (T^*)^{-1}T^{-1} = T^{-1}(T^*)^{-1} \iff T^{-1}$ is normal. $\square$

Problem 19

Suppose $T \in \mathcal{L}(V)$ and $\|T^*v\| \leq \|Tv\|$ for every $v \in V$. Prove that T is normal.

Proof. Note that we have

$$\begin{aligned}
\|T^*v\|^2 \leq \|Tv\|^2 &\iff \langle T^*v, T^*v \rangle \leq \langle Tv, Tv \rangle \\
&\iff \langle TT^*v, v \rangle \leq \langle T^*Tv, v \rangle \\
&\iff \langle (T^*T - TT^*)v, v \rangle \leq 0 \\
&\iff T^*T - TT^* \leq 0
\end{aligned}$$

We want to borrow the fact from later chapters to make the proof substantially easier:

$$\text{tr}(T^*T - TT^*) = 0$$

which means that the sum of the eigenvalues are 0. However, by the fact derived above, let (λ, v) be arbitrary eigenpair of $T^*T - TT^*$, we have that

$$\lambda \langle v, v \rangle = \langle (T^*T - TT^*)v, v \rangle \leq 0$$

meaning that $\lambda \leq 0$ and therefore all $\lambda = 0$. Notice that $T^*T - TT^*$ is self-adjoint as

$$(T^*T - TT^*)^* = (T^*T)^* - (TT^*)^* = T^*T - TT^*$$

Hence, we can claim that $T^*T - TT^* = 0$, and by the result 7.20 T is normal. $\square$

Problem 20

Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that the following are equivalent.

- (a) P is self-adjoint.
- (b) P is normal.
- (c) There is a subspace U of V such that $P = P_U$.

Proof. (a) $\Rightarrow$ (b) Trivial.

(b) $\Rightarrow$ (c) Take arbitrary $v \in V$, then we have that

$$P(v - Pv) = Pv - P^2v = 0$$

which means that $v - Pv \in \text{null } P$. Since P is normal, we know that $V = \text{null } P \oplus \text{range } P$ and thus every v can be written as

$$v = Pv + (v - Pv)$$

which means that $P = P_{\text{range } P}$.

(c) $\Rightarrow$ (a) Take any $v_1, v_2 \in V$, then we know that $v_1 = u_1 + w_1$ and $v_2 = u_2 + w_2$ for $u_1, u_2 \in U$ and $w_1, w_2 \in U^\perp$. It suffices to show that $\langle Pv_1, v_2 \rangle = \langle v_1, Pv_2 \rangle$.

$$\begin{aligned}
\langle Pv_1, v_2 \rangle &= \langle u_1, u_2 + w_2 \rangle \\
&= \langle u_1, u_2 \rangle + \langle u_1, w_2 \rangle \\
&= \langle v_1, Pv_2 \rangle
\end{aligned}$$

$\square$

Problem 23

Suppose T is a normal operator on V . Suppose also that $v, w \in V$ satisfy the equations

$$\|v\| = \|w\| = 2, \quad Tv = 3v, \quad Tw = 4w$$

Show that $\|T(v + w)\| = 10$.

Proof. We note that v and w are distinct eigenvectors with different eigenvalues. As T is normal, v and w are thus orthogonal. We have that

$$\|T(v + w)\|^2 = \|Tv\|^2 + \|Tw\|^2 = 9\|v\|^2 + 16\|w\|^2 = 100$$

which shows the desired result. $\square$

Problem 25

Suppose $T \in \mathcal{L}(V)$. Prove that T is diagonalizable if and only if T^* is diagonalizable.

Proof. We know that (λ, v) is an eigenpair of T if and only if $(\bar{\lambda}, v)$ is an eigenpair of T^* . Since T and T^* share the same eigenvectors, V has an eigenbasis consisting of T 's eigenvectors iff V has an eigenbasis consisting of T^* 's eigenvectors, completing the proof. $\square$

Problem 27

Suppose $T \in \mathcal{L}(V)$ is normal. Prove that

$$\text{null } T^k = \text{null } T \text{ and } \text{range } T^k = \text{range } T$$

for every positive integer k .

Proof. It is obvious that $\text{null } T \subseteq \text{null } T^k$ and $\text{range } T^k \subseteq \text{range } T$. We aim at proving the other direction.

If T is self-adjoint, then the inclusion for the other direction of null space is simpler to prove. Suppose $v \in \text{null } T^k$, then

$$\langle T^k v, T^{k-2} v \rangle = \langle T^{k-1} v, T^{k-1} v \rangle = 0$$

which shows that $T^{k-1} v = 0$. Doing this recursively gives that $Tv = 0$.

Now let's consider the normal operator T , then

$$(T^*T)^k v = (T^*)^k T^k v = 0$$

So $v \in \text{null } (T^*T)^k$. Notice that (T^*T) is a self-adjoint operator and applying the fact proved above yields that $v \in \text{null } T^*T$. Hence we have that

$$\langle T^*T v, v \rangle = \langle Tv, Tv \rangle = 0$$

which shows that $v \in \text{null } T$.

For proving the range-related inclusion, this can relate to the null space, where we have that

$$\text{range } (T^k) = (\text{null } (T^k)^*)^\perp = (\text{null } (T^*)^k)^\perp = (\text{null } T^*)^\perp = \text{range } T$$

□

Problem 30

Suppose that $T \in \mathcal{L}(\mathbb{F}^3)$ is normal and $T(1, 1, 1) = (2, 2, 2)$. Suppose $(z_1, z_2, z_3) \in \text{null } T$. Prove that $z_1 + z_2 + z_3 = 0$.

Proof. We can see that $(1, 1, 1)$ is an eigenvector of T with 2 to be the eigenvalue. Then z_1, z_2, z_3 is orthogonal to $(1, 1, 1)$, so we have that

$$z_1 + z_2 + z_3 = 0$$

by taking the inner product.

□

Problem 32

Suppose $T: V \rightarrow W$ is a linear map. Show that under the standard identification of V with V' and the corresponding identification of W with W' , the adjoint map $T^*: W \rightarrow V$ corresponds to the dual map $T': W' \rightarrow V'$. More precisely, show that

$$T'(\varphi_w) = \varphi_{T^*w}$$

for all $w \in W$, where φ_w and φ_{T^*w} are defined previously in the book.

Proof. We first know that $T'(\varphi_w) = \varphi_w \circ T$, so

$$T'(\varphi_w)(u) = \langle Tu, w \rangle = \langle u, T^*w \rangle = \varphi_{T^*w}(u)$$

□

7B: Spectral Theorem

Lemma 21 (invertible quadratic expressions). *Suppose $T \in \mathcal{L}(V)$ is self-adjoint and $b, c \in \mathbb{R}$ are such that $b^2 < 4c$. Then*

$$T^2 + bT + cI$$

is an invertible operator.

Lemma 22 (minimal polynomial of self-adjoint operator). *Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Then the minimal polynomial of T equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some $\lambda_1, \dots, \lambda_m \in \mathbb{R}$.*

Theorem 23 (**REAL SPECTRAL THEOREM**). *Suppose $\mathbb{F} = \mathbb{R}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent.*

- (a) *T is self-adjoint.*
- (b) *T has a diagonal matrix with respect to some orthonormal basis of V .*
- (c) *V has an orthonormal basis consisting of eigenvectors of T .*

Theorem 24 (**COMPLEX SPECTRAL THEOREM**). *Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent.*

- (a) *T is normal.*
- (b) *T has a diagonal matrix with respect to some orthonormal basis of V .*
- (c) *V has an orthonormal basis consisting of eigenvectors of T .*

Problem 1

Prove that a normal operator on a complex inner product space is self-adjoint if and only if all its eigenvalues are real.

Proof. $\Rightarrow$ Assume the normal operator T is self-adjoint, then we know that $T = T^*$ and their eigenvalue and eigenvectors are the same. However, we know that (λ, v) is an eigenpair of T if and only if $(\bar{\lambda}, v)$ is an eigenpair of T^* . This means that $\lambda = \bar{\lambda}$ and thus all eigenvalues are real.

$\Leftarrow$ Conversely, since T is normal, T has a diagonal entry wrt some orthonormal basis of V where the entries are the eigenvalues. Since the eigenvalues are real, the conjugate transpose of the matrix equals itself, therefore completing the proof. $\square$

Problem 3

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$ is normal. Prove that the set of eigenvalues of T is contained in $\{0, 1\}$ if and only if there is a subspace U of V such that $T = P_U$.

Proof. Since T is normal, there exists a diagonal matrix representation of T wrt. some orthonormal basis $e_1, \dots, e_n$. Then we have that

$$\begin{aligned} \text{eigenvalues of } T \text{ contained in } \{0, 1\} &\iff T e_k = 0 \text{ or } T e_k = 1 \\ &\iff \exists U = \text{span}\{e_i\}_{i \in J} \text{ for some index set} \\ &\quad J \text{ s.t. } T v \in U \text{ for all } v \in V \end{aligned}$$

$\square$

Problem 4

Prove that a normal operator on a complex inner product space is skew (meaning it equals the negative of its adjoint) if and only if all its eigenvalues are purely imaginary (meaning that they have real part equal to 0).

Proof. Suppose T is normal, then we know that λ is an eigenvalue of T if and only if $\bar{\lambda}$ is an eigenvalue of T^* .

$$\begin{aligned} T \text{ is skew} &\iff -\langle T v, v \rangle = \langle T^* v, v \rangle = -\lambda \langle v, v \rangle \\ &\iff -\lambda = \bar{\lambda} \\ &\iff \lambda \text{ is purely imaginary} \end{aligned}$$

$\square$

Problem 6

Suppose V is a complex inner product space and $T \in \mathcal{L}(V)$ is a normal operator such that $T^9 = T^8$. Prove that T is self-adjoint and $T^2 = T$.

Proof. Let (λ, v) be an eigenpair of T . Then we know that

$$T^9 v = \lambda^9 v = T^8 v = \lambda^8 v$$

Therefore $\lambda^9 = \lambda^8$ and thus $\lambda = \{0, 1\}$, then by P1 we know T is self-adjoint and by P3 we know that T is a projection operator and thus $T^2 = T$. $\square$

Problem 8

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Prove that T is normal if and only if every eigenvector of T is also an eigenvector of T^* .

Proof. $\Rightarrow$ This follows by the theorem on the book.

$\Leftarrow$ Since we are concerned with the field $\mathbb{C}$, there exists a matrix of T that is upper-triangular wrt. some orthonormal basis $e_1, \dots, e_n$ such that $Te_j = \sum_{i=1}^j a_{ij} e_i$. In particular, e_1 is an eigenvector of T and it is also an eigenvector of T^* . So the first column of T^* only has the first term nonzero. We can repeat the statement for other eigenvectors of T to that of T^* . Note that since the field is taken over $\mathbb{C}$, T is guaranteed to have n eigenvalues and therefore the corresponding eigenvectors. This means that T and T^* are simultaneously diagonalizable and thus they commute, completing the proof. $\square$

Problem 10

Suppose V is a complex inner product space. Prove that every normal operator on V has a square root. (Note that an operator $S \in \mathcal{L}(V)$ is called a square root of $T \in \mathcal{L}(V)$ if $S^2 = T$).

Proof. Suppose T is a normal operator on V . By complex spectral theorem, $T = \text{diag}(\lambda_1, \dots, \lambda_n)$ wrt. to some orthonormal bases $e_1, \dots, e_n$. We can naturally define an operator S such that

$$Se_k = \sqrt{\lambda_k} e_k$$

Then we have that

$$S^2(v) = S\left(\sum_{i=1}^n a_i S(e_i)\right) = S\left(\sum_{i=1}^n a_i \sqrt{\lambda_i} e_i\right) = \sum_{i=1}^n a_i \lambda_i e_i = \sum_{i=1}^n a_i T(e_i) = T(v)$$

$\square$

Problem 11

Prove that every self-adjoint operator on V has a cube root.

Proof. We can do the similar proof as above. The only difference is that the self-adjoint operators have real eigenvalues and therefore their cube root is also real and uniquely determined. $\square$

Problem 12

Suppose V is a complex vector space and $T \in \mathcal{L}(V)$ is normal. Prove that if S is an operator on V that commutes with T , then S commutes with T^* .

Proof. Let v be an eigenvector of T with eigenvalue of λ . We have that

$$T(Sv) = STv = S(\lambda v) = \lambda(Sv)$$

So Sv is an eigenvector of T with eigenvalue λ . This means that it is also an eigenvector of T^* with eigenvalue $\bar{\lambda}$. Hence, we have

$$T^*(Sv) = \bar{\lambda}Sv = S(\bar{\lambda}v) = ST^*v$$

completing the proof. $\square$

Problem 14

Suppose $\mathbb{F} = \mathbb{R}$ and $T \in \mathcal{L}(V)$. Prove that T is self-adjoint if and only if all pairs of eigenvectors corresponding to distinct eigenvalues of T are orthogonal and $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$, where $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T .

Proof. $\Rightarrow$ Given an self-adjoint operator T , we know that there exists an orthonormal basis of V consisting of eigenvectors of T and thus we know that $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$. Let's say we have arbitrary eigenpairs (λ_1, v_1) and (λ_2, v_2) with $\lambda_1 \neq \lambda_2$. Then

$$0 = \langle Tv_1, v_2 \rangle - \langle v_1, Tv_2 \rangle = (\lambda_1 - \lambda_2)\langle v_1, v_2 \rangle$$

Therefore, we have $\langle v_1, v_2 \rangle = 0$.

$\Leftarrow$ Conversely, Such eigenvectors form a basis of V and thus T is self-adjoint. $\square$

Problem 16

Suppose $\mathbb{F} = \mathbb{C}$ and $\mathcal{E} \subseteq \mathcal{L}(V)$. Prove that there is an orthonormal basis of V with respect to which every element of $\mathcal{E}$ has a diagonal matrix if and only if S and T are commuting normal operators for all $S, T \in \mathcal{E}$.

Proof. $\Rightarrow$ Since $S, T \in \mathcal{E}$ are diagonal, they commute and are normal.

$\Leftarrow$ This side follows as what we've done in ch5E P2. $\square$

Problem 19

Suppose $T \in \mathcal{L}(V)$ is self-adjoint and U is a subspace of V that is invariant under T .

- (a) Prove that $U^\perp$ is invariant under T .
- (b) Prove that $T|_U \in \mathcal{L}(U)$ is self-adjoint.
- (c) Prove that $T|_{U^\perp} \in \mathcal{L}(U^\perp)$ is self-adjoint.

Proof. (a) Let $u \in U$ and $w \in U^\perp$, then

$$\langle u, Tw \rangle = \langle Tu, w \rangle = 0$$

since $Tu \in U$. Thus $Tw \in U^\perp$.

(b) Take $u_1, u_2 \in U$, then we know that $u_1, u_2 \in V$. Since T is self-adjoint, then

$$\langle Tu_1, u_2 \rangle = \langle u_1, Tu_2 \rangle$$

which shows that $T|_U$ is self-adjoint.

(c) This follows similarly. □

Problem 21

Suppose that T is a self-adjoint operator on a finite-dimensional inner product space and that 2 and 3 are the only eigenvalues of T . Prove that

$$T^2 - 5T + 6I = 0$$

Proof. We know that $T - 2I$ and $T - 3I$ equals 0. Combining them to the polynomial of T gives that

$$(T - 2I)(T - 3I) = T^2 - 5T + 6I = 0$$

note that we can do this since T is self-adjoint. □

Problem 23

Suppose $T \in \mathcal{L}(V)$ is self-adjoint, $\lambda \in \mathbb{F}$, and $\epsilon > 0$. Suppose there exists $v \in V$ such that $\|v\| = 1$ and

$$\|Tv - \lambda v\| < \epsilon$$

Prove that T has an eigenvalue λ' such that $|\lambda - \lambda'| < \epsilon$.

Proof. Since T is self-adjoint, there exists an eigenbasis $e_1, \dots, e_n$ with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. WLOG we suppose $\lambda_1 < \dots < \lambda_n$. Then we

know that any $v \in V$ can be expressed as $v = \sum_{i=1}^n a_i e_i$. We now have that

$$Tv = \sum_{i=1}^n a_i \lambda_i e_i$$

Furthermore,

$$\begin{aligned} \|Tv - \lambda v\| &= \left\| \sum_{i=1}^n a_i \lambda_i e_i - \lambda \sum_{i=1}^n a_i e_i \right\| \\ &= \left\| \sum_{i=1}^n a_i e_i (\lambda_i - \lambda) \right\| \\ &= \sum_{i=1}^n \|a_i (\lambda_i - \lambda)\| \\ &\geq \sum_{i=1}^n |a_i| |\lambda' - \lambda| \end{aligned}$$

where we define $\lambda' = \operatorname{argmin}_i |\lambda_i - \lambda|$. This gives that

$$|\lambda' - \lambda| < \epsilon / \left(\sum_{i=1}^n |a_i| \right) = \epsilon / 1 = \epsilon$$

where in the last step we use the fact that $\|v\| = 1$. □

7C: Positive Operators

Definition 25 (positive operator). *An operator $T \in \mathcal{L}(V)$ is called **positive** if T is self-adjoint and*

$$\langle Tv, v \rangle \geq 0$$

for all $v \in V$.

Remark 26. *Positive operators are also known as positive definite operators.*

Definition 27 (square root). *An operator R is called a **squared root** of an operator T if $R^2 = T$.*

Theorem 28 (characterizations of positive operators). *Let $T \in \mathcal{L}(V)$. Then the following are equivalent.*

- (a) *T is a positive operator.*
- (b) *T is self-adjoint and all eigenvalues of T are nonnegative.*
- (c) *With respect to some orthonormal basis of V , the matrix of T is a diagonal matrix with only nonnegative numbers on the diagonal.*
- (d) *T has a positive square root.*
- (e) *T has a self-adjoint square root.*
- (f) *$T = R^*R$ for some $R \in \mathcal{L}(V)$.*

Theorem 29. *Every positive operator on V has a unique positive square root.*

Remark 30. *For T a positive operator, $\sqrt{T}$ denotes the unique positive square root of T .*

Corollary 31. *Suppose T is a positive operator on V and $v \in V$ is such that $\langle Tv, v \rangle = 0$. Then $Tv = 0$.*

Problem 1

Suppose $T \in \mathcal{L}(V)$. Prove that if both T and $-T$ are positive operators, then $T = 0$.

Proof. This means that

$$\langle Tv, v \rangle \geq 0 \quad \langle Tv, v \rangle \leq 0$$

for all $v \in V$, so $T = 0$. □

Problem 3

Suppose n is a positive integer and $T \in \mathcal{L}(\mathbb{F}^n)$ is the operator whose matrix (wrt. the standard basis) consists of all 1's. Show that T is a positive operator.

Proof. Suppose $v = (v_1, \dots, v_n) \in V$, then we have that

$$\langle Tv, v \rangle = \left\langle \left(\sum_{i=1}^n v_i, \dots, \sum_{i=1}^n v_i \right), (v_1, \dots, v_n) \right\rangle = \left(\sum_{i=1}^n v_i \right)^2 \geq 0$$

□

Problem 6

Prove that the sum of two positive operators on V is a positive operator.

Proof. Let S, T be two positive operator and take $v \in V$. Then

$$\langle (S + T)v, v \rangle = \langle Sv, v \rangle + \langle Tv, v \rangle \geq 0$$

□

Problem 7

Suppose $S \in \mathcal{L}(V)$ is an invertible positive operator and $T \in \mathcal{L}(V)$ is a positive operator. Prove that $S + T$ is invertible.

Proof. Take any nonzero $v \in V$, then

$$\langle (S + T)v, v \rangle = \langle Sv, v \rangle + \langle Tv, v \rangle > 0$$

So $\text{null}(S + T) = \{0\}$ and thus $S + T$ is injective and therefore invertible. □

Problem 9

Suppose $T \in \mathcal{L}(V)$ is a positive operator and $S \in \mathcal{L}(W, V)$. Prove that S^*TS is a positive operator on W .

Proof. We have that

$$\langle S^*TSw, w \rangle = \langle T(Sw), (Sw) \rangle \geq 0$$

as T is positive. □

Problem 10

Suppose T is a positive operator on V . Suppose $v, w \in V$ are such that

$$Tv = w \quad Tw = v$$

Prove that $v = w$.

Proof. We have that

$$T^2v = T(Tv) = Tw = v$$

Thus $(T^2 - I)v = 0$. This means that either $v = 0$ or $T = \pm I$. In the first case, $w = Tv = 0 = v$ and $Tw = 0 = v$. In the second case, if $T = I$, then $v = w$. Note that T cannot be $-I$ as T is a positive operator (all eigenvalues nonnegative). □

Problem 12

Suppose $T \in \mathcal{L}(V)$ is a positive operator. Prove that T^k is a positive operator for every positive integer k .

Proof. If k is even, then

$$\langle T^k v, v \rangle = \langle T^{\frac{k}{2}} v, T^{\frac{k}{2}} v \rangle \geq 0$$

If k is odd, then

$$\langle T^k v, v \rangle = \langle T(T^{\frac{k-1}{2}} v), (T^{\frac{k-1}{2}} v) \rangle \geq 0$$

as T is positive. □

Problem 13

Suppose $T \in \mathcal{L}(V)$ is self-adjoint and $\alpha \in \mathbb{R}$.

- (a) Prove that $T - \alpha I$ is a positive operator if and only if α is less than or equal to every eigenvalue of T .
- (b) Prove that $\alpha I - T$ is a positive operator if and only if α is greater than or equal to every eigenvalue of T .

Proof. Let $v_1, \dots, v_n$ be the eigenbasis of V with corresponding sorted eigenvalues $\lambda_1, \dots, \lambda_n$ (λ_1 being smallest). Note that λ is an eigenvalue of T iff $T - \alpha I$ is an eigenvalue of $T - \alpha I$ iff $\alpha - \lambda$ is an eigenvalue of $\alpha I - T$. You may verify this.

(a) $T - \alpha I$ positive $\iff \lambda_1 - \alpha \geq 0 \iff \alpha \leq \lambda_i$ for all i . (b) $\alpha I - T$ positive $\iff \alpha - \lambda_n \geq 0 \iff \alpha \geq \lambda_i$ for all i . □

Problem 14

Suppose T is a positive operator on V and $v_1, \dots, v_m \in V$. Prove that

$$\sum_{j=1}^m \sum_{k=1}^m \langle T v_k, v_j \rangle \geq 0$$

Proof.

$$\sum_{j=1}^m \sum_{k=1}^m \langle T v_k, v_j \rangle = \left\langle T \left(\sum_{i=1}^m v_i \right), \left(\sum_{i=1}^m v_i \right) \right\rangle \geq 0$$

□

Problem 15

Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Prove that there exist positive operators $A, B \in \mathcal{L}(V)$ such that

$$T = A - B \text{ and } \sqrt{T^*T} = A + B \text{ and } AB = BA = 0$$

Proof. Define $|T| = \sqrt{T^2}$ and that

$$A = \frac{|T| + T}{2} \quad B = \frac{|T| - T}{2}$$

So $A - B = T$ and $A + B = |T| = \sqrt{T^*T}$. Finally, we have

$$AB = \frac{|T| + T}{2} \frac{|T| - T}{2} = \frac{|T|^2 - |T|T + T|T| - T^2}{4} = 0$$

As $|T|^2 = T^2$ and $|T|T = T|T|$. Similarly, $BA = 0$.

□

Problem 16

Suppose T is a positive operator on V . Prove that

$$\text{null } \sqrt{T} = \text{null } T \text{ and } \text{range } \sqrt{T} = \text{range } T$$

Proof. We've shown that the unique square root $\sqrt{T}$ is obtained from simply taking the square root of the eigenvalues of T . This means that $\sqrt{T}$ and T share the same eigenbasis, which determines the respective null and range space and completes the proof. □

Problem 18

Suppose S and T are positive operators on V . Prove that ST is a positive operator if and only if S and T commute.

Proof. $\Rightarrow$ Given ST is positive, then

$$ST = (ST)^* = T^*S^* = TS$$

$\Leftarrow$ Given $ST = TS$, then $(ST)^* = T^*S^* = TS = ST$. In addition,

$$\langle STv, v \rangle = \langle S\sqrt{T}v, \sqrt{T}v \rangle \geq 0$$

□

Problem 22

Suppose $T \in \mathcal{L}(V)$ is a positive operator and $u \in V$ is such that $\|u\| = 1$ and $\|Tu\| \geq \|Tv\|$ for all $v \in V$ with $\|v\| = 1$. Show that u is an eigenvector of T corresponding to the largest eigenvalue of T .

Proof. Let $e_1, \dots, e_n$ be the orthogonal eigenbasis of V and the corresponding eigenvalues to be $\lambda_1, \dots, \lambda_n$ with sorted from smallest to largest. Then we have that

$$\|Tu\| = \left\| \sum_{i=1}^n a_i \lambda_i e_i \right\| = \sqrt{\sum_{i=1}^n |a_i|^2 \lambda_i^2}$$

We can take $v = e_n$, then we have that

$$\|Tu\|^2 = \sum_{i=1}^n |a_i|^2 \lambda_i^2 \geq \sum_{i=1}^n |a_i|^2 \lambda_n^2 = \lambda_n^2 = \|Tv\|^2$$

which shows the desired conclusion.

□

7D: Isometries, Unitary Operators, and Matrix Factorization

Definition 32 (isometry). A linear map $S \in \mathcal{L}(V, W)$ is called an **isometry** if

$$\|Sv\| = \|v\|$$

for every $v \in V$. In other words, a linear map is an isometry if it preserves norms.

Remark 33. Every isometry is injective.

Theorem 34 (characterizations of isometries). Suppose $S \in \mathcal{L}(V, W)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $f_1, \dots, f_m$ is an orthonormal basis of W . Then the following are equivalent.

- (a) S is an isometry.
- (b) $S^*S = I$.
- (c) $\langle Su, Sv \rangle = \langle u, v \rangle$ for all $u, v \in V$.
- (d) $Se_1, \dots, Se_n$ is an orthonormal list in W .
- (e) The columns of $\mathcal{M}(S, (e_1, \dots, e_n), (f_1, \dots, f_m))$ form an orthonormal list in $\mathbb{F}^m$ with respect to the Euclidean inner product.

Definition 35 (Unitary Operators). An operator $S \in \mathcal{L}(V)$ is called **unitary** if S is an invertible isometry.

Remark 36. Note that every isometry is injective and therefore invertible on a finite-dimensional vector space. The author makes the distinction simply to avoid confusion in more abstract stages. One might think of unitary operators being equivalent to isometries in finite-dimensional space.

Theorem 37 (characterizations of unitary operators). Suppose $S \in \mathcal{L}(V)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of V . Then the following are equivalent.

- (a) S is a unitary operator.
- (b) $S^*S = SS^* = I$.
- (c) S is invertible and $S^{-1} = S^*$.
- (d) $Se_1, \dots, Se_n$ is an orthonormal list in W .
- (e) The rows of $\mathcal{M}(S, (e_1, \dots, e_n))$ form an orthonormal basis of $\mathbb{F}^m$ with respect to the Euclidean inner product.
- (f) S^* is a unitary operator.

Corollary 38. Suppose λ is an eigenvalue of a unitary operator. Then $|\lambda| = 1$.

Corollary 39. Suppose $\mathbb{F} = \mathbb{C}$ and $S \in \mathcal{L}(V)$. Then the following are equivalent.

- (a) S is a unitary operator.
- (b) There is an orthonormal basis of V consisting of eigenvectors of S whose corresponding eigenvalues all have absolute value 1.

Definition 40 (unitary matrix). An n -by- n matrix is called **unitary** if its columns form an orthonormal list in $\mathbb{F}^n$.

Theorem 41 (characterizations of unitary matrices). Suppose Q is an n -by- n matrix. Then the following are equivalent.

- (a) Q is a unitary matrix.
- (b) The rows of Q form an orthonormal list in $\mathbb{F}^n$.
- (c) $\|Qv\| = \|v\|$ for every $v \in \mathbb{F}^n$.
- (d) $Q^*Q = QQ^* = I_n$.

Theorem 42 (QR factorization). Suppose A is a square matrix with linearly independent columns. Then there exists unique matrices Q and R such that Q is unitary, R is upper triangular with only positive numbers on its diagonal, and

$$A = QR.$$

Lemma 43 (positive invertible operator). A self-adjoint operator $T \in \mathcal{L}(V)$ is a positive invertible operator if and only if $\langle Tv, v \rangle > 0$ for every nonzero $v \in V$.

Definition 44 (positive definite). A matrix $B \in \mathbb{F}^{n,n}$ is called **positive definite** if $B^* = B$ and

$$\langle Bx, x \rangle > 0$$

for every nonzero $x \in \mathbb{F}^n$.

Theorem 45 (Cholesky factorization). Suppose B is a positive definite matrix. Then there exists a unique upper-triangular matrix R with only positive numbers on its diagonal such that

$$B = R^*R$$

Problem 1

Suppose $\dim V \geq 2$ and $S \in \mathcal{L}(V, W)$. Prove that S is an isometry if and only if Se_1, Se_2 is an orthonormal list in W for every orthonormal list e_1, e_2 of length two in V .

Proof. $\Rightarrow$ Let $v \in V$, then

$$\|v\|^2 = \|\langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2\|^2 = |\langle v, e_1 \rangle|^2 + |\langle v, e_2 \rangle|^2 = \|Sv\|^2$$

$\Leftarrow$ we know that

$$\|Sv\|^2 = \|\langle v, e_1 \rangle Se_1 + \langle v, e_2 \rangle Se_2\|^2 = |\langle v, e_1 \rangle|^2 + |\langle v, e_2 \rangle|^2 = \|v\|^2$$

as Se_1, Se_2 are orthonormal basis with norm 1. $\square$

Problem 2

Suppose $T \in \mathcal{L}(V, W)$. Prove that T is a scalar multiple of an isometry if and only if T preserves orthogonality.

Proof. $\Rightarrow$ if $T = aS$ for S to be an isometry, then suppose $\langle u, v \rangle = 0$,

$$\langle Tu, Tv \rangle = |a|^2 \langle Su, Sv \rangle = |a|^2 \langle u, v \rangle = 0$$

$\Leftarrow$ Suppose for all $\langle u, v \rangle = 0$, $\langle Tu, Tv \rangle = 0$. Take an orthonormal basis $\{e_1, \dots, e_n\}$ of V . Then we have that

$$\langle e_i + e_j, e_i - e_j \rangle = \|e_i\|^2 - \langle e_i, e_j \rangle + \langle e_j, e_i \rangle - \|e_j\|^2 = 1 - 1 = 0$$

This means that

$$\langle T(e_i + e_j), T(e_i - e_j) \rangle = \|Te_i\|^2 - \|Te_j\|^2 = 0$$

which means that $\|Te_k\| = a$ for some scalar a . Take any $v \in V$, then

$$\|Tv\|^2 = \sum_{i=1}^n |\langle v, e_i \rangle|^2 \|Te_i\|^2 + \sum_{i \neq j} |\langle v, e_i \rangle \langle v, e_j \rangle| \langle Te_i, Te_j \rangle = \sum_{i=1}^n |\langle v, e_i \rangle|^2 |a|^2 = \|av\|^2$$

This means that

$$\|Tv\| = a\|v\|$$

for some a and thus completes the proof ($\frac{1}{a}T$ is an isometry). $\square$

Problem 3

- (a) Show that the product of two unitary operators on V is a unitary operator.
- (b) Show that the inverse of a unitary operator on V is a unitary operator.

Proof. Let T, S be two unitary operators.

(a) we have

$$(TS)^*(TS) = S^*T^*TS = S^*IS = S^*S = I$$

(b) since $S^{-1} = S^*$ and S^* is a unitary operator, S^{-1} is also unitary. $\square$

Problem 5

Suppose $S \in \mathcal{L}(V)$. Prove that the following are equivalent.

- (a) S is a self-adjoint unitary operator.
- (b) $S = 2P - I$ for some orthogonal projection P on V .
- (c) There exists a subspace U of V such that $Su = u$ for every $u \in U$ and $Sw = -w$ for every $w \in U^\perp$.

Proof. (a) $\Rightarrow$ (b), (c) Since S is unitary, its eigenvalues are either -1 or 1. Since S is self-adjoint, we can decompose V into its eigenspace $E(-1)$ and $E(1)$ such that $V = E(-1) \oplus E(1)$. This means that for all $w \in E(-1)$, $Sw = -w$ and $u \in E(1)$, $Su = u$ where $E(1) = U$. So part (c) is completed. For each $v \in V$, $v = v_1 + v_{-1}$ where $v_1 \in E(1)$, $v_{-1} \in E(-1)$. Then define $P(v) = v_1$ and we have that $Sv = Sv_1 + Sv_{-1} = v_1 - v_{-1} = P(v) - (I - P)(v) = (2P - I)v$. Part(b) is also completed.

(b), (c) $\Rightarrow$ (a) Conversely, first assume (b) holds, then this means that

$$S^* = (2P - I)^* = 2P - I = S$$

so S is self-adjoint. To see that it is unitary,

$$SS^* = S^2 = (2P - I)(2P - I) = 4P^2 - 2P - 2P + I = I$$

Next assume (c) holds, then Let P define to be the projection operator into U and then $(I - P)$ is the projection into $U^\perp$, then we have that $S = P - (I - P) = 2P - I$, so applying the previous proof finishes the problem. $\square$

Problem 6

Suppose T_1, T_2 are both normal operators on $\mathbb{F}^3$ with 2,5,7 as eigenvalues. Prove that there exists a unitary operator $S \in \mathcal{L}(\mathbb{F}^3)$ such that $T_1 = S^*T_2S$.

Proof. By complex spectral theorem, T_1 has $\{a_1, a_2, a_3\}$ as a set of orthonormal eigenvectors that form a basis with eigenvalues 2,5,7. Similarly, T_2 has $\{b_1, b_2, b_3\}$. What we want to do here is simply define a change-of-basis operator that is also unitary. Define $S \in \mathcal{L}(\mathbb{F}^3)$ s.t.

$$Sa_i = b_i$$

Clearly, S is unitary. Now, it suffices to prove that the desired equation holds. We also have that $S^{-1} = S^*$ so $S^*b_i = a_i$. Take any $v = \alpha_1a_1 + \alpha_2a_2 + \alpha_3a_3$, then

$$\begin{aligned}
S^*T_2S(v) &= S^*T_2S\left(\sum_{i=1}^3 \alpha_i a_i\right) \\
&= S^*T_2\left(\sum_{i=1}^3 \alpha_i S(a_i)\right) \\
&= S^*\left(\sum_{i=1}^3 \alpha_i T_2(b_i)\right) \\
&= S^*(\alpha_1 2b_1 + \alpha_2 5b_2 + \alpha_3 7b_3) \\
&= 2\alpha_1 a_1 + 5\alpha_2 a_2 + 7\alpha_3 a_3
\end{aligned}$$

Note that we also have that

$$\begin{aligned}
T_1(v) &= T_1\left(\sum_{i=1}^3 \alpha_i a_i\right) \\
&= 2\alpha_1 a_1 + 5\alpha_2 a_2 + 7\alpha_3 a_3
\end{aligned}$$

□

Problem 9

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Suppose every eigenvalue of T has absolute value 1 and $\|Tv\| \leq \|v\|$ for every $v \in V$. Prove that T is a unitary operator.

Proof. Since T has no eigenvalue 0, it is invertible. We also have that

$$\begin{aligned}
\|Tv\| \leq \|v\| &\iff \langle Tv, Tv \rangle \leq \langle v, v \rangle \\
&\iff \langle T^*Tv, v \rangle \leq \langle v, v \rangle \\
&\iff \langle (T^*T - I)v, v \rangle \leq 0
\end{aligned}$$

which gives that $T^*T \leq I$. Since the eigenvalue of T^*T are all one and it is self-adjoint, there exists eigenbasis all with eigenvalue one. This implies that $T^*T = I$ and thus T is unitary. □

Problem 10

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$ is a self-adjoint operator such that $\|Tv\| \leq \|v\|$ for all $v \in V$.

- (a) Show that $I - T^2$ is a positive operator.
- (b) Show that $T + i\sqrt{I - T^2}$ is a unitary operator.

Proof. (a) take any $v \in V$, then

$$\begin{aligned}\langle (I - T^2)v, v \rangle &= \langle v, v \rangle - \langle Tv, T^*v \rangle \\ &= \langle v, v \rangle - \langle Tv, Tv \rangle \\ &= \langle (I - T)v, (I - T)v \rangle \geq 0\end{aligned}$$

- (b) Let $A = T, B = \sqrt{I - T^2}$. Then we have that

$$(A + Bi)(A + Bi)^* = A^2 - iAB + iBA + B^2$$

We know that A is self-adjoint so A^2 is as well. By (a), B^2 is also self-adjoint. Note that $AB = BA$ algebraically. Thus, we have that

$$(A + Bi)(A + Bi)^* = A^2 + B^2 = I$$

which shows that $A + Bi$ is unitary. $\square$

Problem 11

Suppose $S \in \mathcal{L}(V)$. Prove that S is a unitary operator if and only if

$$\{Sv : v \in V \text{ and } \|v\| \leq 1\} = \{v \in V : \|v\| \leq 1\}$$

Proof. $\Rightarrow$ take any $v \in r.h.s.$, then since S is invertible and thus surjective, there exists $u \in V$ such that $Sv = u$. As $\|v\| \leq 1$ and $\|Su\| = \|u\| = \|v\| \leq 1$, so $v \in l.h.s.$. Conversely, take any $Sv \in l.h.s.$, then $\|Sv\| = \|v\| \leq 1$, so $Sv \in r.h.s.$. Hence we complete this direction.

$\Leftarrow$ Take any $v \in V$ and consider $u := \frac{v}{\|v\|}$. Then we know that there exists w with $\|w\| \leq 1$ such that $Sw = u$. $\square$

Problem 13

Explain why the columns of a square matrix of complex numbers form an orthonormal list in $\mathbb{C}^n$ if and only if the rows of the matrix form an orthonormal list in $\mathbb{C}^n$.

Proof. A square matrix Q is unitary iff Q^* is unitary, completing the proof. $\square$

Problem 18

A square matrix A is called *symmetric* if it equals its transpose. Prove that if A is a symmetric matrix with real entries, then there exists a unitary matrix Q with real entries such that Q^*AQ is a diagonal matrix.

Proof. So we know that A is self-adjoint, so it is orthogonally diagonalizable. There exists eigenbasis $v_1, \dots, v_n$ that are orthogonal. We can thus define Q to be consisting of such eigenvectors as

$$Av_i = \lambda_i v_i$$

Writing this in matrix form gives that

$$AQ = QD$$

for unitary matrix Q and diagonal matrix D with entries to be the corresponding eigenvalues. Hence, we get that

$$Q^*AQ$$

is diagonal. □

7E: Singular Value Decomposition

Lemma 46 (properties of T^*T). Suppose $T \in \mathcal{L}(V, W)$. Then

- (a) T^*T is a positive operator on V ;
- (b) $\text{null } T^*T = \text{null } T$;
- (c) $\text{range } T^*T = \text{range } T^*$;
- (d) $\dim \text{range } T = \dim \text{range } T^* = \dim \text{range } T^*T$.

Definition 47 (singular values). Suppose $T \in \mathcal{L}(V, W)$. The **singular values** of T are the nonnegative square root of the eigenvalues of T^*T , listed in decreasing order, each included as many times as the dimension of the corresponding eigenspace of T^*T .

Theorem 48 (role of positive singular values). Suppose that $T \in \mathcal{L}(V, W)$. Then

- (a) T is injective $\iff 0$ is not a singular value of T ;
- (b) the number of positive singular values of T equals $\dim \text{range } T$;
- (c) T is surjective $\iff$ number of positive singular values of T equals $\dim W$.

Corollary 49. Suppose $S \in \mathcal{L}(V, W)$. Then

$$S \text{ is an isometry} \iff \text{all singular values of } S \text{ equal } 1.$$

Theorem 50 (**SINGULAR VALUE DECOMPOSITION**). Suppose $T \in \mathcal{L}(V, W)$ and the positive singular values of T are $s_1, \dots, s_m$. Then there exist orthonormal lists $e_1, \dots, e_m$ in V and $f_1, \dots, f_m$ in W such that

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \dots + s_m \langle v, e_m \rangle f_m$$

for every $v \in V$.

Remark 51. In the proof, we first let $s_1, \dots, s_n$ to denote the singular values of T (thus $n = \dim V$). Then the spectral theorem implies that there exists an orthonormal list $e_1, \dots, e_n$ of V with

$$T^*Te_k = s_k^2 e_k$$

For each $k = 1, \dots, m$, define

$$f_k = \frac{Te_k}{s_k}$$

Remark 52. Suppose $T \in \mathcal{L}(V, W)$, the positive singular values of T are $s_1, \dots, s_m$ and $e_1, \dots, e_m$ and $f_1, \dots, f_m$ are as in the singular decomposition above. Then the two orthonormal lists can both be extended to basis of the respective vector space. Where we can now define

$$Te_k = \begin{cases} s_k f_k & \text{if } 1 \leq k \leq m, \\ 0 & \text{if } m < k \leq \dim V \end{cases}$$

Definition 53 (diagonal matrix). An M -by- N matrix A is called a **diagonal matrix** if all entries of the matrix are 0 except possibly $A_{k,k}$ for $k = 1, \dots, \min\{M, N\}$.

Theorem 54 (singular value decomposition of adjoint and pseudoinverse). Suppose $T \in \mathcal{L}(V, W)$ and the positive singular values of T are $s_1, \dots, s_m$. Suppose $e_1, \dots, e_m$ and $f_1, \dots, f_m$ are orthonormal list in V and W such that

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \dots + s_m \langle v, e_m \rangle f_m$$

for every $v \in V$. Then

$$T^*w = s_1 \langle w, f_1 \rangle e_1 + s_m \langle w, f_m \rangle e_m$$

and

$$T^\dagger w = \frac{\langle w, f_1 \rangle}{s_1} e_1 + \dots + \frac{\langle w, f_m \rangle}{s_m} e_m$$

for every $w \in W$.

Theorem 55 (matrix version of SVD). Suppose A is a p -by- n matrix of rank $m \geq 1$. Then there exist a p -by- m matrix B with orthonormal columns, an m -by- m diagonal matrix D with positive numbers on the diagonal, and an n -by- m matrix C with orthonormal columns such that

$$A = BDC^*$$

Remark 56. A is a $p \times n$ matrix while BDC^* has a total of $m(p + m + n)$ entries, which could be considerably smaller than A .

Problem 1

Suppose $T \in \mathcal{L}(V, W)$. Show that $T = 0$ if and only if all singular values of T are 0.

Proof. $\Rightarrow$ if $T = 0$, then $T^*T = 0$ and the singular values have to be all 0.

$\Leftarrow$ if all singular values are 0, then $\dim \text{range } T = 0$ and thus $T = 0$. $\square$

Problem 2

Suppose $T \in \mathcal{L}(V, W)$ and $s > 0$. Prove that s is a singular value of T if and only if there exists nonzero vectors $v \in V$ and $w \in W$ such that

$$Tv = sw \text{ and } T^*w = sv$$

Proof. $\Rightarrow$ This means that there exists nonzero eigenvector $v \in V$ such that

$$T^*Tv = s^2v$$

let $w = \frac{Tv}{s}$, then

$$Tv = s \left(\frac{Tv}{s} \right) = sw$$

and

$$T^*w = \frac{T^*Tv}{s} = sv$$

$\Leftarrow$ we have that

$$T^*Tv = T^*(sw) = s^2v$$

Therefore, completing the proof. $\square$

Problem 4

Suppose that $T \in \mathcal{L}(V, W)$, s_1 is the largest singular value of T , and s_n is the smallest value of T . Prove that

$$\{\|Tv\| : v \in V \text{ and } \|v\| = 1\} = [s_n, s_1]$$

Proof. Our proof goes two-fold. First we prove that for any $v \in V$ s.t. $\|v\| = 1$, we have

$$s_1 \leq \|Tv\| \leq s_n$$

Second, we will show that the $A := \{\|Tv\| : v \in V \text{ and } \|v\| = 1\}$ is a closed interval.

To prove the first part, let v be a unit norm vector in V , we know that $v = \sum_{i=1}^n a_i e_i$ and $\|v\|^2 = \sum_{i=1}^n |a_i|^2$ for some orthonormal basis $e_1, \dots, e_n$.

Then we have

$$\begin{aligned}
\|Tv\|^2 &= \langle Tv, Tv \rangle \\
&= \langle v, T^*Tv \rangle \\
&= \left\langle \sum_{i=1}^n a_i e_i, \sum_{i=1}^n s_i^2 a_i e_i \right\rangle \\
&= \sum_{i=1}^n a_i \sum_{j=1}^n \overline{a_j} \overline{s_j^2} \langle e_i, e_j \rangle \\
&= \sum_{i=1}^n |a_i|^2 s_i^2
\end{aligned}$$

Note that the last identity is $\leq \|v\|^2 s_1^2$ and $\geq \|v\|^2 s_n^2$, which gets our desired equality. For the second part, we know that the norm function $\|\cdot\| : v \rightarrow Tv$ is continuous on the unit sphere $\{v \in V : \|v\| = 1\}$, which is compact. The continuous image of the compact set is therefore also compact. Hence, we prove that

$$A = [s_n, s_1]$$

□

Problem 9

Suppose $T \in \mathcal{L}(V, W)$. Show that T and T^* have the same positive singular values.

Proof. We know that T^*T is self-adjoint, and that its eigenvalue equals the eigenvalue of its adjoint. Therefore the squared root of the eigenvalue of $T^*T = TT^*$, and thus T and T^* have the same positive singular values.

Problem 10 follows similarly.

□

Problem 11

Suppose that $T \in \mathcal{L}(V, W)$ and $v_1, \dots, v_n$ is an orthonormal basis of V . Let $s_1, \dots, s_n$ denote the singular values of T .

- (a) Prove that $\|Tv_1\|^2 + \dots + \|Tv_n\|^2 = s_1^2 + \dots + s_n^2$.
- (b) Prove that if $W = V$ and T is a positive operator, then

$$\langle Tv_1, v_1 \rangle + \dots + \langle Tv_n, v_n \rangle = s_1 + \dots + s_n$$

Proof. We first know from the previous problems that for any two orthonormal basis, $e_1, \dots, e_n$ in V and $f_1, \dots, f_m$ in W , then

$$\sum_{i=1}^n \|Te_i\|^2 = \sum_{j=1}^m \|T^*f_j\|^2$$

(a) In the proof of SVD, we have shown that by letting $f_k = \frac{Te_k}{s_k}$, then we have $f_1, \dots, f_n$ also to be an orthonormal list in W . Then applying the fact above yields that

$$\sum_{i=1}^n \|Tv_i\|^2 = \sum_{i=1}^n \|T^*f_i\|^2 = \sum_{i=1}^n \left\| \frac{T^*Te_i}{s_i} \right\|^2 = \sum_{i=1}^n \|s_i e_i\|^2 = \sum_{i=1}^n s_i^2$$

where $e_1, \dots, e_n$ are the orthonormal eigenbasis that diagonalizes T^*T .

(b) Note that since T is positive, the singular value is eigenvalue, as $\lambda(T^*T) = \lambda(T^2)$ and the eigenvalue is positive. Therefore, for each i , $Tv_i = s_i v_i$ and we have that

$$\sum_{i=1}^n \langle Tv_i, v_i \rangle = \sum_{i=1}^n s_i$$

□

Problem 13

Suppose $T_1, T_2 \in \mathcal{L}(V)$. Prove that T_1 and T_2 have the same singular values if and only if there exist unitary operators $S_1, S_2 \in \mathcal{L}(V)$ such that $T_1 = S_1 T_2 S_2$.

Proof. $\Rightarrow$ Let A be the matrix of T_2 and B be the matrix of T_1 then we know that $A = U_1 D V_1^*$ and $B = U_2 D V_2^*$ for unitary matrix U_1, V_1, U_2, V_2 and diagonal matrix D . Then we have

$$B = \underbrace{(U_1 U_2^*)}_{S_1} \underbrace{(U_2 D V_2^*)}_A \underbrace{(V_2 V_1^*)}_{S_2}$$

where we have S_1, S_2 to be unitary.

$\Leftarrow$ Let A be the matrix of T_2 and let $A = U D^*$ be the SVD of A where U and V^* is unitary, then we have that (let B be the matrix of T_1)

$$B = (S_1 U) D (V^* S_2)$$

where $S_1 U$ and $V^* S_2$ are both unitary, therefore B (T_2) also have the singular values as in the diagonal of D . Therefore, T_1 and T_2 shares same eigenvalues. □

Problem 15

Suppose $T \in \mathcal{L}(V)$ and $s_1 \geq \dots \geq s_n$ are the singular values of T . Prove that if λ is an eigenvalue of T , then $s_1 \geq |\lambda| \geq s_n$.

Proof. In the proof of Problem 4, we have shown that for every $v \in V$, we have

$$s_n \|v\| \leq \|Tv\| \leq s_1 \|v\|$$

Since this holds for all vectors, substitute any eigenvector v we can get that

$$s_n \|v\| \leq |\lambda| \|v\| \leq s_1 \|v\|$$

which is the desired result. □

7F: Consequences of Singular Value Decomposition

Theorem 57 (upper bound for $\|Tv\|$). Suppose $T \in \mathcal{L}(V, W)$. Let s_1 be the largest singular value of T . Then

$$\|Tv\| \leq s_1 \|v\|$$

for all $v \in V$.

Definition 58 (norm of a linear map, $\|\cdot\|$). Suppose $T \in \mathcal{L}(V, W)$. Then the **norm** of T , denoted by $\|T\|$, is defined by

$$\|T\| = \max\{\|Tv\| : v \in V \text{ and } \|v\| \leq 1\}$$

Remark 59. For a linear map T , $\|T\| = \sigma_{\max}$ (using the more common notation). Also note that $\|T\| \neq \sqrt{\langle T, T \rangle}$. Now we have two different uses of the word **norm** and the notation $\|\cdot\|$.

Corollary 60 (basic properties of norms of linear maps). Suppose $T \in \mathcal{L}(V, W)$. Then

- (a) $\|T\| \geq 0$;
- (b) $\|T\| = 0 \iff T = 0$;
- (c) $\|\lambda T\| = |\lambda| \|T\|$ for all $\lambda \in \mathbb{F}$;
- (d) $\|S + T\| \leq \|S\| + \|T\|$ for all $S \in \mathcal{L}(V, W)$.

Remark 61. For $S, T \in \mathcal{L}(V, W)$, the quantity $\|S - T\|$ is often called the distance between S and T . Informally, think of the condition that $\|S - T\|$ is a small number as meaning S and T are close together.

Theorem 62 (alternative formulas for $\|T\|$). Suppose $T \in \mathcal{L}(V, W)$. Then

- (a) $\|T\|$ = the largest eigenvalue of T ;
- (b) $\|T\| = \max\{\|Tv\| : v \in V \text{ and } \|v\| = 1\}$;
- (c) $\|T\|$ = the smallest number c such that $\|Tv\| \leq c\|v\|$ for all $v \in V$.

Remark 63. An important inequality during the proof:

$$\|Tv\| \leq \|T\| \|v\|$$

for all $v \in V$ and $v \neq 0$.

Corollary 64 (norm of the adjoint). Suppose $T \in \mathcal{L}(V, W)$. Then $\|T\| = \|T^*\|$.

Theorem 65 (best approximation by linear map whose range has dimension $\leq k$). Suppose $T \in \mathcal{L}(V, W)$ and $s_1 \geq \dots \geq s_m$ are the singular values of T . Suppose $1 \leq k < m$. Then

$$\min\{\|T - S\| : S \in \mathcal{L}(V, W) \text{ and } \dim \text{range } S \leq k\} = s_{k+1}$$

Furthermore, if

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \dots + s_m \langle v, e_m \rangle f_m$$

is a singular value decomposition of T and $T_k \in \mathcal{L}(V, W)$ is defined by

$$T_k v = s_1 \langle v, e_1 \rangle f_1 + \dots + s_k \langle v, e_k \rangle f_k$$

for each $v \in V$, then $\dim \text{range } T_k = k$ and $\|T - T_k\| = s_{k+1}$.

Theorem 66 (polar decomposition). Suppose $T \in \mathcal{L}(V)$. Then there exists a unitary operator $S \in \mathcal{L}(V)$ such that

$$T = S\sqrt{T^*T}$$

Remark 67. This holds for both $\mathbb{C}$ and $\mathbb{R}$.

Definition 68 (ball, B). The **ball** in V of radius 1 centered at 0, is defined by

$$B = \{v \in V : \|v\| \leq 1\}$$

Definition 69 (ellipsoid, $E(s_1 f_1, \dots, s_n f_n)$, principle axes). Suppose that $f_1, \dots, f_n$ is an orthonormal basis of V and $s_1, \dots, s_n$ are positive numbers. The **ellipsoid** $E(s_1 f_1, \dots, s_n f_n)$ with **principle axes** $s_1 f_1, \dots, s_n f_n$ is defined by

$$E(s_1 f_1, \dots, s_n f_n) = \{v \in V : \frac{|\langle v, f_1 \rangle|^2}{s_1} + \dots + \frac{|\langle v, f_n \rangle|^2}{s_n} < 1\}$$

Remark 70. If $\dim V = 2$, the word “disk” is sometimes used to denote ball and the word “ellipse” is sometimes used to denote ellipsoid.

Definition 71 ($T(\Omega)$). For a function T defined on V and $\Omega \subseteq V$, define $T(\Omega)$ by

$$T(\Omega) = \{Tv : v \in \Omega\}$$

Proposition 72 (invertible map takes ball to ellipsoid). Suppose $T \in \mathcal{L}(V)$ is invertible. Then T maps the ball B in V onto an ellipsoid in V .

Proposition 73 (invertible map takes ellipsoid to ellipsoid). Suppose $T \in \mathcal{L}(V)$ is invertible and E is an ellipsoid in V . Then $T(E)$ is an ellipsoid in V .

Definition 74 ($P(v_1, \dots, v_n)$, parallelepiped). Suppose $v_1, \dots, v_n$ is a basis of V . Let

$$P(v_1, \dots, v_n) = \{a_1 v_1 + \dots + a_n v_n : a_1, \dots, a_n \in (0, 1)\}$$

A **parallelepiped** is a set of the form $u + P(v_1, \dots, v_n)$ for some $u \in V$. The vectors $v_1, \dots, v_n$ are called the **edges** of the parallelepiped.

Proposition 75 (invertible operator takes parallelepiped to parallelepiped).
Suppose $u \in V$ and $v_1, \dots, v_n$ is a basis of V . Suppose $T \in \mathcal{L}(V)$ is invertible. Then

$$T(u + P(v_1, \dots, v_n)) = Tu + P(Tv_1, \dots, Tv_n)$$

Definition 76 (box). *A **box** is of the form*

$$u + P(r_1 e_1, \dots, r_n e_n)$$

where $u \in V$, $r_1, \dots, r_n$ are positive numbers and $e_1, \dots, e_n$ is an orthonormal basis of V .

Rest of Notes are omitted.

Problem 1

Prove that if $S, T \in \mathcal{L}(V, W)$, then $|\|S\| - \|T\|| \leq \|S - T\|$.

Proof. We have

$$\begin{aligned}\|S\| &= \|S - T + T\| \leq \|S - T\| + \|T\| \\ \|T\| &= \|T - S + S\| \leq \|T - S\| + \|S\|\end{aligned}$$

This means that

$$\|S - T\| \geq |\|S\| - \|T\||$$

□

Problem 2

Suppose that $T \in \mathcal{L}(V)$ is self-adjoint and or that $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$ is normal, then

$$\|T\| = \{\max |\lambda| : \lambda \text{ is an eigenvalue of } T\}$$

Proof. Given by the conditions we know that the eigenvalues of T are its singular values, therefore $\|T\| = s_1 = \max(\lambda)$ □

Problem 3

Suppose that $T \in \mathcal{L}(V)$ and $v \in V$. Prove that

$$\|Tv\| = \|T\|\|v\| \iff T^*Tv = \|T\|^2 v$$

Proof. $\Rightarrow$ We have that $\|Tv\| = \|Tv/\|v\|\|\|v\| = \|T\|\|v\|$. Therefore $\|Tv/\|v\|\| = \|T\|$ which gives that $v/\|v\|$ being the vector corresponds to the largest singular value of T . This gives that

$$T^*Tv/\|v\| = \|T\|^2 v/\|v\|$$

Multiplying $\|v\|$ on both sides solves the problem.

$\Leftarrow$ Given $T^*Tv = \|T\|^2 v$, then we know that v is the eigenvector of T^*T corresponding to its largest eigenvalue s_1^2 , (i.e. it is the vector that corresponds the largest singular value of T). Therefore, we have $\|Tv\| = \|Tv/\|v\|\|\|v\| = \|T\|\|v\|$. □

Problem 4

Suppose $T \in \mathcal{L}(V, W)$, $v \in V$, and $\|Tv\| = \|T\|\|v\|$. Prove that if $u \in V$ and $\langle u, v \rangle = 0$, then $\langle Tu, Tv \rangle = 0$.

Proof. By P3 we know that $T^*Tv = \|T\|^2 v$, then

$$\langle Tu, Tv \rangle = \langle T^*Tu, v \rangle = \|T\|^2 \langle u, v \rangle = 0$$

□

Problem 5

Suppose U is a finite-dimensional inner product space, $T \in \mathcal{L}(V, U)$, and $S \in \mathcal{L}(U, W)$. Prove that

$$\|ST\| \leq \|S\|\|T\|$$

Proof. Take $v \in V$ with $\|v\| \leq 1$ and $\|ST\| = \|STv\|$, then

$$\|ST\| = \|(ST)v\| = \|S(Tv)\| \leq \|S\|\|Tv\| \leq \|S\|\|T\|\|v\| \leq \|S\|\|T\|$$

□

Problem 7

Show that defining $d(S, T) = \|S - T\|$ for $S, T \in \mathcal{L}(V, W)$ makes d a metric on $\mathcal{L}(V, W)$.

Proof. We will examine the definitions one by one:

- $d(S, S) = \|S - S\| = 0$.
- If $S \neq T$, then $d(S, T) = \|S - T\| = \sigma_{\max}(S - T)$. Since $S - T \neq 0$, its largest singular value is nonzero and therefore $d(S, T) > 0$.
- $d(S, T) = \|S - T\| = \|T - S\| = d(T, S)$.
- $d(S, G) = \|S - G\| = \|(S - T) + (T - G)\| \leq \|S - T\| + \|T - G\| = d(S, T) + d(T, G)$.

□

Problem 8

- (a) Prove that if $T \in \mathcal{L}(V)$ and $\|I - T\| < 1$, then T is invertible.
- (b) Suppose that $S \in \mathcal{L}(V)$ is invertible. Prove that if $T \in \mathcal{L}(V)$ and $\|S - T\| < 1/\|S^{-1}\|$, then T is invertible.

Proof. (a) Let $S = I - T$ and $S_n = \sum_{i=1}^n S^i$. Then

$$(I - S)S_n = I - S^{n+1} \rightarrow I$$

as $n \rightarrow \infty$ because $\|S\| < 1$. Therefore, the operator

$$S_\infty = \sum_{i=1}^{\infty} S^i$$

is also bounded. This gives that

$$(I - S)S_\infty = S_\infty(I - S) = I$$

and thus S is invertible, which leads to that T is also invertible.

(b) Equivalently,

$$\|I - TS^{-1}\| \leq \|S - T\| \|S^{-1}\| < 1$$

So TS^{-1} is invertible. Since we already know that S is invertible, T is therefore also invertible. $\square$

Problem 9

Suppose $T \in \mathcal{L}(V)$. Prove that for every $\epsilon > 0$ there exists an invertible operator $S \in \mathcal{L}(V)$ such that $0 < \|T - S\| < \epsilon$.

Proof. Define $S = T + \delta I$ for some $\epsilon > \delta > 0$. Then we have

$$\|T - S\| = \|\delta I\| = \delta$$

which satisfies the desired condition. Note that if T is invertible, we can simply choose a sufficiently small $\delta < 1/\|T^{-1}\|$; if not, then any δ in $(0, 1)$ can make S invertible. $\square$

Problem 12

Suppose $T \in \mathcal{L}(V)$ is a positive operator. Show that $\|\sqrt{T}\| = \sqrt{\|T\|}$.

Proof. Let $\|T\| = \sigma_1$, then $\|\sqrt{T}\| = \sqrt{\sigma_1} = \sqrt{\|T\|}$. $\square$

Problem 17

Prove that if $u \in V$ and φ_u is the linear functional on V defined by the equation $\varphi_u(v) = \langle v, u \rangle$, then $\|\varphi_u\| = \|u\|$.

Proof. We have that

$$\|\varphi_u\| = \sup_{\|v\|=1} |\langle v, u \rangle| = \left| \left\langle \frac{u}{\|u\|}, u \right\rangle \right| = \|u\|$$

$\square$

Problem 18

Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $T \in \mathcal{L}(V, W)$.

- (a) Prove that $\max\{\|Te_1\|, \dots, \|Te_n\|\} \leq \|T\| \leq (\|Te_1\|^2 + \dots + \|Te_n\|^2)^{1/2}$.
- (b) Prove that $\|T\| = (\|Te_1\|^2 + \dots + \|Te_n\|^2)^{1/2}$ if and only if $\dim \text{range } T \leq 1$.

Proof. (a) We first know that $\|Te_i\| \leq \|T\|$ by definition of $\|\cdot\|$. In the proof of equivalent characterizations, we showed that there exists some orthonormal basis (corresponding to SVD) such that $\|T(f_n)\| = s_1$ for some f_n . This finishes the l.h.s. To see the r.h.s., let $v \in V$ with $\|v\| = 1$ to be such that $\|Tv\| = \|T\|$. Then we have

$$\|T\|^2 = \|Tv\|^2 = \left\| \sum_{i=1}^n a_i Te_i \right\|^2 \leq \left(\sum_{i=1}^n |a_i|^2 \right) \left(\sum_{i=1}^n \|Te_i\|^2 \right) = \sum_{i=1}^n \|Te_i\|^2$$

Taking the square root solves the problem.

(b) $\Rightarrow$ By the Cauchy-Schwarz inequality, we know that this is an equality if and only if Te_i must be proportional to each other. That is, there exists a unit vector w such that $Te_i = a_i w$. So $\dim \text{range } T \leq 1$.

$\Leftarrow$ If $\dim \text{range } T \leq 1$, then there exists a unit vector w and $\beta_i \geq 0$ s.t.

$$Te_i = \beta_i w$$

We now have that for some v with unit norm,

$$\|Tv\| = \left\| \sum_{i=1}^n a_i Te_i \right\| = \left\| \sum_{i=1}^n a_i \beta_i w \right\| \leq \left| \sum_{i=1}^n a_i \beta_i \right|$$

We know that

$$\left| \sum_{i=1}^n a_i \beta_i \right| \leq \left(\sum_{i=1}^n a_i^2 \right)^{1/2} \left(\sum_{i=1}^n \beta_i^2 \right)^{1/2} = \left(\sum_{i=1}^n \beta_i^2 \right)^{1/2}$$

with equality obtained when $\beta_i = ca_i$ for all i . That is, the sup of $\|Tv\|$ occurs under this condition and that we have

$$\|T\| = \left(\sum_{i=1}^n \beta_i^2 \right)^{1/2} = \left(\sum_{i=1}^n \|Te_i\|^2 \right)^{1/2}$$

□

Problem 24

Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

$$\|T^{-1}\| = \|T\|^{-1} \iff \frac{T}{\|T\|} \text{ is a unitary operator}$$

Proof. $\Rightarrow$ Let $U = \frac{T}{\|T\|}$, then $\|U\| = 1$ and $\|U^{-1}\| = 1$. Then

$$\|Ux\| \leq \|U\|\|x\| = \|x\|$$

and

$$\|x\| = \|U^{-1}Ux\| \leq \|U^{-1}\|\|Ux\| = \|Ux\|$$

Therefore, we have $\|Ux\| = \|x\|$ and therefore it is an invertible isometry and thus unitary.

$\Leftarrow$ Conversely, we have $\|U\| = 1 = \|U^{-1}\|$ by it being unitary. Therefore, we have $\|T^{-1}\| = \|T\|^{-1}$ $\square$

Chapter 8: Operators on Complex Vector Spaces

Linear Algebra Done Right (4th Edition), by Sheldon Axler

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Contents

8A: Generalized Eigenvectors and Nilpotent Operators	2
8A Problem Sets	4
8B: Generalized Eigenspace Decomposition	7
8B Problem Sets	9
8C: Consequences of Generalized Eigenspace Decomposition	12
8C Problem Sets	13
8D: Trace: A Connection Between Matrices and Operators	14
8D Problem Sets	15

8A: Generalized Eigenvectors and Nilpotent Operators

Lemma 1 (sequence of increasing null spaces). *Suppose $T \in \mathcal{L}(V)$. Then*

$$\{0\} = \text{null } T^0 \subseteq \text{null } T^1 \subseteq \text{null } T^2 \dots \subseteq \text{null } T^k \subseteq \text{null } T^{k+1} \dots$$

Lemma 2 (equality in the sequence of null spaces). *Suppose $T \in \mathcal{L}(V)$ and m is a nonnegative integer such that*

$$\text{null } T^m = \text{null } T^{m+1}$$

Then

$$\text{null } T^m = \text{null } T^{m+1} = \text{null } T^{m+2} = \dots$$

Lemma 3 (null space stop growing). *Suppose $T \in \mathcal{L}(V)$. Then*

$$\text{null } T^{\dim V} = \text{null } T^{\dim V+1} = \text{null } T^{\dim V+2} = \dots$$

Theorem 4 (V is the direct sum of $\text{null } T^{\dim V}$ and $\text{range } T^{\dim V}$). *Suppose $T \in \mathcal{L}(V)$. Then*

$$V = \text{null } T^{\dim V} \oplus \text{range } T^{\dim V}$$

Definition 5 (generalized eigenvector). *Suppose $T \in \mathcal{L}(V)$ and λ is an eigenvalue of T . A vector $v \in V$ is called a **generalized eigenvector** of T corresponding to λ if $v \neq 0$ and*

$$(T - \lambda I)^k v = 0$$

for some positive integer k .

Remark 6. *There is no notion of “generalized eigenvalues” since we do not create new eigenvalues.*

Remark 7. *A nonzero vector $v \in V$ is a generalized eigenvector of T if and only if $(T - \lambda I)^{\dim V} v = 0$*

Theorem 8 (a basis of generalized eigenvectors). *Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then there is a basis of V consisting of generalized eigenvectors of T .*

Proposition 9. *Suppose $T \in \mathcal{L}(V)$. Then each generalized eigenvector of T only corresponds to one eigenvalue of T .*

Proposition 10. *Suppose that $T \in \mathcal{L}(V)$. Then every list of generalized eigenvectors of T corresponding to distinct eigenvalues are linearly independent.*

Definition 11 (nilpotent). *An operator is called **nilpotent** if some powers of it equals 0.*

Remark 12. *An operator is nilpotent if every nonzero vector in V is a generalized eigenvector of T corresponding to eigenvalue 0.*

Corollary 13. *Suppose $T \in \mathcal{L}(V)$ is nilpotent. Then $T^{\dim V} = 0$.*

Theorem 14 (eigenvalues of nilpotent operator). *Suppose $T \in \mathcal{L}(V)$.*

- (a) *If T is nilpotent, then 0 is an eigenvalue of T and T has no other eigenvalues.*
- (b) *If $\mathbb{F} = \mathbb{C}$ and 0 is the only eigenvalue of T , then T is nilpotent.*

Theorem 15 (minimal polynomial and upper-triangular matrix of nilpotent operator). *Suppose $T \in \mathcal{L}(V)$. Then the following are equivalent.*

- (a) *T is nilpotent.*
- (b) *The minimal polynomial of T is z^m for some positive integer m .*
- (c) *There is a basis of V with respect to which the matrix of T has the form*

$$\begin{pmatrix} 0 & * & \cdots & * \\ 0 & 0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & * \\ 0 & 0 & \cdots & 0 \end{pmatrix},$$

where all entries on and below the diagonal equal 0.

Problem 1

Suppose $T \in \mathcal{L}(V)$. Prove that if $\dim \text{null } T^4 = 8$ and $\dim \text{null } T^6 = 9$, then $\dim \text{null } T^m = 9$ for all integers $m \geq 5$.

Proof. Suppose not, then $\dim \text{null } T^5 = 8 = \dim \text{null } T^6$, forming a contradiction. Therefore, the statement holds. $\square$

Problem 2

Suppose $T \in \mathcal{L}(V)$, m is a positive integer, $v \in V$, and $T^{m-1}v \neq 0$ but $T^m v = 0$. Prove that $v, Tv, T^2v, \dots, T^{m-1}v$ is linearly independent.

Proof. Consider

$$a_0v + a_1Tv + \dots + a_{m-1}T^{m-1}v = 0$$

Apply T^{m-1} on both sides yields that

$$a_0T^{m-1}v = 0$$

and therefore $a_0 = 0$. Note that $v \neq \text{null } T^{m-1}$ and therefore $v \neq \text{null } T^j$ for $j \leq m-1$. Hence, continuing apply the argument above will get that all $a_i = 0$. $\square$

Problem 3

Suppose $T \in \mathcal{L}(V)$. Prove that

$$V = \text{null } T \oplus \text{range } T \iff \text{null } T^2 = \text{null } T$$

Proof. $\Rightarrow$ We know that $\text{null } T \subseteq \text{null } T^2$. Take $v \in \text{null } T^2$, then

$$T^2v = 0 = T(Tv)$$

Therefore $Tv \in \text{null } T$, but $Tv \in \text{range } T$ so $Tv = 0$, which gives that $v \in \text{null } T$.

$\Leftarrow$ Let $v \in (\text{null } T) \cap (\text{range } T)$. Then there exists u s.t. $Tu = v$ and $Tv = 0$. Therefore, we have $T^2u = Tv = 0$ and thus $u \in \text{null } T^2 = \text{null } T$. So $v = T0 = 0$. We've proved the claim. $\square$

Problem 6

Suppose $T \in \mathcal{L}(V)$. Show that

$$V = \text{range } T^0 \supseteq \text{range } T^1 \supseteq \dots \supseteq \text{range } T^k \supseteq T^{k+1} \supseteq \dots$$

Proof. Take $v \in \text{range } T^{k+1}$, then we know that there exists $u \in V$ s.t. $v = T^{k+1}u = T^k(Tu)$. Therefore, we have $v \in \text{range } T^k$. $\square$

Problem 9

Suppose $T \in \mathcal{L}(V)$ and m is a nonnegative integer. Prove that

$$\text{null } T^m = \text{null } T^{m+1} \iff \text{range } T^m = \text{range } T^{m+1}$$

Proof. We know that

$$\dim V = \dim \text{null } T^m + \dim \text{range } T^m = \dim \text{null } T^{m+1} + \dim \text{range } T^{m+1}$$

Therefore, we have

$$\text{null } T^m = \text{null } T^{m+1} \iff \text{range } T^m = \text{range } T^{m+1}$$

□

Problem 12

Suppose $T \in \mathcal{L}(V)$ is such that every vector in V is a generalized eigenvector of T . Prove that there exists $\lambda \in \mathbb{F}$ such that $T - \lambda I$ is nilpotent.

Proof. If T has only one eigenvalue, then it is easy to tell that $T - \lambda I$ is nilpotent for the only eigenvalue λ . Suppose for the contradiction that it has multiple distinct eigenvalues. Then we know that for $v_1 \in G(\lambda_1, T)$ and $v_2 \in G(\lambda_2, T)$ are both invariant under T , but $v = v_1 + v_2 \in G(\lambda, T)$ is also invariant under T . If $\lambda = \lambda_1$ or λ_2 , then this contradicts that $\lambda_1 \neq \lambda_2$. If $\lambda \neq \lambda_1$ and $\lambda \neq \lambda_2$, then this contradicts that $G(\lambda_1, T) \cap G(\lambda, T) = \{0\}$. Therefore, there is only one eigenvalue and thus $T - \lambda I$ is nilpotent for the only eigenvalue λ . □

Problem 13

Suppose $S, T \in \mathcal{L}(V)$ and ST is nilpotent. Prove that TS is nilpotent.

Proof. We know $(ST)^k = 0$ for some k . Then

$$(TS)^{k+1} = T(ST)^k S = 0$$

□

Problem 14

Suppose $T \in \mathcal{L}(V)$ is nilpotent and $T \neq 0$. Prove that T is not diagonalizable.

Proof. 0 is the only eigenvalue of T and any nonzero $v \in V$ cannot be represented by an eigenbasis. □

Problem 22

Suppose $T \in \mathcal{L}(\mathbb{C}^5)$ is such that $\text{range } T^4 \neq \text{range } T^5$. Prove that T is nilpotent.

Proof. By Problem 9, we have that $\text{null } T^4 \neq \text{null } T^5$ and therefore $\dim \text{null } T^4 < \dim \text{null } T^5 = 5$ where $\dim \mathbb{C}^5 = 5$. Hence, T is nilpotent. $\square$

8B: Generalized Eigenspace Decomposition

Definition 16 (generalized eigenspace, $G(\lambda, T)$). Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$. The **generalized eigenspace** of T corresponding to λ , denoted by $G(\lambda, T)$, is defined by

$$G(\lambda, T) = \{v \in V : (T - \lambda I)^k v = 0 \text{ for some positive integer } k\}.$$

Thus $G(\lambda, T)$ is the set of generalized eigenvectors of T corresponding to λ , along with the 0 vector.

Remark 17. $E(\lambda, T) \subseteq G(\lambda, T)$ as each eigenvector is a generalized eigenvector.

Corollary 18 (description of generalized eigenspaces). Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$. Then $G(\lambda, T) = \text{null } (T - \lambda I)^{\dim V}$.

Theorem 19 (generalized eigenspace decomposition). Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T . Then

- (a) $G(\lambda_k, T)$ is invariant under T for each $k = 1, \dots, m$;
- (b) $(T - \lambda_k I)|_{G(\lambda_k, T)}$ is nilpotent for each $k = 1, \dots, m$;
- (c) $V = G(\lambda_1, T) \oplus \dots \oplus G(\lambda_m, T)$.

Definition 20 (multiplicity). Suppose $T \in \mathcal{L}(V)$. The **multiplicity** of an eigenvalue λ of T is defined to be the dimension of the corresponding generalized eigenspace $G(\lambda, T)$. In other words, the multiplicity of an eigenvalue λ of T equals

$$\dim \text{null } (T - \lambda I)^{\dim V}$$

Corollary 21. Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then the sum of the multiplicities of all eigenvalue of T equals $\dim V$.

Remark 22. We may know the term **algebraic multiplicity** and **geometric multiplicity** in some books. We have

$$\begin{aligned} \text{algebraic multiplicity of } \lambda &= \dim \text{null } (T - \lambda I)^{\dim V} = \dim G(\lambda, T). \\ \text{geometric multiplicity of } \lambda &= \dim \text{null } (T - \lambda I) = \dim E(\lambda, T). \end{aligned}$$

Remark 23. If V is an inner product space, $T \in \mathcal{L}(V)$ is normal, and λ is an eigenvalue of T , then the algebraic multiplicity of λ equals the geometric multiplicity of λ (i.e. every eigenvector is a generalized eigenvector).

Definition 24 (**characteristic polynomial**). Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Let $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T , with multiplicities $d_1, \dots, d_m$. The polynomial

$$(z - \lambda_1)^{d_1} \dots (z - \lambda_m)^{d_m}$$

is called the **characteristic polynomial** of T .

Corollary 25. Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then

- (a) the characteristic polynomial of T has degree $\dim V$;
- (b) the zeros of the characteristic polynomial of T are the eigenvalues of T .

Theorem 26 (Cayley-Hamilton theorem). Suppose $\mathbb{F} = \mathbb{C}$, $T \in \mathcal{L}(V)$, and q is the characteristic polynomial of T . Then $q(T) = 0$.

Corollary 27. Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then the characteristic polynomial of T is a polynomial multiple of the minimal polynomial of T .

Theorem 28. Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Suppose $v_1, \dots, v_n$ is a basis of V such that $\mathcal{M}(T, (v_1, \dots, v_n))$ is upper triangular. Then the number of times each eigenvalue λ of T appears on the diagonal of $\mathcal{M}(T, (v_1, \dots, v_n))$ equals the multiplicity of λ as an eigenvalue of T .

Definition 29 (block diagonal matrix). A **block diagonal matrix** is a square matrix of the form

$$\begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{pmatrix}$$

where $A_1, \dots, A_m$ are square matrices lying along the diagonal and all other entries of the matrix equal 0.

Remark 30. wrt. an appropriate basis, every operator on a finite-dimensional complex vector space has a matrix of the form.

Theorem 31 (block diagonal matrix with upper-triangular blocks). Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T , with multiplicities $d_1, \dots, d_m$. Then there is a basis of V with respect to which T has a block diagonal matrix of the form

$$\begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{pmatrix},$$

where each A_k is a d_k -by- d_k upper-triangular matrix of the form

$$A_k = \begin{pmatrix} \lambda_k & & * \\ & \ddots & \\ 0 & & \lambda_k \end{pmatrix}$$

Problem 1

Define $T \in \mathcal{L}(\mathbb{C}^2)$ by $T(w, z) = (-z, w)$. Find the generalized eigenspaces corresponding to the distinct eigenvalues of T .

Proof. We have the matrix of T to be

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

The eigenvalues for T are $\pm i$. For $\lambda_1 = i$, we have $v_1 = (i, 1)$; for $\lambda_2 = -i$, we have $v_2 = (-i, 1)$. There eigenspace is therefore:

$$E_i = \text{span}\{(i, 1)\} \quad E_{-i} = \text{span}\{(-i, 1)\}$$

□

Problem 2

Suppose $T \in \mathcal{L}(V)$. Prove that $G(\lambda, T) = G(\frac{1}{\lambda}, T^{-1})$ for every $\lambda \in \mathbb{F}$ with $\lambda \neq 0$.

Proof. WLOG let $v \in G(\lambda, T)$. Then we have that for some k

$$0 = (T - \lambda I)^k$$

We have

$$(\lambda^{-1})^k (T^{-1})^k (T - \lambda I)^k = (\lambda^{-1} T^{-1} (T - \lambda I))^k = (\lambda^{-1} I - T^{-1})^k = 0$$

which shows that $v \in G(\frac{1}{\lambda}, T)$. The other direction follows accordingly. □

Problem 3

Suppose $T \in \mathcal{L}(V)$. Suppose $S \in \mathcal{L}(V)$ is invertible. Prove that T and $S^{-1}TS$ have the same eigenvalues with the same multiplicities.

Proof. Let λ be an eigenvalue of T with multiplicity d . Then we know

$$(T - \lambda I)^d = 0$$

Therefore, we have

$$(S^{-1})^d (T - \lambda I)^d S^d = (S^{-1} (T - \lambda I) S)^d = (S^{-1} T S - \lambda I)^d = 0$$

The converse is proved identically. □

Problem 5

Suppose $T \in \mathcal{L}(V)$ and 3 and 8 are eigenvalues of T . Let $n = \dim V$. Prove that $V = (\text{null } T^{n-2}) \oplus (\text{range } T^{n-2})$.

Proof. This means that the minimal polynomial of T can be written as

$$m_T(x) = (x - 3)(x - 8)q(x)$$

with $\max \deg q(x) \leq n-2$. Hence, we have that $\text{null } T^n = \text{null } T^{n-1} = \text{null } T^{n-2}$ and $\text{range } T^n = \text{range } T^{n-1} = \text{range } T^{n-2}$. Applying P3 from section 8A solves the problem. $\square$

Problem 10

Suppose V is a complex inner product space, $e_1, \dots, e_n$ is an orthonormal basis of T , and $T \in \mathcal{L}(V)$. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of T , each included as many times as its multiplicity. Prove that

$$|\lambda_1|^2 + \dots + |\lambda_n|^2 \leq \|Te_1\|^2 + \dots + \|Te_n\|^2$$

Proof.

$$\begin{aligned} \sum_{i=1}^n \|Te_i\|^2 &= \sum_{i=1}^n \|U\Sigma V^*e_i\|^2 \\ &= \sum_{i=1}^n \|U\Sigma f_i\|^2 \\ &= \sum_{i=1}^n \|\Sigma f_i\|^2 \\ &= \sum_{i=1}^n \|\lambda_i f_i\|^2 \\ &\geq \sum_{i=1}^n |\lambda_i|^2 \end{aligned}$$

by the Bessel's inequality at the last step. $\square$

Problem 14

Give an example of an operator on $\mathbb{C}^4$ whose characteristic polynomial equals $z(z-1)^2(z-3)$ and whose minimal polynomial equals $z(z-1)(z-3)$.

Proof. Consider

$$J = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

$\square$

Problem 17

Suppose $\mathbb{F} = \mathbb{C}$ and $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that the characteristic polynomial of P is $z^m(z - 1)^n$, where $m = \dim \text{null } P$ and $n = \dim \text{range } P$.

Proof. We know that the projection operator P has eigenvalue 0 and 1 (from definition). By many of our prior exercises, we know that the (generalized eigenspace of) eigenvalue 0 partitions the null space and nonzero ones partitions the range space. You may verify it by yourselves. $\square$

8C: Consequences of Generalized Eigenspace Decomposition

Lemma 32. *Suppose $T \in \mathcal{L}(V)$ is nilpotent. Then $I + T$ has a square root.*

Remark 33. *This lemma holds on both real and complex vector spaces.*

Lemma 34. *Suppose V is a complex vector space and $T \in \mathcal{L}(V)$ is invertible. Then T has a square root.*

Definition 35 (Jordan basis). *Suppose $T \in \mathcal{L}(V)$. A basis of V is called a Jordan basis for T if with respect to this basis T has a block diagonal matrix*

$$\begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_p \end{pmatrix}$$

in which each A_k is an upper-triangular matrix of the form

$$A_k = \begin{pmatrix} \lambda_k & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_k \end{pmatrix}$$

Theorem 36 (every nilpotent operator has a Jordan basis). *Suppose $T \in \mathcal{L}(V)$ is nilpotent. Then there is a basis of V that is a Jordan basis for T .*

Corollary 37 (Jordan form). *Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then there is a basis of V that is a Jordan basis for T .*

Problem 1

Suppose $T \in \mathcal{L}(\mathbb{C}^3)$ is the operator defined by $T(z_1, z_2, z_3) = (z_2, z_3, 0)$. Prove that T does not have a square root.

Proof. Note that for eigenvalue

$$(z_2, z_3, 0) = \lambda(z_1, z_2, z_3)$$

the only solution is $\lambda = 0$ with multiplicity 3. Suppose for contradiction that $S^2 = T$. So any eigenvalue λ of S , λ^2 will be the eigenvalue of T , so S also only has $\lambda = 0$ as its only eigenvalue, indicating that S is nilpotent and $S^3 = 0$. This gives that $T^2 = SS^3 = 0$. However, we in fact have that

$$T^2(z_1, z_2, z_3) = (z_3, 0, 0) \neq 0$$

reaching a contradiction. □

Problem 6

Find a basis of $\mathcal{P}_4(\mathbb{R})$ that is a Jordan basis for the differentiation operator D on $\mathcal{P}_4(\mathbb{R})$ defined by $Dp = p'$.

Proof. Note that the goal here is to find linearly independent $v_1, \dots, v_5$ s.t. $D(v_1) = 0$ and $D(v_i) = v_{i-1}$. This gives that

$$\left\{ 1, x, \frac{1}{2}x^2, \frac{1}{6}x^3, \frac{1}{24}x^4 \right\}$$

□

Skip the rest of questions.

8D: Trace: A Connection Between Matrices and Operators

Definition 38 (trace of a matrix). Suppose A is a square matrix with entries in $\mathbb{F}$. The **trace** of A , denoted by $\text{tr}A$, is defined to be the diagonal entries of A .

Proposition 39 (trace of AB equals trace of BA). Suppose A is an m -by- n matrix and B is an n -by- m matrix. Then

$$\text{tr}(AB) = \text{tr}(BA)$$

Lemma 40. Suppose $T \in \mathcal{L}(V)$. Suppose $u_1, \dots, u_n$ and $v_1, \dots, v_n$ are bases of V . Then

$$\text{tr}\mathcal{M}(T, (u_1, \dots, u_n)) = \text{tr}\mathcal{M}(T, (v_1, \dots, v_n))$$

Definition 41 (trace of an operator). Suppose $T \in \mathcal{L}(V)$. The **trace** of T , denoted by $\text{tr}T$, is defined by

$$\text{tr} T = \text{tr} \mathcal{M}(T, (v_1, \dots, v_n))$$

where $v_1, \dots, v_n$ is any basis of V .

Corollary 42. Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then $\text{tr} T$ equals the sum of the eigenvalues of T , with each eigenvalue included as many times as its multiplicity.

Corollary 43. Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Let $n = \dim V$. Then $\text{tr} T$ equals the negative of the coefficient of z^{n-1} in the characteristic polynomial of T .

Corollary 44. Suppose V is an inner product space, $T \in \mathcal{L}(V)$, and $e_1, \dots, e_n$ is an orthonormal basis of V . Then

$$\text{tr} T = \langle Te_1, e_1 \rangle + \dots + \langle Te_n, e_n \rangle$$

Theorem 45 (trace is linear). The function $\text{tr} : \mathcal{L}(V) \rightarrow \mathbb{F}$ is a linear functional on $\mathcal{L}(V)$ such that

$$\text{tr}(ST) = \text{tr}(TS)$$

for all $S, T \in \mathcal{L}(V)$.

Corollary 46. There do not exist operators $S, T \in \mathcal{L}(V)$ such that $ST - TS = I$.

Problem 1

Suppose V is an inner product space and $v, w \in V$. Define an operator $T \in \mathcal{L}(V)$ by $Tu = \langle u, v \rangle w$. Find a formula for $\text{tr } T$.

Proof. Let $e_1, \dots, e_n$ be the standard orthonormal basis of V . Then we have that

$$\begin{aligned} \text{tr } T &= \sum_{i=1}^n \langle Te_i, e_i \rangle \\ &= \sum_{i=1}^n \langle \langle e_i, v \rangle w, e_i \rangle \\ &= \sum_{i=1}^n \langle e_i, v \rangle \langle w, e_i \rangle \\ &= \sum_{i=1}^n v_i w_i \\ &= v \cdot w \end{aligned}$$

□

Problem 2

Suppose $P \in \mathcal{L}(V)$ satisfies $P^2 = P$. Prove that

$$\text{tr } P = \dim \text{range } P$$

Proof. Note that $\text{tr } P = \sum_{i=1}^n \lambda_i$ where $\lambda_i = 1$ or 0 . The multiplicity of $\lambda_i = 1$ determines the $\dim \text{range } P$ and thus gives the desired conclusion. □

Problem 5

Suppose V is an inner product space. Suppose $T \in \mathcal{L}(V)$ is a positive operator and $\text{tr } T = 0$. Prove that $T = 0$.

Proof. We know that $\lambda_i \geq 0$ for all i for positive T . Since $\text{tr } T = 0$, all eigenvalues are 0 and thus $T = 0$ (as it's self-adjoint by positivity). □

Problem 9

Suppose $T \in \mathcal{L}(V)$ is such that $\text{tr } (ST) = 0$ for all $S \in \mathcal{L}(V)$. Prove that $T = 0$.

Proof. Consider orthonormal basis $e_1, \dots, e_n$ and define S_{ij} to be such that maps e_j to e_i while keep all other zero. Therefore, $\text{tr } (S_{ij}T) = T_{ij} = 0$ for all i, j . Hence, we proved $T = 0$. □

Problem 11

Suppose V and W are inner product spaces and $T \in \mathcal{L}(V, W)$. Prove that if $e_1, \dots, e_n$ is an orthonormal basis of V and $f_1, \dots, f_m$ is an orthonormal basis of W , then

$$\operatorname{tr} (T^*T) = \sum_{k=1}^n \sum_{j=1}^m |\langle Te_k, f_j \rangle|^2$$

Proof. We have that

$$\begin{aligned} \operatorname{tr} (T^*T) &= \sum_{k=1}^n \langle T^*T e_k, e_k \rangle \\ &= \sum_{k=1}^n \langle T e_k, T e_k \rangle \\ &= \sum_{k=1}^n \langle T e_k, \sum_{j=1}^m \langle T e_k, f_j \rangle f_j \rangle \\ &= \sum_{k=1}^n \sum_{j=1}^m |\langle T e_k, f_j \rangle|^2 \end{aligned}$$

□

Problem 12

Suppose V and W are finite-dimensional inner product spaces.

- (a) Prove that $\langle S, T \rangle = \operatorname{tr} (T^*S)$ defines an inner product on $\mathcal{L}(V, W)$.
- (b) Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $f_1, \dots, f_m$ is an orthonormal basis of W . Show that the inner product on $\mathcal{L}(V, W)$ from (a) is the same as the standard inner product on $\mathbb{F}^{mn}$, where we identify each element of $\mathcal{L}(V, W)$ with its matrix (with respect to the bases just mentioned) and then with an element of $\mathbb{F}^{mn}$.

Remark 47. The norm from (a) is called the Frobenius norm or the Hilbert-Schmidt norm.

Proof. (a) We check each condition manually:

- Positivity: $\langle S, S \rangle = \operatorname{tr}(S^*S) = \sum_{i=1}^n (A^*A)_{ii} = \sum_{i=1}^n \sum_{j=1}^m A_{ij}^* A_{ji} = \sum_{i=1}^n \sum_{j=1}^m |A_{ij}|^2 \geq 0$ with equality iff $S = 0$.
- Linearity in the first slot: $\langle \lambda S_1 + S_2, T \rangle = \operatorname{tr}(T^*(\lambda S_1 + S_2)) = \lambda \operatorname{tr}(T^*S_1) + \operatorname{tr}(T^*S_2) = \lambda \langle S_1, T \rangle + \langle S_2, T \rangle$

- Conjugate symmetry: $\overline{\langle S, T \rangle} = \overline{\text{tr}(T^* S)} = \text{tr}(\overline{T^* S}) = \text{tr}(S^* T) = \langle T, S \rangle$
- (b) The standard inner product on $\mathbb{F}^{mn}$ for the two matrices A, B is

$$\langle A, B \rangle = \sum_{i=1}^n \sum_{j=1}^m A_{ij} \overline{B_{ij}}$$

which is exactly how we define in (a).

□

Chapter 9: Multilinear Algebra and Determinants

Linear Algebra Done Right (4th Edition), by Sheldon Axler

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Contents

9A: Bilinear Forms and Quadratic Forms	2
9A Problem Sets	5
9B: Alternating Multilinear Forms	7
9B Problem Sets	9
9C: Determinants	10
9C Problem Sets	14
9D: Tensor Products	17
9D Problem Sets	20

9A: Bilinear Forms and Quadratic Forms

Definition 1 (bilinear form). A **bilinear form** on V is a function $\beta: V \times V \rightarrow \mathbb{F}$ such that

$$v \mapsto \beta(v, u) \text{ and } v \mapsto \beta(u, v)$$

are both linear functionals on V for every $u \in V$.

Remark 2. A better but less popular terminology is “bilinear functional”. If V is real, then the function $(u, v) \mapsto \langle u, v \rangle$ is a bilinear form. If V is complex, then it isn’t.

Remark 3. If $\mathbb{F} = \mathbb{R}$, then a bilinear form differs from an inner product in that it does not require positive definiteness or symmetry.

Remark 4. A bilinear form β on V is a linear map on $V \times V$ only if $\beta = 0$.

Definition 5 ($V^{(2)}$). The set of bilinear forms on V is denoted by $V^{(2)}$.

Definition 6 (matrix of a bilinear form, $\mathcal{M}(\beta)$). Suppose β is a bilinear form on V and $e_1, \dots, e_n$ is a basis of V . The **matrix** of β with respect to this basis is the n -by- n matrix $\mathcal{M}(\beta)$ whose entry $\mathcal{M}(\beta)_{j,k}$ in row j , column k is given by

$$\mathcal{M}(\beta)_{j,k} = \beta(e_j, e_k)$$

If the basis $e_1, \dots, e_n$ is not clear from the context, then the notation $\mathcal{M}(\beta, (e_1, \dots, e_n))$ is used.

Corollary 7 ($\dim V^{(2)} = (\dim V)^2$). Suppose $e_1, \dots, e_n$ is a basis of V . Then the map $\beta \mapsto \mathcal{M}(\beta)$ is an isomorphism of $V^{(2)}$ onto $\mathbb{F}^{n,n}$. Furthermore, $\dim V^{(2)} = (\dim V)^2$.

Lemma 8 (composition of a bilinear form and an operator). Suppose β is a bilinear form on V and $T \in \mathcal{L}(V)$. Define bilinear forms α and ρ on V by

$$\alpha(u, v) = \beta(u, Tv) \text{ and } \rho(u, v) = \beta(Tu, v)$$

Let $e_1, \dots, e_n$ be a basis of V . Then

$$\mathcal{M}(\alpha) = \mathcal{M}(\beta)\mathcal{M}(T) \text{ and } \mathcal{M}(\rho) = \mathcal{M}(T)^\top \mathcal{M}(\beta)$$

Theorem 9 (change-of-basis formula). Suppose $\beta \in V^{(2)}$. Suppose $e_1, \dots, e_n$ and $f_1, \dots, f_n$ are bases of V . Let

$$A = \mathcal{M}(\beta(e_1, \dots, e_n)) \text{ and } B = \mathcal{M}(\beta, (f_1, \dots, f_n))$$

and $C = \mathcal{M}(I, (e_1, \dots, e_n), (f_1, \dots, f_n))$. Then

$$A = C^\top BC$$

Definition 10 (symmetric bilinear form, $V_{sym}^{(2)}$). A bilinear form $\rho \in V^{(2)}$ is called **symmetric** if

$$\rho(u, w) = \rho(w, u)$$

for all $u, w \in V$. The set of symmetric bilinear forms on V is denoted by $V_{sym}^{(2)}$.

Remark 11. For real inner product space, define $\rho(u, w) = \langle u, w \rangle \in V_{sym}^{(2)}$. Additional example include

$$\rho(u, w) = \langle u, Tw \rangle$$

where T is self-adjoint and

$$\rho(S, T) = \text{tr}(ST)$$

where here $\rho: \mathcal{L}(V) \times \mathcal{L}(V) \rightarrow \mathbb{F}$.

Definition 12 (symmetric matrix). A square matrix A is called **symmetric** if it equals its transpose.

Theorem 13 (symmetric bilinear forms are diagonalizable). Suppose $\rho \in V^{(2)}$. Then the following are equivalent.

- (a) ρ is a symmetric bilinear form on V .
- (b) $\mathcal{M}(\rho, (e_1, \dots, e_n))$ is a symmetric matrix for every basis $e_1, \dots, e_n$ of V .
- (c) $\mathcal{M}(\rho, (e_1, \dots, e_n))$ is a symmetric matrix for some basis $e_1, \dots, e_n$ of V .
- (d) $\mathcal{M}(\rho, (e_1, \dots, e_n))$ is a diagonal matrix for some basis $e_1, \dots, e_n$ of V .

Theorem 14. Suppose V is a real inner product space and ρ is a symmetric bilinear form on V . Then ρ has a diagonal matrix with respect to some orthonormal basis of V .

Definition 15 (alternating bilinear form, $V_{alt}^{(2)}$). A bilinear form $\alpha \in V^{(2)}$ is called **alternating** if

$$\alpha(v, v) = 0$$

for all $v \in V$. The set of alternating bilinear forms on V is denoted by $V_{alt}^{(2)}$.

Lemma 16 (characterization of alternating linear forms). A bilinear form α on V is alternating if and only if

$$\alpha(u, w) = -\alpha(w, u)$$

for all $u, w \in V$.

Theorem 17. The sets $V_{sym}^{(2)}$ and $V_{alt}^{(2)}$ are subspaces of $V^{(2)}$. Furthermore,

$$V^{(2)} = V_{sym}^{(2)} \oplus V_{alt}^{(2)}$$

Definition 18 (quadratic form associated with a bilinear form, q_β). For β a bilinear form on V , define a function $q_\beta: V \rightarrow \mathbb{F}$ by $q_\beta(v) = \beta(v, v)$. A function $q: V \rightarrow \mathbb{F}$ is called a **quadratic form** on V if there exists a bilinear form β on V such that $q = q_\beta$.

Corollary 19 (quadratic form on $\mathbb{F}^n$). Suppose n is a positive integer and q is a function from $\mathbb{F}^n$ to $\mathbb{F}$. Then q is a quadratic form on $\mathbb{F}^n$ if and only if there exist numbers $A_{j,k} \in \mathbb{F}$ for $j, k \in \{1, \dots, n\}$ such that

$$q(x_1, \dots, x_n) = \sum_{k=1}^n \sum_{j=1}^n A_{j,k} x_j x_k$$

for all $(x_1, \dots, x_n) \in \mathbb{F}^n$.

Theorem 20 (characterizations of quadratic forms). Suppose $q: V \rightarrow \mathbb{F}$ is a function. Then following are equivalent.

- (a) q is a quadratic form.
- (b) There exists a unique symmetric bilinear form ρ on V such that $q = q_\rho$.
- (c) $q(\lambda v) = \lambda^2 q(v)$ for all $\lambda \in \mathbb{F}$ and all $v \in V$, and the function

$$(u, w) \mapsto q(u + w) - q(u) - q(w)$$

is a symmetric bilinear form on V .

- (d) $q(2v) = 4q(v)$ for all $v \in V$, and the function

$$(u, w) \mapsto q(u + w) - q(u) - q(w)$$

is a symmetric bilinear form on V .

Theorem 21 (diagonalization of quadratic form). Suppose q is a quadratic form on V .

- (a) There exist a basis $e_1, \dots, e_n$ of V and $\lambda_1, \dots, \lambda_n \in \mathbb{F}$ such that

$$q(x_1 e_1 + \dots + x_n e_n) = \lambda_1 x_1^2 + \dots + \lambda_n x_n^2$$

for all $x_1, \dots, x_n \in \mathbb{F}$.

- (b) If $\mathbb{F} = \mathbb{R}$ and V is an inner product space, then the basis in (a) can be chosen to be an orthonormal basis of V .

Remark 22. For each quadratic form we can choose a basis such that the quadratic form looks like a weighted sum of squares of the coordinates.

Problem 1

Prove that if β is a bilinear form on $\mathbb{F}$, then there exists $c \in \mathbb{F}$ such that

$$\beta(x, y) = cxy$$

for all $x, y \in \mathbb{F}$.

Proof. We note that since the input is taken from $\mathbb{F}$, the basis is naturally 1. So we have that

$$\beta(x, y) = x\beta(1, y) = xy\beta(1, 1) = cxy$$

where we take $c = \beta(1, 1)$. □

Problem 2

Let $n = \dim V$. Suppose β is a bilinear form on V . Prove that there exist $\phi_1, \dots, \phi_n, \tau_1, \dots, \tau_n \in V'$ such that

$$\beta(u, v) = \phi_1(u) \cdot \tau_1(v) + \dots + \phi_n(u) \cdot \tau_n(v)$$

for all $u, v \in V$.

Proof.

$$\beta(u, v) = \beta\left(\sum_{i=1}^n u_i e_i, \sum_{j=1}^n v_j e_j\right) = \sum_{i=1}^n \sum_{j=1}^n u_i v_j \beta(e_i, e_j)$$

We can now define the linear function $\phi_i(u) = u_i = e_i^*(u)$ and $\tau_j'(v) = v_j = e_j^*(v)$. Then we have that

$$\beta(u, v) = \sum_{i=1}^n \phi_i(u) \left(\sum_{j=1}^n \beta(e_i, e_j) \tau_j'(v) \right) = \sum_{i=1}^n \phi_i(u) \tau_i(v)$$

□

Problem 3

Suppose $\beta: V \times V \rightarrow \mathbb{F}$ a bilinear form on V and also is a linear functional on $V \times V$. Prove that $\beta = 0$.

Proof. First we show that $\beta \in V_{alt}^{(2)}$. Take any $u \in V$, then we have

$$\begin{aligned} \beta((u, u) + (u, u)) &= 2\beta(u, u) \\ \beta(2u, 2u) &= 4\beta(u, u) \end{aligned}$$

this shows that $\beta(u, u) = 0$ for all u . Next, we show the off-diagonal terms are 0: first,

$$\beta(u, w) = \sum_{i=1}^n \sum_{j=1}^n u_i w_j \beta(e_i, e_j) \quad \text{bilinearity}$$

at the same time,

$$\begin{aligned} \beta(u, w) &= \beta \left(\sum_{i=1}^n u_i e_i, \sum_{j=1}^n w_j e_j \right) \\ &= \beta \left(\sum_{i=1}^n (u_i e_i, w_i e_i) \right) \\ &= \sum_{i=1}^n \beta(u_i e_i, w_i e_i) \quad \text{linearity on } V \times V \\ &= \sum_{i=1}^n u_i w_i \beta(e_i, e_i) \end{aligned}$$

This shows that all off-diagonal terms are 0, i.e., $\beta(e_i, e_j) = 0$ for all $i \neq j$. Therefore, we have $\beta = 0$. $\square$

Problem 6

Prove or give a counterexample: If ρ is a symmetric bilinear form on V , then

$$\{v \in V : \rho(v, v) = 0\}$$

is a subspace of V .

Proof. Consider $V = \mathbb{R}^2$ and $\rho(x, y) = x_1 y_1 - x_2 y_2$. Let $x = (1, 1)$, $y = (-1, 1)$, then we have that $\rho(x, x) = 1 - 1 = 0$, $\rho(y, y) = 1 - 1 = 0$, but $\rho(x + y, x + y) = 0 - 4 = -4 \neq 0$. $\square$

Problem 8

Find formulas for $\dim V_{sym}^{(2)}$ and $\dim V_{alt}^{(2)}$ in terms of $\dim V$.

Proof. Let $\dim V = n$. For $\beta \in V_{sym}^{(2)}$, consider $\mathcal{M}(\beta)$. Its diagonal entries can be chosen arbitrarily. For off-diagonal entries, only half of them can be chosen arbitrarily, therefore the dimension is

$$\frac{n(n-1)}{2} + n = \frac{n(n+1)}{2}$$

For $\beta \in V_{alt}^{(2)}$, consider $\mathcal{M}(\beta)$. The diagonal entries are all 0 and only half of the off-diagonal entries can be chosen arbitrarily. Therefore, the dimension is $\frac{n(n-1)}{2}$. $\square$

9B: Alternating Multilinear Forms

Definition 23 (V^m). For m a positive integer, define V^m by

$$V^m = \underbrace{V \times \cdots \times V}_{m \text{ times}}$$

Definition 24 (m -linear form, $V^{(m)}$, multilinear form). Below we introduce the definitions.

- For m a positive integer, an **m -linear form** on V is a function $\beta: V^m \rightarrow \mathbb{F}$ that is linear in each slot when the other slots are held fixed. This means that for each $k \in \{1, \dots, m\}$ and all $u_1, \dots, u_m \in V$, the function

$$v \mapsto \beta(u_1, \dots, u_{k-1}, v, u_{k+1}, \dots, u_m)$$

is a linear map from V to $\mathbb{F}$.

- The set of m -linear forms on V is denoted by $V^{(m)}$.
- A function β is called a **multilinear form** on V if it is an m -linear form on V for some positive integer m .

Remark 25. A 1-linear form on V is a linear functional on V . A 2-linear form on V is a bilinear form on V . $V^{(m)}$ is a vector space.

Example (m -linear forms). Suppose $\alpha, \beta \in V^{(2)}$. Define a function $\beta: V^4 \rightarrow \mathbb{F}$ by

$$\beta(v_1, v_2, v_3, v_4) = \alpha(v_1, v_2)\beta(v_3, v_4)$$

Then $\beta \in V^{(4)}$.

Example (m -linear forms). Define $\beta: (\mathcal{L}(V))^m \rightarrow \mathbb{F}$ by

$$\beta(T_1, \dots, T_m) = \text{tr}(T_1 \cdots T_m)$$

Then β is an m -linear form on $\mathcal{L}(V)$.

Definition 26 (alternating forms, $V_{\text{alt}}^{(m)}$). Suppose m is a positive integer.

- An m -linear form α on V is called **alternating** if $\alpha(v_1, \dots, v_m) = 0$ whenever $v_1, \dots, v_m$ is a list of vectors in V with $v_j = v_k$ for some two distinct values of j and k in $\{1, \dots, m\}$.
- $V_{\text{alt}}^{(m)} = \{\alpha \in V^{(m)} : \alpha \text{ is an alternating } m\text{-linear form on } V\}$.

Corollary 27. Suppose m is a positive integer and α is an alternating m -linear form on V . If $v_1, \dots, v_m$ is a linearly dependent list in V , then

$$\alpha(v_1, \dots, v_m) = 0$$

Corollary 28. Suppose $m > \dim V$. Then 0 is the only alternating m -linear form on V .

Theorem 29 (swapping input vectors in an alternating multilinear form). Suppose m is a positive integer, α is an alternating m -linear form on V , and $v_1, \dots, v_m$ is a list of vectors in V . Then swapping the vectors in any two slots of $\alpha(v_1, \dots, v_m)$ changes the value of α by a factor of -1 .

Remark 30. An odd number of swaps cause the value of α to change by a factor of -1 and it won't change with an even number of swaps.

Definition 31 (permutation, perm m). Suppose m is a positive integer.

- A **permutation** of $(1, \dots, m)$ is a list $(j_1, \dots, j_m)$ that contains each of the number $1, \dots, m$ exactly once.
- The set of permutations of $(1, \dots, m)$ is denoted by $\text{perm } m$.

Definition 32 (sign of a permutation). The **sign** of a permutation $(j_1, \dots, j_m)$ is defined by

$$\text{sign}(j_1, \dots, j_m) = (-1)^N$$

where N is the number of pairs of integers (k, l) with $1 \leq k < l \leq m$ such that k appears after l in the list $(j_1, \dots, j_m)$.

Lemma 33. Swapping two entries in a permutation multiplies the sign of the permutation by -1 .

Lemma 34 (permutation and alternating multilinear form). Suppose m is a positive integer and $\alpha \in V_{\text{alt}}^{(m)}$. Then

$$\alpha(v_{j_1}, \dots, v_{j_m}) = (\text{sign}(j_1, \dots, j_m)) \alpha(v_1, \dots, v_m)$$

for every list $v_1, \dots, v_m$ of vectors in V and all $(j_1, \dots, j_m) \in \text{perm } m$.

Theorem 35. Let $n = \dim V$. Suppose $e_1, \dots, e_n$ is a basis of V and $v_1, \dots, v_n \in V$. For each $k \in \{1, \dots, n\}$, let $b_{1,k}, \dots, b_{n,k} \in \mathbb{F}$ be such that

$$v_k = \sum_{j=1}^n b_{j,k} e_j$$

Then

$$\alpha(v_1, \dots, v_n) = \alpha(e_1, \dots, e_n) \sum_{(j_1, \dots, j_n) \in \text{perm } n} (\text{sign}(j_1, \dots, j_n)) b_{j_1,1} \cdots b_{j_n,n}$$

for every alternating n -linear form α on V .

Theorem 36. The vector space $V_{\text{alt}}^{(\dim V)}$ has dimension one.

Corollary 37. Let $n = \dim V$. Suppose α is a nonzero alternating n -linear form on V and $e_1, \dots, e_n$ is a list of vectors in V . Then

$$\alpha(e_1, \dots, e_n) \neq 0$$

if and only if $e_1, \dots, e_n$ is linearly independent.

Problem 1

Suppose m is a positive integer. Show that $\dim V^{(m)} = (\dim V)^m$.

Proof. Let $\dim V = n$ with basis $e_1, \dots, e_n$. The basis vector for $V^{(m)}$ can be formed via taking all possible m -tuples $b_{j_1}, \dots, b_{j_m}$ where b_{j_i} is a component of the basis. There are n choices over m positions, so we have that $\dim V^{(m)} = (\dim V)^m$. $\square$

Problem 3

Suppose m is a positive integer and α is an m -linear form on V such that $\alpha(v_1, \dots, v_m) = 0$ whenever $v_1, \dots, v_m$ is a list of vectors in V with $v_j = v_{j+1}$ for some $j \in \{1, \dots, m-1\}$. Prove that α is an alternating m -linear form on V .

Proof. Note that if the list $v_1, \dots, v_n$ comes with consecutive identical numbers, then by definition the output becomes 0. To prove α to be an alternating m -linear form, consider $v_i = v_k$ for $i+1 < k$. Note that then we can now just swap and get the same result:

$$\alpha(v_1, \dots, v_i, v_{i+1}, \dots, v_k, \dots, v_n) = -\alpha(v_1, \dots, v_i, v_k, \dots, v_{i+1}, \dots, v_n) = 0$$

$\square$

Problem 5

Suppose m is a positive integer and β is an m -linear form on V . Define an m -linear form α by

$$\alpha(v_1, \dots, v_m) = \sum_{(j_1, \dots, j_m) \in \text{perm } m} (\text{sign}(j_1, \dots, j_m) \beta(v_{j_1}, \dots, v_{j_m}))$$

for $v_1, \dots, v_m \in V$. Explain why $\alpha \in V_{alt}^{(m)}$.

Proof. If there are two repeating vectors, let's say $v_p = v_q$, then we know that

$$\beta(v_1, \dots, v_p, \dots, v_q, \dots, v_m) = \beta(v_1, \dots, v_q, \dots, v_p, \dots, v_m)$$

However, through swapping, the coefficient differs by (-1) , so we have

$$\begin{aligned} & \text{sign}(1, \dots, p, \dots, q, \dots, m) \beta(v_1, \dots, v_p, \dots, v_q, \dots, v_m) \\ &= -\text{sign}(1, \dots, q, \dots, p, \dots, m) \beta(v_1, \dots, v_q, \dots, v_p, \dots, v_m) \end{aligned}$$

This basically shows the main idea of the proof. To make this more rigorous, we claim that for each permutation $\sigma \in \text{perm } m$, there is a corresponding permutation $\sigma_{pq} \in \text{perm } m$ such that keeps everything unchanged while only swapping the position of p and q . This means that for each permutation, there is a corresponding “cancelling” pair permutation. Since we are summing all permutations, the result is finally 0, finishing the proof. $\square$

9C: Determinants

Definition 38 (α_T). Suppose that m is a positive integer and $T \in \mathcal{L}(V)$. For $\alpha \in V_{\text{alt}}^{(m)}$, define $\alpha_T \in V_{\text{alt}}^{(m)}$ by

$$\alpha_T(v_1, \dots, v_m) = \alpha(Tv_1, \dots, Tv_m)$$

for each list $v_1, \dots, v_m$ of vectors in V .

Remark 39. The function $\alpha \mapsto \alpha_T$ is a linear map of $V_{\text{alt}}^{(m)}$ to itself. We know that $\dim V_{\text{alt}}^{(\dim V)} = 1$, so the linear map is simply a multiplication by some unique scalar. For the linear map $\alpha \mapsto \alpha_T$, we now define $\det T$ to be that scalar.

Definition 40 (determinant of an operator, $\det T$). Suppose $T \in \mathcal{L}(V)$. The **determinant** of T , denoted by $\det T$, is defined to be the unique number in $\mathbb{F}$ such that

$$\alpha_T = (\det T)\alpha$$

for all $\alpha \in V_{\text{alt}}^{(\dim V)}$.

Remark 41. Let $n = \dim V$.

- If I is the identity operator on V , then $\alpha_I = \alpha$ for all $\alpha \in V_{\text{alt}}^{(n)}$. This gives that $\det I = 1$.
- More generally, if $\lambda \in \mathbb{F}$, then $\alpha_{\lambda I} = \lambda^n \alpha$ for all $\alpha \in V_{\text{alt}}^{(n)}$. Thus $\det(\lambda I) = \lambda^n$.
- Since $\alpha_{\lambda T} = \lambda^n \alpha_T = \lambda^n (\det T)\alpha$ for all $\alpha \in V_{\text{alt}}^{(n)}$, $\det(\lambda T) = \lambda^n \det T$.
- Suppose $T \in \mathcal{L}(V)$ and there is a basis $e_1, \dots, e_n$ of V consisting of eigenvectors of T , with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. If $\alpha \in V_{\text{alt}}^{(n)}$, then

$$\alpha_T(e_1, \dots, e_n) = \alpha(\lambda_1 e_1, \dots, \lambda_n e_n) = (\lambda_1 \cdots \lambda_n) \alpha(e_1, \dots, e_n)$$

If $\alpha \neq 0$, then $\alpha(e_1, \dots, e_n) \neq 0$. Thus this means that

$$\det T = \lambda_1 \cdots \lambda_n$$

Definition 42 (determinant of a matrix, $\det A$). Suppose that n is a positive integer and A is an n -by- n matrix square matrix with entries in $\mathbb{F}$. Let $T \in \mathcal{L}(\mathbb{F}^n)$ be the operator whose matrix with respect to the standard basis of $\mathbb{F}^n$ equals A . The **determinant** of A , denoted by $\det A$, is defined by $\det A = \det T$.

Theorem 43 (determinant is an alternating multilinear form). Suppose that n is a positive integer. The map that takes a list $v_1, \dots, v_n$ of vectors in $\mathbb{F}^n$ to $\det(v_1 \cdots v_n)$ is an alternating n -linear form on $\mathbb{F}^n$.

Corollary 44 (formula for determinants of a matrix). *Suppose that n is a positive integer and A is an n -by- n matrix square matrix. Then*

$$\det A = \sum_{(j_1, \dots, j_n) \in \text{perm } n} (\text{sign}(j_1, \dots, j_n)) A_{j_1, 1} \cdots A_{j_n, n}$$

Remark 45. *The sum in the formula above contains $n!$ terms.*

Corollary 46 (determinant of upper-triangular matrix). *Suppose that A is an upper-triangular matrix with $\lambda_1, \dots, \lambda_n$ on the diagonal. Then $\det A = \lambda_1 \cdots \lambda_n$.*

Theorem 47 (determinant is multiplicative). *We have the following result:*

- (a) *Suppose $S, T \in \mathcal{L}(V)$. Then $\det(ST) = \det(S) \det(T)$.*
- (b) *Suppose A and B are square matrices of the same size. Then*

$$\det(AB) = \det(A) \det(B)$$

Corollary 48. *An operator $T \in \mathcal{L}(V)$ is **invertible** if and only if $\det T \neq 0$. Furthermore, if T is invertible, then $\det(T^{-1}) = \frac{1}{\det T}$.*

Corollary 49. *Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$. Then λ is an eigenvalue of T if and only if $\det(\lambda I - T) = 0$.*

Corollary 50. *Suppose $T \in \mathcal{L}(V)$ and $S: W \rightarrow V$ is an invertible linear map. Then*

$$\det(S^{-1}TS) = \det T$$

Corollary 51. *Suppose $T \in \mathcal{L}(V)$ and $e_1, \dots, e_n$ is a basis of V . Then*

$$\det T = \det \mathcal{M}(T, (e_1, \dots, e_n))$$

Corollary 52. *Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then $\det T$ equals the product of the eigenvalues of T , with each eigenvalue included as many times as its multiplicity.*

Corollary 53 (determinant of transpose, dual, or adjoint). *We have the following result:*

- (a) *Suppose A is a square matrix. Then $\det A^\top = \det A$.*
- (b) *Suppose $T \in \mathcal{L}(V)$. Then $\det T' = \det T$.*
- (c) *Suppose V is an inner product space and $T \in \mathcal{L}(V)$. Then*

$$\det(T^*) = \overline{\det T}$$

Corollary 54. *Helpful results in evaluating the determinants:*

- (a) *If either two columns or two rows of a square matrix are equal, then the determinant of the matrix equals 0.*

- (b) Suppose A is a square matrix and B is the matrix obtained from A by swapping either two columns or two rows. Then $\det A = -\det B$.
- (c) If one column or one row of a square matrix is multiplied by a scalar, then the value of the determinant is multiplied by the same scalar.
- (d) If a scalar multiple of one column of a square matrix is added to another column, then the value of the determinant is unchanged.
- (e) If a scalar multiple of one row of a square matrix is added to another row, then the value of the determinant is unchanged.

Corollary 55. Suppose V is an inner product space and $S \in \mathcal{L}(V)$ an unitary operator. Then $|\det S| = 1$.

Corollary 56. Suppose V is an inner product space and $T \in \mathcal{L}(V)$ is a positive operator. Then $\det T \geq 0$.

Corollary 57. Suppose V is an inner product space and $T \in \mathcal{L}(V)$. Then

$$|\det T| = \sqrt{\det(T^*T)} = \text{product of singular values of } T$$

Lemma 58. Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Let $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T , and let $d_1, \dots, d_m$ denote their multiplicities. Then

$$\det(zI - T) = (z - \lambda_1)^{d_1} \cdots (z - \lambda_m)^{d_m}$$

Definition 59 (characteristic polynomial). Suppose $T \in \mathcal{L}(V)$. The polynomial defined by

$$z \mapsto \det(zI - T)$$

is called the **characteristic polynomial** of T .

Theorem 60 (Cayley-Hamilton theorem). Suppose $T \in \mathcal{L}(V)$ and q is the characteristic polynomial of T . Then $q(T) = 0$.

Corollary 61 (characteristic polynomial, trace, and determinant). Suppose $T \in \mathcal{L}(V)$. Let $n = \dim V$. Then the characteristic polynomial of T can be written as

$$z^n - (\operatorname{tr} T)z^{n-1} + \cdots + (-1)^n(\det T)$$

Theorem 62 (Hadamard's inequality). Suppose A is an n -by- n matrix. Let $v_1, \dots, v_n$ denote the columns of A . Then

$$|\det A| \leq \prod_{k=1}^n \|v_k\|$$

Theorem 63 (determinant of Vandermonde matrix). Suppose $n > 1$ and $\beta_1, \dots, \beta_n \in \mathbb{F}$. Then

$$\det \begin{pmatrix} 1 & \beta_1 & \beta_1^2 & \cdots & \beta_1^{n-1} \\ 1 & \beta_2 & \beta_2^2 & \cdots & \beta_2^{n-1} \\ 1 & \beta_3 & \beta_3^2 & \cdots & \beta_3^{n-1} \\ & & & \ddots & \\ 1 & \beta_n & \beta_n^2 & \cdots & \beta_n^{n-1} \end{pmatrix} = \prod_{1 \leq j < k \leq n} (\beta_k - \beta_j).$$

Problem 1

Prove or give a counterexample: $S, T \in \mathcal{L}(V) \Rightarrow \det(S + T) = \det S + \det T$.

Proof. Consider $\mathbb{R}^2$, and that

$$\mathcal{M}(S) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathcal{M}(T) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

Then clearly $\det S = 1$ and $\det T = 2$. However, we have that

$$\mathcal{M}(S) + \mathcal{M}(T) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

which has $\det(S + T) = 6 \neq \det S + \det T$. $\square$

Problem 3

Suppose $T \in \mathcal{L}(V)$ is nilpotent. Prove that $\det(I + T) = 1$.

Proof. We know that 0 is the only eigenvalue of T and thus the only eigenvalue of $I + T$ is 1. Hence $\det(I + T) = 1$. $\square$

Problem 5

Suppose A is a block triangular matrix

$$A = \begin{pmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_m \end{pmatrix}$$

where each A_k along the diagonal is a square matrix. Prove that

$$\det A = (\det A_1) \cdots (\det A_m)$$

Proof. One can show that $\det A = (\det A_1)(\det A_2)$ through direct proof. We use induction on m for solving this problem. The base case is trivial. We assume the statement holds for $m \leq k - 1$. Then for $m = k$, we can partition the matrix into two blocks:

$$\begin{bmatrix} A' & * \\ 0 & A_k \end{bmatrix}$$

where

$$A' = \begin{bmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_{k-1} \end{bmatrix}$$

Then we have that $\det A = (\det A')(\det A_k) = (\det A_1) \cdots (\det A_k)$, finishing the proof. $\square$

Problem 9

Suppose that V is a real vector space of even dimension, $T \in \mathcal{L}(V)$, and $\det T < 0$. Prove that T has at least two distinct eigenvalues.

Proof. Since $\det(T) \neq 0$, T is invertible and thus have n distinct eigenvalues with $n \geq 2$. Another argument could be that for real cases, there have to be at least one negative and one positive eigenvalue to make the determinant negative; for complex cases, there must be two conjugate pairs. $\square$

Problem 11

Prove or give a counter example: If $\mathbb{F} = \mathbb{R}, T \in \mathcal{L}(V)$, and $\det T > 0$, then T has a square root.

Proof. Not necessarily. Consider an operator in $\mathbb{R}^2$ with two negative eigenvalues which is clearly non-positive and therefore does not have a square root. $\square$

Problem 16

Suppose $T \in \mathcal{L}(V)$. Define $g: \mathbb{F} \rightarrow \mathbb{F}$ by $g(x) = \det(I + xT)$. Show that $g'(0) = \text{tr } T$.

Proof.

$$\begin{aligned} g'(x) &= \frac{d}{dx} \det(I + xT) \\ &= \frac{d}{dx} \prod_{i=1}^n (1 + x\lambda_i) \\ &= \sum_{i=1}^n \left(\lambda_i \prod_{j \neq i} (1 + x\lambda_j) \right) \end{aligned}$$

Substitute $x = 0$ yields that

$$g'(0) = \sum_{i=1}^n \lambda_i = \text{tr } T$$

$\square$

Problem 19

Suppose V is an inner product space, $e_1, \dots, e_n$ is an orthonormal basis of V , and $T \in \mathcal{L}(V)$ is a positive operator.

- (a) Prove that $\det T \leq \prod_{k=1}^n \langle Te_k, e_k \rangle$.
- (b) Prove that if T is invertible, then the inequality in (a) is an equality if and only if e_k is an eigenvector of T for each $k = 1, \dots, n$.

Proof. (a) The matrix representation of T wrt. $e_1, \dots, e_n$ is that $A_{ij} = \langle Te_i, e_j \rangle$. Hence the r.h.s of this inequality is simply the product of all the diagonal terms on the matrix of T . We prove this inequality through Cholesky factorization. Note that

$$A = LL^*$$

for lower-triangular matrix L , and thus we have

$$\det T = \det A = (\det L)^2 = \left(\prod_{k=1}^n l_{kk} \right)^2 = \prod_{k=1}^n L_{kk}^2$$

We note that

$$A_{kk} = L_{kk}^2 + \sum_{j=1}^{k-1} L_{kj}^2$$

and thus we have

$$\det A \leq \prod_{k=1}^n A_{kk} = \prod_{k=1}^n \langle Te_k, e_k \rangle$$

- (b) If e_k is an eigenvector of T , then $\langle Te_k, e_k \rangle = \lambda_k$, the k -th eigenvalue of T . Then we know that $\det T$ is the product of all eigenvalues.

Conversely, if (a) is an equality, then we know that L is a diagonal matrix and thus A is also a diagonal matrix. Then the orthonormal basis $e_1, \dots, e_n$ actually diagonalizes T and hence each of them is an eigenvector of T . $\square$

9D: Tensor Products

Definition 64 (bilinear functional on $V \times W$, the vector space $\mathcal{B}(V, W)$). A **bilinear functional** on $V \times W$ is a function $\beta: V \times W \rightarrow \mathbb{F}$ such that $v \mapsto \beta(v, w)$ is a linear functional on V for each $w \in W$ and $w \mapsto \beta(v, w)$ is a linear functional on W for each $v \in V$.

The vector space of bilinear functionals on $V \times W$ is denoted by $\mathcal{B}(V, W)$.

Remark 65. If $V = W$, then a bilinear functional on $V \times W$ is a bilinear form.

Corollary 66. $\dim \mathcal{B}(V, W) = (\dim V)(\dim W)$

Remark 67. We want a basis-free definition of the tensor product.

Definition 68 (tensor product, $V \otimes W, v \otimes w$). The **tensor product** $V \otimes W$ is defined to be $\mathcal{B}(V', W')$.

For $v \in V$ and $w \in W$, the **tensor product** $v \otimes w$ is the element of $V \otimes W$ defined by

$$(v \otimes w)(\varphi, \tau) = \varphi(v)\tau(w)$$

for all $(\varphi, \tau) \in V' \times W'$.

Corollary 69. $\dim(V \otimes W) = (\dim V)(\dim W)$.

Proposition 70 (bilinearity of tensor product). Suppose $v, v_1, v_2 \in V$ and $w, w_1, w_2 \in W$ and $\lambda \in \mathbb{F}$. Then

$$(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w \text{ and } v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2$$

and

$$\lambda(v \otimes w) = (\lambda v) \otimes w = v \otimes (\lambda w)$$

Theorem 71 (basis of $V \otimes W$). Suppose $e_1, \dots, e_m$ is a list of vectors in V and $f_1, \dots, f_n$ is a list of vectors in W .

(a) If $e_1, \dots, e_m$ and $f_1, \dots, f_n$ are both linearly independent list, then

$$\{e_j \otimes f_k\}_{j=1, \dots, m; k=1, \dots, n}$$

is a linearly independent list in $V \otimes W$.

(b) If $e_1, \dots, e_m$ is a basis of V and $f_1, \dots, f_n$ is a basis of W , then the list $\{e_j \otimes f_k\}_{j=1, \dots, m; k=1, \dots, n}$ is a basis of $V \otimes W$.

Definition 72 (bilinear map). A **bilinear map** from $V \times W$ to a vector space U is a function $\Gamma: V \times W \rightarrow U$ such that $v \mapsto \Gamma(v, w)$ is a linear map from V to U for each $w \in W$ and $w \mapsto \Gamma(v, w)$ is a linear map from W to U for each $v \in V$.

Lemma 73 (converting bilinear maps to linear maps). Suppose U is a vector space.

- (a) Suppose $\Gamma: V \times W \rightarrow U$ is a bilinear map. Then there exists a unique linear map $\tilde{\Gamma}: V \otimes W \rightarrow U$ such that

$$\tilde{\Gamma}(v \otimes w) = \Gamma(v, w)$$

for all $(v, w) \in V \times W$.

- (b) Conversely, suppose $T: V \otimes W \rightarrow U$ is a linear map. Then there exists a unique bilinear map $T^\#: V \times W \rightarrow U$ such that

$$T^\#(v, w) = T(v \otimes w)$$

for all $(v, w) \in V \times W$.

Theorem 74 (inner product on tensor product of two inner product spaces). Suppose V and W are inner product spaces. Then there is a unique inner product on $V \otimes W$ such that

$$\langle v \otimes w, u \otimes x \rangle = \langle v, u \rangle \langle w, x \rangle$$

for all $u, v \in V$ and $w, x \in W$.

Remark 75. We have that $\|v \otimes w\| = \|v\| \|w\|$.

Corollary 76. Suppose V and W are inner product spaces, and $e_1, \dots, e_m$ is an orthonormal basis of V and $f_1, \dots, f_n$ is an orthonormal basis of W . Then

$$\{e_j \otimes f_k\}_{j=1, \dots, m; k=1, \dots, n}$$

is an orthonormal basis of $V \otimes W$.

Definition 77. An **m -linear** functional on $V_1 \times \dots \times V_m$ is a function $\beta: V_1 \times \dots \times V_m \rightarrow \mathbb{F}$ that is a linear functional in each slot when the other slots are held fixed.

The vector space of m -linear functionals on $V_1 \times \dots \times V_m$ is denoted by $\mathcal{B}(V_1, \dots, V_m)$.

Corollary 78. $\dim \mathcal{B}(V_1, \dots, V_m) = (\dim V_1) \times \dots \times (\dim V_m)$

Definition 79 (tensor product). The tensor product $V_1 \otimes \dots \otimes V_m$ is defined to be $\mathcal{B}(V'_1, \dots, V'_m)$.

For $v_1 \in V_1, \dots, v_m \in V_m$, the **tensor product** $v_1 \otimes \dots \otimes v_m$ is the element of $V_1 \otimes \dots \otimes V_m$ defined by

$$(v_1 \otimes \dots \otimes v_m)(\varphi_1, \dots, \varphi_m) = \varphi_1(v_1) \dots \varphi_m(v_m)$$

for all $(\varphi_1, \dots, \varphi_m) \in V'_1 \times \dots \times V'_m$.

Corollary 80. Suppose $\dim V_k = n_k$ and $e_1^k, \dots, e_{n_k}^k$ is a basis of V_k for $k = 1, \dots, m$. Then

$$\{e_{j_1}^1 \otimes \dots \otimes e_{j_m}^m\}_{j_1=1, \dots, n_1; \dots; j_m=1, \dots, n_m}$$

is a basis of $V_1 \otimes \dots \otimes V_m$.

Definition 81 (m-linear map). *An m-linear map from $V_1 \times \cdots \times V_m$ to a vector space U is a function $\Gamma: V_1 \times \cdots \times V_m \rightarrow U$ that is a linear map in each slot when the other slots are held fixed.*

Theorem 82 (converting m-linear map to linear maps). *Suppose U is a vector space.*

- (a) *Suppose that $\Gamma: V_1 \times \cdots \times V_m \rightarrow U$ is an m-linear map. Then there exists a unique linear map $\tilde{\Gamma}: V_1 \otimes \cdots \otimes V_m \rightarrow U$ such that*

$$\tilde{\Gamma}(v_1 \otimes \cdots \otimes v_m) = \Gamma(v_1, \dots, v_m)$$

for all $(v_1, \dots, v_m) \in V_1 \times \cdots \times V_m$.

- (b) *Conversely, suppose $T: V_1 \otimes \cdots \otimes V_m \rightarrow U$ is a linear map. Then there exists a unique m-linear map $T^\#: V_1 \times \cdots \times V_m \rightarrow U$ such that*

$$T^\#(v_1, \dots, v_m) = T(v_1 \otimes \cdots \otimes v_m)$$

for all $(v_1, \dots, v_m) \in V_1 \times \cdots \times V_m$.

Problem 1

Suppose $v \in V$ and $w \in W$. Prove that $v \otimes w = 0$ if and only if $v = 0$ or $w = 0$.

Proof. By definition, we have for any $(\varphi, \tau) \in V' \times W'$,

$$(v \otimes w)(\varphi, \tau) = \varphi(v)\tau(w)$$

Then this means that $\varphi(v)\tau(w) = 0$ for arbitrary choice of φ, τ , meaning that either $v = 0$ or $w = 0$. $\square$

Problem 3

Suppose that $v_1, \dots, v_m$ is a linearly independent list in V . Suppose also that $w_1, \dots, w_m$ is a list in W such that

$$v_1 \otimes w_1 + \dots + v_m \otimes w_m = 0$$

Prove that $w_1 = \dots = w_m = 0$.

Proof. By the linear map lemma and the linear independence of $v_1, \dots, v_m$, there exists $\varphi_1, \dots, \varphi_m \in V'$ such that

$$\varphi_j(v_k) = \begin{cases} 1 & \text{if } j = k \\ 0 & \text{if } j \neq k \end{cases}$$

where $j, k \in \{1, \dots, m\}$. Applying such $\{\varphi_i\}_{i=1}^m$ to the list

$$\sum_{i=1}^m v_i \otimes w_i$$

and take $\tau \in W'$ to be the identity map yields that

$$w_1 = \dots = w_m = 0$$

$\square$

Problem 5

Suppose m and n are positive integers. For $v \in \mathbb{F}^m$ and $w \in \mathbb{F}^n$, identify $v \otimes w$ with an m -by- n matrix as in Example 9.76. With that identification, show that the set

$$\{v \otimes w : v \in \mathbb{F}^m \text{ and } w \in \mathbb{F}^n\}$$

is the set of m -by- n matrix matrices (with entries in $\mathbb{F}$) that have rank at most one.

Proof. If one examine the matrices with entries shown on the matrix, it's easy to tell that for row j and row k with $j \neq k$, one can get row k from row j through multiplying v_k/v_j . The same applies to arbitrary pairs of columns. Thus the matrix has at most rank one. $\square$

Problem 8

Suppose $v_1, \dots, v_m \in V$ and $w_1, \dots, w_m \in W$ are such that

$$v_1 \otimes w_1 + \dots + v_m \otimes w_m = 0$$

Suppose that U is a vector space and $\Gamma: V \times W \rightarrow U$ is a bilinear map. Show that

$$\Gamma(v_1, w_1) + \dots + \Gamma(v_m, w_m) = 0$$

Proof. We know there exists a unique “converting” linear map $\tilde{\Gamma}$ such that

$$\Gamma(v \otimes w) = \Gamma(v, w)$$

Hence, applying this gives that

$$\begin{aligned} \sum_{i=1}^m \Gamma(v_i, w_i) &= \sum_{i=1}^m \tilde{\Gamma}(v_i \otimes w_i) \\ &= \tilde{\Gamma} \left(\sum_{i=1}^m v_i \otimes w_i \right) \\ &= 0 \end{aligned}$$

$\square$